

$K = k_1 k_2$



$I = E\,du^2 + 2F\,du\,dv + G\,dv^2$

$II = e\,du^2 + 2f\,du\,dv + g\,dv^2$

POSTGRADUATE LECTURE NOTES

DIFFERENTIAL GEOMETRY

Geometry of Curves and Surfaces

$H = \frac{1}{2}(k_1 + k_2)$



Curves • Surfaces • Curvature • Geodesics

$T' = \kappa N$

$N' = -\kappa T + \tau B$

$B' = -\tau N$

PREPARED AND WRITTEN BY

Dr. AIJAZ

Assistant Professor of Mathematics

JK Higher Education Department



$k_n = \frac{II(v,v)}{I(v,v)}$

Study Materials for UG Mathematics, PG Mathematics and Entrance Examinations

NET–JRF / SET / GATE / PhD / Others

$T^i_{ij} = \frac{1}{2} g^{ik} (\partial_j g_{jl} + \partial_j g_{jl} - \partial_l g_{ij})$

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CHAPTER
1

THEORY OF CURVES

This chapter develops the fundamental theory of plane and space curves, beginning with their parametrization, regularity, and arc length. It introduces curvature, torsion, and the Frenet–Serret frame to describe the local geometric behaviour of curves. The fundamental theorem of space curves, helices, involutes, and evolutes are studied through their geometric properties. The chapter concludes with global properties of curves, including the isoperimetric inequality and the Four Vertex Theorem.

1.1 Differentiable Curves

The study of differential geometry begins with the investigation of curves and surfaces. In elementary geometry, a curve is often described by an equation such as $y = x^2$ or $x^2 + y^2 = a^2$. In differential geometry, however, it is more convenient to describe a curve by specifying the position of a moving point as a function of a real variable. This approach allows us to study not only the shape of a curve but also its tangent vectors, arc length, curvature and other geometric properties.

Definition 1.1.1 (Parametrized Curve)**.** Let I be an interval in $\mathbb{R}$. A parametrized curve in $\mathbb{R}^n$ is a continuous mapping $\alpha : I \to \mathbb{R}^n$ given by

$$\alpha(t) = (x_1(t), x_2(t), \ldots, x_n(t)),$$

where $x_1, x_2, \ldots, x_n$ are real-valued functions defined on I. The variable t is called the parameter of the curve, and the set $\alpha(I) = \{\alpha(t) : t \in I\}$ is called its trace or image.

A parametrized curve describes how a point moves through Euclidean space as the parameter varies. Two different parametrizations may have the same trace, although the corresponding motions may take place at different speeds or in different directions.

Definition 1.1.2 (Differentiable Curve)**.** A parametrized curve $\alpha : I \to \mathbb{R}^n$ is called differentiable if each of its component functions is differentiable. It is said to be of class C^k if its component functions possess continuous derivatives up to order k. If the component functions have continuous derivatives of every order, the curve is called smooth or of class C^∞.

Throughout this chapter, unless otherwise stated, curves will be assumed to be smooth on their parameter intervals.

Notation 1.1.3. If $\alpha(t) = (x_1(t), x_2(t), \ldots, x_n(t))$ is a differentiable curve, its derivative is defined

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componentwise by

$$\alpha'(t) = \frac{d\alpha}{dt} = (x_1'(t), x_2'(t), \ldots, x_n'(t)).$$

The vector $\alpha'(t)$ is called the velocity vector or tangent vector of the parametrized curve at t. Its Euclidean norm

$$\|\alpha'(t)\| = \sqrt{(x_1'(t))^2 + \cdots + (x_n'(t))^2}$$

is called the speed of the curve.

When $n = 2$, the curve is called a plane curve, and when $n = 3$, it is called a space curve. For a plane curve $\alpha(t) = (x(t), y(t))$, its speed is $\sqrt{(x'(t))^2 + (y'(t))^2}$, whereas for a space curve $\alpha(t) = (x(t), y(t), z(t))$, its speed is $\sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}$.

Example 1.1.4 (A Straight Line)**.** Let $P = (a_1, a_2, a_3)$ and $v = (b_1, b_2, b_3)$, where $v \neq 0$. Describe the curve $\alpha : \mathbb{R} \to \mathbb{R}^3$ defined by

$$\alpha(t) = P + tv = (a_1 + b_1 t, a_2 + b_2 t, a_3 + b_3 t).$$

Solution. The trace of α is the straight line passing through P in the direction of v. Differentiating gives $\alpha'(t) = v$, which is constant and nonzero. Thus, the velocity vector has constant magnitude and direction, and the curve is regular for every $t \in \mathbb{R}$. ■

Example 1.1.5 (A Circle)**.** Consider the curve $\alpha(t) = (a\cos t, a\sin t)$, where $a > 0$ and $0 \leq t \leq 2\pi$. Identify its trace and determine its speed.

Solution. Writing $\alpha(t) = (x(t), y(t))$, we have $x(t)^2 + y(t)^2 = a^2\cos^2 t + a^2\sin^2 t = a^2$. Thus, the trace is the circle of radius a centered at the origin. Differentiating, we obtain $\alpha'(t) = (-a\sin t, a\cos t)$, and hence

$$\|\alpha'(t)\| = \sqrt{a^2\sin^2 t + a^2\cos^2 t} = a.$$

Therefore, the curve describes the circle at constant speed a. ■

1.2 Regular Points and Regular Curves

The tangent vector plays a central role in differential geometry. In order to assign a well-defined tangent direction to a parametrized curve, we require its derivative to be nonzero.

Definition 1.2.1 (Regular Point)**.** A parameter value $t_0 \in I$ is called a regular parameter value of a differentiable curve $\alpha : I \to \mathbb{R}^n$ if $\alpha'(t_0) \neq \mathbf{0}$. Otherwise, t_0 is called a singular parameter value.

Definition 1.2.2 (Regular Curve)**.** A differentiable curve $\alpha : I \to \mathbb{R}^n$ is called regular if $\alpha'(t) \neq \mathbf{0}$ for every $t \in I$.

At a regular parameter value t_0, the tangent line to the curve passes through $\alpha(t_0)$ in the direction of $\alpha'(t_0)$. Its equation is

$$X = \alpha(t_0) + \lambda\alpha'(t_0), \qquad \lambda \in \mathbb{R}.$$

Thus, regularity guarantees the existence of a nonzero tangent direction determined by the parametrization. If the parameter represents time, regularity means that the moving point never comes to an

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instantaneous rest.

Example 1.2.3 (A Regular Parabola)**.** Determine whether the curve $\alpha(t) = (t, t^2)$, where $t \in \mathbb{R}$, is regular.

Solution. Differentiating gives $\alpha'(t) = (1, 2t)$. Since the first component is always 1, the derivative can never be the zero vector. Hence α is regular for every $t \in \mathbb{R}$. ■

Example 1.2.4 (A Singular Parametrization of a Straight Line)**.** Examine the regularity of the curve $\alpha(t) = (t^3, 0)$, where $t \in \mathbb{R}$.

Solution. Differentiating gives $\alpha'(t) = (3t^2, 0)$. Thus, $\alpha'(0) = (0, 0)$, while $\alpha'(t) \neq (0, 0)$ whenever $t \neq 0$. Hence the parametrization is singular at $t = 0$ and regular elsewhere. Nevertheless, its trace is the x-axis, which also admits the regular parametrization $\beta(u) = (u, 0)$. Therefore, a singular parameter value does not necessarily correspond to a geometrical cusp. ■

1.3 Parameterization and Reparameterization

The same curve can be represented using different parameters. A change of parameter may alter the speed or direction of traversal without changing the geometric trace.

For instance, the unit circle can be parametrized by $\alpha(t) = (\cos t, \sin t)$ and $\beta(u) = (\cos 2u, \sin 2u)$. Over suitable parameter intervals, both parametrizations describe the same circle, but the second traverses it twice as fast.

Definition 1.3.1 (Reparameterization)**.** Let $\alpha : I \to \mathbb{R}^n$ be a smooth curve, and let $\phi : J \to I$ be a smooth diffeomorphism between intervals. Thus, ϕ is bijective, its inverse is smooth, and $\phi'(u) \neq 0$ for every $u \in J$. The curve $\beta : J \to \mathbb{R}^n$ defined by

$$\beta(u) = \alpha(\phi(u))$$

is called a reparameterization of α.

The requirement that ϕ be a diffeomorphism ensures that the change of parameter is reversible and does not introduce singularities into a regular curve. If $\phi'(u) > 0$, the reparameterization preserves orientation, while if $\phi'(u) < 0$, it reverses orientation.

Proposition 1.3.2 (Regularity under Reparameterization)**.** *A smooth reparameterization of a regular curve is again a regular curve.*

Proof. Let $\alpha : I \to \mathbb{R}^n$ be regular, and suppose that $\beta(u) = \alpha(\phi(u))$, where $\phi : J \to I$ is a smooth diffeomorphism. Since β is the composition of two differentiable mappings, the chain rule gives

$$\beta'(u) = \frac{d}{du}\alpha(\phi(u)) = \alpha'(\phi(u))\phi'(u).$$

The derivative $\alpha'(\phi(u))$ measures how the original curve changes with respect to its parameter, while $\phi'(u)$ measures how that parameter changes with respect to the new parameter.

Since α is regular, $\alpha'(\phi(u)) \neq \mathbf{0}$. Moreover, $\phi'(u) \neq 0$ because ϕ is a diffeomorphism. Consequently, $\beta'(u)$ is a nonzero scalar multiple of a nonzero vector, and hence $\beta'(u) \neq \mathbf{0}$ for every $u \in J$. Thus, β is regular. ■

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Taking norms in the relation $\beta'(u) = \alpha'(\phi(u))\phi'(u)$ gives

$$\|\beta'(u)\| = \|\alpha'(\phi(u))\|\,|\phi'(u)|.$$

Therefore, a reparameterization changes the speed by the factor $|\phi'(u)|$, although the geometric trace remains unchanged.

Example 1.3.3 (Reparameterization of a Circle)**.** Let $\alpha(t) = (a\cos t, a\sin t)$, where $0 < t < 2\pi$, and put $t = 2u$, where $0 < u < \pi$. Find the new parametrization and compare its speed with that of α.

Solution. The change of parameter is $\phi(u) = 2u$, which is a diffeomorphism from $(0,\pi)$ onto $(0,2\pi)$. The new curve is

$$\beta(u) = \alpha(2u) = (a\cos 2u, a\sin 2u).$$

Differentiating gives $\beta'(u) = (-2a\sin 2u, 2a\cos 2u)$, so

$$\|\beta'(u)\| = \sqrt{4a^2\sin^2 2u + 4a^2\cos^2 2u} = 2a.$$

Since $\|\alpha'(t)\| = a$, the new parametrization traverses the circle at twice the original speed. ■

1.4 Arc Length and Unit-Speed Curves

The length of a curve is one of its most basic geometric properties. Although the velocity of a curve depends on its parameter, the total distance travelled along a fixed portion of the curve does not depend on the particular regular parametrization used.

Definition 1.4.1 (Arc Length)**.** Let $\alpha : [a,b] \to \mathbb{R}^n$ be a continuously differentiable curve. The arc length of α from $t = a$ to $t = b$ is defined by

$$L(\alpha) = \int_a^b \|\alpha'(t)\|\,dt.$$

In particular, if $\alpha(t) = (x(t), y(t), z(t))$ is a space curve, then

$$L(\alpha) = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\,dt.$$

The formula for arc length may be understood geometrically by dividing the parameter interval into sufficiently small subintervals. Over a small interval of length Δt, the corresponding portion of the curve is approximately a straight line segment of length $\|\alpha'(t)\|\,\Delta t$. Adding these lengths and passing to the limit gives the arc-length integral.

Definition 1.4.2 (Arc-Length Function)**.** Let $\alpha : I \to \mathbb{R}^n$ be a regular curve and fix $t_0 \in I$. The arc-length function measured from t_0 is defined by

$$s(t) = \int_{t_0}^t \|\alpha'(u)\|\,du.$$

By the fundamental theorem of calculus,

$$\frac{ds}{dt} = \|\alpha'(t)\|.$$

Thus, the derivative of arc length with respect to the parameter is exactly the speed of the curve.

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Definition 1.4.3 (Unit-Speed Curve)**.** A regular curve $\alpha : I \to \mathbb{R}^n$ is called a unit-speed curve if $\|\alpha'(t)\| = 1$ for every $t \in I$. When the parameter itself is arc length, it is customary to denote it by s, so that

$$\|\alpha'(s)\| = 1.$$

Such a curve is also said to be parametrized by arc length.

For a unit-speed curve, the length of the portion corresponding to $s \in [s_1, s_2]$, where $s_1 < s_2$, is

$$L = \int_{s_1}^{s_2} \|\alpha'(s)\| \, ds = \int_{s_1}^{s_2} 1 \, ds = s_2 - s_1.$$

This is the principal advantage of using arc length as a parameter.

Example 1.4.4 (Arc Length of a Circular Helix)**.** (See Figure 1 in the Atlas.) Calculate the length of one turn of the circular helix $\alpha(t) = (a\cos t, a\sin t, ct)$, where $0 \le t \le 2\pi$ and $a, c > 0$.

Solution. Differentiating each component with respect to t, we obtain

$$\alpha'(t) = (-a\sin t, a\cos t, c).$$

Therefore, the speed is

$$\begin{aligned}\|\alpha'(t)\| &= \sqrt{a^2\sin^2 t + a^2\cos^2 t + c^2} \\ &= \sqrt{a^2(\sin^2 t + \cos^2 t) + c^2} \\ &= \sqrt{a^2 + c^2}.\end{aligned}$$

Since the speed is constant, the arc length is

$$L = \int_0^{2\pi} \sqrt{a^2 + c^2}\, dt = 2\pi\sqrt{a^2 + c^2}.$$

Thus, the length of one complete turn of the helix is $2\pi\sqrt{a^2 + c^2}$. ■

The result also has a geometric interpretation. A complete turn of the helix winds once around a cylinder of radius a while rising through a vertical distance $2\pi c$. If the cylinder is unrolled onto a plane, the helix becomes a straight segment of length $\sqrt{(2\pi a)^2 + (2\pi c)^2} = 2\pi\sqrt{a^2 + c^2}$.

Example 1.4.5 (Arc Length of a Semicubical Parabola)**.** (See Figure 2 in the Atlas.) Find the length of the curve $\alpha(t) = (t^2, \frac{2}{3}t^3)$ from $t = 0$ to $t = 2$.

Solution. Differentiating gives $\alpha'(t) = (2t, 2t^2)$. Hence

$$\|\alpha'(t)\| = \sqrt{4t^2 + 4t^4} = 2|t|\sqrt{1 + t^2}.$$

Since $0 \le t \le 2$, we have $|t| = t$. Thus,

$$L = \int_0^2 2t\sqrt{1 + t^2}\, dt.$$

To evaluate the integral, put $u = 1 + t^2$. Then $du = 2t\, dt$, and the limits $t = 0$ and $t = 2$ correspond to $u = 1$ and $u = 5$, respectively. Therefore,

$$\begin{aligned}L &= \int_1^5 u^{1/2}\, du = \left[\frac{2}{3}u^{3/2}\right]_1^5 \\ &= \frac{2}{3}(5\sqrt{5} - 1).\end{aligned}$$

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Hence the required arc length is $\frac{2}{3}(5\sqrt{5}-1)$. ■

Although the derivative vanishes at $t = 0$, the arc length is well-defined because the curve is continuously differentiable. Thus, the integral formula for arc length applies even when a curve is not regular at every parameter value.

Example 1.4.6 (Arc Length of a Cycloid Arch)**.** (See Figure 3 in the Atlas.) Calculate the length of one arch of the cycloid

$$\alpha(t) = (R(t - \sin t), R(1 - \cos t)), \qquad 0 \le t \le 2\pi,$$

where $R > 0$.

Solution. Writing $\alpha(t) = (x(t), y(t))$, differentiation gives $x'(t) = R(1 - \cos t)$ and $y'(t) = R \sin t$. Consequently,

$$\begin{aligned} \|\alpha'(t)\|^2 &= R^2(1 - \cos t)^2 + R^2 \sin^2 t \\ &= R^2(2 - 2\cos t) \\ &= 4R^2 \sin^2(t/2), \end{aligned}$$

where we have used the identity $1 - \cos t = 2\sin^2(t/2)$. Taking square roots gives

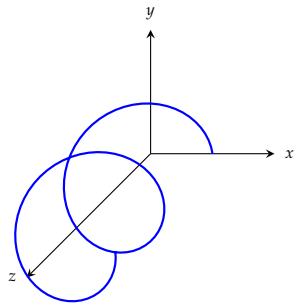
$$\|\alpha'(t)\| = 2R|\sin(t/2)|.$$

Since $\sin(t/2) \ge 0$ for $0 \le t \le 2\pi$, the absolute value may be removed. Hence,

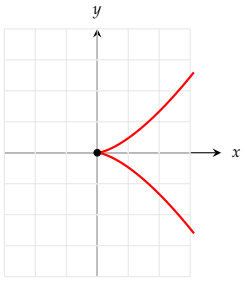
$$\begin{aligned} L &= \int_0^{2\pi} 2R \sin(t/2)\, dt \\ &= [-4R\cos(t/2)]_0^{2\pi} \\ &= -4R\cos\pi + 4R\cos 0 = 8R. \end{aligned}$$

Therefore, the length of one complete arch of the cycloid is $8R$. ■

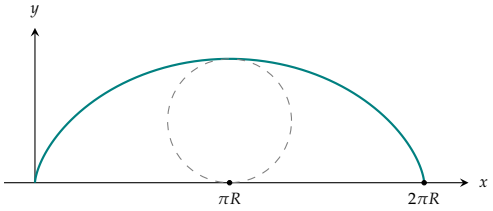
The cycloid has singularities at $t = 0$ and $t = 2\pi$, where its velocity vanishes, but it is regular on $(0, 2\pi)$. These endpoint singularities do not prevent the arc length from being calculated.



(a) Circular Helix
$\alpha(t) = (a\cos t, a\sin t, ct)$



(b) Semicubical Parabola
$\alpha(t) = (t^2, \frac{2}{3}t^3)$



(c) Cycloid Arch
$\alpha(t) = (R(t - \sin t), R(1 - \cos t))$

Figure 1.1: Illustrations of the curves in Examples 1.4.4, 1.4.5 and 1.4.6.

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Theorem 1.4.7 (Existence of Unit-Speed Reparameterization). *Every regular smooth curve admits a smooth reparameterization by arc length.*

Proof. Let $\alpha : I \to \mathbb{R}^n$ be a regular smooth curve, and choose a fixed parameter value $t_0 \in I$. Define the arc-length function $s : I \to \mathbb{R}$ by

$$s(t) = \int_{t_0}^{t} \|\alpha'(u)\| \, du.$$

By the fundamental theorem of calculus,

$$\frac{ds}{dt} = \|\alpha'(t)\|.$$

Since α is regular, $\alpha'(t) \neq \mathbf{0}$ for every $t \in I$, so $ds/dt > 0$. Consequently, $s(t)$ is strictly increasing and therefore one-to-one. Moreover, the inverse function theorem shows that its inverse $t = t(s)$ is smooth on the interval $s(I)$.

We now define a new curve $\beta : s(I) \to \mathbb{R}^n$ by $\beta(s) = \alpha(t(s))$. Differentiating with respect to s and applying the chain rule, we obtain

$$\frac{d\beta}{ds} = \frac{d\alpha}{dt}\frac{dt}{ds}.$$

Since $ds/dt = \|\alpha'(t)\|$, the derivative of the inverse function satisfies

$$\frac{dt}{ds} = \frac{1}{ds/dt} = \frac{1}{\|\alpha'(t)\|}.$$

Substituting into the preceding expression gives

$$\beta'(s) = \frac{\alpha'(t(s))}{\|\alpha'(t(s))\|}.$$

Taking norms of both sides, we obtain

$$\|\beta'(s)\| = \frac{\|\alpha'(t(s))\|}{\|\alpha'(t(s))\|} = 1.$$

Thus, β is a unit-speed curve. Since $t(s)$ is a smooth diffeomorphism, β is a smooth reparameterization of α by arc length. ■

To obtain a unit-speed parametrization in practice, one first calculates the speed $\|\alpha'(t)\|$ and integrates it to obtain $s(t)$. One then solves for t in terms of s and substitutes $t = t(s)$ into the original curve.

Example 1.4.8 (Unit-Speed Reparameterization of a Circle). Find a unit-speed parametrization of the circle $\alpha(t) = (a\cos t, a\sin t)$, where $a > 0$.

Solution. Since $\|\alpha'(t)\| = a$, the arc-length function measured from $t = 0$ is

$$s(t) = \int_0^t a \, du = at.$$

Thus $t = s/a$, and the corresponding reparameterization is

$$\beta(s) = \left(a\cos\frac{s}{a}, a\sin\frac{s}{a}\right).$$

Differentiating gives

$$\beta'(s) = \left(-\sin\frac{s}{a}, \cos\frac{s}{a}\right).$$

Therefore $\|\beta'(s)\| = 1$, as required. ■

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Theorem 1.4.9 (Invariance of Arc Length). *Let $\alpha : [a,b] \to \mathbb{R}^n$ be a regular continuously differentiable curve, and let $\beta : [c,d] \to \mathbb{R}^n$ be a reparameterization given by $\beta(u) = \alpha(\phi(u))$, where $\phi : [c,d] \to [a,b]$ is a continuously differentiable diffeomorphism. Then*

$$L(\beta) = L(\alpha).$$

In particular, arc length is independent of regular reparameterization.

Proof. By the definition of arc length,

$$L(\beta) = \int_c^d \|\beta'(u)\| \, du.$$

Since $\beta(u) = \alpha(\phi(u))$, the chain rule gives

$$\beta'(u) = \alpha'(\phi(u))\phi'(u).$$

Taking norms and using $\|\lambda v\| = |\lambda| \, \|v\|$, we obtain

$$\|\beta'(u)\| = \|\alpha'(\phi(u))\| \, |\phi'(u)|.$$

Consequently,

$$L(\beta) = \int_c^d \|\alpha'(\phi(u))\| \, |\phi'(u)| \, du.$$

We consider the two possible orientations.

Case 1. If $\phi'(u) > 0$, then ϕ is strictly increasing, so $\phi(c) = a$ and $\phi(d) = b$. Since $|\phi'(u)| = \phi'(u)$, we have

$$L(\beta) = \int_c^d \|\alpha'(\phi(u))\| \, \phi'(u) \, du.$$

Putting $t = \phi(u)$ gives $dt = \phi'(u) \, du$, and therefore

$$L(\beta) = \int_a^b \|\alpha'(t)\| \, dt = L(\alpha).$$

Case 2. If $\phi'(u) < 0$, then ϕ is strictly decreasing, so $\phi(c) = b$ and $\phi(d) = a$. Since $|\phi'(u)| = -\phi'(u)$,

$$L(\beta) = -\int_c^d \|\alpha'(\phi(u))\| \, \phi'(u) \, du.$$

Again putting $t = \phi(u)$ gives

$$L(\beta) = -\int_b^a \|\alpha'(t)\| \, dt = \int_a^b \|\alpha'(t)\| \, dt = L(\alpha).$$

Thus, in both cases, $L(\beta) = L(\alpha)$, proving the invariance of arc length. ■

Example 1.4.10 (The Cusp). (See Figure 4 in the Atlas.) Consider the curve $\alpha(t) = (t^3, t^2)$, where $t \in \mathbb{R}$. Determine whether it is regular and identify its singular points.

Solution. Differentiating gives $\alpha'(t) = (3t^2, 2t)$. This vector vanishes precisely when

$$3t^2 = 0, \qquad 2t = 0,$$

which happens only at $t = 0$. Therefore, α is regular for every $t \neq 0$ and singular at $t = 0$.

To examine the geometry, write $x = t^3$ and $y = t^2$. Eliminating t gives $x^2 = y^3$. The two branches meet at the origin, forming a cusp with common tangent along the y-axis. Thus, the origin is a genuine geometric singularity. ■

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Example 1.4.11 (Unit-Speed Reparameterization)**.** Reparameterize $\alpha(t) = (e^t \cos t, e^t \sin t)$, where $t \in \mathbb{R}$, by arc length measured from $t = 0$.

Solution. We first compute

$$\alpha'(t) = \big(e^t(\cos t - \sin t), e^t(\sin t + \cos t)\big).$$

Its squared speed is

$$\begin{aligned} \|\alpha'(t)\|^2 &= e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 \\ &= e^{2t}\big[(\cos t - \sin t)^2 + (\sin t + \cos t)^2\big] \\ &= 2e^{2t}. \end{aligned}$$

Therefore $\|\alpha'(t)\| = \sqrt{2}e^t$. The arc-length function measured from $t = 0$ is

$$s(t) = \int_0^t \sqrt{2}e^u \, du = \sqrt{2}(e^t - 1).$$

Solving for t, we obtain

$$e^t = 1 + \frac{s}{\sqrt{2}}, \qquad t(s) = \log\left(1 + \frac{s}{\sqrt{2}}\right).$$

Since $e^t > 0$, the new parameter satisfies $s > -\sqrt{2}$. Substituting into the original curve gives

$$\beta(s) = \left(\left(1 + \frac{s}{\sqrt{2}}\right)\cos\log\left(1 + \frac{s}{\sqrt{2}}\right), \left(1 + \frac{s}{\sqrt{2}}\right)\sin\log\left(1 + \frac{s}{\sqrt{2}}\right)\right).$$

Finally, the chain rule gives

$$\|\beta'(s)\| = \|\alpha'(t(s))\| \left|\frac{dt}{ds}\right| = \sqrt{2}e^{t(s)} \frac{1}{\sqrt{2}e^{t(s)}} = 1.$$

Hence $\beta(s)$ is the required unit-speed parametrization. ■

Example 1.4.12 (Different Parameterizations)**.** Consider the line segment joining $(0,0)$ and $(1,1)$. Show that the parametrizations

$$\alpha(t) = (t,t), \quad 0 \le t \le 1, \qquad \beta(u) = (u^2, u^2), \quad 0 \le u \le 1$$

have the same arc length.

Solution. For the first parametrization, $\alpha'(t) = (1,1)$, so $\|\alpha'(t)\| = \sqrt{2}$. Hence

$$L(\alpha) = \int_0^1 \sqrt{2}\, dt = \sqrt{2}.$$

For the second parametrization, $\beta'(u) = (2u, 2u)$, giving

$$\|\beta'(u)\| = \sqrt{4u^2 + 4u^2} = 2\sqrt{2}\, u,$$

where $u \ge 0$. Therefore,

$$L(\beta) = \int_0^1 2\sqrt{2}\, u \, du = \sqrt{2}\,\big[u^2\big]_0^1 = \sqrt{2}.$$

Thus, $L(\alpha) = L(\beta) = \sqrt{2}$. Both parametrizations describe the same line segment, although their speeds are different. Notice that the change of parameter $t = u^2$ has derivative zero at $u = 0$, so it is not a regular diffeomorphism at that endpoint. Nevertheless, the equality of arc lengths follows directly from the integral formula. ■

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1.5 Curvature of Plane Curves

Curvature measures how fast the unit tangent vector $T(s)$ rotates as we move along the curve.

Definition 1.5.1 (Signed Curvature). Let $\alpha(s)$ be a unit-speed plane curve. The **signed curvature** $\kappa(s)$ is defined by the relation:

$$\kappa(s) = |T'(s)| = |\beta''(s)|.$$

Theorem 1.5.2. *Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$. Then, its curvature is $\kappa = \frac{|\dot{\alpha}\times\ddot{\alpha}|}{|\dot{\alpha}|^3}$, where $\times$ denotes the cross product and a dot denotes differentiation with respect to t.*

Proof. Let $\hat{\alpha}$ (with parameter s) be a unit-speed reparameterization of α, and let us denote d/ds by a dash. Then, by the chain rule,

$$\dot{\alpha} = \frac{d\alpha}{dt} = \frac{d\alpha}{ds}\frac{ds}{dt}, \quad \text{that is,} \quad \hat{\alpha}'\frac{ds}{dt} = \dot{\alpha},$$

so

$$\kappa = |\hat{\alpha}''| = \left|\frac{d}{ds}\left(\frac{\dot{\alpha}}{ds/dt}\right)\right| = \left|\frac{\frac{d}{dt}\left(\frac{\dot{\alpha}}{ds/dt}\right)}{ds/dt}\right| = \left|\frac{\ddot{\alpha}\frac{ds}{dt} - \dot{\alpha}\frac{d^2s}{dt^2}}{(ds/dt)^3}\right|.$$

Now,

$$\left(\frac{ds}{dt}\right)^2 = |\dot{\alpha}|^2 = \dot{\alpha}\cdot\dot{\alpha},$$

and differentiating with respect to t gives $\frac{ds}{dt}\frac{d^2s}{dt^2} = \dot{\alpha}\cdot\ddot{\alpha}$. Using this and the expression for κ, we obtain

$$\kappa = \left|\frac{\ddot{\alpha}\left(\frac{ds}{dt}\right)^2 - \dot{\alpha}\frac{d^2s}{dt^2}\frac{ds}{dt}}{(ds/dt)^4}\right| = \frac{|\ddot{\alpha}(\dot{\alpha}\cdot\dot{\alpha}) - \dot{\alpha}(\dot{\alpha}\cdot\ddot{\alpha})|}{|\dot{\alpha}|^4}.$$

Using the vector triple product identity

$$\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$$

(where $\mathbf{a},\mathbf{b},\mathbf{c}\in\mathbb{R}^3$), we obtain

$$\dot{\alpha}\times(\ddot{\alpha}\times\dot{\alpha}) = \ddot{\alpha}(\dot{\alpha}\cdot\dot{\alpha}) - \dot{\alpha}(\dot{\alpha}\cdot\ddot{\alpha}).$$

Further, $\dot{\alpha}$ and $\ddot{\alpha}\times\dot{\alpha}$ are perpendicular vectors, so

$$|\dot{\alpha}\times(\ddot{\alpha}\times\dot{\alpha})| = |\dot{\alpha}||\ddot{\alpha}\times\dot{\alpha}|.$$

Hence,

$$\begin{aligned}\frac{|\ddot{\alpha}(\dot{\alpha}\cdot\dot{\alpha}) - \dot{\alpha}(\dot{\alpha}\cdot\ddot{\alpha})|}{|\dot{\alpha}|^4} &= \frac{|\dot{\alpha}\times(\ddot{\alpha}\times\dot{\alpha})|}{|\dot{\alpha}|^4}\\ &= \frac{|\dot{\alpha}||\ddot{\alpha}\times\dot{\alpha}|}{|\dot{\alpha}|^4}\\ &= \frac{|\dot{\alpha}\times\ddot{\alpha}|}{|\dot{\alpha}|^3}.\end{aligned}$$

If α is a non-regular curve, its curvature is not defined in general. Note, however, that the formula shows that the curvature is defined at all regular points of the curve (where $\dot{\alpha}$ is non-zero). ■

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Theorem 1.5.3 (General Curvature Formula). *Let $\alpha(t) = (x(t), y(t))$ be a regular plane curve. The signed curvature $\kappa(t)$ is given by*

$$\kappa(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{((x'(t))^2 + (y'(t))^2)^{3/2}}$$

Proof. Let $s(t)$ be the arc length and $\beta(s) = \alpha(t(s))$ be the reparameterization by arc length. The curvature is defined as $\kappa(s) = |\beta''(s)|$. Using the chain rule:

$$\beta'(s) = \alpha'(t)\frac{dt}{ds} \quad \text{and} \quad \beta''(s) = \alpha''(t)\left(\frac{dt}{ds}\right)^2 + \alpha'(t)\frac{d^2t}{ds^2}$$

Given that $\frac{ds}{dt} = \sqrt{x'^2 + y'^2}$, it follows that $\frac{dt}{ds} = (x'^2 + y'^2)^{-1/2}$. Squaring both sides gives,

$$\left(\frac{dt}{ds}\right)^2 = \frac{1}{x'^2 + y'^2}$$

Differentiating with respect to s gives $2\frac{dt}{ds}\frac{d^2t}{ds^2} = -2\frac{x'x'' + y'y''}{(x'^2 + y'^2)^2}\frac{dt}{ds}$, which simplifies to

$$\frac{d^2t}{ds^2} = -\frac{x'x'' + y'y''}{(x'^2 + y'^2)^2}$$

Substituting these expressions into $\beta''(s)$:

$$\begin{aligned}
\beta''(s) &= \frac{(x'', y'')}{x'^2 + y'^2} - \frac{(x'x'' + y'y'')(x', y')}{(x'^2 + y'^2)^2} \\
&= \frac{1}{(x'^2 + y'^2)^2}\left((x'', y'')(x'^2 + y'^2) - (x'x'' + y'y'')(x', y')\right) \\
&= \frac{1}{(x'^2 + y'^2)^2}\left((x'^2x'' + y'^2x'', x'^2y'' + y'^2y'') - (x'^2x'' + x'y'y'', x'y'x'' + y'^2y'')\right) \\
&= \frac{1}{(x'^2 + y'^2)^2}\left(y'^2x'' - x'y'y'', x'^2y'' - x'y'x''\right) \\
&= \frac{1}{(x'^2 + y'^2)^2}\left(-y'(x'y'' - y'x''), x'(x'y'' - y'x'')\right) \\
&= \frac{x'y'' - y'x''}{(x'^2 + y'^2)^2}(-y', x')
\end{aligned}$$

Taking norms gives

$$|\beta''(s)| = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^2}\sqrt{x'^2 + y'^2} = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}}$$

■

2nd Proof. Let s be the arc length parameter and $v(t) = \frac{ds}{dt} = \|\alpha'(t)\|$. We have $\alpha'(t) = vT$. Differentiating with respect to t using the chain rule:

$$\alpha''(t) = \frac{dv}{dt}T + v\frac{dT}{dt} = v'T + v\left(\frac{dT}{ds}\frac{ds}{dt}\right) = v'T + v^2\kappa N$$

We use the determinant (scalar triple product in 2D) to isolate κ. For vectors $a = (a_1, a_2)$ and $b = (b_1, b_2)$, let $[a, b] = a_1b_2 - a_2b_1$. Then:

$$[\alpha'(t), \alpha''(t)] = [vT, v'T + v^2\kappa N] = vv'[T, T] + v^3\kappa[T, N]$$

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Since $[T, T] = 0$ and $[T, N] = 1$ (by the positive orientation of the frame), we have

$$[\alpha'(t), \alpha''(t)] = v^3\kappa \implies \kappa = \frac{x'y'' - y'x''}{v^3}$$

Substituting $v = \sqrt{x'^2 + y'^2}$ completes the proof. ■

Example 1.5.4 (Curvature of a Parabola)**.** Find the signed curvature $\kappa(t)$ of the parabola parameterized by $\alpha(t) = (t, t^2)$ for $t \in \mathbb{R}$. Determine the point where the curvature is maximal.

Solution. The curve is given by $x(t) = t$ and $y(t) = t^2$. We compute the first and second derivatives with respect to t:

$$x'(t) = 1, \quad x''(t) = 0$$

$$y'(t) = 2t, \quad y''(t) = 2$$

The formula for the signed curvature of a plane curve is $\kappa(t) = \frac{x'y'' - y'x''}{((x')^2 + (y')^2)^{3/2}}$. Substituting these derivatives gives

$$\kappa(t) = \frac{(1)(2) - (2t)(0)}{(1^2 + (2t)^2)^{3/2}} = \frac{2}{(1 + 4t^2)^{3/2}}$$

To find where the curvature is maximal, we examine the denominator. Since the numerator is constant, $\kappa(t)$ is maximized when the denominator $(1 + 4t^2)^{3/2}$ is minimized. This occurs at $t = 0$. The maximum curvature is $\kappa(0) = \frac{2}{1^{3/2}} = 2$. This point $(0, 0)$ is the vertex of the parabola. ■

Example 1.5.5 (Curvature of the Logarithmic Spiral)**.** (See Figure 5 in the Atlas.) Calculate the curvature of the logarithmic spiral $\alpha(t) = (e^t \cos t, e^t \sin t)$.

Solution. We have $x(t) = e^t \cos t$ and $y(t) = e^t \sin t$. Differentiating gives

$$\begin{aligned}
x' &= e^t(\cos t - \sin t) \\
y' &= e^t(\sin t + \cos t) \\
x'' &= e^t(\cos t - \sin t) + e^t(-\sin t - \cos t) = -2e^t \sin t \\
y'' &= e^t(\sin t + \cos t) + e^t(\cos t - \sin t) = 2e^t \cos t
\end{aligned}$$

First compute the speed squared $v^2 = (x')^2 + (y')^2$:

$$v^2 = e^{2t}(\cos^2 t - 2\cos t \sin t + \sin^2 t + \sin^2 t + 2\sin t \cos t + \cos^2 t) = 2e^{2t}$$

Thus, $v^3 = (2e^{2t})^{3/2} = 2\sqrt{2}e^{3t}$. Next compute the numerator $x'y'' - y'x''$:

$$\begin{aligned}
x'y'' - y'x'' &= [e^t(\cos t - \sin t)][2e^t \cos t] - [e^t(\sin t + \cos t)][-2e^t \sin t] \\
&= 2e^{2t}(\cos^2 t - \sin t \cos t + \sin^2 t + \cos t \sin t) \\
&= 2e^{2t}(\cos^2 t + \sin^2 t) = 2e^{2t}
\end{aligned}$$

Therefore, the curvature is

$$\kappa(t) = \frac{2e^{2t}}{2\sqrt{2}e^{3t}} = \frac{1}{\sqrt{2}e^t}$$

■

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1.6 The Osculating Circle

The tangent line provides a first-order approximation to a regular curve at a given point. A closer approximation is obtained by considering a circle that has the same tangent direction and curvature as the curve at that point. Such a circle is called the **osculating circle**. The term *osculating* is derived from the Latin word *osculum*, meaning "kiss", and describes the close contact between the circle and the curve.

Let $\alpha(s)$ be a unit-speed plane curve, and let $T(s)$ and $N(s)$ denote its unit tangent and oriented unit normal vectors, respectively. Recall that the signed curvature $\kappa(s)$ is defined by $T'(s) = \kappa(s)N(s)$. The magnitude $|\kappa(s)|$ measures how rapidly the tangent direction changes with respect to arc length. Consequently, a circle of large radius has small curvature, while a circle of small radius has large curvature.

Definition 1.6.1 (Osculating Circle)**.** Let $\alpha(s)$ be a smooth unit-speed plane curve with signed curvature $\kappa(s) \neq 0$. The **radius of curvature** is defined by $\rho(s) = 1/|\kappa(s)|$, and the **center of curvature** is given by

$$C(s) = \alpha(s) + \frac{1}{\kappa(s)}N(s).$$

The circle with center $C(s)$ and radius $\rho(s)$ is called the **osculating circle** of α at the point $\alpha(s)$.

The osculating circle lies on the side toward which the curve bends. If $\kappa(s) > 0$, its center lies in the direction of $N(s)$, whereas if $\kappa(s) < 0$, it lies in the opposite direction. At a point where $\kappa(s) = 0$, there is no finite osculating circle determined by the curvature.

Proposition 1.6.2 (Second-Order Contact)**.** *Let $\alpha(s)$ be a unit-speed plane curve of class C^2 with $\kappa(s_0) \neq 0$. Then the osculating circle at $P = \alpha(s_0)$ has the same position, unit tangent and second derivative as the curve at $s = s_0$. Thus, it provides a second-order approximation to the curve near P.*

Proof. Write $P = \alpha(s_0)$, $T_0 = T(s_0)$, $N_0 = N(s_0)$ and $\kappa_0 = \kappa(s_0)$. The center of the osculating circle is $C = P + \frac{1}{\kappa_0}N_0$, and its radius is $\rho = 1/|\kappa_0|$.

Consider the parametrization of this circle given by

$$\gamma(s) = C - \frac{1}{\kappa_0}\left[N_0\cos\big(\kappa_0(s-s_0)\big) - T_0\sin\big(\kappa_0(s-s_0)\big)\right].$$

Since T_0 and N_0 are perpendicular unit vectors, we have $\|\gamma(s) - C\| = 1/|\kappa_0| = \rho$. Hence γ parametrizes the required circle.

At $s = s_0$, we obtain

$$\gamma(s_0) = C - \frac{1}{\kappa_0}N_0 = P = \alpha(s_0).$$

Differentiating gives

$$\gamma'(s) = N_0\sin\big(\kappa_0(s-s_0)\big) + T_0\cos\big(\kappa_0(s-s_0)\big),$$

and therefore $\gamma'(s_0) = T_0 = \alpha'(s_0)$. Differentiating again, we obtain

$$\gamma''(s) = \kappa_0 N_0\cos\big(\kappa_0(s-s_0)\big) - \kappa_0 T_0\sin\big(\kappa_0(s-s_0)\big).$$

Consequently, $\gamma''(s_0) = \kappa_0 N_0 = \alpha''(s_0)$, where the last equality follows from $\alpha''(s) = T'(s) = \kappa(s)N(s)$. Thus, the curve and its osculating circle have the same position, first derivative and second derivative at $s = s_0$, proving the result. ∎

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For practical calculations, a curve is often given in terms of an arbitrary regular parameter t rather than arc length. If $\alpha(t) = (x(t), y(t))$, then its signed curvature and oriented unit normal are

$$\kappa(t) = \frac{x'(t)y''(t) - y'(t)x''(t)}{\left((x'(t))^2 + (y'(t))^2\right)^{3/2}}, \qquad N(t) = \frac{(-y'(t), x'(t))}{\sqrt{(x'(t))^2 + (y'(t))^2}}.$$

These expressions may be substituted directly into $C(t) = \alpha(t) + N(t)/\kappa(t)$ to determine the center of curvature whenever $\kappa(t) \neq 0$.

Example 1.6.3 (Osculating Circle of an Ellipse)**.** (See Figure 7 in the Atlas.) Find the center and equation of the osculating circle to the ellipse $\alpha(t) = (2\cos t, 3\sin t)$ at $t = 0$.

Solution. At $t = 0$, the point on the ellipse is $\alpha(0) = (2, 0)$. Differentiating, we obtain

$$\alpha'(t) = (-2\sin t, 3\cos t), \qquad \alpha''(t) = (-2\cos t, -3\sin t).$$

Thus, $\alpha'(0) = (0, 3)$ and $\alpha''(0) = (-2, 0)$. The signed curvature at this point is

$$\kappa(0) = \frac{x'(0)y''(0) - y'(0)x''(0)}{\left((x'(0))^2 + (y'(0))^2\right)^{3/2}} = \frac{(0)(0) - (3)(-2)}{(0^2 + 3^2)^{3/2}} = \frac{2}{9}.$$

Hence the radius of curvature is $\rho = 1/|\kappa(0)| = 9/2$. The unit tangent vector is

$$T(0) = \frac{\alpha'(0)}{\|\alpha'(0)\|} = (0, 1),$$

so the oriented unit normal is $N(0) = (-1, 0)$. Therefore, the center of curvature is

$$C(0) = \alpha(0) + \frac{1}{\kappa(0)}N(0) = (2, 0) + \frac{9}{2}(-1, 0) = \left(-\frac{5}{2}, 0\right).$$

Thus, the osculating circle has center $(-5/2, 0)$ and radius $9/2$, and its equation is

$$\boxed{\left(x + \frac{5}{2}\right)^2 + y^2 = \frac{81}{4}}.$$

■

Example 1.6.4 (Osculating Circle of a Parabola)**.** (See Figure 6 in the Atlas.) Find the center and equation of the osculating circle to the parabola $y = x^2$ at its vertex $(0, 0)$.

Solution. Parametrize the parabola by $\alpha(t) = (t, t^2)$. Then

$$\alpha'(t) = (1, 2t), \qquad \alpha''(t) = (0, 2).$$

At $t = 0$, we have $\alpha(0) = (0, 0)$, $\alpha'(0) = (1, 0)$ and $\alpha''(0) = (0, 2)$. The signed curvature is

$$\kappa(0) = \frac{(1)(2) - (0)(0)}{(1^2 + 0^2)^{3/2}} = 2.$$

The radius of curvature is therefore $\rho = 1/2$. Since $T(0) = (1, 0)$, the oriented unit normal is $N(0) = (0, 1)$. Hence the center of curvature is

$$C(0) = \alpha(0) + \frac{1}{\kappa(0)}N(0) = (0, 0) + \frac{1}{2}(0, 1) = \left(0, \frac{1}{2}\right).$$

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The osculating circle has center $(0, 1/2)$ and radius $1/2$, so its equation is

$$\boxed{x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}}.$$

Expanding gives the equivalent equation $x^2 + y^2 - y = 0$. ■

Example 1.6.5 (Osculating Circle of $y = e^x$)**.** (See Figure 8 in the Atlas.) Find the radius, center and equation of the osculating circle to the curve $y = e^x$ at the point $(0, 1)$.

Solution. Parametrize the curve by $\alpha(t) = (t, e^t)$. Differentiating gives

$$\alpha'(t) = (1, e^t), \qquad \alpha''(t) = (0, e^t).$$

At $t = 0$, we have $\alpha(0) = (0, 1)$, $\alpha'(0) = (1, 1)$ and $\alpha''(0) = (0, 1)$. Thus,

$$\kappa(0) = \frac{(1)(1) - (1)(0)}{(1^2 + 1^2)^{3/2}} = \frac{1}{2\sqrt{2}}.$$

The radius of curvature is $\rho = 2\sqrt{2}$. The unit tangent and oriented unit normal at $t = 0$ are

$$T(0) = \frac{1}{\sqrt{2}}(1, 1), \qquad N(0) = \frac{1}{\sqrt{2}}(-1, 1).$$

Therefore,

$$C(0) = \alpha(0) + \frac{1}{\kappa(0)} N(0) = (0, 1) + 2\sqrt{2}\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (-2, 3).$$

Hence the osculating circle has center $(-2, 3)$ and radius $2\sqrt{2}$, and its equation is

$$\boxed{(x + 2)^2 + (y - 3)^2 = 8}.$$

■

The concept of an osculating circle extends naturally to regular space curves having nonzero curvature. In this case, the circle lies in the osculating plane spanned by the principal unit tangent and normal vectors. If κ denotes the ordinary curvature of the space curve, then its center is $\alpha(s) + N(s)/\kappa(s)$ and its radius is $1/\kappa(s)$.

Example 1.6.6 (The Twisted Cubic)**.** (See Figure 9 in the Atlas.) For the twisted cubic $\alpha(t) = (t, t^2, t^3)$, find the unit tangent vector T, curvature κ, principal unit normal N and unit binormal B at the origin. Also determine the center and radius of its osculating circle.

Solution. The curve passes through the origin at $t = 0$. Differentiating gives

$$\alpha'(t) = (1, 2t, 3t^2), \qquad \alpha''(t) = (0, 2, 6t).$$

Therefore, $\alpha'(0) = (1, 0, 0)$ and $\alpha''(0) = (0, 2, 0)$. Since $\|\alpha'(0)\| = 1$, the unit tangent vector is

$$T(0) = \frac{\alpha'(0)}{\|\alpha'(0)\|} = (1, 0, 0).$$

For a regular space curve, the curvature is

$$\kappa(t) = \frac{\|\alpha'(t) \times \alpha''(t)\|}{\|\alpha'(t)\|^3}.$$

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At $t = 0$, we have

$$\alpha'(0) \times \alpha''(0) = (1,0,0) \times (0,2,0) = (0,0,2).$$

Consequently, $\kappa(0) = 2$.

To determine the principal unit normal, we use $N = (dT/ds)/\kappa$, where s denotes arc length. Differentiating $T(t) = \alpha'(t)/\|\alpha'(t)\|$ and using $ds/dt = \|\alpha'(t)\|$, we obtain

$$\left.\frac{dT}{ds}\right|_{t=0} = \frac{1}{\|\alpha'(t)\|}\frac{d}{dt}\left(\frac{\alpha'(t)}{\|\alpha'(t)\|}\right)\Bigg|_{t=0} = (0,2,0).$$

Thus,

$$N(0) = \frac{1}{\kappa(0)}\left.\frac{dT}{ds}\right|_{t=0} = \frac{1}{2}(0,2,0) = (0,1,0).$$

The unit binormal is obtained from $B = T \times N$, giving

$$B(0) = (1,0,0) \times (0,1,0) = (0,0,1).$$

Hence,

$$\boxed{T(0) = (1,0,0), \quad \kappa(0) = 2, \quad N(0) = (0,1,0), \quad B(0) = (0,0,1)}.$$

The center of the osculating circle is

$$C(0) = \alpha(0) + \frac{1}{\kappa(0)}N(0) = (0,0,0) + \frac{1}{2}(0,1,0) = \left(0, \frac{1}{2}, 0\right).$$

Its radius is $\rho = 1/2$. Since the osculating plane is spanned by $T(0)$ and $N(0)$, it is the xy-plane, whose equation is $z = 0$. Thus, the osculating circle is described by

$$\boxed{x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}, \qquad z = 0}.$$

■

1.7 The Tangent Line to a Plane Curve

Let $\alpha : I \to \mathbb{R}^2$ be a regular differentiable plane curve given by $\alpha(t) = (x(t), y(t))$. The derivative $\alpha'(t)$ determines the direction of the curve at the corresponding point. The straight line through a point of the curve in this direction is called the tangent line, while the line through the same point perpendicular to the tangent is called the normal line.

Definition 1.7.1 (Tangent Vector). Let $\alpha : I \to \mathbb{R}^2$ be a differentiable curve. The derivative $\alpha'(t_0) = (x'(t_0), y'(t_0))$ is called the tangent vector or velocity vector at the point $P = \alpha(t_0)$. If $\alpha'(t_0) \neq 0$, the corresponding unit tangent vector is

$$T(t_0) = \frac{\alpha'(t_0)}{\|\alpha'(t_0)\|}.$$

Proposition 1.7.2 (Equation of the Tangent Line). *Let $\alpha(t) = (x(t), y(t))$ be a regular differentiable plane curve. The tangent line at $P = \alpha(t_0)$ has the vector equation*

$$L(u) = \alpha(t_0) + u\alpha'(t_0), \qquad u \in \mathbb{R},$$

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and its Cartesian equation is

$$y'(t_0)(X - x(t_0)) - x'(t_0)(Y - y(t_0)) = 0.$$

Proof. The tangent line passes through $P = \alpha(t_0)$ and has direction vector $\alpha'(t_0)$. Therefore, its vector equation is $L(u) = \alpha(t_0) + u\alpha'(t_0)$, where $u \in \mathbb{R}$. Writing this in coordinates, we obtain

$$X = x(t_0) + ux'(t_0), \qquad Y = y(t_0) + uy'(t_0).$$

Subtracting the coordinates of P gives

$$X - x(t_0) = ux'(t_0), \qquad Y - y(t_0) = uy'(t_0).$$

Multiplying the first equation by $y'(t_0)$ and the second by $x'(t_0)$, and then subtracting, we obtain

$$y'(t_0)(X - x(t_0)) - x'(t_0)(Y - y(t_0)) = 0.$$

This is the required Cartesian equation of the tangent line. ■

When $x'(t_0) \neq 0$, the equation may be written in point–slope form as

$$Y - y(t_0) = \frac{y'(t_0)}{x'(t_0)}(X - x(t_0)).$$

Here $y'(t_0)/x'(t_0) = dy/dx$ is the slope of the tangent line, obtained by the chain rule. If $x'(t_0) = 0$, regularity implies $y'(t_0) \neq 0$, and the tangent line is vertical, with equation $X = x(t_0)$.

Definition 1.7.3 (Normal Line)**.** Let $\alpha : I \to \mathbb{R}^2$ be a regular differentiable plane curve. The **normal line** at $P = \alpha(t_0)$ is the straight line passing through P and perpendicular to the tangent line at that point.

A vector perpendicular to the tangent vector $\alpha'(t_0) = (x'(t_0), y'(t_0))$ is $(-y'(t_0), x'(t_0))$. Consequently, the oriented unit normal vector is

$$N(t_0) = \frac{(-y'(t_0), x'(t_0))}{\sqrt{(x'(t_0))^2 + (y'(t_0))^2}}.$$

Since $T(t_0) \cdot N(t_0) = 0$, the vectors $T(t_0)$ and $N(t_0)$ are perpendicular unit vectors.

Proposition 1.7.4 (Equation of the Normal Line)**.** *Let $\alpha(t) = (x(t), y(t))$ be a regular differentiable plane curve. The normal line at $P = \alpha(t_0)$ has the vector equation*

$$L_N(u) = \alpha(t_0) + uN(t_0), \qquad u \in \mathbb{R},$$

and its Cartesian equation is

$$x'(t_0)(X - x(t_0)) + y'(t_0)(Y - y(t_0)) = 0.$$

Proof. Let (X, Y) be an arbitrary point on the normal line. The vector joining $P = (x(t_0), y(t_0))$ to (X, Y) is

$$(X - x(t_0), Y - y(t_0)).$$

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Since the normal line is perpendicular to the tangent vector $\alpha'(t_0) = (x'(t_0), y'(t_0))$, the dot product of these two vectors must vanish. Hence,

$$(X - x(t_0), Y - y(t_0)) \cdot (x'(t_0), y'(t_0)) = 0.$$

Expanding gives

$$x'(t_0)(X - x(t_0)) + y'(t_0)(Y - y(t_0)) = 0,$$

which is the required Cartesian equation. ■

When $y'(t_0) \neq 0$, the normal line can also be written in point–slope form as

$$Y - y(t_0) = -\frac{x'(t_0)}{y'(t_0)}(X - x(t_0)).$$

Thus, whenever both slopes are defined and the tangent slope is nonzero, the tangent and normal slopes are negative reciprocals.

Example 1.7.5 (Normal Line to a Parabola)**.** Consider the parabola $\alpha(t) = (t, t^2)$, where $t \in \mathbb{R}$. Find the equation of the normal line at $t = 1$.

Solution. At $t = 1$, the point on the curve is $P = \alpha(1) = (1, 1)$. Differentiating gives $\alpha'(t) = (1, 2t)$, and therefore $\alpha'(1) = (1, 2)$.

Using the Cartesian equation of the normal line, we obtain

$$x'(1)(X - 1) + y'(1)(Y - 1) = 0.$$

Substituting $x'(1) = 1$ and $y'(1) = 2$, we have

$$(X - 1) + 2(Y - 1) = 0.$$

Simplifying gives

$$\boxed{X + 2Y - 3 = 0}.$$

Equivalently, the normal line may be written in slope–intercept form as

$$\boxed{Y = -\frac{1}{2}X + \frac{3}{2}}.$$

■

Example 1.7.6 (Tangent and Normal Lines to a Circle)**.** Find the equations of the tangent and normal lines to the circle $\alpha(t) = (a\cos t, a\sin t)$ at $t = \pi/4$, where $a > 0$.

Solution. At $t = \pi/4$, the point on the circle is

$$P = \left(\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right).$$

Differentiating gives $\alpha'(t) = (-a\sin t, a\cos t)$, and hence

$$\alpha'\left(\frac{\pi}{4}\right) = \left(-\frac{a}{\sqrt{2}}, \frac{a}{\sqrt{2}}\right).$$

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Using the Cartesian equation of the tangent line, we obtain

$$\frac{a}{\sqrt{2}}\left(X-\frac{a}{\sqrt{2}}\right)+\frac{a}{\sqrt{2}}\left(Y-\frac{a}{\sqrt{2}}\right)=0.$$

Simplifying gives the tangent line

$$\boxed{X+Y=a\sqrt{2}}.$$

Similarly, the normal line satisfies

$$-\frac{a}{\sqrt{2}}\left(X-\frac{a}{\sqrt{2}}\right)+\frac{a}{\sqrt{2}}\left(Y-\frac{a}{\sqrt{2}}\right)=0,$$

which reduces to

$$\boxed{Y=X}.$$

Thus, the normal line passes through the center of the circle, as expected. ■

1.8 The Frenet–Serret Formulas

The local geometry of a unit-speed curve $\alpha : I \to \mathbb{R}^3$ is entirely determined by its curvature $\kappa(s)$ and its torsion $\tau(s)$. The Frenet–Serret formulas describe how the moving orthonormal frame $\{T, N, B\}$ evolves along the curve.

In three-dimensional space, a regular curve $\alpha : I \to \mathbb{R}^3$ is described locally by three mutually orthogonal unit vectors known as the **Frenet–Serret Frame**. These vectors—the tangent, the normal, and the binormal—characterize the bending and twisting of the curve in space.

Definition 1.8.1 (The Unit Tangent Vector T)**.** Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$. The **unit tangent vector** $T(t)$ is defined as the normalized velocity vector:

$$T(t)=\frac{\alpha'(t)}{\|\alpha'(t)\|}$$

If the curve is parameterized by arc length s, then $T(s)=\alpha'(s)$.

Definition 1.8.2 (The Principal Unit Normal Vector N)**.** The **principal unit normal vector** $N(t)$ points in the direction in which the curve is bending. It is defined as

$$N(t)=\frac{T'(t)}{\|T'(t)\|}$$

The magnitude $\kappa(t)=\frac{\|T'(t)\|}{\|\alpha'(t)\|}$ is the **curvature** of the curve. N is only defined at points where $\kappa\neq 0$. Specifically, if $T(s)=(x(s),y(s))$, then $N(s)=(-y(s),x(s))$.

Proposition 1.8.3. *Let* $\mathbf{a}$ *and* $\mathbf{b}$ *be two unit vectors in* $\mathbb{R}^3$*. Their cross product* $\mathbf{a}\times\mathbf{b}$ *is a unit vector if and only if* $\mathbf{a}\perp\mathbf{b}$*.*

Proof. By the definition of the magnitude of a cross product, for any two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$:

$$\|\mathbf{a}\times\mathbf{b}\|=\|\mathbf{a}\|\,\|\mathbf{b}\|\sin\theta$$

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where θ is the angle between the two vectors ($0 \leq \theta \leq \pi$). Since $\mathbf{a}$ and $\mathbf{b}$ are unit vectors, we have $\|\mathbf{a}\| = 1$ and $\|\mathbf{b}\| = 1$. Substituting these values into the equation:

$$\|\mathbf{a} \times \mathbf{b}\| = (1)(1)\sin\theta = \sin\theta$$

For the cross product $\mathbf{a} \times \mathbf{b}$ to be a unit vector, its magnitude must be equal to 1:

$$\sin\theta = 1$$

In the interval $[0, \pi]$, the sine function equals 1 only when $\theta = \frac{\pi}{2}$ (or $90°$). Thus, the cross product of two unit vectors is a unit vector only if the vectors are orthogonal. ■

Definition 1.8.4 (The Binormal Vector B)**.** The **binormal vector** $B(t)$ is defined as the cross product of the tangent and the normal vectors:

$$B(t) = T(t) \times N(t)$$

The triple $\{T, N, B\}$ forms a right-handed orthonormal basis at each point along the curve.

Proposition 1.8.5. *For any regular differentiable curve $\alpha(s)$, the unit tangent vector $T(s)$ and the principal unit normal vector $N(s)$ are orthogonal; that is, $\langle T(s), N(s)\rangle = 0$.*

Proof. By definition, $T(s)$ is a unit vector, which means its magnitude is constant:

$$\|T(s)\| = 1$$

Taking the square of the norm and expressing it as a dot product, we have $\langle T(s), T(s)\rangle = 1$. We differentiate both sides of this equation with respect to the parameter s. Using the product rule for dot products:

$$\frac{d}{ds}\langle T(s), T(s)\rangle = \frac{d}{ds}(1)$$
$$\langle T'(s), T(s)\rangle + \langle T(s), T'(s)\rangle = 0$$
$$2\langle T'(s), T(s)\rangle = 0$$

Dividing by 2, we obtain $\langle T'(s), T(s)\rangle = 0$. Thus the derivative vector $T'(s)$ is always orthogonal to $T(s)$. Since the principal unit normal vector $N(s)$ is defined as the normalized derivative of the tangent vector:

$$N(s) = \frac{T'(s)}{\|T'(s)\|}$$

Since $N(s)$ is simply a scalar multiple of $T'(s)$, and $T'(s)$ is orthogonal to $T(s)$, it follows that

$$\langle T(s), N(s)\rangle = 0$$

Thus, the principal unit normal vector is perpendicular to the tangent vector at every point where it is defined. ■

Example 1.8.6 (The Circular Helix)**.** Find the Frenet frame $\{T, N, B\}$ for the circular helix:

$$\alpha(t) = (a\cos t, a\sin t, bt)$$

where $a^2 + b^2 = c^2$.

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Solution. First compute the velocity: $\alpha'(t) = (-a\sin t, a\cos t, b)$. The speed is

$$v = \|\alpha'(t)\| = \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2} = \sqrt{a^2+b^2} = c.$$

Therefore,

$$T(t) = \frac{\alpha'(t)}{c} = \left(-\frac{a}{c}\sin t, \frac{a}{c}\cos t, \frac{b}{c}\right)$$

Differentiate $T(t)$: $T'(t) = \left(-\frac{a}{c}\cos t, -\frac{a}{c}\sin t, 0\right)$. Its norm is

$$\|T'(t)\| = \sqrt{\frac{a^2}{c^2}(\cos^2 t + \sin^2 t)} = \frac{a}{c}.$$

Thus,

$$N(t) = \frac{T'(t)}{a/c} = (-\cos t, -\sin t, 0)$$

We compute the cross product $T \times N = B(t)$:

$$B(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a}{c}\sin t & \frac{a}{c}\cos t & \frac{b}{c} \\ -\cos t & -\sin t & 0 \end{vmatrix}$$

$$B(t) = \left(\frac{b}{c}\sin t, -\frac{b}{c}\cos t, \frac{a}{c}\right)$$

■

Definition 1.8.7 (Torsion)**.** The **torsion** $\tau(s)$ of the curve α measures the departure of the curve from planarity and is defined by $B'(s) = -\tau(s)N(s)$.

Since $B(s)$ is a unit vector, $B'(s)$ is perpendicular to $B(s)$. Also, differentiating $\langle B, T\rangle = 0$ and using $T' = \kappa N$ shows that $B'(s)$ is perpendicular to $T(s)$. Hence $B'(s)$ must be parallel to $N(s)$.

The minus sign is a conventional choice that gives the Frenet–Serret formulas the standard form

$$T' = \kappa N, \qquad N' = -\kappa T + \tau B, \qquad B' = -\tau N.$$

Thus $|\tau(s)|$ measures the rate at which the binormal, and hence the osculating plane, rotates along the curve.

Example 1.8.8 (Calculating Torsion for a General Parameterization)**.** Calculate the torsion of the twisted cubic $\alpha(t) = (t, t^2, t^3)$.

Solution. For a regular curve $\alpha(t)$ not necessarily parameterized by arc length, the torsion is

$$\tau(t) = \frac{(\alpha'(t) \times \alpha''(t)) \cdot \alpha'''(t)}{\|\alpha'(t) \times \alpha''(t)\|^2}.$$

We have

$$\begin{aligned} \alpha'(t) &= (1, 2t, 3t^2), \\ \alpha''(t) &= (0, 2, 6t), \\ \alpha'''(t) &= (0, 0, 6). \end{aligned}$$

The cross product is

$$\alpha'(t) \times \alpha''(t) = (6t^2, -6t, 2).$$

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Therefore,

$$\|\alpha'(t) \times \alpha''(t)\|^2 = 36t^4 + 36t^2 + 4,$$

while $(\alpha'(t) \times \alpha''(t)) \cdot \alpha'''(t) = 12$.

Hence,

$$\tau(t) = \frac{12}{36t^4 + 36t^2 + 4} = \frac{3}{9t^4 + 9t^2 + 1}.$$

Since $\tau(t) \neq 0$ for all t, the twisted cubic is not planar. ■

Theorem 1.8.9 (Frenet–Serret Formulas). *Let $\alpha(s)$ be a unit-speed curve in $\mathbb{R}^3$ of class C^3 with curvature $\kappa(s) > 0$. Then the derivatives of the Frenet frame $\{T, N, B\}$ satisfy*

$$\begin{aligned} T' &= \kappa N, \\ N' &= -\kappa T + \tau B, \\ B' &= -\tau N, \end{aligned}$$

where κ and τ denote the curvature and torsion of the curve, respectively. Equivalently,

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}.$$

Proof. Since α is parametrized by arc length, its unit tangent vector is $T = \alpha'$, and its curvature is $\kappa = \|T'\|$. As $\kappa > 0$, the principal unit normal is defined by $N = T'/\kappa$. Thus, the first formula follows immediately:

$$T' = \kappa N.$$

To derive the remaining formulas, recall that $B = T \times N$ and that $\{T, N, B\}$ is an orthonormal basis of $\mathbb{R}^3$ at every point of the curve. Consequently,

$$\langle T, T\rangle = \langle N, N\rangle = \langle B, B\rangle = 1, \qquad \langle T, N\rangle = \langle N, B\rangle = \langle B, T\rangle = 0.$$

We shall differentiate these identities to determine the components of N' and B'.

First, consider the derivative of the binormal vector B. Differentiating $\langle B, B\rangle = 1$ with respect to s and using the product rule for the inner product, we obtain

$$\langle B', B\rangle + \langle B, B'\rangle = 0.$$

Since the inner product is symmetric, this gives $2\langle B', B\rangle = 0$, and hence $\langle B', B\rangle = 0$. Thus, B' is perpendicular to B.

Next, differentiating $\langle B, T\rangle = 0$, we obtain

$$\langle B', T\rangle + \langle B, T'\rangle = 0.$$

Using $T' = \kappa N$ and $\langle B, N\rangle = 0$, we get

$$\langle B', T\rangle = -\kappa \langle B, N\rangle = 0.$$

Therefore, B' is also perpendicular to T. Since $\{T, N, B\}$ is an orthonormal basis, a vector perpendicular to both T and B must be parallel to N. Hence there exists a scalar function $\lambda(s)$ such that $B' = \lambda N$.

The torsion of the curve is defined by $\tau = -\langle B', N\rangle$. Taking the inner product of $B' = \lambda N$ with N, we obtain $\lambda = \langle B', N\rangle = -\tau$. Consequently,

$$B' = -\tau N.$$

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It remains to determine N'. Since $\{T, N, B\}$ is an orthonormal basis, we can express N' uniquely as

$$N' = \langle N', T \rangle T + \langle N', N \rangle N + \langle N', B \rangle B.$$

We now calculate the three coefficients.

Differentiating $\langle N, N \rangle = 1$ gives $2 \langle N', N \rangle = 0$, so $\langle N', N \rangle = 0$.

Similarly, differentiating $\langle N, T \rangle = 0$ gives

$$\langle N', T \rangle + \langle N, T' \rangle = 0.$$

Since $T' = \kappa N$, it follows that

$$\langle N', T \rangle = - \langle N, \kappa N \rangle = -\kappa.$$

Finally, differentiating $\langle N, B \rangle = 0$, we obtain

$$\langle N', B \rangle + \langle N, B' \rangle = 0.$$

Substituting $B' = -\tau N$, we find

$$\langle N', B \rangle = - \langle N, -\tau N \rangle = \tau.$$

Substituting these three coefficients into the expansion of N' yields

$$N' = -\kappa T + \tau B.$$

We have therefore established the three Frenet–Serret formulas

$$T' = \kappa N, \qquad N' = -\kappa T + \tau B, \qquad B' = -\tau N,$$

which describe how the tangent, principal normal and binormal vectors vary along the curve. ■

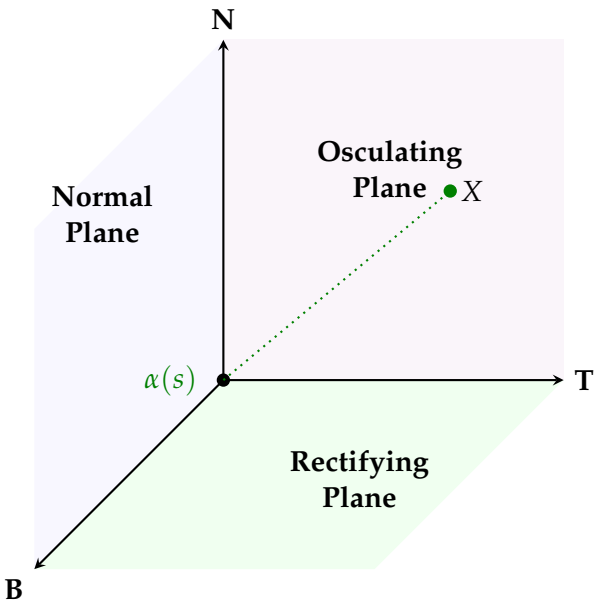


Figure 1.2: The normal, osculating, and rectifying planes of a space curve.

Remark 1.8.10. The skew-symmetry of the matrix in the Frenet–Serret formulas is a general property of the derivatives of a moving orthonormal frame. It reflects the fact that the frame undergoes an instantaneous rotation without changing the lengths of the basis vectors.

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The orthonormal triad $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ is referred to as the **Frenet–Serret frame** of the curve. These three vectors naturally determine three mutually perpendicular coordinate planes at each point along the curve, which describe its local spatial orientation. (See Figure 10 in the Atlas.)

Each pair of vectors in the Frenet–Serret frame spans a unique plane passing through the point $\alpha(s)$ defined below.

Definition 1.8.11.

- The **normal plane** to the curve α at the point $\alpha(s)$ is the plane spanned by the principal normal $\mathbf{N}(s)$ and the binormal $\mathbf{B}(s)$. Since it is perpendicular to the tangent vector $\mathbf{T}(s)$, its equation in terms of an arbitrary point $\mathbf{x} = (X, Y, Z)$ on the plane is given by $(\mathbf{x} - \alpha(s)) \cdot \mathbf{T}(s) = 0$.
- The **osculating plane** (literally, the "kissing plane") is the plane spanned by the tangent vector $\mathbf{T}(s)$ and the principal normal vector $\mathbf{N}(s)$. This plane represents the flat surface that is locally closest to containing the curve. Its normal vector is the binormal $\mathbf{B}(s)$, and its equation is $(\mathbf{x} - \alpha(s)) \cdot \mathbf{B}(s) = 0$.
- The **rectifying plane** is the plane spanned by the tangent vector $\mathbf{T}(s)$ and the binormal vector $\mathbf{B}(s)$. It is perpendicular to the principal normal vector $\mathbf{N}(s)$. Its equation is given by $(\mathbf{x} - \alpha(s)) \cdot \mathbf{N}(s) = 0$.

Remark 1.8.12. For a curve $\alpha(t)$ with a general parameter t, we can formulate these equations without converting to arc length. Letting $\dot{\alpha}$ and $\ddot{\alpha}$ denote derivatives with respect to t:

- The normal vector of the **normal plane** is parallel to $\dot{\alpha}(t)$.
- The normal vector of the **osculating plane** is parallel to $\dot{\alpha}(t) \times \ddot{\alpha}(t)$.
- The normal vector of the **rectifying plane** is parallel to $(\dot{\alpha}(t) \times \ddot{\alpha}(t)) \times \dot{\alpha}(t)$.

Example 1.8.13 (The Circular Helix)**.** Let $\mathbf{r}(t) = (a\cos t,\ a\sin t,\ bt)$, with $a > 0$ and $b \neq 0$. Derive the Frenet frame $(\mathbf{T}, \mathbf{N}, \mathbf{B})$ and the equations of the three fundamental planes at an arbitrary point of the curve.

Solution. The derivative is $\mathbf{r}'(t) = (-a\sin t,\ a\cos t,\ b)$, and the speed is constant:

$$\|\mathbf{r}'(t)\| = \sqrt{a^2\sin^2 t + a^2\cos^2 t + b^2} = \sqrt{a^2 + b^2} =: c.$$

Hence the unit tangent vector is $\mathbf{T}(t) = \frac{1}{c}(-a\sin t,\ a\cos t,\ b)$.

Differentiating $\mathbf{T}$ gives $\mathbf{T}'(t) = \frac{1}{c}(-a\cos t,\ -a\sin t,\ 0)$, so

$$\|\mathbf{T}'(t)\| = \frac{a}{c}.$$

Therefore the principal normal is

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = (-\cos t,\ -\sin t,\ 0).$$

The binormal vector is obtained from the cross product:

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\frac{a}{c}\sin t & \frac{a}{c}\cos t & \frac{b}{c} \\ -\cos t & -\sin t & 0 \end{vmatrix}.$$

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Expanding the determinant yields $\mathbf{B}(t) = \frac{1}{c}(b\sin t,\ -b\cos t,\ a)$.

We now determine the fundamental planes at the point

$$P = \mathbf{r}(t) = (a\cos t,\ a\sin t,\ bt).$$

Let $\mathbf{X} = (x, y, z)$.

Osculating plane. Its normal is $\mathbf{B}(t)$, hence

$$\mathbf{B}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0.$$

Multiplying by c and simplifying gives $b\sin t\,(x - a\cos t) - b\cos t\,(y - a\sin t) + a(z - bt) = 0$.

Normal plane. Its normal is $\mathbf{T}(t)$, so

$$\mathbf{T}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0,$$

which expands to

$$-a\sin t\,(x - a\cos t) + a\cos t\,(y - a\sin t) + b(z - bt) = 0.$$

Rectifying plane. Its normal is $\mathbf{N}(t)$, thus

$$\mathbf{N}(t) \cdot (\mathbf{X} - \mathbf{r}(t)) = 0.$$

Substituting $\mathbf{N}(t)$ and $\mathbf{r}(t)$ gives $-\cos t\,(x - a\cos t) - \sin t\,(y - a\sin t) = 0$, which simplifies to

$$\cos t\, x + \sin t\, y = a.$$

Thus the Frenet frame and all three fundamental planes are explicitly determined for every point of the helix. ■

Example 1.8.14 (The Twisted Cubic)**.** Consider the twisted cubic $\mathbf{r}(t) = (t,\ t^2,\ t^3)$. Determine the Frenet frame and the fundamental planes at the point corresponding to $t = 0$.

Solution. The curve is regular since $\mathbf{r}'(t) = (1, 2t, 3t^2) \neq \mathbf{0}$ for all t. At $t = 0$, the point is $P = \mathbf{r}(0) = (0, 0, 0)$.

The derivative at $t = 0$ is $\mathbf{r}'(0) = (1, 0, 0), \qquad \|\mathbf{r}'(0)\| = 1$, so the unit tangent vector is $\mathbf{T}(0) = (1, 0, 0)$.

To find the principal normal, we compute $\mathbf{T}'(0)$. Since the speed $\|\mathbf{r}'(t)\|$ equals 1 at $t = 0$, the derivative of the denominator in the definition of $\mathbf{T}$ vanishes at that point; hence $\mathbf{T}'(0) = \mathbf{r}''(0)$. Now

$$\mathbf{r}''(t) = (0, 2, 6t), \qquad \mathbf{r}''(0) = (0, 2, 0),$$

so

$$\|\mathbf{T}'(0)\| = 2.$$

Therefore

$$\mathbf{N}(0) = \frac{\mathbf{T}'(0)}{\|\mathbf{T}'(0)\|} = \frac{(0, 2, 0)}{2} = (0, 1, 0).$$

The binormal vector is the cross product:

$$\mathbf{B}(0) = \mathbf{T}(0) \times \mathbf{N}(0) = (1, 0, 0) \times (0, 1, 0) = (0, 0, 1).$$

We now write the equations of the three fundamental planes at $P = (0, 0, 0)$. Let $\mathbf{X} = (x, y, z)$.

Osculating plane. Since the normal is $\mathbf{B}(0) = (0, 0, 1)$, we have

$$(0, 0, 1) \cdot (x, y, z) = 0 \quad \Longrightarrow \quad z = 0.$$

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Normal plane. The normal is $\mathbf{T}(0) = (1,0,0)$, so

$$(1,0,0)\cdot(x,y,z) = 0 \quad \Longrightarrow \quad x = 0.$$

Rectifying plane. The normal is $\mathbf{N}(0) = (0,1,0)$, hence

$$(0,1,0)\cdot(x,y,z) = 0 \quad \Longrightarrow \quad y = 0.$$

Thus, at the origin, the osculating plane is the xy-plane, the normal plane is the yz-plane, and the rectifying plane is the xz-plane. These are precisely the three coordinate planes. ■

The following theorem gives the curvature of space curves in any regular parametrization.

Theorem 1.8.15. *Let $\alpha(t)$ be a regular curve in $\mathbb{R}^3$ with nowhere vanishing curvature. Then, denoting d/dt by a dot, its torsion is given by*

$$\tau = \frac{(\dot\alpha \times \ddot\alpha)\cdot \dddot\alpha}{|\dot\alpha \times \ddot\alpha|^2}$$

Proof. **Case 1:** We first treat the case in which α is unit-speed. Using Frenet Frame,

$$\tau = -\mathbf{N}\cdot\dot{\mathbf{B}} = -\mathbf{N}\cdot(\mathbf{T}\times\mathbf{N})^{\cdot} = -\mathbf{N}\cdot(\dot{\mathbf{T}}\times\mathbf{N} + \mathbf{T}\times\dot{\mathbf{N}}) = -\mathbf{N}\cdot(\mathbf{T}\times\dot{\mathbf{N}}).$$

Now, $\mathbf{N} = \frac{1}{\kappa}\dot{\mathbf{T}} = \frac{1}{\kappa}\ddot\alpha$, so

$$\begin{aligned}
\tau &= -\frac{1}{\kappa}\ddot\alpha\cdot\left(\dot\alpha\times\frac{d}{dt}\left(\frac{1}{\kappa}\ddot\alpha\right)\right)\\
&= -\frac{1}{\kappa}\ddot\alpha\cdot\left(\dot\alpha\times\left(\frac{1}{\kappa}\dddot\alpha - \frac{\dot\kappa}{\kappa^2}\ddot\alpha\right)\right)\\
&= \frac{1}{\kappa^2}\dddot\alpha\cdot(\dot\alpha\times\ddot\alpha),
\end{aligned}$$

since $\ddot\alpha\cdot(\dot\alpha\times\ddot\alpha) = 0$ and $\ddot\alpha\cdot(\dot\alpha\times\dddot\alpha) = -\dddot\alpha\cdot(\dot\alpha\times\ddot\alpha)$. Again since α is unit-speed, $\dot\alpha$ and $\ddot\alpha$ are perpendicular, so

$$|\dot\alpha\times\ddot\alpha| = |\dot\alpha||\ddot\alpha| = |\ddot\alpha| = \kappa.$$

Case 2: In the general case, let s be arc length along α and denote d/ds by a dash. Then,

$$\begin{aligned}
\frac{d\alpha}{dt} = \dot\alpha &= \frac{d\alpha}{ds}\frac{ds}{dt} = \frac{ds}{dt}\alpha',\\
\ddot\alpha &= \left(\frac{ds}{dt}\right)^2\alpha'' + \frac{d^2s}{dt^2}\alpha',\\
\dddot\alpha &= \left(\frac{ds}{dt}\right)^3\alpha''' + 3\frac{ds}{dt}\frac{d^2s}{dt^2}\alpha'' + \frac{d^3s}{dt^3}\alpha'.
\end{aligned}$$

Hence,

$$\begin{aligned}
\dot\alpha\times\ddot\alpha &= \left(\frac{ds}{dt}\right)^3\alpha'\times\alpha'',\\
\dddot\alpha\cdot(\dot\alpha\times\ddot\alpha) &= \left(\frac{ds}{dt}\right)^6\alpha'''\cdot(\alpha'\times\alpha''),
\end{aligned}$$

and so

$$\frac{\dddot\alpha\cdot(\dot\alpha\times\ddot\alpha)}{|\dot\alpha\times\ddot\alpha|^2} = \frac{\alpha'''\cdot(\alpha'\times\alpha'')}{|\alpha'\times\alpha''|^2}.$$

■

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Theorem 1.8.16. *A regular unit-speed curve $\alpha : I \to \mathbb{R}^3$ is a straight line if and only if its curvature $\kappa(s) = 0$ for all $s \in I$.*

Proof. Let $\mathbf{T}(s) = \alpha'(s)$. By definition, $\kappa(s) = |\mathbf{T}'(s)| = |\alpha''(s)|$.

If α is a straight line, it can be parameterized as $\alpha(s) = \mathbf{a}s + \mathbf{b}$ for constant vectors $\mathbf{a}, \mathbf{b}$ with $|\mathbf{a}| = 1$. Then $\alpha''(s) = \mathbf{0}$, hence $\kappa(s) = |\mathbf{0}| = 0$.

If $\kappa(s) = 0$ for all $s \in I$, then $|\alpha''(s)| = 0 \implies \alpha''(s) = \mathbf{0}$. Integrating twice with respect to s yields $\alpha(s) = \mathbf{a}s + \mathbf{b}$ for some constant vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$, which is a straight line. ■

Theorem 1.8.17. *A regular unit-speed curve $\alpha : I \to \mathbb{R}^3$ with $\kappa(s) > 0$ is a plane curve if and only if its torsion $\tau(s) = 0$ for all $s \in I$.*

Proof. Suppose $\alpha(s)$ is a plane curve. Then there exists a constant unit vector $\mathbf{k}$ and a fixed point P_0 such that the curve satisfies the plane equation:

$$(\alpha(s) - P_0) \cdot \mathbf{k} = 0.$$

Differentiating with respect to s yields $\alpha'(s) \cdot \mathbf{k} = 0$, which implies:

$$\mathbf{T}(s) \cdot \mathbf{k} = 0$$

Differentiating a second time gives $\mathbf{T}'(s) \cdot \mathbf{k} = 0$. Applying the Frenet relation $\mathbf{T}' = \kappa\mathbf{N}$:

$$\kappa(s)\mathbf{N}(s) \cdot \mathbf{k} = 0$$

Given $\kappa > 0$, it follows that $\mathbf{N}(s) \cdot \mathbf{k} = 0$. Since $\mathbf{k}$ is a constant unit vector orthogonal to both $\mathbf{T}$ and $\mathbf{N}$, the binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ must be:

$$\mathbf{B}(s) = \pm\mathbf{k}$$

Because $\mathbf{k}$ is constant, its derivative is zero, meaning $\mathbf{B}'(s) = \mathbf{0}$. Finally, using the Frenet–Serret formula $\mathbf{B}' = -\tau\mathbf{N}$, we have

$$-\tau(s)\mathbf{N}(s) = \mathbf{0} \implies \tau(s) = 0 \text{ for all } s.$$

Suppose $\tau = 0$ for all s. From $\mathbf{B}' = -\tau\mathbf{N}$, we have $\mathbf{B}' = \mathbf{0}$, which implies $\mathbf{B}$ is a constant vector, $\mathbf{B}_0$. Consider the scalar function $f(s) = (\alpha(s) - \alpha(s_0)) \cdot \mathbf{B}_0$. Its derivative is

$$f'(s) = \alpha'(s) \cdot \mathbf{B}_0 = \mathbf{T}(s) \cdot \mathbf{B}_0 = 0$$

Thus $f(s)$ is a constant. Since $f(s_0) = 0$, we conclude that $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{B}_0 = 0$ for all s, which is the equation of a plane. ■

2nd Proof. Suppose the image of α lies in a plane with constant unit normal $\mathbf{u}$, meaning $(\alpha(s) - \mathbf{x}_0) \cdot \mathbf{u} = 0$. Differentiating once yields $\mathbf{T}(s) \cdot \mathbf{u} = 0$. Differentiating again yields $\mathbf{T}'(s) \cdot \mathbf{u} = \kappa(s)\mathbf{N}(s) \cdot \mathbf{u} = 0$. Since $\kappa(s) > 0$, we have $\mathbf{N}(s) \cdot \mathbf{u} = 0$. Because $\mathbf{u}$ is orthogonal to both $\mathbf{T}$ and $\mathbf{N}$, the binormal vector $\mathbf{B} = \mathbf{T} \times \mathbf{N}$ must be parallel to $\mathbf{u}$. Thus, $\mathbf{B}(s) = \pm\mathbf{u}$ is constant, meaning $\mathbf{B}'(s) = \mathbf{0}$. By the Frenet–Serret formula $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s)$, we conclude $\tau(s) = 0$ since $\mathbf{N}(s) \neq \mathbf{0}$.

If $\tau(s) = 0$, then $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s) = \mathbf{0}$, so $\mathbf{B}(s)$ is a constant vector $\mathbf{b}$. Consider the derivative:

$$\frac{d}{ds}\left[(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b}\right] = \alpha'(s) \cdot \mathbf{b} = \mathbf{T}(s) \cdot \mathbf{B}(s) = 0$$

Integrating this expression yields $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b} = C$ for some constant C. Evaluating at $s = s_0$ gives $C = 0$. Therefore, $(\alpha(s) - \alpha(s_0)) \cdot \mathbf{b} = 0$ for all $s \in I$, which is the equation of a plane. ■

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1.9 Characterization of Special Space Curves

In the differential geometry of space curves, the intrinsic properties—curvature (κ) and torsion (τ)—completely determine the shape of a curve up to a rigid motion. Certain geometric constraints, such as lying on a cylinder or a sphere, manifest as specific algebraic or differential relationships between κ and τ.

1.9.1 General Helices

A **general helix** is a curve whose tangent vector makes a constant angle with a fixed direction in space. (See Figure 11 in the Atlas.)

Theorem 1.9.1 (Lancret's Theorem). *A regular curve $\alpha(s)$ with $\kappa(s) > 0$ is a general helix if and only if the ratio of its torsion to its curvature is constant:*

$$\frac{\tau(s)}{\kappa(s)} = c, \quad \text{for some constant } c.$$

Proof. Suppose α is a general helix. By definition, there exists a constant unit vector $\mathbf{u}$ and a constant angle θ such that:

$$\langle T, \mathbf{u} \rangle = \cos\theta$$

Differentiating with respect to arc length s and using the Frenet formula $T' = \kappa N$:

$$\langle \kappa N, \mathbf{u} \rangle = 0$$

Since $\kappa > 0$, we must have $\langle N, \mathbf{u} \rangle = 0$. This implies that the fixed vector $\mathbf{u}$ lies in the plane spanned by T and B (the rectifying plane). We can write:

$$\mathbf{u} = \cos\theta T + \sin\theta B$$

Differentiating this constant vector $\mathbf{u}$ again:

$$\mathbf{0} = \cos\theta T' + \sin\theta B' = \cos\theta(\kappa N) + \sin\theta(-\tau N) = (\kappa\cos\theta - \tau\sin\theta)N$$

Since $N \neq \mathbf{0}$, the coefficient must be zero:

$$\kappa\cos\theta = \tau\sin\theta \implies \frac{\tau}{\kappa} = \cot\theta = \text{constant}.$$

Conversely, suppose $\kappa/\tau = c$, where c is a constant. Then $\kappa = c\tau$. Multiplying by $\mathbf{N}$:

$$\kappa\mathbf{N} = c\tau\mathbf{N}$$

This gives

$$\mathbf{T}' = -c\mathbf{B}' \quad (\text{since } \mathbf{T}' = \kappa\mathbf{N} \text{ and } \mathbf{B}' = -\tau\mathbf{N})$$

or,

$$\mathbf{T}' + c\mathbf{B}' = 0.$$

This implies

$$\frac{d}{ds}(\mathbf{T} + c\mathbf{B}) = 0,$$

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which further implies $\mathbf{T} + c\mathbf{B} = \mathbf{a}$, where $\mathbf{a}$ is a constant vector. Taking the dot product of both sides with $\mathbf{T}$:

$$\begin{aligned} \mathbf{T} \cdot (\mathbf{T} + c\mathbf{B}) &= \mathbf{a} \cdot \mathbf{T} \\ 1 + 0 &= |\mathbf{a}||\mathbf{T}| \cos\theta \\ 1 &= |\mathbf{a}| \cos\theta \\ \implies \cos\theta &= \frac{1}{|\mathbf{a}|} = \text{constant} \end{aligned}$$

where θ is the angle between $\mathbf{T}$ and $\mathbf{a}$. Since the tangent vector $\mathbf{T}$ of the curve α makes a constant angle with the fixed vector $\mathbf{a}$, the curve α is a helix. ■

A curve is called **spherical** if it lies entirely on the surface of a sphere of radius R centered at some point $\mathbf{C}$.

Theorem 1.9.2 (Characterization of Spherical Curves)**.** *(See Figure 12 in the Atlas.) A curve $\alpha(s)$ with $\kappa > 0$ and $\tau \neq 0$ lies on a sphere if and only if:*

$$\frac{\tau}{\kappa} + \frac{d}{ds}\left(\frac{1}{\tau}\frac{d}{ds}\left(\frac{1}{\kappa}\right)\right) = 0$$

Alternatively, in terms of the radius of curvature $\rho = 1/\kappa$ and the radius of torsion $\sigma = 1/\tau$, the condition is $\rho^2 + (\sigma\rho')^2 = R^2$ (a constant).

Proof. Suppose $\alpha(s)$ lies on a sphere of radius R centered at $\mathbf{C}$. Then $\|\alpha(s) - \mathbf{C}\|^2 = R^2$. Differentiating three times with respect to s:

1. $2\langle \alpha - \mathbf{C}, T\rangle = 0 \implies \alpha - \mathbf{C}$ is in the NB-plane.
2. $\langle T, T\rangle + \langle \alpha - \mathbf{C}, \kappa N\rangle = 0 \implies 1 + \kappa\langle \alpha - \mathbf{C}, N\rangle = 0 \implies \langle \alpha - \mathbf{C}, N\rangle = -1/\kappa$.
3. Differentiating $\langle \alpha - \mathbf{C}, N\rangle = -1/\kappa$:

$$\langle T, N\rangle + \langle \alpha - \mathbf{C}, -\kappa T + \tau B\rangle = (1/\kappa)'$$

$$0 + 0 + \tau\langle \alpha - \mathbf{C}, B\rangle = (1/\kappa)' \implies \langle \alpha - \mathbf{C}, B\rangle = \frac{1}{\tau}(1/\kappa)'$$

Thus, we can express the vector from the center to the curve as:

$$\alpha - \mathbf{C} = 0T - \frac{1}{\kappa}N + \frac{1}{\tau}\left(\frac{1}{\kappa}\right)' B$$

Since the distance $\|\alpha - \mathbf{C}\|$ is the constant radius R, we have

$$\left(\frac{1}{\kappa}\right)^2 + \left[\frac{1}{\tau}\left(\frac{1}{\kappa}\right)'\right]^2 = R^2$$

Differentiating this constant expression with respect to s yields the differential equation mentioned in the theorem. ■

Example 1.9.3 (The Circular Helix)**.** Show that the circular helix $\alpha(t) = (a\cos t, a\sin t, bt)$ is a general helix using Lancret's Theorem.

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Solution. From previous calculations for the circular helix, we found:

$$\kappa = \frac{a}{a^2+b^2} \quad \text{and} \quad \tau = \frac{b}{a^2+b^2}$$

Both κ and τ are constants. Therefore, their ratio is

$$\frac{\tau}{\kappa} = \frac{b/(a^2+b^2)}{a/(a^2+b^2)} = \frac{b}{a}$$

Since a and b are constants, the ratio τ/κ is constant. By Lancret's Theorem, the circular helix is a general helix. (Note: Because κ and τ are both constant, it is a very specific type of helix that lies on a circular cylinder). ■

Example 1.9.4 (A Non-Spherical Curve)**.** Determine if a curve with constant curvature $\kappa = 1$ and constant torsion $\tau = 1$ can lie on a sphere.

Solution. Let us check the spherical condition: $\rho^2 + (\sigma\rho')^2 = R^2$. Given $\kappa = 1$, we have $\rho = 1/\kappa = 1$. The derivative is $\rho' = \frac{d}{ds}(1) = 0$. Substituting into the formula:

$$(1)^2 + (\sigma \cdot 0)^2 = 1$$

This results in $1 = R^2$, which is a constant. This suggests the curve *could* be spherical. However, we must also check the second derivative condition:

$$\frac{\tau}{\kappa} + \frac{d}{ds}(\sigma\rho') = 0$$

$$\frac{1}{1} + \frac{d}{ds}(1 \cdot 0) = 1 + 0 = 1$$

Since $1 \neq 0$, the differential equation for a spherical curve is not satisfied. A curve with constant κ and τ (which is a circular helix) does *not* lie on a sphere. It lies on a cylinder. ■

1.10 Involutes and Evolutes

The geometric relationship between a curve and its "unwrapped" counterpart is described through the concepts of involutes and evolutes. Originally developed by Christiaan Huygens for the design of isochronous pendulum clocks, these constructions remain a beautiful application of the Frenet–Serret frame.

1.10.1 The Involute of a Curve

Intuitively, an involute is the path traced by the end of a taut string as it is unwrapped from a curve.

Definition 1.10.1 (Involute)**.** (See Figure 14 in the Atlas.) Let $\alpha : I \to \mathbb{R}^2$ be a unit-speed planar curve. The curve $\bar{\alpha}$ defined by

$$\bar{\alpha}(s) = \alpha(s) + (c - s)\mathbf{T}(s)$$

for a constant c, is called an **involute** of α.

The term $(c - s)$ represents the length of the "unwrapped" portion of the string at arc length s. The following theorem establishes the fundamental geometric link between the two curves.

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Theorem 1.10.2. *The tangent line to the original curve α at any point s is the normal line to its involute $\bar{\alpha}$ at the corresponding point.*

Proof. Differentiating the expression for $\bar{\alpha}(s)$ with respect to s, we obtain

$$\bar{\alpha}'(s) = \alpha'(s) - \mathbf{T}(s) + (c - s)\mathbf{T}'(s)$$

Since α is unit-speed, $\alpha' = \mathbf{T}$. Applying the Frenet formula $\mathbf{T}' = \kappa\mathbf{N}$, we have

$$\bar{\alpha}'(s) = \mathbf{T}(s) - \mathbf{T}(s) + (c - s)\kappa\mathbf{N}(s) = (c - s)\kappa\mathbf{N}(s)$$

This shows that the velocity vector $\bar{\alpha}'(s)$ is parallel to the normal vector $\mathbf{N}(s)$ of the original curve. Consequently, the tangent of the involute is perpendicular to the tangent of the original curve: $\bar{\alpha}'(s) \cdot \mathbf{T}(s) = 0$. ■

1.10.2 The Evolute of a Curve

While the involute is formed by unwrapping a string, the evolute is the "source" curve from which such a string would be unwrapped.

Definition 1.10.3 (Evolute)**.** (See Figure 13 in the Atlas.) Let $\alpha(s)$ be a regular planar curve with curvature $\kappa(s) \neq 0$. The **evolute** $\beta(s)$ of α is the locus of the centers of curvature of α. It is given by $\beta(s) = \alpha(s) + \rho(s)\mathbf{N}(s)$. where $\rho(s) = 1/\kappa(s)$ is the radius of curvature.

There is a reciprocal relationship between these two constructions: a curve β is the evolute of α if and only if α is an involute of β.

Theorem 1.10.4 (Tangents of the Evolute)**.** *The tangent vector to the evolute $\beta(s)$ is always parallel to the principal normal vector $\mathbf{N}(s)$ of the original curve.*

Proof. Differentiating $\beta(s) = \alpha(s) + \rho(s)\mathbf{N}(s)$ with respect to arc length s:

$$\beta'(s) = \alpha'(s) + \rho'(s)\mathbf{N}(s) + \rho(s)\mathbf{N}'(s)$$

Substituting the Frenet–Serret relations $\alpha' = \mathbf{T}$ and $\mathbf{N}' = -\kappa\mathbf{T}$ (noting that $\tau = 0$ for planar curves):

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) + \rho(s)(-\kappa(s)\mathbf{T}(s))$$

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) - (\rho(s)\kappa(s))\mathbf{T}(s)$$

Using the property $\rho = 1/\kappa$, the tangent terms cancel:

$$\beta'(s) = \mathbf{T}(s) + \rho'(s)\mathbf{N}(s) - \mathbf{T}(s) = \rho'(s)\mathbf{N}(s)$$

Thus, the velocity of the evolute $\beta'(s)$ is a scalar multiple of the normal vector $\mathbf{N}(s)$. This confirms that the tangent to the evolute is precisely the normal to the original curve. ■

Summary

- To find an **Involute**, we move away from the curve along its **Tangent** vector.
- To find the **Evolute**, we move away from the curve along its **Normal** vector.

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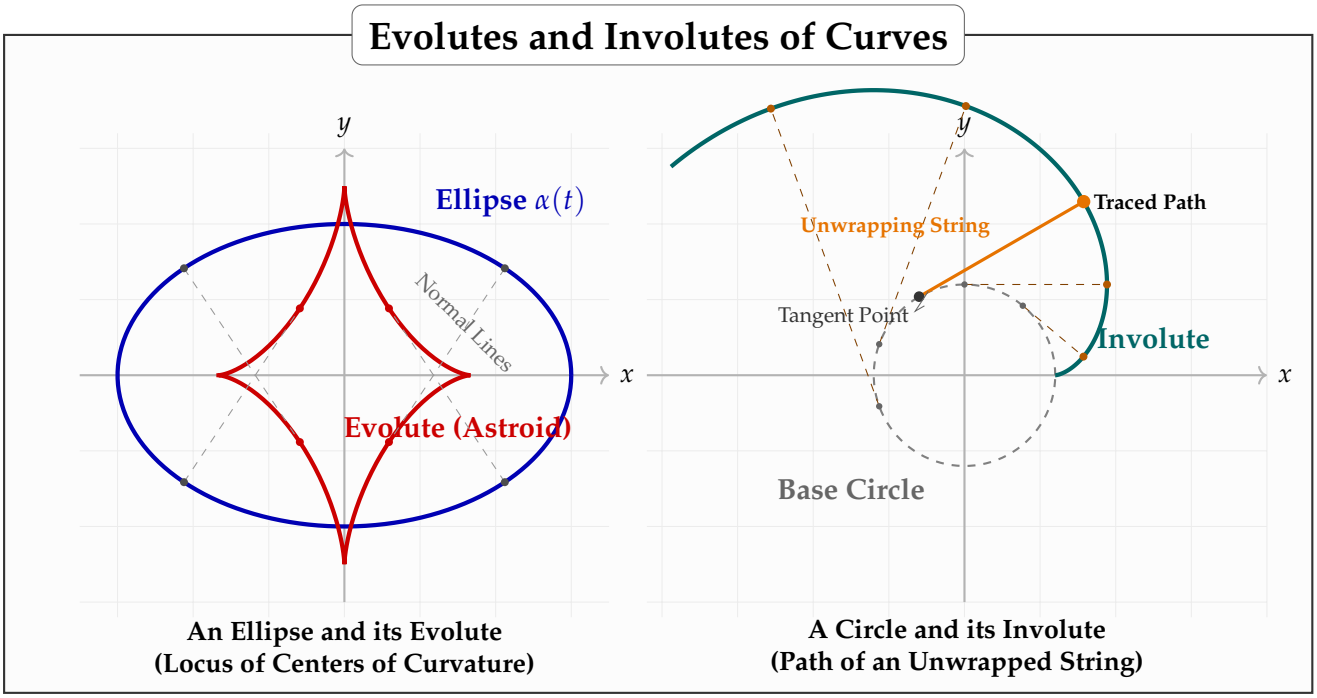


Figure 1.3: Caption

1.11 The Fundamental Theorem of Space Curves

In the derivation of the *Fundamental Theorem of Space Curves*, we encounter a system of linear differential equations of the form $\mathbf{F}'(s) = \mathbf{A}(s)\mathbf{F}(s)$. For such linear systems, if the coefficients (the curvature κ and torsion τ) are continuous on an interval $[0, L]$, the Lipschitz condition is automatically satisfied on any compact sub-interval. Consequently, the Picard–Lindelöf Theorem guarantees that once an initial orientation (frame) is chosen, the entire geometry of the curve is uniquely determined for all s. The Picard–Lindelöf Theorem provides the sufficient conditions under which a first-order initial value problem (IVP) is guaranteed to have a unique solution in a local neighborhood of the initial point.
Theorem: [Picard–Lindelöf] Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be an open set, and consider the initial value problem:

$$\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y}), \quad \mathbf{y}(t_0) = \mathbf{y}_0$$

Suppose that the function $\mathbf{f}(t, \mathbf{y})$ satisfies the following two conditions:

1. $\mathbf{f}$ is continuous in t and $\mathbf{y}$ on a closed rectangle $R \subset D$ centered at $(t_0, \mathbf{y}_0)$.
2. $\mathbf{f}$ is **Lipschitz continuous** in $\mathbf{y}$ on R. That is, there exists a constant $L \geq 0$ such that for all $(t, \mathbf{y}_1)$ and $(t, \mathbf{y}_2)$ in R:
$$\|\mathbf{f}(t, \mathbf{y}_1) - \mathbf{f}(t, \mathbf{y}_2)\| \leq L \|\mathbf{y}_1 - \mathbf{y}_2\|$$

Then, there exists some $\epsilon > 0$ such that the initial value problem has a **unique solution** $\mathbf{y}(t)$ on the interval $[t_0 - \epsilon, t_0 + \epsilon]$. The local geometry of a space curve is completely characterized by two scalar functions: its curvature $\kappa(s)$ and its torsion $\tau(s)$. The following theorem, often referred to as the existence and uniqueness theorem for space curves, asserts that these functions serve as a complete set of invariants.

Theorem 1.11.1 (Picard–Lindelöf Theorem). *Consider the initial value problem*

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0.$$

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Suppose that f is continuous in a neighborhood of (t_0, y_0) and locally Lipschitz continuous with respect to y. Then there exists an interval containing t_0 on which the initial value problem has a unique solution.

The theorem also applies to a system of differential equations $Y'(t) = F(t, Y(t))$, with $Y(t_0) = Y_0$, provided that F is continuous and locally Lipschitz continuous with respect to Y. In particular, a linear system $Y' = A(t)Y$ with continuous coefficient matrix $A(t)$ has a unique solution on the entire interval on which A is defined.

Theorem 1.11.2 (Fundamental Theorem of Space Curves). *Let $\kappa(s)$ and $\tau(s)$ be continuous functions on $I = [0, L]$, with $\kappa(s) > 0$ for every $s \in I$. Then there exists a unit-speed curve $\alpha : I \to \mathbb{R}^3$ having curvature $\kappa(s)$ and torsion $\tau(s)$. Moreover, any two such curves differ by an orientation-preserving rigid motion of $\mathbb{R}^3$.*

Proof. We first prove existence. Let $\kappa : I \to (0, \infty)$ and $\tau : I \to \mathbb{R}$ be the prescribed continuous functions. Consider the Frenet–Serret system

$$T' = \kappa N, \qquad N' = -\kappa T + \tau B, \qquad B' = -\tau N.$$

This is a linear system of nine first-order differential equations for the components of the three unknown vector-valued functions T, N and B. It can be written in matrix form as

$$\frac{d}{ds}\begin{pmatrix} T \\ N \\ B \end{pmatrix} = A(s)\begin{pmatrix} T \\ N \\ B \end{pmatrix}, \qquad A(s) = \begin{pmatrix} 0 & \kappa(s) & 0 \\ -\kappa(s) & 0 & \tau(s) \\ 0 & -\tau(s) & 0 \end{pmatrix}.$$

Since κ and τ are continuous, the matrix $A(s)$ is continuous on I. Moreover, the right-hand side is linear in the unknown vectors and hence is locally Lipschitz continuous in them. Therefore, the hypotheses of the Picard–Lindelöf theorem are satisfied.

Choose a positively oriented orthonormal basis $\{T_0, N_0, B_0\}$ of $\mathbb{R}^3$ and impose the initial conditions

$$T(0) = T_0, \qquad N(0) = N_0, \qquad B(0) = B_0.$$

By the Picard–Lindelöf theorem, the system has a unique solution in a neighborhood of $s = 0$. Since it is a linear system with continuous coefficients on the compact interval I, the solution extends uniquely throughout I.

We now show that the vectors T, N and B remain orthonormal. Let $M(s)$ be the matrix whose rows are $T(s), N(s)$ and $B(s)$. Then $M' = AM$. Since $A^T = -A$, differentiating MM^T gives

$$(MM^T)' = AMM^T + MM^T A^T = AMM^T - MM^T A.$$

Thus, the matrix $Q(s) = MM^T$ satisfies the linear differential equation $Q' = AQ - QA$. Since the initial frame is orthonormal, $Q(0) = I_3$. The constant matrix $Q(s) = I_3$ also satisfies this equation. By uniqueness of solutions to linear differential equations, we obtain $Q(s) = I_3$ for every $s \in I$. Hence T, N and B remain mutually perpendicular unit vectors.

Furthermore, $M(s)$ is continuous and orthogonal, so $\det M(s) \in \{1, -1\}$. Since the initial frame is positively oriented, $\det M(0) = 1$. By continuity, $\det M(s) = 1$ throughout I. Therefore, $B = T \times N$.

Choose an arbitrary point $P_0 \in \mathbb{R}^3$ and define

$$\alpha(s) = P_0 + \int_0^s T(u)\, du.$$

By the fundamental theorem of calculus, $\alpha'(s) = T(s)$. Since $\|T(s)\| = 1$, we have $\|\alpha'(s)\| = 1$, so α is parametrized by arc length. Differentiating once more and using the first Frenet–Serret equation gives

$$\alpha''(s) = T'(s) = \kappa(s)N(s).$$

Consequently, the curvature of α is

$$\left\|\alpha''(s)\right\| = \kappa(s)\,\|N(s)\| = \kappa(s).$$

Since $\kappa(s) > 0$, the principal unit normal of α is precisely $N(s)$, and its binormal is $B(s) = T(s) \times N(s)$. The third Frenet–Serret equation gives $B' = -\tau N$, so

$$-\left\langle B'(s), N(s)\right\rangle = \tau(s).$$

Thus, the torsion of α is the prescribed function $\tau(s)$, proving existence.

We next prove uniqueness. Suppose that α and $\widetilde{\alpha}$ are two unit-speed curves having the same curvature κ and torsion τ. By applying a suitable orientation-preserving rigid motion to $\widetilde{\alpha}$, we may assume that

$$\alpha(0) = \widetilde{\alpha}(0)$$

and that their Frenet frames agree initially:

$$T(0) = \widetilde{T}(0), \qquad N(0) = \widetilde{N}(0), \qquad B(0) = \widetilde{B}(0).$$

Both frames satisfy the same Frenet–Serret system with the same initial conditions. By the uniqueness part of the Picard–Lindelöf theorem, together with uniqueness of continuation for linear systems, their solutions agree throughout I. In particular,

$$T(s) = \widetilde{T}(s) \qquad \text{for every } s \in I.$$

Since $\alpha'(s) = T(s)$ and $\widetilde{\alpha}'(s) = \widetilde{T}(s)$, it follows that $\alpha'(s) = \widetilde{\alpha}'(s)$. Integrating from 0 to s and using $\alpha(0) = \widetilde{\alpha}(0)$, we obtain $\alpha(s) = \widetilde{\alpha}(s)$ for every $s \in I$.

Hence any two unit-speed curves with the same curvature and torsion differ only by an orientation-preserving rigid motion of $\mathbb{R}^3$. ■

1.12 Global Properties of Curves

While local differential geometry focuses on the behavior of a curve in the neighborhood of a point (such as curvature and torsion), global differential geometry examines the curve as a whole. Here, we consider properties that depend on the entire trace of the curve and its topological nature.

A curve that returns to its starting point and does not intersect itself is a fundamental object in global geometry.

Definition 1.12.1 (Simple Closed Curve)**.** A parametrized curve $\alpha : [a, b] \to \mathbb{R}^n$ is said to be **closed** if $\alpha(a) = \alpha(b)$ and all its derivatives match at the endpoints; that is, $\alpha^{(k)}(a) = \alpha^{(k)}(b)$ for all $k \geq 1$. A closed curve is **simple** if it is injective on the half-open interval $[a, b)$.

Remark 1.12.2 (The Jordan Curve Theorem). A famous result by Camille Jordan states that every simple closed curve in the plane divides $\mathbb{R}^2$ into exactly two disjoint regions: a bounded "interior" and an unbounded "exterior." Although intuitively obvious, the proof is famously non-trivial and forms a cornerstone of topology.

1.12.1 The Isoperimetric Inequality

Green's Theorem establishes a fundamental relationship between a line integral around a simple closed curve and a double integral over the plane region enclosed by the curve.

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Theorem 1.12.3 (Green's Theorem). *Let C be a positively oriented, piecewise smooth, simple closed curve in the plane, and let D be the region bounded by C. If $P(x,y)$ and $Q(x,y)$ have continuous partial derivatives on an open region containing D, then:*

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

In the context of the Isoperimetric Inequality, we are interested in the case where the double integral represents the area A of the region D. By choosing specific functions for P and Q such that $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1$, we obtain several equivalent formulas for the area:

1. Let $P = 0$ and $Q = x$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - 0 = 1$. Thus

$$A = \oint_C x\,dy$$

Proof. Consider Green's Theorem in its general form, which relates a line integral around a simple closed curve C to a double integral over the region D it encloses:

$$\oint_C (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

To derive the specific area formula $A = \oint_C x\,dy$, we define the functions $P(x,y) = 0$ and $Q(x,y) = x$. We then evaluate the required partial derivatives:

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(x) = 1, \quad \frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(0) = 0$$

Substituting these derivatives into the right-hand side of the theorem yields $\iint_D (1-0)\,dA = \iint_D 1\,dA = A$. where A represents the area of the region D. Substituting the definitions of P and Q into the left-hand side line integral gives $\oint_C (0 \cdot dx + x \cdot dy) = \oint_C x\,dy$. Equating the simplified left-hand and right-hand sides, we arrive at the area identity:

$$A = \oint_C x\,dy$$

This demonstrates that the area of a region D can be calculated by integrating the x-coordinate with respect to y along its boundary C. ■

2. Let $P = -y$ and $Q = 0$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0 - (-1) = 1$. Thus

$$A = -\oint_C y\,dx$$

3. Let $P = -\frac{1}{2}y$ and $Q = \frac{1}{2}x$. Then $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{1}{2} - (-\frac{1}{2}) = 1$. Thus

$$A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

One of the oldest problems in geometry is determining which closed curve of a fixed length encloses the maximum area. The answer is provided by the Isoperimetric Inequality.

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Theorem 1.12.4 (Isoperimetric Inequality). *(See Figure 16 in the Atlas.) If L is the length of a simple closed plane curve α and A is the area of the region it encloses, then*

$$L^2 \geq 4\pi A$$

Furthermore, equality holds if and only if the curve is a circle.

Proof. Let $\alpha(s) = (x(s), y(s))$ be a simple closed curve parameterized by arc length $s \in [0, L]$. We begin by placing the curve between two parallel lines, say $x = -r$ and $x = r$, such that the lines are tangent to the curve and the curve lies entirely within the strip they define. This implies that the maximum width of the curve in the x-direction is $2r$.

Consider a circle γ of radius r centered at the origin, which can be represented as $\gamma(s) = (x(s), \bar{y}(s))$. Observe that while $x(s)$ is the same coordinate function as in $\alpha(s)$, the function $\bar{y}(s)$ is defined such that $(x(s), \bar{y}(s))$ lies on the circle, meaning $x(s)^2 + \bar{y}(s)^2 = r^2$. Using Green's Theorem, the area A enclosed by α and the area of the circle can be expressed as

$$A = \int_0^L x(s)y'(s)\,ds \quad \text{and} \quad \pi r^2 = -\int_0^L \bar{y}(s)x'(s)\,ds$$

Adding these two integrals allows us to apply the Cauchy-Schwarz inequality to the integrand. We have

$$A + \pi r^2 = \int_0^L (xy' - \bar{y}x')\,ds = \int_0^L \langle (x, -\bar{y}), (y', x') \rangle\,ds$$

By the Cauchy-Schwarz inequality, $\langle (x, -\bar{y}), (y', x') \rangle \leq \sqrt{x^2 + \bar{y}^2}\sqrt{(y')^2 + (x')^2}$. Since $(x, \bar{y})$ lies on a circle of radius r, we know $\sqrt{x^2 + \bar{y}^2} = r$. Furthermore, since α is a unit-speed curve, $\sqrt{(x')^2 + (y')^2} = 1$. Substituting these values into the integral yields $A + \pi r^2 \leq \int_0^L r \cdot 1\,ds = Lr$. Using the inequality between the arithmetic and geometric means, specifically that $2\sqrt{A \cdot \pi r^2} \leq A + \pi r^2$, we arrive at:

$$2\sqrt{A\pi r^2} \leq Lr$$

Squaring both sides gives $4\pi A r^2 \leq L^2 r^2$, and dividing by r^2 (which is non-zero) results in the desired isoperimetric inequality:

$$4\pi A \leq L^2$$

The condition for equality $L^2 = 4\pi A$ requires that $A = \pi r^2$ and that the Cauchy-Schwarz inequality holds as an equality at every point. This occurs only if the vectors $(x, -\bar{y})$ and (y', x') are proportional. Specifically, for the magnitudes to match, we must have $x = ry'$ and $-\bar{y} = rx'$. Squaring and adding these relations, we find $x^2 + \bar{y}^2 = r^2((y')^2 + (x')^2)$, which simplifies to $r^2 = r^2(1)$. This system of differential equations characterizes a circle of radius r, thus proving that the circle is the unique minimizer of the perimeter for a fixed area. ∎

1.12.2 The Four Vertex Theorem

The curvature of a plane curve measures how rapidly its tangent direction changes as we move along the curve. For a closed curve, the curvature generally varies from point to point, attaining local maxima and minima. The Four Vertex Theorem is a remarkable result which shows that the curvature of a simple closed curve cannot vary arbitrarily: it must have at least four stationary points.

Definition 1.12.5 (Vertex of a Plane Curve). Let $\alpha(s)$ be a smooth regular plane curve parametrized by arc length, with curvature $\kappa(s)$. A point $\alpha(s_0)$ is called a **vertex** of the curve if $\kappa'(s_0) = 0$.

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In particular, every local maximum or minimum of curvature is a vertex. However, a vertex need not always be a local extremum, since the derivative of curvature may also vanish at a stationary point where the curvature continues to increase or decrease.

For example, a non-circular ellipse has exactly four vertices, occurring at the endpoints of its major and minor axes. The Four Vertex Theorem establishes that four is the smallest possible number of vertices for a smooth simple closed curve whose curvature is not constant.

Theorem 1.12.6 (Four Vertex Theorem). *(See Figure 15 in the Atlas.) Every simple closed convex plane curve of class C^3, parametrized regularly, has at least four vertices.*

Proof. Let $\alpha(s) = (x(s), y(s))$, $0 \le s \le L$, be a simple closed convex plane curve parametrized by arc length, where L is its total length. We choose the counterclockwise orientation, so that its signed curvature $\kappa(s)$ is nonnegative. Since the curve is smooth and closed, κ is a differentiable periodic function satisfying $\kappa(0) = \kappa(L)$.

If κ is constant, then $\kappa'(s) = 0$ at every point, and hence every point of the curve is a vertex. Thus, the theorem is immediate in this case. We may therefore assume that κ is not constant.

Since κ is continuous on the compact interval $[0, L]$, it attains a maximum and a minimum. Let these occur at the distinct points $P = \alpha(s_1)$ and $Q = \alpha(s_2)$. As the curve is smooth and closed, the curvature is periodic, and hence

$$\kappa'(s_1) = \kappa'(s_2) = 0.$$

Thus, the curve has at least two vertices.

Suppose, contrary to the theorem, that the curve has fewer than four vertices. It must then have either two or three vertices. In either case, the two arcs joining P and Q contain no additional local maximum or minimum of curvature. Consequently, κ is monotone along each of these arcs, increasing from its minimum to its maximum along one arc and decreasing from its maximum to its minimum along the other.

Let

$$f(s) = ax(s) + by(s) + c,$$

where $ax + by + c = 0$ is the equation of the straight line passing through P and Q.

Since the curve is convex, this line divides it into two arcs lying on opposite sides of the line. In the case under consideration, the curve contains no straight segment, for otherwise its curvature would vanish identically along that segment and every interior point of the segment would be a vertex. Hence the curve is strictly convex, and the line meets it only at P and Q.

It follows that $f(s)$ has one fixed sign on one open arc and the opposite sign on the other. Similarly, $\kappa'(s)$ has opposite signs on the two arcs, apart from possible isolated zeros. Therefore, the product $\kappa'(s)f(s)$ has the same sign throughout the curve.

Moreover, this product is not identically zero, because the curvature is not constant. Consequently,

$$\int_0^L \kappa'(s) f(s)\, ds \neq 0.$$

We shall now show that this integral is actually zero, obtaining a contradiction.

Substituting the expression for $f(s)$, we have

$$\int_0^L \kappa'(s) f(s)\, ds = a \int_0^L \kappa'(s) x(s)\, ds + b \int_0^L \kappa'(s) y(s)\, ds + c \int_0^L \kappa'(s)\, ds.$$

Since the curvature is periodic,

$$\int_0^L \kappa'(s)\, ds = \kappa(L) - \kappa(0) = 0.$$

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To evaluate the remaining integrals, let $\theta(s)$ denote a continuous angle made by the unit tangent vector with the positive x-axis. Since the curve is parametrized by arc length,

$$x'(s) = \cos\theta(s), \qquad y'(s) = \sin\theta(s).$$

By the definition of signed curvature, $\theta'(s) = \kappa(s)$.

Applying integration by parts, we obtain

$$\int_0^L \kappa'(s)x(s)\,ds = [\kappa(s)x(s)]_0^L - \int_0^L \kappa(s)x'(s)\,ds.$$

Because the curve is closed and its curvature is periodic, the boundary term vanishes. Hence

$$\begin{aligned}\int_0^L \kappa'(s)x(s)\,ds &= -\int_0^L \theta'(s)\cos\theta(s)\,ds \\ &= -[\sin\theta(s)]_0^L = 0.\end{aligned}$$

Here we have used the fact that the tangent vectors at $s = 0$ and $s = L$ are equal, so $\sin\theta(L) = \sin\theta(0)$.

Similarly, integration by parts gives

$$\begin{aligned}\int_0^L \kappa'(s)y(s)\,ds &= -\int_0^L \kappa(s)y'(s)\,ds \\ &= -\int_0^L \theta'(s)\sin\theta(s)\,ds \\ &= [\cos\theta(s)]_0^L = 0.\end{aligned}$$

Therefore,

$$\int_0^L \kappa'(s)f(s)\,ds = 0,$$

contradicting the fact that this integral is nonzero.

Thus, a smooth simple closed convex plane curve cannot have only two or three vertices. Consequently, it must have at least four vertices. ■

The theorem illustrates an important connection between the local and global properties of a curve. The condition $\kappa'(s) = 0$ describes the local behaviour of curvature, whereas the requirement that the curve be simple and closed imposes a global restriction. The Four Vertex Theorem shows that these properties are closely related.

The theorem was originally established for convex curves and was subsequently extended to smooth simple closed plane curves without the assumption of convexity.

The Circle as a Special Case

A circle of radius $R > 0$ has constant curvature

$$\kappa(s) = \frac{1}{R}.$$

Consequently, $\kappa'(s) = 0$ for every s. Thus, every point on the circle is a vertex, and the circle has infinitely many vertices.

This does not contradict the Four Vertex Theorem, since the theorem asserts only the existence of at least four vertices. The circle is a special case in which the curvature does not vary at all.

Example 1.12.7 (Vertices of a Circle)**.** Find all the vertices of the circle $\alpha(t) = (R\cos t, R\sin t)$, where $0 \le t \le 2\pi$ and $R > 0$.

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Solution. Differentiating, we obtain

$$\alpha'(t) = (-R\sin t, R\cos t), \qquad \alpha''(t) = (-R\cos t, -R\sin t).$$

The curvature of a regular plane curve is

$$\kappa(t) = \frac{|x'(t)y''(t) - y'(t)x''(t)|}{\big((x'(t))^2 + (y'(t))^2\big)^{3/2}}.$$

Substituting the derivatives gives

$$\kappa(t) = \frac{R^2(\sin^2 t + \cos^2 t)}{(R^2\sin^2 t + R^2\cos^2 t)^{3/2}} = \frac{1}{R}.$$

Thus, the curvature is constant, and hence $d\kappa/dt = 0$ for every t. Since $ds/dt = \|\alpha'(t)\| = R > 0$, the chain rule gives

$$\frac{d\kappa}{ds} = \frac{d\kappa/dt}{ds/dt} = 0.$$

Therefore, every point of the circle is a vertex, and the circle has infinitely many vertices. ■

Vertices of an Ellipse

An ellipse provides a particularly useful illustration of the Four Vertex Theorem. Unlike a circle, its curvature is not constant. It bends more sharply near the endpoints of its major axis and more gently near the endpoints of its minor axis.

Example 1.12.8 (Vertices of an Ellipse)**.** Find all the vertices of the ellipse

$$\alpha(t) = (a\cos t, b\sin t), \qquad 0 \le t \le 2\pi,$$

where $a > b > 0$. Determine which vertices correspond to maximum curvature and which correspond to minimum curvature.

Solution. Differentiating the parametrization, we obtain

$$\alpha'(t) = (-a\sin t, b\cos t), \qquad \alpha''(t) = (-a\cos t, -b\sin t).$$

The curvature is

$$\kappa(t) = \frac{|x'(t)y''(t) - y'(t)x''(t)|}{\big((x'(t))^2 + (y'(t))^2\big)^{3/2}}.$$

Since

$$x'y'' - y'x'' = ab(\sin^2 t + \cos^2 t) = ab,$$

we obtain

$$\kappa(t) = \frac{ab}{(a^2\sin^2 t + b^2\cos^2 t)^{3/2}}.$$

To determine the vertices, we differentiate the curvature with respect to t. Using the chain rule, we obtain

$$\begin{aligned}\kappa'(t) &= -\frac{3ab}{2}(a^2\sin^2 t + b^2\cos^2 t)^{-5/2} \\ &\quad \cdot (2a^2\sin t\cos t - 2b^2\sin t\cos t) \\ &= -\frac{3ab(a^2 - b^2)\sin t\cos t}{(a^2\sin^2 t + b^2\cos^2 t)^{5/2}}.\end{aligned}$$

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Because the ellipse is regular, $ds/dt > 0$, so $d\kappa/ds = 0$ if and only if $d\kappa/dt = 0$. Since $a > b > 0$, the denominator is positive and $a^2 - b^2 \neq 0$. Therefore, the vertex condition reduces to

$$\sin t \cos t = 0.$$

Thus,

$$t = 0, \qquad t = \frac{\pi}{2}, \qquad t = \pi, \qquad t = \frac{3\pi}{2}.$$

The corresponding points are

$$(a, 0), \qquad (0, b), \qquad (-a, 0), \qquad (0, -b).$$

Hence the ellipse has exactly four vertices.

At $t = 0$ and $t = \pi$, the curvature is

$$\kappa(0) = \kappa(\pi) = \frac{ab}{b^3} = \frac{a}{b^2}.$$

At $t = \pi/2$ and $t = 3\pi/2$, we have

$$\kappa\left(\frac{\pi}{2}\right) = \kappa\left(\frac{3\pi}{2}\right) = \frac{ab}{a^3} = \frac{b}{a^2}.$$

Since $a > b$, it follows that $a/b^2 > b/a^2$. Therefore, the vertices $(a, 0)$ and $(-a, 0)$ are points of maximum curvature, while $(0, b)$ and $(0, -b)$ are points of minimum curvature.

Thus, the four vertices occur precisely at the endpoints of the major and minor axes, showing that the lower bound in the Four Vertex Theorem is attained. ■

Example 1.12.9 (Vertices of a Particular Ellipse)**.** Find the vertices and the corresponding radii of curvature of the ellipse

$$\frac{x^2}{9} + \frac{y^2}{4} = 1.$$

Solution. The ellipse has semiaxes $a = 3$ and $b = 2$, and it may be parametrized by $\alpha(t) = (3\cos t, 2\sin t)$. From the preceding example, its vertices occur at

$$(3, 0), \qquad (0, 2), \qquad (-3, 0), \qquad (0, -2).$$

At the endpoints of the major axis, the curvature is $\kappa_{\max} = a/b^2 = 3/4$, so the corresponding radius of curvature is

$$\rho_{\min} = \frac{1}{\kappa_{\max}} = \frac{4}{3}.$$

At the endpoints of the minor axis, the curvature is $\kappa_{\min} = b/a^2 = 2/9$, giving

$$\rho_{\max} = \frac{1}{\kappa_{\min}} = \frac{9}{2}.$$

Thus, the ellipse bends most sharply at $(\pm 3, 0)$, where its radius of curvature is $4/3$, and is flattest at $(0, \pm 2)$, where its radius of curvature is $9/2$. ■

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A Convex Curve with Six Vertices

The Four Vertex Theorem gives a minimum number of vertices, but it does not assert that every convex curve has exactly four vertices. Some convex curves have six, eight, or more vertices.

One convenient way to construct such curves is through their support functions. For a strictly convex plane curve, let θ denote the angle of its outward unit normal with the positive x-axis. Its support function $h(\theta)$ is the signed distance from the origin to the tangent line having outward normal $(\cos\theta, \sin\theta)$.

The corresponding curve can be parametrized by

$$\alpha(\theta) = \big(h(\theta)\cos\theta - h'(\theta)\sin\theta,\, h(\theta)\sin\theta + h'(\theta)\cos\theta\big).$$

Differentiation gives

$$\alpha'(\theta) = \big(h(\theta) + h''(\theta)\big)(-\sin\theta, \cos\theta).$$

If $h + h'' > 0$, the curve is regular and strictly convex, and its radius of curvature is $\rho(\theta) = h(\theta) + h''(\theta)$. Consequently,

$$\kappa(\theta) = \frac{1}{h(\theta) + h''(\theta)}.$$

Example 1.12.10 (A Convex Curve with Six Vertices)**.** Let a plane curve be defined by the support function

$$h(\theta) = R + \varepsilon\cos 3\theta,$$

where $R > 8\varepsilon > 0$. Show that the corresponding closed convex curve has exactly six vertices.

Solution. Differentiating the support function, we obtain

$$h'(\theta) = -3\varepsilon\sin 3\theta, \qquad h''(\theta) = -9\varepsilon\cos 3\theta.$$

Therefore,

$$h(\theta) + h''(\theta) = R - 8\varepsilon\cos 3\theta.$$

Since $R > 8\varepsilon > 0$, this expression is positive for every θ. Hence the corresponding parametrized curve is regular and strictly convex.

Its curvature is

$$\kappa(\theta) = \frac{1}{R - 8\varepsilon\cos 3\theta}.$$

Differentiating gives

$$\frac{d\kappa}{d\theta} = -\frac{24\varepsilon\sin 3\theta}{(R - 8\varepsilon\cos 3\theta)^2}.$$

Since $ds/d\theta = R - 8\varepsilon\cos 3\theta > 0$, the vertex condition $d\kappa/ds = 0$ is equivalent to $d\kappa/d\theta = 0$. Thus,

$$\sin 3\theta = 0.$$

For $0 \le \theta < 2\pi$, the solutions are

$$\theta = 0, \quad \frac{\pi}{3}, \quad \frac{2\pi}{3}, \quad \pi, \quad \frac{4\pi}{3}, \quad \frac{5\pi}{3}.$$

These correspond to six distinct points of the curve because its outward normal direction is unique at each point of a strictly convex curve.

Therefore, the curve has exactly six vertices. Three of these are points of maximum curvature, and the remaining three are points of minimum curvature. ■

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Geometric Interpretation of the Four Vertex Theorem

The circle and ellipse illustrate two important possibilities. For a circle, the curvature is constant, so every point is a vertex. When the circle is stretched into a non-circular ellipse, the curvature becomes variable and has exactly two maxima and two minima, producing four vertices.

More complicated smooth convex deformations may produce additional maxima and minima of curvature, as illustrated by the six-vertex curve above. Nevertheless, the Four Vertex Theorem guarantees that the number of vertices can never fall below four.

Thus, the theorem expresses a fundamental geometric restriction: the curvature of a smooth simple closed convex plane curve cannot have only one maximum and one minimum. The closed nature of the curve forces a richer variation in its bending.

CHAPTER

2

DIFFERENTIAL GEOMETRY OF SURFACES

2.1 Introduction

The theory of surfaces is a fundamental part of classical differential geometry. While a curve is described locally by one parameter, a surface requires two parameters to specify the position of its points in three-dimensional space. Familiar examples include planes, spheres, cylinders, cones and surfaces obtained by rotating plane curves about an axis. The principal aim of the theory is to investigate the geometric properties of such surfaces using the methods of differential calculus.

Intuitively, a smooth surface is a two-dimensional object in $\mathbb{R}^3$ which, near each of its points, resembles a portion of a plane. This means that every sufficiently small part of the surface can be described using two independent coordinates. The precise mathematical formulation of this idea leads to the notion of a regular surface.

Definition 2.1.1 (Regular Surface)**.** A subset $S \subset \mathbb{R}^3$ is called a **regular surface** if, for every point $p \in S$, there exist an open neighbourhood V of p in $\mathbb{R}^3$, an open set $U \subset \mathbb{R}^2$, and a mapping $\mathbf{x} : U \to S \cap V$ satisfying the following conditions:

(i) The mapping $\mathbf{x}$ is smooth, that is, each of its component functions possesses continuous partial derivatives of every order.

(ii) The mapping $\mathbf{x}$ is a homeomorphism onto $S \cap V$, that is, it is bijective and both $\mathbf{x}$ and its inverse $\mathbf{x}^{-1}$ are continuous.

(iii) The differential $d\mathbf{x}_q : \mathbb{R}^2 \to \mathbb{R}^3$ is injective for every $q \in U$, or equivalently, the partial derivative vectors $\mathbf{x}_u(q)$ and $\mathbf{x}_v(q)$ are linearly independent.

The mapping $\mathbf{x}$ is called a **local parametrization** or a **coordinate chart** of the surface.

To understand the definition, write

$$\mathbf{x}(u, v) = (x(u, v), y(u, v), z(u, v)).$$

For each pair $(u, v) \in U$, the mapping assigns a point $\mathbf{x}(u, v)$ on the surface. Thus, the two variables u and v serve as local coordinates on S.

The first condition ensures that the surface can be studied using differentiation. The second ensures that each point of the surface patch corresponds to exactly one pair of parameters and that nearby points correspond to nearby parameter values. The third ensures that the two coordinate directions are independent, so the parametrization genuinely describes a two-dimensional surface rather than collapsing it locally into a curve or a point.

The partial derivatives of $\mathbf{x}$ are

$$\mathbf{x}_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right), \qquad \mathbf{x}_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right).$$

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These vectors represent the directions in which the surface changes when one parameter varies while the other remains fixed.

The differential of $\mathbf{x}$ at (u, v) is the linear mapping

$$d\mathbf{x}_{(u,v)}(a, b) = a\mathbf{x}_u(u, v) + b\mathbf{x}_v(u, v).$$

Therefore, $d\mathbf{x}_{(u,v)}$ is injective precisely when $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent. Since two vectors in $\mathbb{R}^3$ are linearly independent if and only if their cross product is nonzero, the regularity condition becomes

$$\mathbf{x}_u \times \mathbf{x}_v \neq \mathbf{0}.$$

Equivalently, the Jacobian matrix

$$D\mathbf{x} = \begin{pmatrix} x_u & x_v \\ y_u & y_v \\ z_u & z_v \end{pmatrix}$$

must have rank 2 at every point of U.

For practical purposes, checking the cross product is usually the simplest method of verifying that a given parametrization is regular. However, the nonvanishing of the cross product establishes only the third condition of the definition. To obtain a regular surface patch, the homeomorphism condition must also be satisfied.

We now examine some important examples of regular surfaces.

Example 2.1.2 (The Plane)**.** (See Figure 17 in the Atlas.) Show that the xy-plane is a regular surface by considering the parametrization

$$\mathbf{x}(u, v) = (u, v, 0), \qquad (u, v) \in \mathbb{R}^2.$$

Solution. The mapping $\mathbf{x}$ is smooth because all its component functions are polynomials. Moreover, it is bijective onto the xy-plane, and its inverse is given by $\mathbf{x}^{-1}(x, y, 0) = (x, y)$, which is continuous. Thus, the first two conditions in the definition are satisfied.

Differentiating with respect to u and v, we obtain

$$\mathbf{x}_u = (1, 0, 0), \qquad \mathbf{x}_v = (0, 1, 0).$$

Their cross product is

$$\mathbf{x}_u \times \mathbf{x}_v = (0, 0, 1) \neq \mathbf{0}.$$

Hence the regularity condition holds at every point. Therefore, the xy-plane is a regular surface.

More generally, any plane passing through a point $P \in \mathbb{R}^3$ and spanned by two linearly independent vectors $a, b \in \mathbb{R}^3$ admits the parametrization $\mathbf{x}(u, v) = P + ua + vb$. Since $\mathbf{x}_u \times \mathbf{x}_v = a \times b \neq \mathbf{0}$, every plane is a regular surface. ■

Example 2.1.3 (Graph of a Smooth Function)**.** (See Figure 18 in the Atlas.) Let $f : U \to \mathbb{R}$ be a smooth function defined on an open set $U \subset \mathbb{R}^2$. Show that its graph

$$S = \{(x, y, f(x, y)) : (x, y) \in U\}$$

is a regular surface.

Solution. Consider the parametrization $\mathbf{x}(u, v) = (u, v, f(u, v))$, where $(u, v) \in U$. Since f is smooth, the mapping $\mathbf{x}$ is smooth.

The mapping is one-to-one because its first two components are precisely u and v. Its inverse on the graph is $\mathbf{x}^{-1}(x, y, f(x, y)) = (x, y)$, which is continuous. Hence $\mathbf{x}$ is a homeomorphism onto its image.

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Differentiating, we obtain

$$\mathbf{x}_u = (1, 0, f_u), \qquad \mathbf{x}_v = (0, 1, f_v).$$

Taking the cross product gives

$$\mathbf{x}_u \times \mathbf{x}_v = (-f_u, -f_v, 1).$$

The third component of this vector is always 1, so the cross product can never vanish. Therefore, the graph of every smooth function of two variables is a regular surface. ■

The preceding example is particularly important because many surfaces can be represented locally as graphs of smooth functions. This provides a useful connection between the geometry of surfaces and the calculus of functions of two variables.

Example 2.1.4 (The Sphere). (See Figure 19 in the Atlas.) Show that the sphere of radius $R > 0$,

$$S^2(R) = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = R^2\},$$

is a regular surface.

Solution. Consider the spherical-coordinate parametrization

$$\mathbf{x}(\theta, \phi) = (R \sin\theta \cos\phi, R \sin\theta \sin\phi, R \cos\theta),$$

where $0 < \theta < \pi$ and $0 < \phi < 2\pi$. The mapping is smooth, and on this parameter domain it is a homeomorphism onto its image.

Differentiating with respect to the two parameters, we obtain

$$\mathbf{x}_\theta = (R \cos\theta \cos\phi, R \cos\theta \sin\phi, -R \sin\theta)$$

and

$$\mathbf{x}_\phi = (-R \sin\theta \sin\phi, R \sin\theta \cos\phi, 0).$$

Their cross product is

$$\mathbf{x}_\theta \times \mathbf{x}_\phi = R^2 \sin\theta (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta).$$

The vector inside the parentheses has unit length. Consequently,

$$\|\mathbf{x}_\theta \times \mathbf{x}_\phi\| = R^2 \sin\theta.$$

Since $0 < \theta < \pi$, we have $\sin\theta > 0$, so the cross product never vanishes on the chosen domain.

This parametrization does not cover the poles or the meridian corresponding to $\phi = 0$. To cover the entire sphere, we use additional local parametrizations. For example, near the north pole the sphere is the graph

$$z = \sqrt{R^2 - x^2 - y^2},$$

defined for $x^2 + y^2 < R^2$. Similarly, the lower hemisphere can be represented locally by $z = -\sqrt{R^2 - x^2 - y^2}$. Near points where the surface cannot be written as a graph over the xy-plane, we may instead solve the sphere equation for x or y.

Thus, every point of the sphere admits a smooth regular local parametrization, and hence the sphere is a regular surface. ■

Example 2.1.5 (Circular Cylinder). (See Figure 20 in the Atlas.) Show that the circular cylinder

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = R^2\}, \qquad R > 0,$$

is a regular surface.

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Solution. Consider the parametrization

$$\mathbf{x}(\theta, v) = (R\cos\theta, R\sin\theta, v),$$

where $0 < \theta < 2\pi$ and $v \in \mathbb{R}$. The mapping is smooth and is a homeomorphism onto the portion of the cylinder obtained by excluding the generator corresponding to $\theta = 0$.

Differentiating gives

$$\mathbf{x}_\theta = (-R\sin\theta, R\cos\theta, 0), \qquad \mathbf{x}_v = (0, 0, 1).$$

Thus,

$$\mathbf{x}_\theta \times \mathbf{x}_v = (R\cos\theta, R\sin\theta, 0).$$

Taking its norm, we obtain

$$\|\mathbf{x}_\theta \times \mathbf{x}_v\| = \sqrt{R^2\cos^2\theta + R^2\sin^2\theta} = R > 0.$$

Hence the parametrization is regular.

The excluded generator can be covered by another angular interval, such as $\pi < \theta < 3\pi$. Together, these local parametrizations cover the entire cylinder. Therefore, the circular cylinder is a regular surface. ■

Example 2.1.6 (Torus)**.** (See Figure 21 in the Atlas.) Show that the torus obtained by rotating a circle of radius r about an axis at distance R from its center is a regular surface, where $0 < r < R$.

Solution. A standard parametrization of the torus is

$$\mathbf{x}(\theta, \phi) = \big((R + r\cos\theta)\cos\phi, (R + r\cos\theta)\sin\phi, r\sin\theta\big).$$

Here θ describes a point on the generating circle, while ϕ describes the rotation about the z-axis.

Differentiating, we obtain

$$\mathbf{x}_\theta = (-r\sin\theta\cos\phi, -r\sin\theta\sin\phi, r\cos\theta)$$

and

$$\mathbf{x}_\phi = (-(R + r\cos\theta)\sin\phi, (R + r\cos\theta)\cos\phi, 0).$$

Taking the cross product gives

$$\mathbf{x}_\theta \times \mathbf{x}_\phi = -r(R + r\cos\theta)(\cos\theta\cos\phi, \cos\theta\sin\phi, \sin\theta).$$

Since the vector in parentheses has unit length,

$$\big\|\mathbf{x}_\theta \times \mathbf{x}_\phi\big\| = r(R + r\cos\theta).$$

Now $r > 0$ and $R + r\cos\theta \geq R - r > 0$. Therefore,

$$\big\|\mathbf{x}_\theta \times \mathbf{x}_\phi\big\| > 0$$

for every θ and ϕ.

On suitable open intervals of length less than 2π for both angular variables, the parametrization is a homeomorphism onto its image. By using overlapping angular intervals, the entire torus can be covered by such regular surface patches. Hence the torus is a regular surface. ■

Example 2.1.7 (Surfaces of Revolution)**.** (See Figure 22 in the Atlas.) Let $\alpha(u) = (f(u), 0, g(u))$, where u belongs to an open interval I, be a smooth regular curve in the xz-plane with $f(u) > 0$.

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Show that the surface obtained by rotating α about the z-axis is regular.

Solution. Rotating the generating curve about the z-axis gives the parametrization

$$\mathbf{x}(u,\theta) = (f(u)\cos\theta, f(u)\sin\theta, g(u)).$$

The mapping is smooth because f and g are smooth. Differentiating with respect to u and θ, we obtain

$$\mathbf{x}_u = (f'(u)\cos\theta, f'(u)\sin\theta, g'(u))$$

and

$$\mathbf{x}_\theta = (-f(u)\sin\theta, f(u)\cos\theta, 0).$$

Their cross product is

$$\mathbf{x}_u \times \mathbf{x}_\theta = f(u)(-g'(u)\cos\theta, -g'(u)\sin\theta, f'(u)).$$

Consequently,

$$\begin{aligned}\|\mathbf{x}_u \times \mathbf{x}_\theta\| &= f(u)\sqrt{(g'(u))^2\cos^2\theta + (g'(u))^2\sin^2\theta + (f'(u))^2} \\ &= f(u)\sqrt{(f'(u))^2 + (g'(u))^2}.\end{aligned}$$

Since the generating curve is regular, $(f'(u), g'(u)) \neq (0,0)$. Moreover, $f(u) > 0$. Hence the cross product is nonzero at every parameter value.

To complete the verification, observe that the generating curve is locally one-to-one because it is regular. By restricting u to a sufficiently small interval and θ to an angular interval of length less than 2π, the resulting parametrization is a homeomorphism onto a surface patch. Since every point of the surface is covered by such a patch, the surface of revolution is regular. ■

The assumption $f(u) > 0$ ensures that the generating curve does not meet the axis of revolution. If the generating curve meets the axis, the above parametrization becomes singular there because $\mathbf{x}_\theta = \mathbf{0}$. This does not necessarily mean that the resulting surface is singular. For example, a sphere admits regular parametrizations at its poles even though the usual spherical-coordinate parametrization fails there.

Example 2.1.8 (Enneper's Surface)**.** (See Figure 23 in the Atlas.) Consider the parametrized surface

$$\mathbf{x}(u,v) = \left(u - \frac{u^3}{3} + uv^2,\ v - \frac{v^3}{3} + u^2 v,\ u^2 - v^2\right).$$

Show that the parametrization is regular for every $(u,v) \in \mathbb{R}^2$.

Solution. Differentiating with respect to u and v, we obtain

$$\mathbf{x}_u = (1 - u^2 + v^2, 2uv, 2u)$$

and

$$\mathbf{x}_v = (2uv, 1 + u^2 - v^2, -2v).$$

Taking the cross product, we find

$$\mathbf{x}_u \times \mathbf{x}_v = \big(-2u(1 + u^2 + v^2),\ 2v(1 + u^2 + v^2),\ 1 - (u^2 + v^2)^2\big).$$

Put $r^2 = u^2 + v^2$. Then

$$\begin{aligned}\|\mathbf{x}_u \times \mathbf{x}_v\|^2 &= 4u^2(1 + r^2)^2 + 4v^2(1 + r^2)^2 + (1 - r^4)^2 \\ &= 4r^2(1 + r^2)^2 + (1 - r^4)^2 \\ &= (1 + r^2)^4.\end{aligned}$$

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Taking square roots gives

$$\|\mathbf{x}_u \times \mathbf{x}_v\| = (1 + u^2 + v^2)^2 > 0.$$

Therefore, $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent at every point of $\mathbb{R}^2$.

In particular, at $(u, v) = (0, 0)$, we have

$$\mathbf{x}_u(0,0) = (1,0,0), \qquad \mathbf{x}_v(0,0) = (0,1,0),$$

so their cross product is $(0, 0, 1) \neq \mathbf{0}$. Hence the origin is not a singular parameter value.

Thus, Enneper's parametrization is a regular immersion of $\mathbb{R}^2$ into $\mathbb{R}^3$. By restricting it to sufficiently small neighbourhoods on which it is one-to-one, we obtain regular parametrized surface patches. ■

Remark 2.1.9. There is an important distinction between a regular parametrization and a regular surface as a subset of $\mathbb{R}^3$. A smooth mapping may have linearly independent partial derivatives everywhere but still map different parameter values to the same point, producing self-intersections. Such a mapping is called an immersion. Enneper's surface provides an example of a regular immersion whose full image is not an embedded regular surface at its self-intersection points. The nonvanishing of $\mathbf{x}_u \times \mathbf{x}_v$ guarantees local regularity of the parametrization, but the homeomorphism condition must also be considered when studying the image as a regular surface.

The examples above show that regular surfaces may be described in several ways: as planes, graphs of functions, level sets of equations, or images of parametrizations. In every case, the essential local condition is that two independent tangent directions exist at each point. These directions determine the tangent plane and allow us to introduce the fundamental geometric concepts studied in the subsequent sections.

Throughout the remainder of this chapter, $S \subset \mathbb{R}^3$ will denote a regular surface, and $\mathbf{x} : U \to S$ will denote a smooth local parametrization with $\mathbf{x}_u \times \mathbf{x}_v \neq \mathbf{0}$.

Theorem 2.1.10 (Tangent Plane). *(See Figure 32 in the Atlas.) Let S be a regular surface and let $\mathbf{x} : U \subset \mathbb{R}^2 \to S$ be a local parametrization such that $p = \mathbf{x}(u_0, v_0)$. Then the set T_pS of all tangent vectors to smooth curves on S passing through p is a two-dimensional vector subspace of $\mathbb{R}^3$, given by*

$$T_pS = \operatorname{span}\{\mathbf{x}_u(u_0, v_0), \mathbf{x}_v(u_0, v_0)\}.$$

Equivalently,

$$T_pS = \{a\mathbf{x}_u(u_0, v_0) + b\mathbf{x}_v(u_0, v_0) : a, b \in \mathbb{R}\}.$$

Moreover, the tangent plane to S at p is the affine plane $p + T_pS$, which is perpendicular to the unit normal vector

$$\mathbf{N}(p) = \frac{\mathbf{x}_u(u_0, v_0) \times \mathbf{x}_v(u_0, v_0)}{\|\mathbf{x}_u(u_0, v_0) \times \mathbf{x}_v(u_0, v_0)\|}.$$

Proof. By definition, a vector $\mathbf{v} \in \mathbb{R}^3$ is tangent to S at p if there exists a smooth curve $\alpha : (-\varepsilon, \varepsilon) \to S$ such that $\alpha(0) = p$ and $\alpha'(0) = \mathbf{v}$. We shall prove that the set of all such vectors is precisely the span of $\mathbf{x}_u(u_0, v_0)$ and $\mathbf{x}_v(u_0, v_0)$.

First, let $\mathbf{v}$ be an arbitrary linear combination of these two vectors. Then there exist $a, b \in \mathbb{R}$ such that

$$\mathbf{v} = a\mathbf{x}_u(u_0, v_0) + b\mathbf{x}_v(u_0, v_0).$$

Since U is open and $(u_0, v_0) \in U$, the point $(u_0 + at, v_0 + bt)$ belongs to U for all sufficiently small values of t. We may therefore define a curve on the surface by

$$\alpha(t) = \mathbf{x}(u_0 + at, v_0 + bt).$$

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At $t = 0$, we have $\alpha(0) = \mathbf{x}(u_0, v_0) = p$. Differentiating with respect to t and applying the chain rule, we obtain

$$\alpha'(t) = a\mathbf{x}_u(u_0 + at, v_0 + bt) + b\mathbf{x}_v(u_0 + at, v_0 + bt).$$

Putting $t = 0$ gives

$$\alpha'(0) = a\mathbf{x}_u(u_0, v_0) + b\mathbf{x}_v(u_0, v_0) = \mathbf{v}.$$

Thus, $\mathbf{v}$ is the tangent vector of a curve on S passing through p, and hence $\mathbf{v} \in T_pS$. Consequently,

$$\operatorname{span}\{\mathbf{x}_u(u_0, v_0), \mathbf{x}_v(u_0, v_0)\} \subseteq T_pS.$$

Conversely, let $\mathbf{v} \in T_pS$. By the definition of a tangent vector, there exists a smooth curve $\beta : (-\varepsilon, \varepsilon) \to S$ such that $\beta(0) = p$ and $\beta'(0) = \mathbf{v}$.

Since $\mathbf{x}$ is a regular local parametrization, its inverse is smooth on the corresponding surface patch. Hence, by restricting the parameter interval if necessary, we can express β in local coordinates as

$$\beta(t) = \mathbf{x}(u(t), v(t)),$$

where $u(t)$ and $v(t)$ are smooth functions satisfying $u(0) = u_0$ and $v(0) = v_0$.

Differentiating this expression with respect to t, the chain rule gives

$$\beta'(t) = \mathbf{x}_u(u(t), v(t))u'(t) + \mathbf{x}_v(u(t), v(t))v'(t).$$

Evaluating at $t = 0$, we obtain

$$\mathbf{v} = \beta'(0) = u'(0)\mathbf{x}_u(u_0, v_0) + v'(0)\mathbf{x}_v(u_0, v_0).$$

Thus, $\mathbf{v}$ is a linear combination of $\mathbf{x}_u(u_0, v_0)$ and $\mathbf{x}_v(u_0, v_0)$. Therefore,

$$T_pS \subseteq \operatorname{span}\{\mathbf{x}_u(u_0, v_0), \mathbf{x}_v(u_0, v_0)\}.$$

Combining the two inclusions, we conclude that

$$T_pS = \operatorname{span}\{\mathbf{x}_u(u_0, v_0), \mathbf{x}_v(u_0, v_0)\}.$$

Since S is regular, the vectors $\mathbf{x}_u(u_0, v_0)$ and $\mathbf{x}_v(u_0, v_0)$ are linearly independent. Hence T_pS is a two-dimensional vector subspace of $\mathbb{R}^3$.

Finally, the cross product $\mathbf{x}_u \times \mathbf{x}_v$ is perpendicular to both $\mathbf{x}_u$ and $\mathbf{x}_v$, and therefore to every vector in T_pS. Since this cross product is nonzero, it determines a unit normal vector $\mathbf{N}(p)$. Thus, the affine tangent plane through p consists of all points $X \in \mathbb{R}^3$ satisfying

$$(X - p) \cdot \mathbf{N}(p) = 0.$$

This proves the result. ■

The theorem gives a practical method for determining the tangent plane to a regular surface. If $\mathbf{x}(u, v)$ is a local parametrization and $p = \mathbf{x}(u_0, v_0)$, then the tangent plane is spanned by the two coordinate tangent vectors $\mathbf{x}_u(u_0, v_0)$ and $\mathbf{x}_v(u_0, v_0)$. Its Cartesian equation is

$$(X - p) \cdot \big(\mathbf{x}_u(u_0, v_0) \times \mathbf{x}_v(u_0, v_0)\big) = 0.$$

For a surface given explicitly as the graph $z = f(x, y)$, the tangent plane at $p = (x_0, y_0, f(x_0, y_0))$ is

$$z - f(x_0, y_0) = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

Similarly, if a surface is defined implicitly by $F(x, y, z) = c$, where $\nabla F(p) \neq \mathbf{0}$, the gradient $\nabla F(p)$ is normal to the surface, and the tangent plane has equation

$$\nabla F(p) \cdot (X - p) = 0.$$

These formulas are different representations of the same geometric object.

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Example 2.1.11 (Tangent Plane to a Sphere)**.** Find the tangent plane to the sphere $x^2 + y^2 + z^2 = 9$ at the point $p = (1, 2, 2)$.

Solution. The sphere is defined implicitly by $F(x, y, z) = x^2 + y^2 + z^2 = 9$. Its gradient is

$$\nabla F(x, y, z) = (2x, 2y, 2z).$$

At $p = (1, 2, 2)$, we obtain $\nabla F(p) = (2, 4, 4)$. Since this vector is nonzero, it is normal to the tangent plane. Therefore, the required equation is

$$(2, 4, 4) \cdot (x - 1, y - 2, z - 2) = 0.$$

Expanding and simplifying gives

$$x + 2y + 2z = 9.$$

■

Example 2.1.12 (Tangent Plane to a Paraboloid)**.** Find the tangent plane to the paraboloid $z = x^2 + y^2$ at the point $(1, 1, 2)$.

Solution. Write $z = f(x, y)$, where $f(x, y) = x^2 + y^2$. Then $f_x(x, y) = 2x$ and $f_y(x, y) = 2y$. At $(x_0, y_0) = (1, 1)$, we have $f_x(1, 1) = 2$, $f_y(1, 1) = 2$ and $f(1, 1) = 2$.

Using the tangent plane formula for the graph of a function, we obtain

$$z - 2 = 2(x - 1) + 2(y - 1).$$

Simplifying gives

$$z = 2x + 2y - 2.$$

■

Example 2.1.13 (Plane)**.** Consider the surface S given by $\mathbf{x}(u, v) = (2u + v,\ u - 3v,\ 4u + 5v)$. Find the tangent plane at $p = \mathbf{x}(0, 0)$.

Solution. Compute the partial derivatives:

$$\mathbf{x}_u = (2, 1, 4), \qquad \mathbf{x}_v = (1, -3, 5).$$

These are linearly independent (their cross product is $(2, 1, 4) \times (1, -3, 5) = (17, -6, -7) \neq \mathbf{0}$). At $p = \mathbf{x}(0, 0) = (0, 0, 0)$, the tangent plane is simply $\operatorname{span}\{\mathbf{x}_u, \mathbf{x}_v\}$, i.e. the plane through the origin spanned by these two vectors. Its equation can be obtained using the normal vector $\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = (17, -6, -7)$:

$$17x - 6y - 7z = 0.$$

■

Example 2.1.14 (Sphere)**.** Let S be the sphere $x^2 + y^2 + z^2 = 25$. Find the tangent plane at the point $p = (3, 4, 0)$.

Solution. We use the implicit method. Define $F(x, y, z) = x^2 + y^2 + z^2$. Then $\nabla F = (2x, 2y, 2z)$. At $p = (3, 4, 0)$, $\nabla F(p) = (6, 8, 0)$. The tangent plane consists of points (x, y, z) such that

$$(6, 8, 0) \cdot (x - 3,\ y - 4,\ z - 0) = 0 \implies 6(x - 3) + 8(y - 4) = 0.$$

Simplify: $6x - 18 + 8y - 32 = 0$, i.e. $6x + 8y = 50$, or $3x + 4y = 25$. ■

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Example 2.1.15 (Elliptic Paraboloid)**.** (See Figure 24 in the Atlas.) Parametrize the surface $z = x^2 + y^2$ as $\mathbf{x}(x, y) = (x, y, x^2 + y^2)$. Find the tangent plane at $p = (1, 1, 2)$.

Solution. Compute $\mathbf{x}_x = (1, 0, 2x)$, $\mathbf{x}_y = (0, 1, 2y)$. At $(x, y) = (1, 1)$:

$$\mathbf{x}_x = (1, 0, 2), \qquad \mathbf{x}_y = (0, 1, 2).$$

A normal vector is

$$\mathbf{n} = \mathbf{x}_x \times \mathbf{x}_y = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 2 \\ 0 & 1 & 2 \end{vmatrix} = (-2,\ -2,\ 1).$$

The tangent plane equation at p is

$$(-2, -2, 1) \cdot (x - 1,\ y - 1,\ z - 2) = 0 \implies -2(x - 1) - 2(y - 1) + (z - 2) = 0.$$

Simplify: $z - 2 = 2(x - 1) + 2(y - 1)$, so $z = 2x + 2y - 2$. ■

Example 2.1.16 (Cylinder)**.** Consider the cylinder $x^2 + y^2 = 1$. Find the tangent plane at the point $p = (0, 1, 5)$.

Solution. We can parametrize by $\mathbf{x}(\theta, z) = (\cos\theta, \sin\theta, z)$. At $p = (0, 1, 5)$ we have $\theta = \pi/2$, $z = 5$. Then

$$\mathbf{x}_\theta = (-\sin\theta, \cos\theta, 0) = (-1, 0, 0), \qquad \mathbf{x}_z = (0, 0, 1).$$

Thus T_pS spans $(-1, 0, 0)$ and $(0, 0, 1)$; it is the xz-plane shifted to pass through p. The equation is $y = 1$. Alternatively, using the implicit form $F(x, y, z) = x^2 + y^2$, $\nabla F = (2x, 2y, 0)$; at $p = (0, 1, 5)$, $\nabla F = (0, 2, 0)$. Then $(0, 2, 0) \cdot (x - 0, y - 1, z - 5) = 0$ gives $2(y - 1) = 0$, i.e. $y = 1$. ■

Example 2.1.17 (Hyperbolic Paraboloid)**.** (See Figure 25 in the Atlas.) Let S be the surface $z = xy$. Find the tangent plane at $(2, 3, 6)$.

Solution. Parametrize by $\mathbf{x}(x, y) = (x, y, xy)$. Then $\mathbf{x}_x = (1, 0, y)$, $\mathbf{x}_y = (0, 1, x)$. At $(x, y) = (2, 3)$:

$$\mathbf{x}_x = (1, 0, 3), \qquad \mathbf{x}_y = (0, 1, 2).$$

Normal vector:

$$\mathbf{n} = (1, 0, 3) \times (0, 1, 2) = (-3,\ -2,\ 1).$$

The tangent plane equation at $(2, 3, 6)$ is

$$(-3, -2, 1) \cdot (x - 2,\ y - 3,\ z - 6) = 0 \implies -3(x - 2) - 2(y - 3) + (z - 6) = 0.$$

Simplify: $-3x + 6 - 2y + 6 + z - 6 = 0 \implies z = 3x + 2y - 6$. ■

2.1 What is NOT a Regular Surface?

Example 2.1.18 (Cone without vertex)**.** (See Figure 26 in the Atlas.) The set $\{(x, y, z) \mid z = \sqrt{x^2 + y^2},\ (x, y) \neq (0, 0)\}$ is not a regular surface because the function $\sqrt{x^2 + y^2}$ is not differentiable at $(0, 0)$. If we remove the apex, the remainder is actually a regular surface (it is a graph of a smooth function on $\mathbb{R}^2 \setminus \{(0, 0)\}$). The problem is that the apex itself cannot be included.

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Example 2.1.19 (Self-intersecting surface)**.** (See Figure 27 in the Atlas.) The Whitney umbrella given by $\mathbf{x}(u,v) = (uv, u, v^2)$ is not a regular surface because it is not a homeomorphism onto its image; points with $v = 0$ and $u \neq 0$ map to the same x coordinate? Actually it self-intersects along the x-axis, violating the homeomorphism condition.

Example 2.1.20 (Surface with a cusp)**.** (See Figure 28 in the Atlas.) The set $z = (x^2+y^2)^{1/3}$ is not C^1 at the origin, so it fails the smoothness condition.

Many surfaces arise as level sets $F(x,y,z) = c$ where $\nabla F \neq 0$ on the set. The implicit function theorem guarantees that such a level set is a regular surface.

Proposition 2.1.21. *Let $F : \mathbb{R}^3 \to \mathbb{R}$ be a smooth function and let $c \in \mathbb{R}$. If $\nabla F(p) \neq \mathbf{0}$ for every p with $F(p) = c$, then $S = F^{-1}(c)$ is a regular surface.*

Example 2.1.22. The sphere $x^2+y^2+z^2 = R^2$ has $\nabla F = (2x, 2y, 2z)$ which vanishes only at $(0,0,0)$, not on the sphere. Hence the sphere is regular.

Example 2.1.23. The cylinder $x^2+y^2 = R^2$ has $\nabla F = (2x, 2y, 0)$, never zero on the cylinder.

Example 2.1.24. The hyperboloid $x^2+y^2-z^2 = 1$ has $\nabla F = (2x, 2y, -2z)$; if this were zero then $x = y = z = 0$, but $(0,0,0)$ does not satisfy the equation. Therefore the hyperboloid is regular. (See Figure 29 in the Atlas.)

Table of Examples for Quick Reference

Surface	Parametrization	Regularity condition
Plane	$(u, v, 0)$	Always
Graph $z = f(x,y)$	$(x, y, f(x,y))$	Always
Sphere	spherical coordinates	$\sin\theta \neq 0$ (charts cover all)
Cylinder	$(R\cos\theta, R\sin\theta, z)$	$R > 0$
Torus	$((R + r\cos\theta)\cos\phi, \dots)$	$R > r > 0$
Surface of revolution	$(f(u)\cos\theta, f(u)\sin\theta, g(u))$	$f(u) > 0$, $(f', g') \neq (0,0)$
Enneper's surface	$(u - u^3/3 + uv^2,\ v - v^3/3 + u^2v,\ u^2 - v^2)$	$(u,v) \neq (0,0)$

2.1 Exercises

1. Show that the helicoid $\mathbf{x}(u,v) = (u\cos v, u\sin v, v)$ is a regular surface for $u > 0$.

2. Prove that the hyperboloid $x^2+y^2-z^2 = 1$ is a regular surface using the implicit function theorem.

3. Find all points where the parametrization $\mathbf{x}(u,v) = (u^3, v^3, uv)$ fails to be regular.

4. Give an example of a subset of $\mathbb{R}^3$ that is a regular surface but cannot be represented as the graph of a function of the form $z = f(x,y)$ over any open set.

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2.1.1 Change of Coordinates

If $\mathbf{x} : U \to S$ and $\mathbf{y} : V \to S$ are two parametrizations with overlapping images, then the transition map $\mathbf{x}^{-1} \circ \mathbf{y} : \mathbf{y}^{-1}(\mathbf{x}(U) \cap \mathbf{y}(V)) \to \mathbf{x}^{-1}(\mathbf{x}(U) \cap \mathbf{y}(V))$ is a smooth diffeomorphism between open sets of $\mathbb{R}^2$. This ensures that geometric quantities (such as curvature) are independent of the chosen parametrization.

Theorem: [Inverse Function Theorem in $\mathbb{R}^3$] Let $F : \mathcal{U} \subseteq \mathbb{R}^3 \to \mathbb{R}^3$ be a smooth map. If at a point $q = (u_0, v_0, t_0) \in \mathcal{U}$, the differential dF_q is invertible (i.e., $\det(dF_q) \neq 0$), then there exist open neighborhoods U of q and V of $F(q)$ such that $F : U \to V$ is a diffeomorphism.

Theorem 2.1.25 (Change of Coordinates). *Let $S \subset \mathbb{R}^3$ be a regular surface. Let $\mathbf{x} : U \to S$ and $\mathbf{y} : V \to S$ be two coordinate patches such that $W = \mathbf{x}(U) \cap \mathbf{y}(V)$ is non-empty. Then the transition map*

$$h = \mathbf{x}^{-1} \circ \mathbf{y} : \mathbf{y}^{-1}(W) \longrightarrow \mathbf{x}^{-1}(W)$$

is a diffeomorphism.

Proof. Let $p \in W$. Since $\mathbf{x}$ and $\mathbf{y}$ are coordinate patches, they are, by definition, homeomorphisms from open sets in $\mathbb{R}^2$ onto open sets in S. Consequently, $h = \mathbf{x}^{-1} \circ \mathbf{y}$ is a composition of homeomorphisms and is itself a homeomorphism. It remains to show that h is smooth (of class C^∞).

Let $q \in \mathbf{y}^{-1}(W)$ such that $\mathbf{y}(q) = p$. We represent the coordinates in U as (u, v). Since $\mathbf{x}$ is a regular patch, the partial derivatives $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent at $\mathbf{x}^{-1}(p)$. We define the unit normal vector field $\mathbf{n}(u, v) = \dfrac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}$.

Consider the map $F : U \times \mathbb{R} \to \mathbb{R}^3$ defined by:

$$F(u, v, t) = \mathbf{x}(u, v) + t\,\mathbf{n}(u, v) \tag{2.1}$$

The map F is differentiable because $\mathbf{x}$ and $\mathbf{n}$ are smooth. The Jacobian of F at $(u, v, 0)$ is

$$\det(dF) = \det(\mathbf{x}_u, \mathbf{x}_v, \mathbf{n}) = \langle \mathbf{x}_u \times \mathbf{x}_v, \mathbf{n} \rangle = |\mathbf{x}_u \times \mathbf{x}_v|$$

Since $\mathbf{x}$ is regular, $|\mathbf{x}_u \times \mathbf{x}_v| \neq 0$. Thus, dF is non-singular. By the **Inverse Function Theorem**, there exists an open neighborhood $\mathcal{O}$ of p in $\mathbb{R}^3$ such that F^{-1} exists and is differentiable on $\mathcal{O}$.

Observe that for any point $p' \in S \cap \mathcal{O}$, the t-coordinate of $F^{-1}(p')$ is zero. Therefore, the restriction of F^{-1} to the surface S yields the first two components (u, v), which is precisely the map $\mathbf{x}^{-1}$. Formally, if $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ is the projection $\pi(u, v, t) = (u, v)$, then:

$$\mathbf{x}^{-1} = \pi \circ F^{-1}|_{S \cap \mathcal{O}}$$

Since F^{-1} and $\mathbf{y}$ are differentiable, the transition map h in a neighborhood of q is given by $h = (\pi \circ F^{-1}) \circ \mathbf{y}$. As a composition of differentiable maps, h is differentiable at q. By an identical argument, $h^{-1} = \mathbf{y}^{-1} \circ \mathbf{x}$ is also differentiable. Thus, h is a diffeomorphism. ■

Explanation of the Geometry Used in the Proof. The proof is structured into two main parts: verifying that the map is a bijection and then proving that it is smooth (differentiable) using an extension into 3D space.

Establishing Bijectivity

By the definition of a regular surface, a coordinate patch $\mathbf{x}$ is a *homeomorphism* onto its image. This means $\mathbf{x}$ is continuous and has a continuous inverse. Since $\mathbf{y}$ is a bijection from $\mathbf{y}^{-1}(W)$ to W, and $\mathbf{x}^{-1}$ is a bijection from W to $\mathbf{x}^{-1}(W)$, their composition $h = \mathbf{x}^{-1} \circ \mathbf{y}$ is necessarily a bijection. Geometrically, this means every point in one coordinate system corresponds uniquely to exactly one point in the other.

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The Differentiability Problem
We know $\mathbf{y}$ is differentiable, but we cannot immediately say $\mathbf{x}^{-1}$ is differentiable because its domain is the surface $S \subset \mathbb{R}^3$, which is not an open set in $\mathbb{R}^3$. To use the standard rules of calculus, we must "extend" our coordinate patch from the 2D surface into a 3D neighborhood.

Extending the Patch to 3D Space
Let $q \in \mathbf{y}^{-1}(W)$ and $p = \mathbf{y}(q) \in W$. We define a new map $F : U \times \mathbb{R} \to \mathbb{R}^3$ by "thickening" the surface in the direction of the normal vector:

$$F(u, v, t) = \mathbf{x}(u, v) + t\,\mathbf{n}(u, v)$$

Here, $(u, v) \in U$ are the surface coordinates, $\mathbf{n}$ is the unit normal vector to the surface at $\mathbf{x}(u, v)$, and t represents the distance from the surface.

Applying the Inverse Function Theorem
F is a map between two open sets in $\mathbb{R}^3$. Its derivative at a point $(u, v, 0)$ is a 3×3 matrix with columns $(\mathbf{x}_u, \mathbf{x}_v, \mathbf{n})$. The determinant of this Jacobian is

$$\det(dF) = (\mathbf{x}_u \times \mathbf{x}_v) \cdot \mathbf{n} = |\mathbf{x}_u \times \mathbf{x}_v|$$

Because the surface is **regular**, the vectors $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent, so their cross product is non-zero. Thus, $\det(dF) \neq 0$. By the **Inverse Function Theorem**, F has a differentiable inverse F^{-1} in a small 3D neighborhood of the point p.

Linking F^{-1} back to $\mathbf{x}^{-1}$
Observe that for points lying exactly on the surface S, the t coordinate is 0. Therefore, $F(u, v, 0) = \mathbf{x}(u, v)$. This implies that the inverse map F^{-1}, when restricted to the surface S, behaves exactly like $\mathbf{x}^{-1}$ (returning the u and v coordinates).

We can now write the transition map h as a composition of smooth functions:

$$h = \pi \circ F^{-1} \circ \mathbf{y}$$

where $\pi(u, v, t) = (u, v)$ is the standard projection map. Since $\mathbf{y}$, F^{-1}, and π are all differentiable, their composition h is differentiable at q.

By swapping the roles of $\mathbf{x}$ and $\mathbf{y}$, we can show that $h^{-1} = \mathbf{y}^{-1} \circ \mathbf{x}$ is also differentiable. Thus, h is a diffeomorphism, proving that the change of coordinates is smooth. ■

A regular surface is locally smooth, meaning it can be approximated by a plane at any given point. This plane is known as the tangent plane. In differential geometry, we distinguish between the **Tangent Space** (a set of vectors) and the **Tangent Plane** (the geometric set of points in $\mathbb{R}^3$).

Definition 2.1.26 (Tangent Vector)**.** Let S be a regular surface and $p \in S$. A vector $w \in \mathbb{R}^3$ is a **tangent vector** to S at p if there exists a differentiable parameterized curve $\alpha : (-\epsilon, \epsilon) \to S$ such that $\alpha(0) = p$ and $\alpha'(0) = w$. The **tangent space** to S at p, denoted by T_pS, is the set of all tangent vectors to S at p.

Proposition 2.1.27 (Basis of the Tangent Space)**.** *Let $\mathbf{x} : U \subset \mathbb{R}^2 \to S$ be a local parametrization of a regular surface S such that $\mathbf{x}(q) = p$ for some $q = (u_0, v_0) \in U$. The tangent space T_pS is a two-dimensional vector subspace of $\mathbb{R}^3$ spanned by the partial derivative vectors:*

$$\mathbf{x}_u(q) = \frac{\partial \mathbf{x}}{\partial u}(q) \quad \textit{and} \quad \mathbf{x}_v(q) = \frac{\partial \mathbf{x}}{\partial v}(q)$$

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Proof. Let $w \in T_pS$. By definition, $w = \alpha'(0)$ for some curve $\alpha(t)$ on S. We can write the curve in the coordinate neighborhood as $\alpha(t) = \mathbf{x}(u(t), v(t))$. By the chain rule,

$$w = \alpha'(0) = \frac{d}{dt}\left[\mathbf{x}(u(t), v(t))\right]_{t=0} = \mathbf{x}_u \frac{du}{dt}(0) + \mathbf{x}_v \frac{dv}{dt}(0)$$

This shows that w is a linear combination of $\mathbf{x}_u$ and $\mathbf{x}_v$. Since the surface is regular, $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent, thus they form a basis for T_pS. ■

Definition 2.1.28 (Normal Vector)**.** Let $\mathbf{x}(u, v)$ be a parametrization of S at p. The **unit normal vector** to the surface at p is defined as

$$\mathbf{N}(p) = \frac{\mathbf{x}_u \times \mathbf{x}_v}{|\mathbf{x}_u \times \mathbf{x}_v|}$$

Theorem 2.1.29 (Equation of the Tangent Plane)**.** *Let $S \subset \mathbb{R}^3$ be a regular surface and $\mathbf{x} : U \subset \mathbb{R}^2 \to S$ be a local parametrization of S such that $p = \mathbf{x}(u_0, v_0)$ for some $(u_0, v_0) \in U$. The tangent plane to S at $p = (x_0, y_0, z_0)$ is the set of points $P = (x, y, z) \in \mathbb{R}^3$ satisfying:*

$$\langle (P - p), \mathbf{N}(p) \rangle = 0$$

Furthermore, if the cross product of the partial derivatives is given by $\mathbf{x}_u \times \mathbf{x}_v = (A, B, C)$, then the Cartesian equation of the plane is $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$.

Proof. By the definition of a regular surface, the tangent space T_pS is the 2D vector subspace of $\mathbb{R}^3$ spanned by the basis $\{\mathbf{x}_u, \mathbf{x}_v\}$.

Let $\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v$. By the properties of the cross product, $\mathbf{n}$ is orthogonal to T_pS. The regularity condition ($\mathbf{x}_u \times \mathbf{x}_v \neq \mathbf{0}$) ensures that $\mathbf{n}$ defines a unique normal direction.

A point P lies in the tangent plane at p if and only if the displacement vector $P - p$ is a tangent vector (i.e., $P - p \in T_pS$). This geometric condition is satisfied if and only if $P - p$ is orthogonal to the normal vector $\mathbf{n}$:

$$\langle P - p, \mathbf{n} \rangle = 0$$

Expanding this inner product with $P - p = (x - x_0, y - y_0, z - z_0)$ and $\mathbf{n} = (A, B, C)$ yields the Cartesian equation:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

■

Proposition 2.1.30 (Tangent Plane of a Level Set)**.** *If a regular surface S is defined implicitly by $f(x, y, z) = a$, where a is a constant and $\nabla f(p) \neq 0$, then the tangent space T_pS is the kernel of the linear map df_p. That is, a vector w is tangent to S at p if and only if:*

$$\langle \nabla f(p), w \rangle = 0$$

Example 2.1.31 (The Sphere - Implicit Method)**.** Find the equation of the tangent plane to the sphere $x^2 + y^2 + z^2 = 14$ at the point $P(1, 2, 3)$.

Solution. The surface is given implicitly by $f(x, y, z) = x^2 + y^2 + z^2 - 14 = 0$. The gradient vector ∇f acts as the normal vector to the surface:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right) = (2x, 2y, 2z)$$

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Evaluating the gradient at the point $P(1,2,3)$:

$$\mathbf{n} = \nabla f(1,2,3) = (2(1), 2(2), 2(3)) = (2,4,6)$$

The equation of the tangent plane at (x_0, y_0, z_0) is $A(x-x_0) + B(y-y_0) + C(z-z_0) = 0$:

$$2(x-1) + 4(y-2) + 6(z-3) = 0$$

Dividing by 2 and simplifying:

$$(x-1) + 2(y-2) + 3(z-3) = 0 \implies x + 2y + 3z = 14$$

■

Example 2.1.32 (The Hyperbolic Paraboloid - Parametric Method)**.** Find the tangent plane to the surface $\mathbf{x}(u,v) = (u, v, u^2 - v^2)$ at the point where $u = 2, v = 1$.

Solution. First determine the point p on the surface:

$$p = \mathbf{x}(2,1) = (2, 1, 2^2 - 1^2) = (2,1,3)$$

Next, compute the partial derivative vectors (the basis of the tangent space):

$$\mathbf{x}_u = (1, 0, 2u) \implies \mathbf{x}_u(2,1) = (1,0,4)$$

$$\mathbf{x}_v = (0, 1, -2v) \implies \mathbf{x}_v(2,1) = (0,1,-2)$$

The normal vector $\mathbf{n}$ is the cross product of these basis vectors:

$$\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 0 & 4 \\ 0 & 1 & -2 \end{vmatrix} = (0-4)\mathbf{i} - (-2-0)\mathbf{j} + (1-0)\mathbf{k} = (-4,2,1)$$

The equation of the plane is

$$-4(x-2) + 2(y-1) + 1(z-3) = 0 \implies -4x + 8 + 2y - 2 + z - 3 = 0$$

Simplifying, we obtain $4x - 2y - z = 3$. ■

Example 2.1.33 (The Ellipsoid - Implicit Method)**.** (See Figure 30 in the Atlas.) Find the tangent plane to the ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$ at the point $P(-2,1,3)$.

Solution. Define the implicit function $f(x,y,z) = \frac{x^2}{4} + y^2 + \frac{z^2}{9} - 3 = 0$. The normal vector is the gradient:

$$\nabla f = \left(\frac{x}{2}, 2y, \frac{2z}{9}\right)$$

At the point $P(-2,1,3)$:

$$\mathbf{n} = \nabla f(-2,1,3) = \left(\frac{-2}{2}, 2(1), \frac{2(3)}{9}\right) = \left(-1, 2, \frac{2}{3}\right)$$

The tangent plane equation is $-1(x+2) + 2(y-1) + \frac{2}{3}(z-3) = 0$. Multiply by 3 to clear the fraction:

$$-3(x+2) + 6(y-1) + 2(z-3) = 0 \implies -3x - 6 + 6y - 6 + 2z - 6 = 0$$

Resulting in: $3x - 6y - 2z = -18$ or $3x - 6y - 2z + 18 = 0$. ■

Example 2.1.34 (The Helicoid - Parametric Method)**.** (See Figure 31 in the Atlas.) Find the tangent plane to the helicoid $\mathbf{x}(u,v) = (u\cos v, u\sin v, v)$ at $(u,v) = (1, \pi/2)$.

Solution. First, determine the point coordinates:

$$p = \mathbf{x}(1,\pi/2) = (1\cos(\pi/2), 1\sin(\pi/2), \pi/2) = (0,1,\pi/2)$$

Compute the partial derivatives:

$$\mathbf{x}_u = (\cos v, \sin v, 0) \implies \mathbf{x}_u(1,\pi/2) = (0,1,0)$$

$$\mathbf{x}_v = (-u\sin v, u\cos v, 1) \implies \mathbf{x}_v(1,\pi/2) = (-1,0,1)$$

Compute the normal vector via cross product:

$$\mathbf{n} = \mathbf{x}_u \times \mathbf{x}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = (1)\mathbf{i} - (0)\mathbf{j} + (1)\mathbf{k} = (1,0,1)$$

The equation of the tangent plane is

$$1(x-0) + 0(y-1) + 1(z-\pi/2) = 0 \implies x + z = \frac{\pi}{2}$$

■

Remark 2.1.35 (Geometric Summary). The regularity of a surface ensures that the tangent plane is well-defined. If $\mathbf{x}_u \times \mathbf{x}_v = 0$, the "plane" would collapse into a line or a point, which happens at singularities (like the tip of a cone). Therefore, regularity is a requirement for the existence of a unique tangent plane.

A curvilinear coordinate system is described by a set of coordinates $(q_1, q_2, \ldots, q_n)$. The local basis vectors at any point are tangent to the coordinate curves passing through that point. We can find these vectors by first defining the position vector $\mathbf{r}$ in terms of Cartesian coordinates:

$$\mathbf{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}.$$

We then express $\mathbf{r}$ as a function of the curvilinear coordinates, $\mathbf{r}(q_1, q_2, q_3)$.

The (unnormalized) basis vectors $\mathbf{b}_i$ are obtained by taking the partial derivative of the position vector with respect to each coordinate:

$$\mathbf{b}_i = \frac{\partial \mathbf{r}}{\partial q_i}.$$

To obtain the orthonormal basis vectors $\hat{\mathbf{e}}_i$, we normalize these vectors:

$$\hat{\mathbf{e}}_i = \frac{\mathbf{b}_i}{\|\mathbf{b}_i\|}.$$

The magnitude $\|\mathbf{b}_i\|$ is known as the scale factor, often denoted h_i.

Example 2.1.36 (Unit Sphere (Spherical Coordinates))**.** A unit sphere is a 2-dimensional surface embedded in 3-dimensional space. Any point on its surface can be described using two coordinates: the polar angle θ (measured from the positive z-axis, $0 \le \theta \le \pi$) and the azimuthal angle ϕ (measured in the xy-plane from the positive x-axis, $0 \le \phi < 2\pi$). The radius is fixed at $r = 1$.

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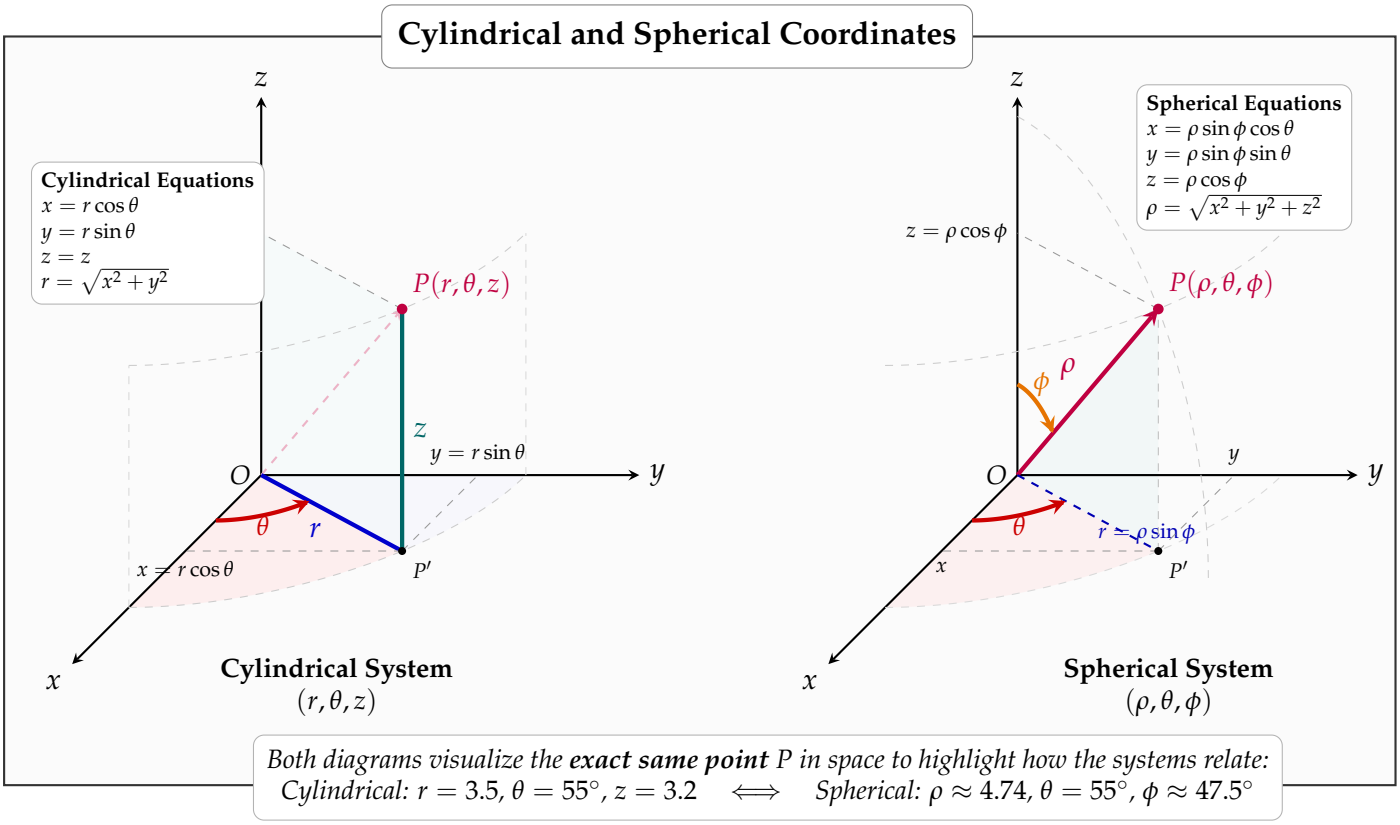


Figure 2.1: Caption

The transformation from spherical to Cartesian coordinates is

$$\begin{aligned} x &= r\sin\theta\cos\phi, \\ y &= r\sin\theta\sin\phi, \\ z &= r\cos\theta. \end{aligned}$$

With $r = 1$, the position vector for a point on the unit sphere becomes

$$\mathbf{r}(\theta,\phi) = (\sin\theta\cos\phi)\,\hat{\mathbf{i}} + (\sin\theta\sin\phi)\,\hat{\mathbf{j}} + (\cos\theta)\,\hat{\mathbf{k}}.$$

Solution. **Basis vector $\hat{\mathbf{e}}_\theta$:** We take the partial derivative with respect to θ:

$$\frac{\partial \mathbf{r}}{\partial\theta} = (\cos\theta\cos\phi)\,\hat{\mathbf{i}} + (\cos\theta\sin\phi)\,\hat{\mathbf{j}} - (\sin\theta)\,\hat{\mathbf{k}}.$$

Its magnitude (the scale factor h_θ) is

$$\begin{aligned} \left\|\frac{\partial \mathbf{r}}{\partial\theta}\right\|^2 &= (\cos\theta\cos\phi)^2 + (\cos\theta\sin\phi)^2 + (-\sin\theta)^2 \\ &= \cos^2\theta\cos^2\phi + \cos^2\theta\sin^2\phi + \sin^2\theta \\ &= \cos^2\theta(\cos^2\phi + \sin^2\phi) + \sin^2\theta \\ &= \cos^2\theta(1) + \sin^2\theta = 1. \end{aligned}$$

Since the magnitude is 1, the normalized basis vector is simply the partial derivative itself:

$$\hat{\mathbf{e}}_\theta = (\cos\theta\cos\phi)\,\hat{\mathbf{i}} + (\cos\theta\sin\phi)\,\hat{\mathbf{j}} - (\sin\theta)\,\hat{\mathbf{k}}.$$

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Basis vector $\hat{\mathbf{e}}_\phi$: We take the partial derivative with respect to ϕ:

$$\frac{\partial \mathbf{r}}{\partial \phi} = (-\sin\theta\sin\phi)\,\hat{\mathbf{i}} + (\sin\theta\cos\phi)\,\hat{\mathbf{j}} + 0\,\hat{\mathbf{k}}.$$

Its magnitude (the scale factor h_ϕ) is

$$\begin{aligned}\left\|\frac{\partial \mathbf{r}}{\partial \phi}\right\|^2 &= (-\sin\theta\sin\phi)^2 + (\sin\theta\cos\phi)^2 + 0^2 \\ &= \sin^2\theta\sin^2\phi + \sin^2\theta\cos^2\phi \\ &= \sin^2\theta(\sin^2\phi + \cos^2\phi) = \sin^2\theta(1) = \sin^2\theta.\end{aligned}$$

Thus $\left\|\frac{\partial \mathbf{r}}{\partial \phi}\right\| = \sin\theta$. Normalizing gives

$$\hat{\mathbf{e}}_\phi = \frac{1}{\sin\theta}\left((-\sin\theta\sin\phi)\,\hat{\mathbf{i}} + (\sin\theta\cos\phi)\,\hat{\mathbf{j}}\right) = (-\sin\phi)\,\hat{\mathbf{i}} + (\cos\phi)\,\hat{\mathbf{j}}.$$

Result: The orthonormal basis for the tangent space of the unit sphere is $\{\hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_\phi\}$, where:

$$\begin{aligned}\hat{\mathbf{e}}_\theta &= (\cos\theta\cos\phi)\,\hat{\mathbf{i}} + (\cos\theta\sin\phi)\,\hat{\mathbf{j}} - (\sin\theta)\,\hat{\mathbf{k}}, \\ \hat{\mathbf{e}}_\phi &= (-\sin\phi)\,\hat{\mathbf{i}} + (\cos\phi)\,\hat{\mathbf{j}}.\end{aligned}$$

Observe that a third basis vector, $\hat{\mathbf{e}}_r$, points radially outward and is normal to the surface; it is not part of the basis for the surface itself. ■

Example 2.1.37 (Cylindrical Coordinates)**.** The cylindrical coordinate system is a three-dimensional coordinate system described by the coordinates (ρ, ϕ, z), where ρ is the radial distance from the z-axis, ϕ is the azimuthal angle, and z is the height.

The transformation from cylindrical to Cartesian coordinates is given by

$$\begin{aligned}x &= \rho\cos\phi, \\ y &= \rho\sin\phi, \\ z &= z.\end{aligned}$$

The position vector of a point is represented as:

$$\mathbf{r}(\rho, \phi, z) = (\rho\cos\phi)\,\hat{\mathbf{i}} + (\rho\sin\phi)\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}.$$

Solution. To find the orthonormal basis, we compute the partial derivative of the position vector with respect to each coordinate and normalize the resulting tangent vector.

1. Radial Direction (ρ): The tangent vector, its scale factor (magnitude), and the corresponding normalized unit vector are

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial \rho} &= (\cos\phi)\,\hat{\mathbf{i}} + (\sin\phi)\,\hat{\mathbf{j}}, \\ \left\|\frac{\partial \mathbf{r}}{\partial \rho}\right\| &= \sqrt{\cos^2\phi + \sin^2\phi} = 1, \\ \hat{\mathbf{e}}_\rho &= (\cos\phi)\,\hat{\mathbf{i}} + (\sin\phi)\,\hat{\mathbf{j}}.\end{aligned}$$

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2. Azimuthal Direction (ϕ): Similarly, along the azimuthal direction, we compute:

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial \phi} &= (-\rho \sin\phi)\,\hat{\mathbf{i}} + (\rho\cos\phi)\,\hat{\mathbf{j}},\\ \left\|\frac{\partial \mathbf{r}}{\partial \phi}\right\| &= \sqrt{(-\rho\sin\phi)^2 + (\rho\cos\phi)^2} = \rho,\\ \hat{\mathbf{e}}_\phi &= \frac{1}{\rho}\frac{\partial \mathbf{r}}{\partial \phi} = (-\sin\phi)\,\hat{\mathbf{i}} + (\cos\phi)\,\hat{\mathbf{j}}.\end{aligned}$$

3. Vertical Direction (z): Finally, along the vertical direction, the derivation yields

$$\begin{aligned}\frac{\partial \mathbf{r}}{\partial z} &= \hat{\mathbf{k}},\\ \left\|\frac{\partial \mathbf{r}}{\partial z}\right\| &= 1,\\ \hat{\mathbf{e}}_z &= \hat{\mathbf{k}}.\end{aligned}$$

Result Summary: The final orthonormal basis $\{\hat{\mathbf{e}}_\rho, \hat{\mathbf{e}}_\phi, \hat{\mathbf{e}}_z\}$ for the cylindrical coordinate system is

$$\begin{aligned}\hat{\mathbf{e}}_\rho &= (\cos\phi)\,\hat{\mathbf{i}} + (\sin\phi)\,\hat{\mathbf{j}},\\ \hat{\mathbf{e}}_\phi &= (-\sin\phi)\,\hat{\mathbf{i}} + (\cos\phi)\,\hat{\mathbf{j}},\\ \hat{\mathbf{e}}_z &= \hat{\mathbf{k}}.\end{aligned}$$

■

2.2 The First Fundamental Form

Let a surface S in the Euclidean space $\mathbb{R}^3$ be given by a regular parametrization $\vec{r} : D \to \mathbb{R}^3$, where D is an open domain in the uv-plane.

$$\vec{r}(u,v) = (x(u,v), y(u,v), z(u,v))$$

At any point $P = \vec{r}(u_0, v_0)$ on the surface, the partial derivative vectors

$$\vec{r}_1(u,v) = \frac{\partial \vec{r}}{\partial u} \quad \text{and} \quad \vec{r}_2(u,v) = \frac{\partial \vec{r}}{\partial v}$$

are tangent to the coordinate curves passing through P. Since the parametrization is regular, the vectors $\vec{r}_1$ and $\vec{r}_2$ are linearly independent. They form a basis for the *tangent plane* to the surface at P, denoted $T_P(S)$. Any tangent vector $\vec{w} \in T_P(S)$ can be expressed as a unique linear combination $\vec{w} = \alpha\vec{r}_1 + \beta\vec{r}_2$.

Our goal is to measure geometric quantities like length and area on the surface S itself. We begin by considering the infinitesimal displacement vector $d\vec{r}$ along a curve on the surface. This vector can be expressed in terms of the basis vectors $\vec{r}_1, \vec{r}_2$:

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u}du + \frac{\partial \vec{r}}{\partial v}dv = \vec{r}_1 du + \vec{r}_2 dv$$

The square of the length of this infinitesimal vector, denoted ds^2, is found by taking its dot product with itself.

$$\begin{aligned}ds^2 &= d\vec{r}\cdot d\vec{r}\\ &= (\vec{r}_1 du + \vec{r}_2 dv)\cdot(\vec{r}_1 du + \vec{r}_2 dv)\\ &= (\vec{r}_1\cdot\vec{r}_1)du^2 + 2(\vec{r}_1\cdot\vec{r}_2)du dv + (\vec{r}_2\cdot\vec{r}_2)dv^2\end{aligned}$$

This quadratic form is fundamental to the study of surface geometry.

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Definition 2.2.1 (Coefficients of the First Fundamental Form)**.** The coefficients of the first fundamental form are three functions of (u, v) denoted by E, F, G:

$$\begin{aligned} E(u,v) &= \vec{r}_1 \cdot \vec{r}_1 = |\vec{r}_1|^2 \\ F(u,v) &= \vec{r}_1 \cdot \vec{r}_2 \\ G(u,v) &= \vec{r}_2 \cdot \vec{r}_2 = |\vec{r}_2|^2 \end{aligned}$$

Definition 2.2.2 (The First Fundamental Form)**.** The **first fundamental form**, denoted by I, is the quadratic form on the tangent plane that gives the squared infinitesimal arc length ds^2:

$$I = ds^2 = E(u,v)du^2 + 2F(u,v)dudv + G(u,v)dv^2$$

In matrix notation, this is written as

$$ds^2 = \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

The symmetric matrix $g = (g_{ij}) = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$ is called the **metric tensor** of the surface.

Observe that since $\vec{r}_1$ and $\vec{r}_2$ are linearly independent, the determinant of the metric tensor, $\det(g) = EG - F^2$, is always positive.

We establish two fundamental properties of the first fundamental form. These properties confirm its role as a proper metric on the surface.

Theorem 2.2.3. *The first fundamental form $I = ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2$ has the following properties:*

1. ***Positive Definiteness:*** *The form is positive definite. That is, $ds^2 \geq 0$, and $ds^2 = 0$ if and only if $du = 0$ and $dv = 0$.*
2. ***Invariance:*** *The value of the first fundamental form is invariant under a regular change of parameters (reparametrization).*

Proof. **(1):** By definition, the first fundamental form is the squared magnitude of the infinitesimal displacement vector $d\vec{r} = \vec{r}_1 du + \vec{r}_2 dv$.

$$ds^2 = d\vec{r} \cdot d\vec{r} = |\vec{r}_1 du + \vec{r}_2 dv|^2$$

The square of the magnitude of any real vector is always non-negative. Therefore, $ds^2 \geq 0$.

Now, we must show that $ds^2 = 0$ if and only if $du = 0$ and $dv = 0$. The equality $ds^2 = 0$ holds if and only if the vector $d\vec{r}$ is the zero vector:

$$\vec{r}_1 du + \vec{r}_2 dv = \vec{0}$$

The surface $\vec{r}(u, v)$ is assumed to be regular. By the definition of a regular surface, the tangent vectors $\vec{r}_1 = \partial\vec{r}/\partial u$ and $\vec{r}_2 = \partial\vec{r}/\partial v$ are linearly independent at every point.

The definition of linear independence states that the only linear combination of $\vec{r}_1$ and $\vec{r}_2$ that equals the zero vector is the trivial one, where all scalar coefficients are zero. In our case, the scalars are du and dv.

Therefore, the equation $\vec{r}_1 du + \vec{r}_2 dv = \vec{0}$ implies that $du = 0$ and $dv = 0$.

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Conversely, if $du = 0$ and $dv = 0$, it is trivial to see that $ds^2 = E(0)^2 + 2F(0)(0) + G(0)^2 = 0$.

Thus, $ds^2 > 0$ for any non-zero pair of differentials (du, dv), which proves that the first fundamental form is positive definite.

(2): Let the surface be described by two different regular parametrizations:

1. $\vec{r}(u, v)$, with first fundamental form $I = E\,du^2 + 2F\,du\,dv + G\,dv^2$.
2. $\vec{r}(u', v')$, with first fundamental form $I' = E'\,(du')^2 + 2F'\,du'\,dv' + G'\,(dv')^2$.

Let the change of parameters be given by a regular transformation:

$$u' = \phi(u, v) \quad \text{and} \quad v' = \psi(u, v)$$

This implies we can also write $u = u(u', v')$ and $v = v(u', v')$. The infinitesimal displacement vector on the surface, $d\vec{r}$, is a geometric entity. In the (u, v) coordinate system, it is expressed as

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u} du + \frac{\partial \vec{r}}{\partial v} dv = \vec{r}_1 du + \vec{r}_2 dv$$

In the (u', v') coordinate system, the same vector $d\vec{r}$ is expressed as

$$d\vec{r} = \frac{\partial \vec{r}}{\partial u'} du' + \frac{\partial \vec{r}}{\partial v'} dv' = \vec{r}_{1'} du' + \vec{r}_{2'} dv'$$

Using the chain rule, we have the differential relations:

$$du = \frac{\partial u}{\partial u'} du' + \frac{\partial u}{\partial v'} dv'$$
$$dv = \frac{\partial v}{\partial u'} du' + \frac{\partial v}{\partial v'} dv'$$

Substituting these into the first expression for $d\vec{r}$:

$$\begin{aligned} d\vec{r} &= \vec{r}_1 \left(\frac{\partial u}{\partial u'} du' + \frac{\partial u}{\partial v'} dv' \right) + \vec{r}_2 \left(\frac{\partial v}{\partial u'} du' + \frac{\partial v}{\partial v'} dv' \right) \\ &= \left(\vec{r}_1 \frac{\partial u}{\partial u'} + \vec{r}_2 \frac{\partial v}{\partial u'} \right) du' + \left(\vec{r}_1 \frac{\partial u}{\partial v'} + \vec{r}_2 \frac{\partial v}{\partial v'} \right) dv' \end{aligned}$$

By the chain rule for vector functions, the terms in the parentheses are precisely the definitions of the new tangent vectors:

$$\frac{\partial \vec{r}}{\partial u'} = \frac{\partial \vec{r}}{\partial u}\frac{\partial u}{\partial u'} + \frac{\partial \vec{r}}{\partial v}\frac{\partial v}{\partial u'} = \vec{r}_1 \frac{\partial u}{\partial u'} + \vec{r}_2 \frac{\partial v}{\partial u'} = \vec{r}_{1'}$$
$$\frac{\partial \vec{r}}{\partial v'} = \frac{\partial \vec{r}}{\partial u}\frac{\partial u}{\partial v'} + \frac{\partial \vec{r}}{\partial v}\frac{\partial v}{\partial v'} = \vec{r}_1 \frac{\partial u}{\partial v'} + \vec{r}_2 \frac{\partial v}{\partial v'} = \vec{r}_{2'}$$

Substituting these back, we obtain $d\vec{r} = \vec{r}_{1'} du' + \vec{r}_{2'} dv'$. This confirms that our two expressions for $d\vec{r}$ represent the same vector in $\mathbb{R}^3$.

The first fundamental form is simply the squared magnitude of this vector, $ds^2 = |d\vec{r}|^2$. Since both coordinate systems produce the exact same vector $d\vec{r}$, their squared magnitudes must be identical.

$$I = |\vec{r}_1 du + \vec{r}_2 dv|^2 = |d\vec{r}|^2 = |\vec{r}_{1'} du' + \vec{r}_{2'} dv'|^2 = I'$$

Thus, the value of the first fundamental form is independent of the chosen parametrization, proving its invariance. ■

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Example 2.2.4 (The Sphere)**.** Consider a sphere of radius a with the standard spherical parametrization:

$$\vec{r}(u,v) = (a\sin u\cos v, a\sin u\sin v, a\cos u)$$

where $u \in [0,\pi]$ is the polar angle and $v \in [0,2\pi)$ is the azimuthal angle. Find the first fundamental form.

Solution. First compute the partial derivatives $\vec{r}_1$ (w.r.t u) and $\vec{r}_2$ (w.r.t v):

$$\vec{r}_1 = (a\cos u\cos v, a\cos u\sin v, -a\sin u)$$
$$\vec{r}_2 = (-a\sin u\sin v, a\sin u\cos v, 0)$$

Next compute the coefficients E, F, G:

$$\begin{aligned} E &= \vec{r}_1\cdot\vec{r}_1 = a^2\cos^2 u\cos^2 v + a^2\cos^2 u\sin^2 v + a^2\sin^2 u \\ &= a^2\cos^2 u(\cos^2 v + \sin^2 v) + a^2\sin^2 u \\ &= a^2(\cos^2 u + \sin^2 u) = a^2 \\ F &= \vec{r}_1\cdot\vec{r}_2 = -a^2\cos u\sin u\cos v\sin v + a^2\cos u\sin u\sin v\cos v + 0 = 0 \\ G &= \vec{r}_2\cdot\vec{r}_2 = a^2\sin^2 u\sin^2 v + a^2\sin^2 u\cos^2 v + 0 \\ &= a^2\sin^2 u(\sin^2 v + \cos^2 v) = a^2\sin^2 u \end{aligned}$$

The first fundamental form for the sphere is $ds^2 = a^2du^2 + a^2\sin^2 u\,dv^2$. Since $F = 0$, the lines of constant latitude ($u = \text{const}$) and longitude ($v = \text{const}$) are orthogonal everywhere except at the poles where the parametrization is singular. The area element is $dA = \sqrt{EG - F^2}\,du\,dv = \sqrt{a^2\cdot a^2\sin^2 u}\,du\,dv = a^2\sin u\,du\,dv$. ■

Example 2.2.5 (The General Plane)**.** Let a plane in $\mathbb{R}^3$ be parametrized by $\vec{r}(u,v) = \vec{p}_0 + u\vec{a} + v\vec{b}$, where $\vec{p}_0$ is a point on the plane and $\{\vec{a},\vec{b}\}$ is a basis of linearly independent (but not necessarily orthogonal) constant vectors. Find the first fundamental form.

Solution. Differentiating, we obtain the partial derivative vectors, which are constant for this parametrization.

$$\vec{r}_1 = \frac{\partial}{\partial u}(\vec{p}_0 + u\vec{a} + v\vec{b}) = \vec{a}$$
$$\vec{r}_2 = \frac{\partial}{\partial v}(\vec{p}_0 + u\vec{a} + v\vec{b}) = \vec{b}$$

Now, we compute the coefficients E, F, G by taking the dot products of these basis vectors.

$$E = \vec{r}_1\cdot\vec{r}_1 = \vec{a}\cdot\vec{a} = |\vec{a}|^2$$
$$F = \vec{r}_1\cdot\vec{r}_2 = \vec{a}\cdot\vec{b}$$
$$G = \vec{r}_2\cdot\vec{r}_2 = \vec{b}\cdot\vec{b} = |\vec{b}|^2$$

Since the vectors $\vec{a}$ and $\vec{b}$ are constant, the coefficients E, F, G are constants for the entire surface. The first fundamental form is

$$ds^2 = |\vec{a}|^2du^2 + 2(\vec{a}\cdot\vec{b})dudv + |\vec{b}|^2dv^2$$

Remark: If we choose an orthonormal basis, where $|\vec{a}| = 1$, $|\vec{b}| = 1$, and $\vec{a}\cdot\vec{b} = 0$, then $E = 1, F = 0, G = 1$. The metric becomes $ds^2 = du^2 + dv^2$, which is the familiar Pythagorean theorem in the Euclidean plane. This example shows how a non-zero F coefficient arises from using a non-orthogonal coordinate system. ■

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Example 2.2.6 (The Right Circular Cone)**.** Find the first fundamental form for a right circular cone parametrized by:

$$\vec{r}(u,v) = (u\cos v, u\sin v, ku), \quad u > 0, \quad k = \cot\alpha > 0$$

where α is the half-angle at the vertex.

Solution. First obtain the tangent vectors by differentiating with respect to u and v.

$$\vec{r}_1 = \frac{\partial \vec{r}}{\partial u} = (\cos v, \sin v, k)$$
$$\vec{r}_2 = \frac{\partial \vec{r}}{\partial v} = (-u\sin v, u\cos v, 0)$$

Next compute the dot products to find E, F, G.

$$\begin{aligned} E &= \vec{r}_1 \cdot \vec{r}_1 = (\cos v)^2 + (\sin v)^2 + k^2 = \cos^2 v + \sin^2 v + k^2 = 1 + k^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = (\cos v)(-u\sin v) + (\sin v)(u\cos v) + (k)(0) \\ &= -u\sin v\cos v + u\sin v\cos v = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-u\sin v)^2 + (u\cos v)^2 + 0^2 \\ &= u^2\sin^2 v + u^2\cos^2 v = u^2(\sin^2 v + \cos^2 v) = u^2 \end{aligned}$$

The first fundamental form for the cone is $ds^2 = (1+k^2)du^2 + u^2dv^2$. **Remark:** Since $F = 0$, the coordinate curves are orthogonal. The u-curves are straight lines along the generators of the cone, and the v-curves are circles of constant height. This metric shows that the cone is locally Euclidean (or "flat") because it can be unrolled into a sector of a plane without stretching, a property shared by all surfaces with zero Gaussian curvature. ■

Example 2.2.7 (The Helicoid)**.** Compute the first fundamental form of the helicoid, given by the parametrization:

$$\vec{r}(u,v) = (u\cos v, u\sin v, cv), \quad u \in \mathbb{R}, \quad c \neq 0$$

Solution. We compute the partial derivatives of $\vec{r}(u,v)$.

$$\vec{r}_1 = \frac{\partial \vec{r}}{\partial u} = (\cos v, \sin v, 0)$$
$$\vec{r}_2 = \frac{\partial \vec{r}}{\partial v} = (-u\sin v, u\cos v, c)$$

Now we calculate the coefficients E, F, G.

$$\begin{aligned} E &= \vec{r}_1 \cdot \vec{r}_1 = \cos^2 v + \sin^2 v + 0^2 = 1 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = (\cos v)(-u\sin v) + (\sin v)(u\cos v) + (0)(c) = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-u\sin v)^2 + (u\cos v)^2 + c^2 \\ &= u^2\sin^2 v + u^2\cos^2 v + c^2 = u^2 + c^2 \end{aligned}$$

The first fundamental form for the helicoid is $ds^2 = du^2 + (u^2 + c^2)dv^2$. **Remark:** The coordinate curves are again orthogonal because $F = 0$. The u-curves ($v = \text{const}$) are straight lines radiating horizontally from the z-axis. The v-curves ($u = \text{const}$) are helices of radius u. ■

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Example 2.2.8 (Surface of Revolution). Let a regular plane curve (the profile curve) be given by $\vec{\alpha}(u) = (f(u), 0, g(u))$ with $f(u) > 0$. Find the first fundamental form of the surface generated by revolving this curve about the z-axis.

Solution. The parametrization of the surface of revolution is obtained by rotating the profile curve by an angle v around the z-axis.

$$\vec{r}(u, v) = (f(u)\cos v, f(u)\sin v, g(u))$$

We compute the tangent vectors $\vec{r}_1$ and $\vec{r}_2$. Let $f' = df/du$ and $g' = dg/du$.

$$\vec{r}_1 = \frac{\partial \vec{r}}{\partial u} = (f'(u)\cos v, f'(u)\sin v, g'(u))$$
$$\vec{r}_2 = \frac{\partial \vec{r}}{\partial v} = (-f(u)\sin v, f(u)\cos v, 0)$$

Next obtain the coefficients E, F, G.

$$\begin{aligned} E &= \vec{r}_1 \cdot \vec{r}_1 = (f'(u))^2\cos^2 v + (f'(u))^2\sin^2 v + (g'(u))^2 \\ &= (f'(u))^2(\cos^2 v + \sin^2 v) + (g'(u))^2 = (f'(u))^2 + (g'(u))^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = (f'(u)\cos v)(-f(u)\sin v) + (f'(u)\sin v)(f(u)\cos v) + (g'(u))(0) \\ &= -f(u)f'(u)\sin v\cos v + f(u)f'(u)\sin v\cos v = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (-f(u)\sin v)^2 + (f(u)\cos v)^2 + 0^2 \\ &= (f(u))^2\sin^2 v + (f(u))^2\cos^2 v = (f(u))^2 \end{aligned}$$

The first fundamental form for a general surface of revolution is $ds^2 = \left((f'(u))^2 + (g'(u))^2\right)du^2 + (f(u))^2dv^2$. **Remark:** This is a powerful general result. For example, for a sphere of radius a, we can use the profile curve $f(u) = a\sin u, g(u) = a\cos u$. Then $f' = a\cos u, g' = -a\sin u$. $E = (a\cos u)^2 + (-a\sin u)^2 = a^2$ and $G = (a\sin u)^2 = a^2\sin^2 u$. This yields $ds^2 = a^2du^2 + a^2\sin^2 u\, dv^2$, recovering the result from Example 2.2.4. ■

The first fundamental form allows us to compute all metric properties of the surface (those depending on measurements made along the surface) using only the coefficients E, F, G.

Theorem 2.2.9 (Length of a Curve). *Let C be a curve on the surface S given by $\vec{c}(t) = \vec{r}(u(t), v(t))$ for $t \in [a, b]$. The arc length L of C is given by*

$$L = \int_a^b \sqrt{E\left(\frac{du}{dt}\right)^2 + 2F\left(\frac{du}{dt}\right)\left(\frac{dv}{dt}\right) + G\left(\frac{dv}{dt}\right)^2}\, dt$$

Proof. The tangent vector to the curve is $\vec{c}'(t) = \frac{d\vec{r}}{dt} = \frac{\partial r}{\partial u}\frac{\partial u}{\partial t} + \frac{\partial r}{\partial v}\frac{\partial v}{\partial t} = \vec{r}_1 u'(t) + \vec{r}_2 v'(t)$. The speed along the curve is $|\vec{c}'(t)|$. The square of the speed is

$$\begin{aligned} |\vec{c}'(t)|^2 &= \vec{c}'(t) \cdot \vec{c}'(t) \\ &= (\vec{r}_1 u' + \vec{r}_2 v') \cdot (\vec{r}_1 u' + \vec{r}_2 v') \\ &= (\vec{r}_1 \cdot \vec{r}_1)(u')^2 + 2(\vec{r}_1 \cdot \vec{r}_2)u'v' + (\vec{r}_2 \cdot \vec{r}_2)(v')^2 \\ &= E(u')^2 + 2Fu'v' + G(v')^2 \end{aligned}$$

The arc length is the integral of the speed, $L = \int_a^b |\vec{c}'(t)|dt$. Taking the square root of the expression above yields the desired formula. ■

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Theorem 2.2.10 (Angle Between Tangent Vectors). *Let $\vec{w}_1 = a_1\vec{r}_1 + b_1\vec{r}_2$ and $\vec{w}_2 = a_2\vec{r}_1 + b_2\vec{r}_2$ be two tangent vectors at a point P on the surface. The angle θ between them is given by*

$$\cos\theta = \frac{\vec{w}_1 \cdot \vec{w}_2}{|\vec{w}_1||\vec{w}_2|} = \frac{Ea_1a_2 + F(a_1b_2 + a_2b_1) + Gb_1b_2}{\sqrt{Ea_1^2 + 2Fa_1b_1 + Gb_1^2}\sqrt{Ea_2^2 + 2Fa_2b_2 + Gb_2^2}}$$

Proof. The proof is a direct computation using definition. ■

Corollary 2.2.11 (Orthogonality of Coordinate Curves). *The coordinate curves (u-curves and v-curves) are orthogonal at a point P if and only if $F(u, v) = 0$ at that point.*

Proof. The tangent vectors to the coordinate curves are $\vec{r}_1$ and $\vec{r}_2$. They are orthogonal if and only if their dot product is zero, which is precisely $F = \vec{r}_1 \cdot \vec{r}_2 = 0$. ■

Note: Since for any two vectors $\vec{a}, \vec{b} \in \mathbb{R}^3$, the magnitudes of their dot and cross products are related by the angle θ between them:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$$
$$|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$$

From these, we derive the celebrated **Lagrange's Identity**:

$$\begin{aligned} |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 &= |\vec{a}|^2|\vec{b}|^2 - (|\vec{a}||\vec{b}| \cos\theta)^2 \\ &= |\vec{a}|^2|\vec{b}|^2(1 - \cos^2\theta) \\ &= |\vec{a}|^2|\vec{b}|^2 \sin^2\theta \\ &= (|\vec{a}||\vec{b}| \sin\theta)^2 = |\vec{a} \times \vec{b}|^2 \end{aligned}$$

Thus, we have the identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$.

Theorem 2.2.12. *Let S be a regular surface parametrized by $\vec{r}(u, v)$. Let $\vec{r}_1 = \partial\vec{r}/\partial u$ and $\vec{r}_2 = \partial\vec{r}/\partial v$ be the tangent vectors at a point on S. The squared magnitude of the cross product of these vectors is given by $|\vec{r}_1 \times \vec{r}_2|^2 = EG - F^2$. where E, F, and G are the coefficients of the first fundamental form.*

Proof using Lagrange's Identity. We apply the Lagrange's identity to the tangent vectors $\vec{r}_1$ and $\vec{r}_2$.

$$|\vec{r}_1 \times \vec{r}_2|^2 = |\vec{r}_1|^2|\vec{r}_2|^2 - (\vec{r}_1 \cdot \vec{r}_2)^2$$

By the definitions of the first fundamental coefficients, we have

- $E = \vec{r}_1 \cdot \vec{r}_1 = |\vec{r}_1|^2$
- $F = \vec{r}_1 \cdot \vec{r}_2$
- $G = \vec{r}_2 \cdot \vec{r}_2 = |\vec{r}_2|^2$

Substituting these definitions into the identity yields the desired result:

$$|\vec{r}_1 \times \vec{r}_2|^2 = EG - F^2$$

This proves the result. ■

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Proof by Direct Computation. Let the surface be given in component form by

$$\vec{r}(u,v) = (x(u,v), y(u,v), z(u,v)).$$

The partial derivative vectors are

$$\vec{r}_1 = (x_u, y_u, z_u) \quad \text{and} \quad \vec{r}_2 = (x_v, y_v, z_v)$$

The cross product $\vec{r}_1 \times \vec{r}_2$ is given by

$$\vec{r}_1 \times \vec{r}_2 = (y_u z_v - z_u y_v,\ z_u x_v - x_u z_v,\ x_u y_v - y_u x_v)$$

The squared magnitude of this vector is the sum of the squares of its components:

$$|\vec{r}_1 \times \vec{r}_2|^2 = (y_u z_v - z_u y_v)^2 + (z_u x_v - x_u z_v)^2 + (x_u y_v - y_u x_v)^2 \tag{2.2}$$

Next compute the right-hand side, $EG - F^2$. The coefficients are

$$\begin{aligned} E &= x_u^2 + y_u^2 + z_u^2 \\ G &= x_v^2 + y_v^2 + z_v^2 \\ F &= x_u x_v + y_u y_v + z_u z_v \end{aligned}$$

Expanding the expression $EG - F^2$ gives

$$\begin{aligned} EG - F^2 &= (x_u^2 + y_u^2 + z_u^2)(x_v^2 + y_v^2 + z_v^2) - (x_u x_v + y_u y_v + z_u z_v)^2 \\ &= (x_u^2 y_v^2 - 2x_u y_v y_u x_v + y_u^2 x_v^2) + (x_u^2 z_v^2 - 2x_u z_v z_u x_v + z_u^2 x_v^2) \\ &\quad + (y_u^2 z_v^2 - 2y_u z_v z_u y_v + z_u^2 y_v^2) \\ &= (x_u y_v - y_u x_v)^2 + (x_u z_v - z_u x_v)^2 + (y_u z_v - z_u y_v)^2 \end{aligned}$$

Upon rearranging the terms, this becomes

$$EG - F^2 = (y_u z_v - z_u y_v)^2 + (z_u x_v - x_u z_v)^2 + (x_u y_v - y_u x_v)^2 \tag{2.3}$$

A direct comparison of equations (2.2) and (2.3) shows they are identical. ■

Theorem 2.2.13 (Surface Area). *Let $R \subset S$ be a region on the surface corresponding to a domain D in the uv-plane. The area of R, denoted $A(R)$, is given by $A(R) = \iint_D \sqrt{EG - F^2}\, du\, dv$.*

Proof. An infinitesimal parallelogram on the surface is spanned by the vectors $d\vec{u} = \vec{r}_1 du$ and $d\vec{v} = \vec{r}_2 dv$. The area of this parallelogram, dA, is the magnitude of their cross product:

$$dA = |d\vec{u} \times d\vec{v}| = |\vec{r}_1 du \times \vec{r}_2 dv| = |\vec{r}_1 \times \vec{r}_2|\, du\, dv$$

By Lagrange's identity, $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2$. Applying this, we obtain

$$|\vec{r}_1 \times \vec{r}_2|^2 = |\vec{r}_1|^2 |\vec{r}_2|^2 - (\vec{r}_1 \cdot \vec{r}_2)^2 = EG - F^2$$

Thus, the area element is $dA = \sqrt{EG - F^2}\, du\, dv$. Integrating this over the domain D gives the total area of the region R. ■

Theorem 2.2.14. *The area of a regular surface is a geometric invariant, independent of its parametrization.*

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Proof. Let R be a region on a regular surface S. Suppose this region can be described by two different regular parametrizations: $\vec{r}(u,v)$ over a domain D in the uv-plane and $\vec{s}(u',v')$ over a domain D' in the $u'v'$-plane. The area of R can be computed using either parametrization:

$$A_1 = \iint_D |\vec{r}_u \times \vec{r}_v|\, du\, dv \quad \text{and} \quad A_2 = \iint_{D'} |\vec{s}_{u'} \times \vec{s}_{v'}|\, du'\, dv'$$

Our goal is to prove that $A_1 = A_2$. We will achieve this by transforming the first integral into the second using the change of variables formula. The heart of the proof lies in understanding how the area element transforms between the two coordinate systems.

Let the coordinate transformation be given by $u = u(u',v')$ and $v = v(u',v')$. By the chain rule, the tangent vectors of the two systems are related as follows:

$$\vec{s}_{u'} = \frac{\partial \vec{r}}{\partial u'} = \frac{\partial \vec{r}}{\partial u}\frac{\partial u}{\partial u'} + \frac{\partial \vec{r}}{\partial v}\frac{\partial v}{\partial u'} = \vec{r}_u \frac{\partial u}{\partial u'} + \vec{r}_v \frac{\partial v}{\partial u'}$$
$$\vec{s}_{v'} = \frac{\partial \vec{r}}{\partial v'} = \frac{\partial \vec{r}}{\partial u}\frac{\partial u}{\partial v'} + \frac{\partial \vec{r}}{\partial v}\frac{\partial v}{\partial v'} = \vec{r}_u \frac{\partial u}{\partial v'} + \vec{r}_v \frac{\partial v}{\partial v'}$$

The key step is to compute the cross product $\vec{s}_{u'} \times \vec{s}_{v'}$. Using the bilinearity of the cross product and the fact that $\vec{v} \times \vec{w} = -\vec{w} \times \vec{v}$, we obtain

$$\begin{aligned}
\vec{s}_{u'} \times \vec{s}_{v'} &= \left(\vec{r}_u \frac{\partial u}{\partial u'} + \vec{r}_v \frac{\partial v}{\partial u'}\right) \times \left(\vec{r}_u \frac{\partial u}{\partial v'} + \vec{r}_v \frac{\partial v}{\partial v'}\right) \\
&= (\vec{r}_u \times \vec{r}_v)\frac{\partial u}{\partial u'}\frac{\partial v}{\partial v'} + (\vec{r}_v \times \vec{r}_u)\frac{\partial v}{\partial u'}\frac{\partial u}{\partial v'} \\
&= (\vec{r}_u \times \vec{r}_v)\left(\frac{\partial u}{\partial u'}\frac{\partial v}{\partial v'} - \frac{\partial u}{\partial v'}\frac{\partial v}{\partial u'}\right)
\end{aligned}$$

We recognize the scalar term in the parentheses as the Jacobian determinant of the transformation, denoted $\frac{\partial(u,v)}{\partial(u',v')}$. This gives the fundamental relationship between the normal vectors of the two parametrizations:

$$\vec{s}_{u'} \times \vec{s}_{v'} = (\vec{r}_u \times \vec{r}_v)\frac{\partial(u,v)}{\partial(u',v')}$$

Taking the magnitude of both sides yields the transformation rule for the area distortion factor:

$$|\vec{s}_{u'} \times \vec{s}_{v'}| = |\vec{r}_u \times \vec{r}_v| \left|\frac{\partial(u,v)}{\partial(u',v')}\right|$$

Now we apply the standard change of variables formula to the first area integral:

$$A_1 = \iint_D |\vec{r}_u \times \vec{r}_v|\, du\, dv = \iint_{D'} |\vec{r}_u \times \vec{r}_v| \left|\frac{\partial(u,v)}{\partial(u',v')}\right| du'\, dv'$$

Substituting the transformation rule we just derived, we obtain $A_1 = \iint_{D'} |\vec{s}_{u'} \times \vec{s}_{v'}|\, du'\, dv' = A_2$. Since the two expressions yield the same value, we conclude that the surface area is an intrinsic property of the surface, independent of the particular parametrization chosen to describe it. ■

Example 2.2.15. Use the first fundamental form to compute the surface area of a sphere of radius a.

Solution. We begin by choosing the standard spherical coordinate parametrization for a sphere of radius a:

$$\vec{r}(u,v) = (a\sin u\cos v, a\sin u\sin v, a\cos u)$$

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The domain of the parameters that covers the entire sphere is $D = [0, \pi] \times [0, 2\pi]$, where u is the polar angle and v is the azimuthal angle. The area A is given by the integral of the area element over this domain:

$$A = \iint_D \sqrt{EG - F^2}\, du\, dv$$

Our first task is to compute the coefficients E, F, and G for this parametrization. First obtain the partial derivatives of $\vec{r}(u, v)$ with respect to u and v:

$$\vec{r}_1 = \frac{\partial \vec{r}}{\partial u} = (a\cos u\cos v, a\cos u\sin v, -a\sin u)$$
$$\vec{r}_2 = \frac{\partial \vec{r}}{\partial v} = (-a\sin u\sin v, a\sin u\cos v, 0)$$

From these, we compute the coefficients of the first fundamental form by taking their dot products:

$$\begin{aligned} E = \vec{r}_1 \cdot \vec{r}_1 &= (a\cos u\cos v)^2 + (a\cos u\sin v)^2 + (-a\sin u)^2 \\ &= a^2\cos^2 u(\cos^2 v + \sin^2 v) + a^2\sin^2 u = a^2(\cos^2 u + \sin^2 u) = \boldsymbol{a^2} \end{aligned}$$

$$F = \vec{r}_1 \cdot \vec{r}_2 = -a^2\sin u\cos u\sin v\cos v + a^2\sin u\cos u\sin v\cos v = \mathbf{0}$$

$$\begin{aligned} G = \vec{r}_2 \cdot \vec{r}_2 &= (-a\sin u\sin v)^2 + (a\sin u\cos v)^2 + (0)^2 \\ &= a^2\sin^2 u(\sin^2 v + \cos^2 v) = \boldsymbol{a^2\sin^2 u} \end{aligned}$$

With the coefficients determined, we obtain the term under the square root:

$$EG - F^2 = (a^2)(a^2\sin^2 u) - (0)^2 = a^4\sin^2 u$$

The area element is therefore

$$dA = \sqrt{EG - F^2}\, du\, dv = \sqrt{a^4\sin^2 u}\, du\, dv = a^2|\sin u|\, du\, dv$$

Since $u \in [0, \pi]$, $\sin u \geq 0$, so we may drop the absolute value. We now integrate the area element over the domain $D = [0, \pi] \times [0, 2\pi]$:

$$\begin{aligned} A &= \iint_D a^2\sin u\, du\, dv \\ &= \int_{v=0}^{2\pi} \left(\int_{u=0}^{\pi} a^2\sin u\, du \right) dv \end{aligned}$$

The inner integral with respect to u evaluates to:

$$\int_0^{\pi} a^2\sin u\, du = a^2\left[-\cos u\right]_0^{\pi} = a^2\left(-(-1) - (-1)\right) = 2a^2$$

Substituting this constant result into the outer integral, we obtain the total area:

$$A = \int_0^{2\pi} 2a^2\, dv = 2a^2\left[v\right]_0^{2\pi} = 2a^2(2\pi) = 4\pi a^2$$

■

Example 2.2.16 (Area of a Cylinder)**.** Find the lateral surface area of a right circular cylinder with radius a and height h.

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Solution. We can parametrize the cylinder as follows:

$$\vec{r}(u,v) = (a\cos u, a\sin u, v)$$

The domain for the lateral surface is $D = [0, 2\pi] \times [0, h]$. First compute the tangent vectors $\vec{r}_1 = \partial\vec{r}/\partial u$ and $\vec{r}_2 = \partial\vec{r}/\partial v$:

$$\vec{r}_1 = (-a\sin u, a\cos u, 0)$$
$$\vec{r}_2 = (0, 0, 1)$$

Next determine the coefficients of the first fundamental form:

$$E = \vec{r}_1 \cdot \vec{r}_1 = (-a\sin u)^2 + (a\cos u)^2 + 0^2 = a^2$$
$$F = \vec{r}_1 \cdot \vec{r}_2 = 0$$
$$G = \vec{r}_2 \cdot \vec{r}_2 = 0^2 + 0^2 + 1^2 = 1$$

The area element is therefore $\sqrt{EG - F^2} = \sqrt{(a^2)(1) - 0^2} = \sqrt{a^2} = a$. The surface area is found by integrating this constant value over the domain D:

$$\begin{aligned} A &= \iint_D a\, du\, dv = \int_{v=0}^{h}\int_{u=0}^{2\pi} a\, du\, dv \\ &= a\left(\int_0^{2\pi} du\right)\left(\int_0^{h} dv\right) \\ &= a(2\pi)(h) = 2\pi a h \end{aligned}$$

This is the well-known formula for the lateral area of a cylinder. ■

Example 2.2.17 (Area of a Cone)**.** Find the lateral surface area of a right circular cone with base radius R and height H.

Solution. A cone can be parametrized by letting u be the distance from the apex and v be the angle of revolution. The relationship between height and radius is linear, so we set $z = (H/R)u$.

$$\vec{r}(u,v) = \left(u\cos v, u\sin v, \frac{H}{R}u\right)$$

The domain is $D = [0, R] \times [0, 2\pi]$, where u represents the radius of a circular cross-section. The tangent vectors are

$$\vec{r}_1 = \left(\cos v, \sin v, \frac{H}{R}\right)$$
$$\vec{r}_2 = (-u\sin v, u\cos v, 0)$$

The first fundamental coefficients are

$$E = \vec{r}_1 \cdot \vec{r}_1 = \cos^2 v + \sin^2 v + \left(\frac{H}{R}\right)^2 = 1 + \frac{H^2}{R^2} = \frac{R^2 + H^2}{R^2}$$
$$F = \vec{r}_1 \cdot \vec{r}_2 = 0$$
$$G = \vec{r}_2 \cdot \vec{r}_2 = (-u\sin v)^2 + (u\cos v)^2 = u^2$$

The area element integrand is

$$\sqrt{EG - F^2} = \sqrt{\left(\frac{R^2 + H^2}{R^2}\right)u^2 - 0} = \sqrt{\frac{L^2}{R^2}u^2} = \frac{L}{R}u$$

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where $L = \sqrt{R^2 + H^2}$ is the slant height of the cone. Now we integrate over the domain:

$$\begin{aligned} A &= \iint_D \frac{L}{R} u \, du \, dv = \int_{v=0}^{2\pi} \int_{u=0}^{R} \frac{L}{R} u \, du \, dv \\ &= \frac{L}{R} \int_0^{2\pi} \left[\frac{u^2}{2} \right]_0^R dv \\ &= \frac{L}{R} \int_0^{2\pi} \frac{R^2}{2} \, dv = \frac{LR}{2} \int_0^{2\pi} dv \\ &= \frac{LR}{2} (2\pi) = \pi R L \end{aligned}$$

This gives the formula for the lateral surface area of a cone. ■

Example 2.2.18 (Area of a Torus)**.** Find the surface area of a torus with major radius R and minor radius r ($R > r$).

Solution. A torus is generated by revolving a circle of radius r around an axis at a distance R from its center. A standard parametrization is $\vec{r}(u, v) = ((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u)$. The domain that covers the entire surface is $D = [0, 2\pi] \times [0, 2\pi]$. We compute the tangent vectors:

$$\begin{aligned} \vec{r}_1 &= (-r\sin u \cos v, -r\sin u \sin v, r\cos u) \\ \vec{r}_2 &= (-(R + r\cos u)\sin v, (R + r\cos u)\cos v, 0) \end{aligned}$$

Next calculate the fundamental coefficients:

$$\begin{aligned} E &= \vec{r}_1 \cdot \vec{r}_1 = (-r\sin u)^2(\cos^2 v + \sin^2 v) + (r\cos u)^2 = r^2 \sin^2 u + r^2 \cos^2 u = r^2 \\ F &= \vec{r}_1 \cdot \vec{r}_2 = r\sin u \cos v (R + r\cos u)\sin v - r\sin u \sin v (R + r\cos u)\cos v + 0 = 0 \\ G &= \vec{r}_2 \cdot \vec{r}_2 = (R + r\cos u)^2(\sin^2 v + \cos^2 v) = (R + r\cos u)^2 \end{aligned}$$

The area element is given by $\sqrt{EG - F^2} = \sqrt{r^2(R + r\cos u)^2 - 0} = r(R + r\cos u)$. (The absolute value is not needed since $R > r$ ensures $R + r\cos u > 0$). We integrate this over the domain:

$$\begin{aligned} A &= \iint_D r(R + r\cos u) \, du \, dv = \int_{v=0}^{2\pi} \int_{u=0}^{2\pi} (rR + r^2 \cos u) \, du \, dv \\ &= \int_0^{2\pi} \left[rRu + r^2 \sin u \right]_0^{2\pi} dv \\ &= \int_0^{2\pi} (rR(2\pi) + r^2 \sin(2\pi) - 0) \, dv \\ &= \int_0^{2\pi} 2\pi r R \, dv = 2\pi r R \left[v \right]_0^{2\pi} \\ &= 2\pi r R (2\pi) = 4\pi^2 R r \end{aligned}$$

The surface area of the torus is $4\pi^2 Rr$. This result is consistent with Pappus's second centroid theorem: Area = (arc length of generating curve) $\times$ (distance travelled by centroid) = $(2\pi r)(2\pi R)$. ■

2.3 Orientability of Surfaces

Orientability is an important property of surfaces that describes whether a consistent choice of normal direction can be made throughout a surface. Intuitively, an orientable surface has two distinguishable sides. For example, a sphere has an inside and an outside, while a plane and a torus also admit two consistent choices of normal direction. In contrast, the Möbius strip is non-orientable because a

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normal vector transported continuously around its central closed curve returns pointing in the opposite direction.

This geometric distinction is made precise by the existence of a differentiable unit normal vector field. On an orientable surface, one can choose a unit normal at every point so that its direction varies smoothly across the entire surface. On a non-orientable surface, no such global choice is possible.

2.3.1 The Differentiable Normal Field

Let $S \subset \mathbb{R}^3$ be a regular surface, and let $\mathbf{x} : U \to S$ be a local parametrization given by $\mathbf{x}(u, v)$. The vectors $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent and span the tangent plane at each point of the parametrized surface. Consequently, their cross product is a nonzero vector perpendicular to the tangent plane.

We may therefore define a unit normal vector on this surface patch by

$$N(u, v) = \frac{\mathbf{x}_u(u, v) \times \mathbf{x}_v(u, v)}{\|\mathbf{x}_u(u, v) \times \mathbf{x}_v(u, v)\|}.$$

Since $\mathbf{x}$ is regular, the denominator is nonzero. Moreover, because $\mathbf{x}$ is smooth, this formula defines a smooth unit normal vector field on the parameter domain.

At every point of a regular surface, there are exactly two unit normal vectors, namely N and $-N$. Although either choice can always be made locally, it may not be possible to choose one consistently throughout the entire surface. This leads to the following definition.

Definition 2.3.1 (Differentiable Normal Field)**.** Let $S \subset \mathbb{R}^3$ be a regular surface. A mapping $N : S \to \mathbb{R}^3$ is called a **differentiable unit normal vector field** if $\|N(p)\| = 1$ and $N(p)$ is perpendicular to T_pS for every $p \in S$, and if, for every local parametrization $\mathbf{x} : U \to S$, the composite mapping $N \circ \mathbf{x} : U \to \mathbb{R}^3$ is smooth.

Definition 2.3.2 (Orientable Surface)**.** A regular surface $S \subset \mathbb{R}^3$ is called **orientable** if it admits a differentiable unit normal vector field defined on the entire surface. Such a field determines an **orientation** of S. A regular surface that does not admit a global differentiable unit normal vector field is called **non-orientable**.

Thus, orientability concerns the possibility of choosing normal vectors globally, rather than merely at individual points. For a connected orientable surface, the two fields N and $-N$ determine its two possible orientations.

Theorem 2.3.3 (Orientability of Graph Surfaces)**.** *Every surface that can be represented globally as the graph of a smooth function of two variables is orientable.*

Proof. Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$ be a smooth function defined on an open set U, and consider its graph

$$S = \{(x, y, f(x, y)) : (x, y) \in U\}.$$

A natural parametrization of S is

$$\mathbf{x}(u, v) = (u, v, f(u, v)).$$

Differentiating with respect to u and v, we obtain

$$\mathbf{x}_u = (1, 0, f_u), \qquad \mathbf{x}_v = (0, 1, f_v).$$

Their cross product is

$$\mathbf{x}_u \times \mathbf{x}_v = (-f_u, -f_v, 1).$$

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Since its third component is always 1, this vector is nonzero at every point of the surface.

We can therefore define a unit normal vector field by

$$N(u,v) = \frac{(-f_u(u,v), -f_v(u,v), 1)}{\sqrt{1 + (f_u(u,v))^2 + (f_v(u,v))^2}}.$$

Because f is smooth and the denominator is strictly positive, N is smooth. Moreover, it is defined at every point of the graph and is perpendicular to both tangent vectors $\mathbf{x}_u$ and $\mathbf{x}_v$.

Thus, N defines a global differentiable unit normal vector field on S. Hence S is orientable. ■

The normal field constructed in the proof has a positive third component and is therefore called the **upward-pointing unit normal**. Its negative gives the downward-pointing orientation.

Example 2.3.4 (Orientability of a Paraboloid)**.** Show that the paraboloid $S = \{(x,y,z) \in \mathbb{R}^3 : z = x^2 + y^2\}$ is orientable, and find a unit normal vector field.

Solution. The paraboloid is the graph of the smooth function $f(x,y) = x^2 + y^2$. A parametrization is $\mathbf{x}(u,v) = (u, v, u^2 + v^2)$. Differentiating gives

$$\mathbf{x}_u = (1, 0, 2u), \qquad \mathbf{x}_v = (0, 1, 2v).$$

Hence,

$$\mathbf{x}_u \times \mathbf{x}_v = (-2u, -2v, 1).$$

Normalizing this vector, we obtain

$$N(u,v) = \frac{(-2u, -2v, 1)}{\sqrt{1 + 4u^2 + 4v^2}}.$$

The denominator is positive everywhere, so N is a smooth unit normal vector field defined on the entire paraboloid. Therefore, the paraboloid is orientable. ■

2.3.2 Regular Level Surfaces

Another important class of orientable surfaces consists of regular level surfaces of smooth functions. A level surface is the set of points at which a real-valued function of three variables takes a fixed value.

For example, if $F(x,y,z)$ represents temperature at a point in space, then the equation $F(x,y,z) = c$ describes the set of points having temperature c. Similarly, the equation $x^2 + y^2 + z^2 = R^2$ describes a sphere of radius R.

Definition 2.3.5 (Level Surface)**.** Let $U \subset \mathbb{R}^3$ be open and let $F : U \to \mathbb{R}$ be a smooth function. For a constant $c \in \mathbb{R}$, the set

$$S_c = F^{-1}(c) = \{(x,y,z) \in U : F(x,y,z) = c\}$$

is called the **level set** of F corresponding to c. When this set is a regular surface, it is called a **regular level surface**.

The gradient of F is defined by

$$\nabla F = \left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right).$$

At any point of a regular level surface where $\nabla F \neq \mathbf{0}$, the gradient is perpendicular to the tangent plane.

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Indeed, if $\alpha(t)$ is any smooth curve contained in S_c, then $F(\alpha(t)) = c$. Differentiating with respect to t and applying the chain rule gives

$$\nabla F(\alpha(t)) \cdot \alpha'(t) = 0.$$

Thus, the gradient is perpendicular to every tangent vector to the level surface.

Definition 2.3.6 (Regular Value)**.** Let $F : U \subset \mathbb{R}^3 \to \mathbb{R}$ be a smooth function. A number $c \in \mathbb{R}$ is called a **regular value** of F if $\nabla F(p) \neq \mathbf{0}$ for every $p \in F^{-1}(c)$.

By the implicit function theorem, if c is a regular value of F, then every point of the nonempty level set $F^{-1}(c)$ has a neighbourhood in which the level set can be represented as the graph of a smooth function of two variables. Consequently, $F^{-1}(c)$ is a regular surface.

Example 2.3.7 (The Sphere as a Regular Level Surface)**.** Show that the sphere $x^2 + y^2 + z^2 = a^2$, where $a > 0$, is a regular level surface.

Solution. Consider the smooth function $F : \mathbb{R}^3 \to \mathbb{R}$ defined by $F(x, y, z) = x^2 + y^2 + z^2$. The sphere is the level set $F^{-1}(a^2)$.

The gradient is

$$\nabla F(x, y, z) = (2x, 2y, 2z).$$

This vector vanishes only at the origin. Since $a > 0$, the origin does not lie on the sphere $x^2 + y^2 + z^2 = a^2$. Therefore, $\nabla F(p) \neq \mathbf{0}$ at every point of the sphere.

Hence a^2 is a regular value of F, and the sphere is a regular level surface. ■

Example 2.3.8 (A Plane as a Regular Level Surface)**.** Show that the plane $Ax + By + Cz = D$, where A, B, C are not all zero, is a regular level surface.

Solution. Let $F(x, y, z) = Ax + By + Cz$. The plane is the level set $F^{-1}(D)$, and

$$\nabla F(x, y, z) = (A, B, C).$$

Since A, B, C are not all zero, ∇F is nonzero at every point. Hence D is a regular value of F, and the plane is a regular level surface. ■

Theorem 2.3.9 (Orientability of Regular Level Surfaces)**.** *Let $F : U \subset \mathbb{R}^3 \to \mathbb{R}$ be a smooth function, and let c be a regular value of F. Then every nonempty level surface $S = F^{-1}(c)$ is orientable.*

Proof. Since c is a regular value of F, we have $\nabla F(p) \neq \mathbf{0}$ for every $p \in S$. Moreover, the gradient $\nabla F(p)$ is perpendicular to the tangent plane T_pS.

We may therefore define a unit normal vector field on S by

$$N(p) = \frac{\nabla F(p)}{\|\nabla F(p)\|}.$$

This field is defined at every point of S because the denominator never vanishes. Since F is smooth, its gradient is smooth, and hence N is also smooth.

Thus, N is a global differentiable unit normal vector field on S. By the definition of orientability, S is orientable. ■

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Corollary 2.3.10. *Every sphere $x^2 + y^2 + z^2 = a^2$, where $a > 0$, is orientable.*

Proof. The sphere is the regular level surface of $F(x, y, z) = x^2 + y^2 + z^2$ corresponding to the regular value a^2. By the preceding theorem, it is orientable.

In particular, since $\nabla F = (2x, 2y, 2z)$ and $\sqrt{x^2 + y^2 + z^2} = a$ on the sphere, a global unit normal field is

$$N(x, y, z) = \frac{(x, y, z)}{a}.$$

This field points radially outward and defines the outward orientation of the sphere. The field $-N$ defines the inward orientation. ■

Example 2.3.11 (Orientability of a Circular Cylinder)**.** Show that the circular cylinder $x^2 + y^2 = R^2$, where $R > 0$, is orientable and determine a global unit normal vector field.

Solution. Let $F(x, y, z) = x^2 + y^2$. The cylinder is the level surface $F^{-1}(R^2)$, and its gradient is

$$\nabla F(x, y, z) = (2x, 2y, 0).$$

On the cylinder, $x^2 + y^2 = R^2 > 0$, so ∇F never vanishes. Hence the cylinder is a regular level surface and is therefore orientable.

Normalizing the gradient gives

$$N(x, y, z) = \frac{(2x, 2y, 0)}{\sqrt{4x^2 + 4y^2}} = \frac{(x, y, 0)}{R}.$$

This field is smooth and defined everywhere on the cylinder. It points away from the axis and determines the outward orientation. ■

Example 2.3.12 (Orientability of a Torus)**.** Consider the torus defined by

$$\left(x^2 + y^2 + z^2 + R^2 - r^2\right)^2 = 4R^2(x^2 + y^2),$$

where $R > r > 0$. Show that the torus is orientable.

Solution. Define

$$F(x, y, z) = \left(x^2 + y^2 + z^2 + R^2 - r^2\right)^2 - 4R^2(x^2 + y^2).$$

The torus is the level set $F^{-1}(0)$. Put $A = x^2 + y^2 + z^2 + R^2 - r^2$. Differentiating, we obtain

$$\nabla F = (4x(A - 2R^2), 4y(A - 2R^2), 4zA).$$

On the torus, $A^2 = 4R^2(x^2 + y^2)$. Since $A = x^2 + y^2 + z^2 + R^2 - r^2 > 0$, this implies $A = 2R\sqrt{x^2 + y^2}$. In particular, $x^2 + y^2 > 0$.

Suppose that $\nabla F = \mathbf{0}$ at a point of the torus. Since x and y are not both zero, the first two components imply $A = 2R^2$. The third component then implies $z = 0$. Substituting $A = 2R^2$ into $A^2 = 4R^2(x^2 + y^2)$ gives $x^2 + y^2 = R^2$.

But the definition of A would then give

$$A = R^2 + 0 + R^2 - r^2 = 2R^2 - r^2,$$

contradicting $A = 2R^2$ because $r > 0$. Therefore, ∇F never vanishes on the torus.

Hence 0 is a regular value of F, and the torus is orientable. ■

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The preceding results provide two useful methods for establishing orientability. If a surface is globally the graph of a smooth function, a unit normal can be obtained from its partial derivatives. If it is a regular level surface, a unit normal can be obtained by normalizing the gradient of its defining function. In both cases, the essential idea is the construction of a smooth unit normal field defined consistently on the entire surface.

2.3.3 The Oriented Atlas Criterion

We have defined an orientable surface as a regular surface admitting a globally defined differentiable unit normal vector field. There is another equivalent characterization of orientability which uses the changes of coordinates between overlapping parametrizations.

Let $\mathbf{x}_\alpha : U_\alpha \to S$ and $\mathbf{x}_\beta : U_\beta \to S$ be two local parametrizations whose images overlap. A point belonging to both patches has two sets of coordinates, related by the transition map

$$\phi = \mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha.$$

Since both parametrizations are regular, ϕ is a smooth diffeomorphism between the corresponding open subsets of $\mathbb{R}^2$. The sign of its Jacobian determinant indicates whether the change of coordinates preserves or reverses orientation.

Definition 2.3.13 (Oriented Atlas)**.** An atlas $\{\mathbf{x}_\alpha : U_\alpha \to S\}$ of a regular surface S is called an **oriented atlas** if, whenever two parametrizations have overlapping images, the Jacobian determinant of their transition map is positive:

$$\det D(\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha) > 0.$$

A regular surface is said to be orientable if it admits such an atlas.

The geometric meaning of this condition follows from the behaviour of cross products under a change of coordinates. Suppose $\mathbf{x}(u, v) = \mathbf{y}(\phi(u, v), \psi(u, v))$. By the chain rule,

$$\mathbf{x}_u = \mathbf{y}_{\tilde{u}}\phi_u + \mathbf{y}_{\tilde{v}}\psi_u, \qquad \mathbf{x}_v = \mathbf{y}_{\tilde{u}}\phi_v + \mathbf{y}_{\tilde{v}}\psi_v.$$

Taking the cross product and using its bilinearity gives

$$\mathbf{x}_u \times \mathbf{x}_v = (\phi_u\psi_v - \phi_v\psi_u)(\mathbf{y}_{\tilde{u}} \times \mathbf{y}_{\tilde{v}}).$$

Thus,

$$\mathbf{x}_u \times \mathbf{x}_v = \det D(\phi, \psi)\,(\mathbf{y}_{\tilde{u}} \times \mathbf{y}_{\tilde{v}}).$$

Consequently, a positive Jacobian determinant means that the two parametrizations determine the same normal direction on their common surface patch.

Theorem 2.3.14 (Equivalent Characterizations of Orientability)**.** *A regular surface $S \subset \mathbb{R}^3$ is orientable if and only if it admits an oriented atlas.*

Proof. Suppose first that S is orientable. Then there exists a global differentiable unit normal vector field N on S.

Let $\mathbf{x} : U \to S$ be a local parametrization. Its associated unit normal field is

$$n(u, v) = \frac{\mathbf{x}_u \times \mathbf{x}_v}{\|\mathbf{x}_u \times \mathbf{x}_v\|}.$$

At every point of the patch, both n and N are unit normal vectors, so $n = \pm N$. Since both fields are continuous, the sign is constant on each connected component of U. By restricting to connected

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parameter domains, we may therefore assume that either $n = N$ throughout the patch or $n = -N$ throughout the patch.

If $n = N$, we retain the parametrization. If $n = -N$, we reverse the order of its parameters. The resulting parametrization has the opposite cross product and therefore agrees with N. Repeating this procedure for all the coordinate patches gives an atlas in which every parametrization determines the same normal direction as the global field N.

Now consider any two overlapping parametrizations $\mathbf{x}_\alpha$ and $\mathbf{x}_\beta$ from this atlas. On their common surface patch, the change-of-coordinates formula gives

$$(\mathbf{x}_\alpha)_u \times (\mathbf{x}_\alpha)_v = \det D(\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha)\big((\mathbf{x}_\beta)_u \times (\mathbf{x}_\beta)_v\big).$$

Since both cross products point in the same direction as N, the Jacobian determinant must be positive. Hence the atlas is oriented.

Conversely, suppose that S admits an oriented atlas $\{\mathbf{x}_\alpha\}$. On each coordinate patch, define

$$N_\alpha = \frac{(\mathbf{x}_\alpha)_u \times (\mathbf{x}_\alpha)_v}{\|(\mathbf{x}_\alpha)_u \times (\mathbf{x}_\alpha)_v\|}.$$

Each N_α is a smooth unit normal field on its corresponding patch.

On the overlap of two patches, the cross-product transformation formula gives

$$N_\alpha = \frac{\det D(\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha)}{|\det D(\mathbf{x}_\beta^{-1} \circ \mathbf{x}_\alpha)|} N_\beta.$$

Since the atlas is oriented, the Jacobian determinant is positive. Therefore, $N_\alpha = N_\beta$ on every overlap.

Thus, the local normal fields agree wherever their domains intersect and consequently define a single smooth unit normal vector field N on the entire surface. Hence S is orientable. ■

This theorem shows that orientability may be studied either geometrically, by constructing a global unit normal field, or analytically, by examining the Jacobian determinants of changes of coordinates.

2.3.4 The Möbius Strip

The Möbius strip is a classical example of a non-orientable surface. It is obtained by taking a rectangular strip, giving one end a half-twist, and joining the two ends. Unlike a cylinder, which has two distinct sides, the Möbius strip has only one side.

Theorem 2.3.15. *The Möbius strip is non-orientable.*

Proof. Suppose, on the contrary, that the Möbius strip admits a continuous unit normal vector field N.

Consider its standard parametrization

$$\mathbf{x}(u, v) = \left(\left(R + v\cos\frac{u}{2}\right)\cos u,\ \left(R + v\cos\frac{u}{2}\right)\sin u,\ v\sin\frac{u}{2}\right),$$

where $0 \le u \le 2\pi$, $|v| < a$, and $0 < a < R$. The parameters satisfy the identification

$$\mathbf{x}(0, v) = \mathbf{x}(2\pi, -v),$$

which produces the half-twist characteristic of the Möbius strip.

We examine the unit normal along the central closed curve $\gamma(u) = \mathbf{x}(u, 0)$. Since

$$\gamma(u) = (R\cos u, R\sin u, 0),$$

we have $\gamma(0) = \gamma(2\pi) = (R, 0, 0)$.

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The partial derivatives along this curve are

$$\mathbf{x}_u(u,0) = (-R\sin u, R\cos u, 0)$$

and

$$\mathbf{x}_v(u,0) = \left(\cos\frac{u}{2}\cos u,\ \cos\frac{u}{2}\sin u,\ \sin\frac{u}{2}\right).$$

Taking their cross product gives

$$\mathbf{x}_u(u,0)\times\mathbf{x}_v(u,0) = R\left(\sin u\sin\frac{u}{2},\ -\cos u\sin\frac{u}{2},\ -\cos\frac{u}{2}\right).$$

Its norm is R, so a continuous unit normal along the central curve is

$$n(u) = \left(\sin u\sin\frac{u}{2},\ -\cos u\sin\frac{u}{2},\ -\cos\frac{u}{2}\right).$$

At the endpoints,

$$n(0) = (0,0,-1), \qquad n(2\pi) = (0,0,1).$$

Hence $n(2\pi) = -n(0)$, although $\gamma(0) = \gamma(2\pi)$.

Since N is assumed to be a continuous unit normal field, at every point of the central curve we must have

$$N(\gamma(u)) = \varepsilon(u)n(u),$$

where $\varepsilon(u) \in \{1,-1\}$. The function ε is continuous, and since $[0,2\pi]$ is connected, it must be constant.

It follows that

$$N(\gamma(2\pi)) = -N(\gamma(0)).$$

But $\gamma(0) = \gamma(2\pi)$, so a single-valued normal field must assign the same vector to these two parameter values. This is a contradiction.

Therefore, no continuous global unit normal vector field exists on the Möbius strip, and hence it is non-orientable. ■

The essential feature of this proof is that the normal direction reverses after one complete traversal of the central closed curve. Thus, although a smooth normal can be chosen locally at every point, these local choices cannot be made consistent throughout the strip.

2.3.5 The Gauss Map

Let $S \subset \mathbb{R}^3$ be an oriented regular surface with a chosen differentiable unit normal vector field N. Since every vector $N(p)$ has length one, its endpoint lies on the unit sphere

$$S^2 = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 1\}.$$

Thus, the normal field naturally defines a mapping from the surface S to the unit sphere. This mapping is called the Gauss map.

Definition 2.3.16 (Gauss Map)**.** (See Figure 47 in the Atlas.) Let S be an oriented regular surface with unit normal vector field N. The **Gauss map** is the smooth mapping

$$N : S \to S^2, \qquad p \mapsto N(p),$$

which assigns to each point $p \in S$ the endpoint of its chosen unit normal vector on the unit sphere.

The Gauss map describes how the normal direction changes as we move along a surface. If the normal vector is constant on a surface patch, the Gauss map sends the entire patch to a single point on the unit sphere. If the normal direction varies, the image of the patch also varies. Thus, the Gauss map provides a way to study the bending of a surface in $\mathbb{R}^3$.

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Example 2.3.17 (Gauss Map of a Plane)**.** Determine the Gauss map of the plane $z = 0$ with the upward orientation.

Solution. A parametrization of the plane is $\mathbf{x}(u, v) = (u, v, 0)$. Its partial derivatives are $\mathbf{x}_u = (1, 0, 0)$ and $\mathbf{x}_v = (0, 1, 0)$, so

$$\mathbf{x}_u \times \mathbf{x}_v = (0, 0, 1).$$

Therefore, the upward unit normal is $N(u, v) = (0, 0, 1)$ at every point. Hence the Gauss map is constant and maps the entire plane to the north pole of the unit sphere. ■

Example 2.3.18 (Gauss Map of a Sphere)**.** Find the Gauss map of the sphere $x^2 + y^2 + z^2 = R^2$, where $R > 0$, with its outward orientation.

Solution. At a point $p = (x, y, z)$ on the sphere, the outward unit normal is

$$N(p) = \frac{(x, y, z)}{R},$$

because $x^2 + y^2 + z^2 = R^2$. Thus, the Gauss map is

$$N(x, y, z) = \left(\frac{x}{R}, \frac{y}{R}, \frac{z}{R}\right).$$

It maps each point of the sphere of radius R to the corresponding point on the unit sphere. In particular, the Gauss map of the unit sphere with outward orientation is the identity map. ■

2.3.6 The Differential of the Gauss Map (The Shape Operator)

Since the Gauss map is differentiable, we can study how the unit normal vector changes in a given tangent direction. This leads to the differential of the Gauss map, which plays a central role in the theory of surface curvature.

Let $p \in S$ and let $v \in T_pS$. The differential of the Gauss map at p is the linear mapping

$$dN_p : T_pS \to T_{N(p)}S^2.$$

To understand its action, choose a smooth curve $\alpha(t)$ on S satisfying $\alpha(0) = p$ and $\alpha'(0) = v$. Then

$$dN_p(v) = \frac{d}{dt} N(\alpha(t))\Big|_{t=0}.$$

Thus, $dN_p(v)$ measures the instantaneous rate of change of the unit normal as we move along the surface in the direction v.

Because $\|N(p)\| = 1$, we have $N(p) \cdot N(p) = 1$. Differentiating this identity along a curve on S gives

$$dN_p(v) \cdot N(p) = 0.$$

Hence $dN_p(v)$ is perpendicular to $N(p)$. Since

$$T_pS = \{w \in \mathbb{R}^3 : w \cdot N(p) = 0\},$$

we conclude that $dN_p(v) \in T_pS$, after identifying tangent vectors with vectors in $\mathbb{R}^3$. Therefore, the differential of the Gauss map may be regarded as a linear operator on T_pS.

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Definition 2.3.19 (Shape Operator or Weingarten Map)**.** Let S be an oriented regular surface with Gauss map N. The **shape operator**, or **Weingarten map**, at $p \in S$ is the linear operator

$$S_p : T_pS \to T_pS$$

defined by

$$S_p(v) = -dN_p(v), \qquad v \in T_pS.$$

The negative sign is a standard convention in differential geometry. With this convention, the shape operator of a sphere of radius R with outward orientation is $S_p = -\frac{1}{R}I$, while with inward orientation it is $S_p = \frac{1}{R}I$. Reversing the orientation therefore changes the sign of the shape operator but does not change the underlying surface.

Example 2.3.20 (Shape Operator of a Sphere)**.** Find the shape operator of the sphere $x^2 + y^2 + z^2 = R^2$ with outward orientation.

Solution. The outward unit normal field is $N(p) = p/R$. Since R is constant, differentiation in any tangent direction $v \in T_pS$ gives

$$dN_p(v) = \frac{1}{R}v.$$

By the definition of the shape operator,

$$S_p(v) = -dN_p(v) = -\frac{1}{R}v.$$

Thus, the shape operator is

$$S_p = -\frac{1}{R}I.$$

Consequently, every tangent direction is an eigenvector with eigenvalue $-1/R$. ■

One of the fundamental properties of the shape operator is that it is self-adjoint with respect to the Euclidean inner product on the tangent plane.

Theorem 2.3.21 (Self-Adjointness of the Shape Operator)**.** *Let S be an oriented regular surface, and let $S_p : T_pS \to T_pS$ be its shape operator at $p \in S$. Then S_p is self-adjoint; that is,*

$$\langle S_p(v), w \rangle = \langle v, S_p(w) \rangle$$

for all $v, w \in T_pS$.

Proof. Let $\mathbf{x}(u, v)$ be a local parametrization of S near p, and write $N(u, v)$ for the corresponding chosen unit normal field. Since $\mathbf{x}_u$ and $\mathbf{x}_v$ form a basis of T_pS, it is enough to establish the symmetry relation for these basis vectors.

By the definition of the shape operator,

$$S_p(\mathbf{x}_u) = -N_u, \qquad S_p(\mathbf{x}_v) = -N_v.$$

Thus, we need to show that

$$\langle N_u, \mathbf{x}_v \rangle = \langle N_v, \mathbf{x}_u \rangle .$$

Since N is normal to the surface, we have

$$\langle N, \mathbf{x}_u \rangle = 0, \qquad \langle N, \mathbf{x}_v \rangle = 0.$$

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Differentiating the first identity with respect to v and applying the product rule gives

$$\langle N_v, \mathbf{x}_u \rangle + \langle N, \mathbf{x}_{uv} \rangle = 0,$$

so

$$\langle N_v, \mathbf{x}_u \rangle = - \langle N, \mathbf{x}_{uv} \rangle .$$

Similarly, differentiating the second identity with respect to u gives

$$\langle N_u, \mathbf{x}_v \rangle = - \langle N, \mathbf{x}_{vu} \rangle .$$

Since $\mathbf{x}$ is smooth, its mixed partial derivatives are equal:

$$\mathbf{x}_{uv} = \mathbf{x}_{vu}.$$

Consequently,

$$\langle N_u, \mathbf{x}_v \rangle = \langle N_v, \mathbf{x}_u \rangle .$$

Therefore,

$$\langle S_p(\mathbf{x}_u), \mathbf{x}_v \rangle = \langle \mathbf{x}_u, S_p(\mathbf{x}_v) \rangle .$$

The corresponding identities for pairs of equal basis vectors hold automatically because the inner product is symmetric. Hence, by bilinearity, the required identity holds for all $v, w \in T_pS$, proving that S_p is self-adjoint. ■

Corollary 2.3.22 (Existence of Principal Curvatures and Directions). *At every point p of an oriented regular surface S, the shape operator S_p has two real eigenvalues k_1 and k_2, and there exists an orthonormal basis $\{e_1, e_2\}$ of T_pS consisting of corresponding eigenvectors.*

*The eigenvalues k_1, k_2 are called the **principal curvatures**, and the unit eigenvectors e_1, e_2 are called the **principal directions**. The Gaussian curvature and mean curvature are defined by*

$$K = k_1 k_2, \qquad H = \frac{1}{2}(k_1 + k_2).$$

Proof. Since S_p is a self-adjoint linear operator on the two-dimensional Euclidean vector space T_pS, the spectral theorem for self-adjoint operators guarantees that all its eigenvalues are real and that T_pS has an orthonormal basis consisting of eigenvectors of S_p. Denoting these eigenvalues by k_1 and k_2 proves the result. ■

At a point where $k_1 = k_2$, every tangent direction is a principal direction. Such a point is called an **umbilic point**. The sphere provides a familiar example, since its principal curvatures are equal at every point.

2.4 The Second Fundamental Form

The first fundamental form describes the intrinsic geometry of a surface, including lengths, angles and areas. However, it does not directly describe how the surface bends in the surrounding space $\mathbb{R}^3$. To study this bending, we introduce the **second fundamental form**, which measures the variation of the unit normal vector along the surface.

Let $S \subset \mathbb{R}^3$ be an oriented regular surface, and let $\vec{r} : U \subset \mathbb{R}^2 \to S$ be a local parametrization given by $\vec{r}(u, v)$. The partial derivatives $\vec{r}_u$ and $\vec{r}_v$ are linearly independent and span the tangent space T_pS at $p = \vec{r}(u, v)$. Recall that the first fundamental form is

$$I = E\,du^2 + 2F\,du\,dv + G\,dv^2,$$

where $E = \langle \vec{r}_u, \vec{r}_u \rangle$, $F = \langle \vec{r}_u, \vec{r}_v \rangle$ and $G = \langle \vec{r}_v, \vec{r}_v \rangle$.

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The unit normal vector field, chosen consistently with the orientation, is locally given by

$$\mathbb{N} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}.$$

As we move along the surface, the direction of $\mathbb{N}$ generally changes. This variation is described by the shape operator.

Definition 2.4.1 (Shape Operator)**.** Let $p \in S$ and $\vec{v} \in T_pS$. The **shape operator**, or **Weingarten map**, is the linear mapping $S_p : T_pS \to T_pS$ defined by

$$S_p(\vec{v}) = -d\mathbb{N}_p(\vec{v}).$$

Equivalently, if $\alpha(t)$ is a smooth curve on S satisfying $\alpha(0) = p$ and $\alpha'(0) = \vec{v}$, then

$$S_p(\vec{v}) = -\left.\frac{d}{dt}\mathbb{N}(\alpha(t))\right|_{t=0}.$$

The shape operator is self-adjoint with respect to the Euclidean inner product on T_pS. This property allows us to associate with it a symmetric bilinear form.

Definition 2.4.2 (Second Fundamental Form)**.** Let S be an oriented regular surface. The **second fundamental form** at $p \in S$ is the symmetric bilinear form $II_p : T_pS \times T_pS \to \mathbb{R}$ defined by

$$II_p(\vec{v}, \vec{w}) = \langle S_p(\vec{v}), \vec{w} \rangle = -\langle d\mathbb{N}_p(\vec{v}), \vec{w} \rangle.$$

In particular, the value $II_p(\vec{v}, \vec{v})$ measures the normal component of the bending of the surface in the tangent direction $\vec{v}$.

Local Coordinate Expression

We now express the second fundamental form in terms of the partial derivatives of a local parametrization.

Let $p = \vec{r}(u, v)$ and consider two tangent vectors

$$\vec{v} = a\vec{r}_u + b\vec{r}_v, \qquad \vec{w} = c\vec{r}_u + d\vec{r}_v.$$

Since the differential of the Gauss map is linear, we have

$$d\mathbb{N}_p(\vec{v}) = a\mathbb{N}_u + b\mathbb{N}_v.$$

Substituting into the definition of the second fundamental form gives

$$\begin{aligned} II_p(\vec{v}, \vec{w}) &= -\langle a\mathbb{N}_u + b\mathbb{N}_v, c\vec{r}_u + d\vec{r}_v \rangle \\ &= -ac\langle \mathbb{N}_u, \vec{r}_u \rangle - ad\langle \mathbb{N}_u, \vec{r}_v \rangle \\ &\quad - bc\langle \mathbb{N}_v, \vec{r}_u \rangle - bd\langle \mathbb{N}_v, \vec{r}_v \rangle. \end{aligned}$$

To simplify this expression, recall that $\mathbb{N}$ is perpendicular to the tangent vectors $\vec{r}_u$ and $\vec{r}_v$. Thus,

$$\langle \mathbb{N}, \vec{r}_u \rangle = 0, \qquad \langle \mathbb{N}, \vec{r}_v \rangle = 0.$$

Differentiating the first identity with respect to u and applying the product rule, we obtain

$$\langle \mathbb{N}_u, \vec{r}_u \rangle + \langle \mathbb{N}, \vec{r}_{uu} \rangle = 0,$$

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so

$$\langle \mathbb{N}_u, \vec{r}_u \rangle = - \langle \mathbb{N}, \vec{r}_{uu} \rangle .$$

Similarly, differentiating the second identity with respect to u gives

$$\langle \mathbb{N}_u, \vec{r}_v \rangle = - \langle \mathbb{N}, \vec{r}_{uv} \rangle .$$

Differentiating the first identity with respect to v, we obtain

$$\langle \mathbb{N}_v, \vec{r}_u \rangle = - \langle \mathbb{N}, \vec{r}_{vu} \rangle .$$

Since the parametrization is smooth, its mixed partial derivatives are equal, that is, $\vec{r}_{uv} = \vec{r}_{vu}$. Finally, differentiating the second identity with respect to v gives

$$\langle \mathbb{N}_v, \vec{r}_v \rangle = - \langle \mathbb{N}, \vec{r}_{vv} \rangle .$$

Substituting these relations into the expression for $II_p(\vec{v}, \vec{w})$, we obtain

$$\begin{aligned} II_p(\vec{v}, \vec{w}) = ac \langle \mathbb{N}, \vec{r}_{uu} \rangle + (ad + bc) \langle \mathbb{N}, \vec{r}_{uv} \rangle \\ + bd \langle \mathbb{N}, \vec{r}_{vv} \rangle . \end{aligned}$$

We therefore introduce the following coefficients.

Definition 2.4.3 (Coefficients of the Second Fundamental Form)**.** Let $\vec{r}(u, v)$ be a local parametrization of an oriented regular surface with unit normal field $\mathbb{N}$. The coefficients of the second fundamental form are defined by

$$L = \langle \vec{r}_{uu}, \mathbb{N} \rangle , \qquad M = \langle \vec{r}_{uv}, \mathbb{N} \rangle , \qquad N = \langle \vec{r}_{vv}, \mathbb{N} \rangle .$$

Equivalently,

$$L = II(\vec{r}_u, \vec{r}_u), \qquad M = II(\vec{r}_u, \vec{r}_v), \qquad N = II(\vec{r}_v, \vec{r}_v).$$

With these coefficients, the bilinear form becomes

$$II_p(\vec{v}, \vec{w}) = acL + (ad + bc)M + bdN.$$

In particular, taking $\vec{w} = \vec{v}$, we obtain

$$II_p(\vec{v}, \vec{v}) = a^2 L + 2abM + b^2 N.$$

Since $du(\vec{v}) = a$ and $dv(\vec{v}) = b$, the second fundamental form is written in local coordinates as

$$II = L\, du^2 + 2M\, du\, dv + N\, dv^2.$$

Thus, the second fundamental form is represented by the symmetric matrix

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix}$$

relative to the coordinate basis $\{\vec{r}_u, \vec{r}_v\}$. The symmetry of this matrix follows from the equality $\vec{r}_{uv} = \vec{r}_{vu}$.

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Explicit Formulas for the Coefficients

The coefficients L, M, N can be computed directly from the first and second partial derivatives of the parametrization.

Using

$$\mathbb{N} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|},$$

we obtain

$$L = \frac{\langle \vec{r}_{uu}, \vec{r}_u \times \vec{r}_v \rangle}{\|\vec{r}_u \times \vec{r}_v\|},$$

$$M = \frac{\langle \vec{r}_{uv}, \vec{r}_u \times \vec{r}_v \rangle}{\|\vec{r}_u \times \vec{r}_v\|},$$

and

$$N = \frac{\langle \vec{r}_{vv}, \vec{r}_u \times \vec{r}_v \rangle}{\|\vec{r}_u \times \vec{r}_v\|}.$$

These expressions can also be written as scalar triple products. Recall that for any vectors $a, b, c \in \mathbb{R}^3$,

$$\langle a, b \times c \rangle = \det(a, b, c).$$

Consequently,

$$L = \frac{\det(\vec{r}_{uu}, \vec{r}_u, \vec{r}_v)}{\|\vec{r}_u \times \vec{r}_v\|}, \qquad M = \frac{\det(\vec{r}_{uv}, \vec{r}_u, \vec{r}_v)}{\|\vec{r}_u \times \vec{r}_v\|},$$

and

$$N = \frac{\det(\vec{r}_{vv}, \vec{r}_u, \vec{r}_v)}{\|\vec{r}_u \times \vec{r}_v\|}.$$

These formulas are particularly useful when the surface is given parametrically, since the coefficients can be determined directly by differentiation.

If the opposite orientation is chosen, the unit normal $\mathbb{N}$ is replaced by $-\mathbb{N}$. Consequently, all three coefficients L, M, N change sign, and the second fundamental form becomes $-II$. The first fundamental form, however, remains unchanged because it depends only on the tangent vectors and their inner products.

2.5 Weingarten Equations (Shape Operator in Coordinates)

The partial derivatives of $\mathbb{N}$ can be expressed in the basis $\{\vec{r}_u, \vec{r}_v\}$.

Theorem 2.5.1 (Weingarten Equations). *Let $S \subset \mathbb{R}^3$ be a regular surface parametrized by $\mathbf{r}(u, v)$ with unit normal $\mathbf{N}$. Denote the coefficients of the first and second fundamental forms by E, F, G and L, M, N respectively. Then the partial derivatives of $\mathbf{N}$ are given by*

$$\mathbf{N}_u = \frac{FM - GL}{EG - F^2}\,\mathbf{r}_u + \frac{FL - EM}{EG - F^2}\,\mathbf{r}_v,$$

$$\mathbf{N}_v = \frac{FN - GM}{EG - F^2}\,\mathbf{r}_u + \frac{FM - EN}{EG - F^2}\,\mathbf{r}_v.$$

Proof. Since $\mathbf{N}$ is a unit normal, we have $\mathbf{N} \cdot \mathbf{r}_u = 0$ and $\mathbf{N} \cdot \mathbf{r}_v = 0$. Differentiating the first equality with respect to u and v yields

$$\mathbf{N}_u \cdot \mathbf{r}_u + \mathbf{N} \cdot \mathbf{r}_{uu} = 0 \quad \Longrightarrow \quad \mathbf{N}_u \cdot \mathbf{r}_u = -\mathbf{N} \cdot \mathbf{r}_{uu} = -L,$$

$$\mathbf{N}_v \cdot \mathbf{r}_u + \mathbf{N} \cdot \mathbf{r}_{uv} = 0 \quad \Longrightarrow \quad \mathbf{N}_v \cdot \mathbf{r}_u = -\mathbf{N} \cdot \mathbf{r}_{uv} = -M.$$

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Similarly, from $\mathbf{N} \cdot \mathbf{r}_v = 0$ we obtain

$$\mathbf{N}_u \cdot \mathbf{r}_v = -\mathbf{N} \cdot \mathbf{r}_{uv} = -M, \qquad \mathbf{N}_v \cdot \mathbf{r}_v = -\mathbf{N} \cdot \mathbf{r}_{vv} = -N.$$

Now, because $\{\mathbf{r}_u, \mathbf{r}_v\}$ is a basis of the tangent plane, we can write

$$\mathbf{N}_u = \alpha\, \mathbf{r}_u + \beta\, \mathbf{r}_v,$$

for some functions α, β on the surface. Taking the dot product of this expression with $\mathbf{r}_u$ and $\mathbf{r}_v$ gives the linear system

$$\begin{cases} \alpha\, E + \beta\, F = \mathbf{N}_u \cdot \mathbf{r}_u = -L, \\ \alpha\, F + \beta\, G = \mathbf{N}_u \cdot \mathbf{r}_v = -M. \end{cases}$$

We solve for α and β. The determinant of the coefficient matrix is $EG - F^2 \neq 0$ (regularity). Using Cramer's rule:

$$\alpha = \frac{\det \begin{pmatrix} -L & F \\ -M & G \end{pmatrix}}{\det \begin{pmatrix} E & F \\ F & G \end{pmatrix}} = \frac{(-L)G - F(-M)}{EG - F^2} = \frac{-LG + FM}{EG - F^2} = \frac{FM - GL}{EG - F^2},$$

$$\beta = \frac{\det \begin{pmatrix} E & -L \\ F & -M \end{pmatrix}}{EG - F^2} = \frac{E(-M) - (-L)F}{EG - F^2} = \frac{-EM + LF}{EG - F^2} = \frac{FL - EM}{EG - F^2}.$$

Thus the first Weingarten equation is proved.

For $\mathbf{N}_v$, write $\mathbf{N}_v = \gamma\, \mathbf{r}_u + \delta\, \mathbf{r}_v$. Dotting with $\mathbf{r}_u$ and $\mathbf{r}_v$ yields

$$\gamma\, E + \delta\, F = \mathbf{N}_v \cdot \mathbf{r}_u = -M, \qquad \gamma\, F + \delta\, G = \mathbf{N}_v \cdot \mathbf{r}_v = -N.$$

Solving similarly,

$$\gamma = \frac{(-M)G - F(-N)}{EG - F^2} = \frac{-MG + FN}{EG - F^2} = \frac{FN - GM}{EG - F^2},$$

$$\delta = \frac{E(-N) - (-M)F}{EG - F^2} = \frac{-EN + MF}{EG - F^2} = \frac{FM - EN}{EG - F^2}.$$

This proves the result. ■

Let $S \subset \mathbb{R}^3$ be a regular surface with local parametrization $\mathbf{r}(u, v)$. The shape operator $S_p : T_pS \to T_pS$ is defined by $S_p(\mathbf{v}) = -d\mathbf{N}_p(\mathbf{v})$, where $\mathbf{N}$ is a unit normal field. In the basis $\{\mathbf{r}_u, \mathbf{r}_v\}$ of the tangent plane, the matrix of S_p is obtained from the Weingarten equations:

$$\mathbf{N}_u = \frac{FM - GL}{EG - F^2}\, \mathbf{r}_u + \frac{FL - EM}{EG - F^2}\, \mathbf{r}_v,$$

$$\mathbf{N}_v = \frac{FN - GM}{EG - F^2}\, \mathbf{r}_u + \frac{FM - EN}{EG - F^2}\, \mathbf{r}_v.$$

Since $S(\mathbf{r}_u) = -\mathbf{N}_u$ and $S(\mathbf{r}_v) = -\mathbf{N}_v$, the matrix of S with respect to $\{\mathbf{r}_u, \mathbf{r}_v\}$ is

$$[S] = \frac{1}{EG - F^2} \begin{pmatrix} GL - FM & GM - FN \\ EM - FL & EN - FM \end{pmatrix}.$$

Note: Let $M_1 = \begin{bmatrix} E & F \\ F & G \end{bmatrix}$ and $M_2 = \begin{bmatrix} L & M \\ M & N \end{bmatrix}$ be the matrices of the First and the Second fundamental forms, then the Weingarten equations can be written as

$$\begin{bmatrix} N_u \\ N_v \end{bmatrix} = M_1^{-1} M_2 = \frac{1}{EG - F^2} \begin{bmatrix} G & -F \\ -F & E \end{bmatrix} \begin{bmatrix} L & M \\ M & N \end{bmatrix} \begin{bmatrix} r_u \\ r_v \end{bmatrix}$$

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Theorem 2.5.2 (Principal Curvatures). *At any point $p \in S$, the eigenvalues of the shape operator S_p are the **principal curvatures** k_1 and k_2. They are given explicitly by*

$$k_{1,2} = H \pm \sqrt{H^2 - K},$$

where

$$H = \frac{EN - 2FM + GL}{2(EG - F^2)}, \qquad K = \frac{LN - M^2}{EG - F^2}$$

are the mean curvature and Gaussian curvature, respectively.

Proof. The characteristic polynomial of $[S]$ is $\det([S] - kI) = 0$. The trace and determinant of $[S]$ are

$$\operatorname{tr}(S) = \frac{GL - FM + EN - FM}{EG - F^2} = \frac{EN + GL - 2FM}{EG - F^2} = 2H,$$

$$\det(S) = \frac{(GL - FM)(EN - FM) - (GM - FN)(EM - FL)}{(EG - F^2)^2}.$$

Expanding the numerator:

$$\begin{aligned}
&(GL - FM)(EN - FM) - (GM - FN)(EM - FL) \\
&= (GL)(EN) - (GL)(FM) - (FM)(EN) + (FM)^2 \\
&\quad - \big[(GM)(EM) - (GM)(FL) - (FN)(EM) + (FN)(FL)\big] \\
&= GL\,EN - GL\,FM - FM\,EN + F^2M^2 \\
&\quad - GM\,EM + GM\,FL + FN\,EM - FN\,FL.
\end{aligned}$$

Using $EG - F^2 = \|\mathbf{r}_u \times \mathbf{r}_v\|^2 > 0$, one can simplify to

$$\det([S]) = \frac{(LN - M^2)(EG - F^2)}{(EG - F^2)^2} = \frac{LN - M^2}{EG - F^2} = K.$$

Thus the characteristic equation is $k^2 - (\operatorname{tr} S)k + \det S = 0$, i.e.

$$k^2 - 2Hk + K = 0.$$

Solving the quadratic:

$$k = \frac{2H \pm \sqrt{4H^2 - 4K}}{2} = H \pm \sqrt{H^2 - K}.$$

Hence, the eigenvalues (principal curvatures) are as stated. ∎

Example 2.5.3. Show that:

(i) $L = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{11}]$

(ii) $M = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{12}]$

(iii) $N = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{22}]$

where $H = |\mathbf{r}_1 \times \mathbf{r}_2|$ and $[a, b, c]$ denotes the scalar triple product.

Solution. The coefficients of the second fundamental form are defined by the projection of the second-order partial derivatives onto the unit normal vector $\mathbf{n}$:

$$L = \mathbf{r}_{11} \cdot \mathbf{n}, \quad M = \mathbf{r}_{12} \cdot \mathbf{n}, \quad N = \mathbf{r}_{22} \cdot \mathbf{n}$$

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The unit normal vector $\mathbf{n}$ to the surface $\mathbf{r}(u, v)$ is defined as

$$\mathbf{n} = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{|\mathbf{r}_1 \times \mathbf{r}_2|} = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{H}$$

Substituting this expression for $\mathbf{n}$ into the definitions:

1. For L:
$$L = \mathbf{r}_{11} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{11} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{11}]$$
2. For M:
$$M = \mathbf{r}_{12} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{12} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{12}]$$
3. For N:
$$N = \mathbf{r}_{22} \cdot \left(\frac{\mathbf{r}_1 \times \mathbf{r}_2}{H} \right) = \frac{1}{H}(\mathbf{r}_{22} \cdot (\mathbf{r}_1 \times \mathbf{r}_2)) = \frac{1}{H}[\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_{22}]$$

This confirms the required identities. ■

Definition 2.5.4 (Gaussian and Mean Curvature)**.** Let k_1, k_2 be the principal curvatures at a point P.

- The **Gaussian curvature** is $K = k_1 k_2 = \det(S_P)$.
- The **Mean curvature** is $H = \frac{1}{2}(k_1 + k_2) = \frac{1}{2}\mathrm{tr}(S_P)$.

These can be computed directly from the coefficients of the two fundamental forms:

$$K = \frac{LN - M^2}{EG - F^2} \quad \text{and} \quad H = \frac{EN - 2FM + GL}{2(EG - F^2)}$$

Corollary 2.5.5. *The principal curvatures are real because $H^2 - K \geq 0$ for any surface (the discriminant of the characteristic polynomial is non-negative for a self-adjoint operator). Moreover, the principal directions are the corresponding eigenvectors of $[S]$, obtained by solving $(S - k_i I)\mathbf{v} = 0$.*

Example 2.5.6 (The Sphere)**.** Find the second fundamental form, Gaussian curvature, and mean curvature of a sphere of radius a.

Solution. Parametrize the sphere by $\vec{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u)$. We have previously found:

- $\vec{r}_1 = (a \cos u \cos v, a \cos u \sin v, -a \sin u)$
- $\vec{r}_2 = (-a \sin u \sin v, a \sin u \cos v, 0)$
- The outward unit normal is $\vec{N} = (\sin u \cos v, \sin u \sin v, \cos u)$.
- $E = a^2$, $F = 0$, $G = a^2 \sin^2 u$.

We compute the second partial derivatives:

$$\vec{r}_{11} = (-a \sin u \cos v, -a \sin u \sin v, -a \cos u)$$
$$\vec{r}_{12} = (-a \cos u \sin v, a \cos u \cos v, 0)$$
$$\vec{r}_{22} = (-a \sin u \cos v, -a \sin u \sin v, 0)$$

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Now we compute L, M, N by taking dot products with $\vec{N}$:

$$L = \vec{r}_{11} \cdot \vec{N} = -a\sin^2 u\cos^2 v - a\sin^2 u\sin^2 v - a\cos^2 u = -a(\sin^2 u + \cos^2 u) = \boldsymbol{-a}$$
$$M = \vec{r}_{12} \cdot \vec{N} = -a\cos u\sin u\sin v\cos v + a\cos u\sin u\cos v\sin v = \mathbf{0}$$
$$N = \vec{r}_{22} \cdot \vec{N} = -a\sin^2 u\cos^2 v - a\sin^2 u\sin^2 v = -a\sin^2 u(\cos^2 v + \sin^2 v) = \boldsymbol{-a\sin^2 u}$$

The second fundamental form is $II = -a\,du^2 - a\sin^2 u\,dv^2$. The curvatures are

$$K = \frac{LN - M^2}{EG - F^2} = \frac{(-a)(-a\sin^2 u) - 0^2}{(a^2)(a^2\sin^2 u) - 0^2} = \frac{a^2\sin^2 u}{a^4\sin^2 u} = \frac{1}{a^2}$$
$$H = \frac{EN - 2FM + GL}{2(EG - F^2)} = \frac{a^2(-a\sin^2 u) - 0 + a^2\sin^2 u(-a)}{2(a^4\sin^2 u)} = \frac{-2a^3\sin^2 u}{2a^4\sin^2 u} = -\frac{1}{a}$$

The negative sign for H depends on the choice of the outward normal. The magnitudes are constant, as expected. Principal curvatures are $k_1 = k_2 = -1/a$. ■

Example 2.5.7 (Circular Cylinder of Radius a)**.** (See Figure 33 in the Atlas.) Compute the curvatures of the cylinder $x^2 + y^2 = a^2$.

Solution. We parametrize the cylinder by $\vec{r}(u, v) = (a\cos u, a\sin u, v)$.

First Fundamental Form: The tangent vectors are $\vec{r}_1 = (-a\sin u, a\cos u, 0)$ and $\vec{r}_2 = (0, 0, 1)$.

$$E = a^2, \quad F = 0, \quad G = 1$$

Normal Vector and Second Derivatives: For a more intuitive result where curvature is positive, we choose the inward-pointing normal:

$$\vec{N} = (-\cos u, -\sin u, 0)$$

The second derivatives are

$$\vec{r}_{11} = (-a\cos u, -a\sin u, 0)$$
$$\vec{r}_{12} = (0, 0, 0)$$
$$\vec{r}_{22} = (0, 0, 0)$$

Second Fundamental Form: The coefficients are

$$L = \vec{r}_{11} \cdot \vec{N} = (-a\cos u)(-\cos u) + (-a\sin u)(-\sin u) = a$$
$$M = \vec{r}_{12} \cdot \vec{N} = 0$$
$$N = \vec{r}_{22} \cdot \vec{N} = 0$$

The second fundamental form is $II = a\,du^2$.

Curvatures:

$$K = \frac{LN - M^2}{EG - F^2} = \frac{a(0) - 0^2}{a^2(1) - 0^2} = 0$$
$$H = \frac{EN - 2FM + GL}{2(EG - F^2)} = \frac{a^2(0) - 0 + 1(a)}{2(a^2)} = \frac{1}{2a}$$

The zero Gaussian curvature confirms that the cylinder is a developable surface. The principal curvatures are $k_1 = 1/a$ (the curvature of the circular cross-section) and $k_2 = 0$ (the curvature of the straight rulings). ■

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Example 2.5.8 (Elliptic Paraboloid $z = x^2 + y^2$)**.** Compute the curvatures of the elliptic paraboloid.

Solution. We use the graph parametrization $\vec{r}(x, y) = (x, y, x^2 + y^2)$.

First Fundamental Form: The tangent vectors are $\vec{r}_1 = (1, 0, 2x)$ and $\vec{r}_2 = (0, 1, 2y)$.

$$E = 1 + 4x^2, \quad F = 4xy, \quad G = 1 + 4y^2$$

Normal Vector and Second Derivatives: The upward-pointing normal is $\vec{r}_1 \times \vec{r}_2 = (-2x, -2y, 1)$. Normalizing gives $\vec{N} = \frac{(-2x,-2y,1)}{\sqrt{1+4x^2+4y^2}}$. The second derivatives are constant:

$$\vec{r}_{11} = (0, 0, 2), \quad \vec{r}_{12} = (0, 0, 0), \quad \vec{r}_{22} = (0, 0, 2)$$

Second Fundamental Form: The coefficients are

$$L = \vec{r}_{11} \cdot \vec{N} = \frac{2}{\sqrt{1 + 4x^2 + 4y^2}}$$
$$M = \vec{r}_{12} \cdot \vec{N} = 0$$
$$N = \vec{r}_{22} \cdot \vec{N} = \frac{2}{\sqrt{1 + 4x^2 + 4y^2}}$$

Curvatures: We first compute the denominators and numerators. Let $r^2 = x^2 + y^2$.

$$EG - F^2 = (1 + 4x^2)(1 + 4y^2) - (4xy)^2 = 1 + 4x^2 + 4y^2 + 16x^2y^2 - 16x^2y^2 = 1 + 4r^2$$
$$LN - M^2 = \left(\frac{2}{\sqrt{1 + 4r^2}}\right)\left(\frac{2}{\sqrt{1 + 4r^2}}\right) - 0 = \frac{4}{1 + 4r^2}$$

The Gaussian curvature is

$$K = \frac{LN - M^2}{EG - F^2} = \frac{4/(1 + 4r^2)}{1 + 4r^2} = \frac{4}{(1 + 4r^2)^2}$$

The mean curvature is

$$H = \frac{EN + GL}{2(EG - F^2)} = \frac{(1 + 4x^2)\frac{2}{\sqrt{1+4r^2}} + (1 + 4y^2)\frac{2}{\sqrt{1+4r^2}}}{2(1 + 4r^2)} = \frac{2(2 + 4r^2)}{2(1 + 4r^2)^{3/2}} = \frac{2 + 4r^2}{(1 + 4r^2)^{3/2}}$$

Both curvatures are highest at the origin ($r = 0$) and decrease as one moves away from the origin. ■

The second fundamental form describes how a surface bends in the surrounding Euclidean space. Its coefficients L, M, N are determined by the second derivatives of a local parametrization and the unit normal vector field. Through the shape operator, the second fundamental form provides the basis for studying normal curvature, principal curvatures, Gaussian curvature and mean curvature. Although these quantities arise from the geometry of the surface in $\mathbb{R}^3$, Gauss's Theorema Egregium shows that Gaussian curvature is an intrinsic invariant, determined entirely by the first fundamental form. This remarkable result is one of the foundations of modern Riemannian geometry.

To understand normal curvature geometrically, let P be a point on a regular surface S. A plane passing through P generally intersects the surface locally in a curve, called a **plane section** of the surface, provided the intersection is regular near P. If the plane contains the normal vector $N(P)$, the resulting curve is called a **normal section**; otherwise, it is called an **oblique section**.

Every tangent direction at P determines a unique plane containing that direction and the surface normal $N(P)$. Consequently, there are infinitely many normal sections through P, one for each unoriented tangent direction. The tangent line to a normal section lies in its defining plane, and hence its principal normal, whenever defined, is parallel to the surface normal at P.

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The **normal curvature** of a surface at P in a given tangent direction measures how the surface bends in that direction. Geometrically, it is the signed curvature of the corresponding normal section, measured with respect to the chosen unit normal $N(P)$. Equivalently, if $\alpha(s)$ is a unit-speed curve on S passing through P with unit tangent vector T at P, its normal curvature is

$$\kappa_n = \langle \alpha''(s), N(P) \rangle = II_P(T, T).$$

Thus, normal curvature depends only on the tangent direction at P and not on the particular curve chosen on the surface with that tangent direction.

Definition 2.5.9 (Normal Curvature)**.** Let S be an oriented surface, $P \in S$, and $\vec{t}$ be a unit tangent vector in the tangent plane $T_P(S)$. The **normal curvature** of S at P in the direction $\vec{t}$, denoted $k_n(\vec{t})$, is defined as the second fundamental form evaluated on $\vec{t}$:

$$k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t} \rangle$$

where L_P is the shape operator at P. This value is equivalent to the projection of the acceleration of any curve $\vec{\alpha}(s)$ with tangent $\vec{t}$ onto the surface normal:

$$k_n(\vec{t}) = \left\langle \vec{\alpha}''(s), \vec{N} \right\rangle$$

Remark 2.5.10. A fundamental property is that k_n depends only on the direction $\vec{t}$ at the point P, not on the specific curve passing through P with that tangent.

Theorem 2.5.11. *Prove that the normal curvature of a surface $\vec{r}(u, v)$ at any point P is*

$$\kappa_n = \frac{II_P}{I_P} = \frac{Ldu^2 + 2Mdudv + Ndv^2}{Edu^2 + 2Fdudv + Gdv^2}$$

Proof. The proof begins with the definition of normal curvature and expresses it in terms of the two fundamental forms.

The normal curvature k_n at a point P is defined for a direction specified by a *unit* tangent vector $\vec{t} \in T_P(S)$, and is given by the action of the shape operator L_P:

$$k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t} \rangle$$

A general tangent direction can be represented by an arbitrary non-zero tangent vector $\vec{v}$, which corresponds to an infinitesimal displacement $\vec{v} = \vec{r}_1 du + \vec{r}_2 dv$. To use the definition, we normalize this vector to get the corresponding unit tangent vector $\vec{t} = \vec{v}/|\vec{v}|$.

Substituting this into the definition of k_n, and using the linearity of the shape operator and the bilinearity of the inner product, we have

$$k_n = \left\langle L_P\left(\frac{\vec{v}}{|\vec{v}|}\right), \frac{\vec{v}}{|\vec{v}|} \right\rangle = \frac{1}{|\vec{v}|^2} \langle L_P(\vec{v}), \vec{v} \rangle$$

We now identify the numerator and denominator of this expression.

The numerator, $\langle L_P(\vec{v}), \vec{v} \rangle$, is by definition the second fundamental form II_P evaluated on the vector $\vec{v}$. Expanding this gives

$$\begin{aligned} \langle L_P(\vec{r}_1 du + \vec{r}_2 dv), \vec{r}_1 du + \vec{r}_2 dv \rangle &= \langle L_P(\vec{r}_1), \vec{r}_1 \rangle\, du^2 + 2\langle L_P(\vec{r}_1), \vec{r}_2 \rangle\, du\, dv + \langle L_P(\vec{r}_2), \vec{r}_2 \rangle\, dv^2 \\ &= L\, du^2 + 2M\, du\, dv + N\, dv^2 = II_P \end{aligned}$$

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The denominator, $|\vec{v}|^2 = \langle \vec{v}, \vec{v} \rangle$, is the first fundamental form I_P evaluated on $\vec{v}$:

$$\begin{aligned}\langle \vec{r}_1 du + \vec{r}_2 dv, \vec{r}_1 du + \vec{r}_2 dv \rangle &= \langle \vec{r}_1, \vec{r}_1 \rangle \, du^2 + 2 \langle \vec{r}_1, \vec{r}_2 \rangle \, du \, dv + \langle \vec{r}_2, \vec{r}_2 \rangle \, dv^2 \\ &= E \, du^2 + 2F \, du \, dv + G \, dv^2 = I_P\end{aligned}$$

Substituting these expressions back into our formula for k_n yields the desired result:

$$k_n = \frac{\langle L_P(\vec{v}), \vec{v} \rangle}{|\vec{v}|^2} = \frac{II_P}{I_P} = \frac{L \, du^2 + 2M \, du \, dv + N \, dv^2}{E \, du^2 + 2F \, du \, dv + G \, dv^2}$$

This proves the result. ■

2nd Proof. This proof begins with the geometric definition of normal curvature and derives the formula by analyzing the acceleration of a curve on the surface.

Let $\vec{\alpha}(s)$ be a curve on the surface S parametrized by arc length s, passing through the point P. The curve can be expressed in terms of the surface parametrization as $\vec{\alpha}(s) = \vec{r}(u(s), v(s))$. The normal curvature at P is defined as

$$k_n = \left\langle \vec{\alpha}''(s), \vec{N} \right\rangle$$

where $\vec{N}$ is the unit normal to the surface at P. Our goal is to express $\vec{\alpha}''(s)$ in terms of the parametrization $\vec{r}$.

First compute the tangent vector $\vec{\alpha}'(s)$ using the chain rule. We use the notation $u' = \frac{du}{ds}$ and $v' = \frac{dv}{ds}$.

$$\vec{\alpha}'(s) = \frac{\partial \vec{r}}{\partial u}\frac{du}{ds} + \frac{\partial \vec{r}}{\partial v}\frac{dv}{ds} = \vec{r}_1 u' + \vec{r}_2 v'$$

Next differentiate $\vec{\alpha}'(s)$ with respect to s to find the acceleration vector $\vec{\alpha}''(s)$. This requires both the product rule and the chain rule, as the basis vectors $\vec{r}_1$ and $\vec{r}_2$ change with position $(u(s), v(s))$:

$$\begin{aligned}\vec{\alpha}''(s) &= \frac{d}{ds}(\vec{r}_1 u' + \vec{r}_2 v') \\ &= \left(\frac{d\vec{r}_1}{ds}\right) u' + \vec{r}_1 u'' + \left(\frac{d\vec{r}_2}{ds}\right) v' + \vec{r}_2 v''\end{aligned}$$

Applying the chain rule to the derivatives of the basis vectors:

$$\begin{aligned}\frac{d\vec{r}_1}{ds} &= \frac{\partial \vec{r}_1}{\partial u}\frac{du}{ds} + \frac{\partial \vec{r}_1}{\partial v}\frac{dv}{ds} = \vec{r}_{11} u' + \vec{r}_{12} v' \\ \frac{d\vec{r}_2}{ds} &= \frac{\partial \vec{r}_2}{\partial u}\frac{du}{ds} + \frac{\partial \vec{r}_2}{\partial v}\frac{dv}{ds} = \vec{r}_{21} u' + \vec{r}_{22} v'\end{aligned}$$

Substituting these back into the expression for $\vec{\alpha}''(s)$ and using the equality of mixed partials ($\vec{r}_{12} = \vec{r}_{21}$), we obtain

$$\vec{\alpha}''(s) = \vec{r}_{11}(u')^2 + 2\vec{r}_{12} u'v' + \vec{r}_{22}(v')^2 + \vec{r}_1 u'' + \vec{r}_2 v''$$

Now we take the inner product of $\vec{\alpha}''(s)$ with the surface normal $\vec{N}$ to find k_n:

$$k_n = \left\langle \vec{r}_{11}(u')^2 + 2\vec{r}_{12} u'v' + \vec{r}_{22}(v')^2 + \vec{r}_1 u'' + \vec{r}_2 v'', \vec{N} \right\rangle$$

By definition, the tangent vectors $\vec{r}_1$ and $\vec{r}_2$ are orthogonal to the surface normal $\vec{N}$. Therefore, $\left\langle \vec{r}_1, \vec{N} \right\rangle = 0$ and $\left\langle \vec{r}_2, \vec{N} \right\rangle = 0$. The last two terms of the expression vanish, leaving:

$$k_n = \left\langle \vec{r}_{11}, \vec{N} \right\rangle (u')^2 + 2 \left\langle \vec{r}_{12}, \vec{N} \right\rangle u'v' + \left\langle \vec{r}_{22}, \vec{N} \right\rangle (v')^2$$

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We recognize the inner products as the coefficients of the second fundamental form: $L = \left\langle \vec{r}_{11}, \vec{N} \right\rangle$, $M = \left\langle \vec{r}_{12}, \vec{N} \right\rangle$, and $N = \left\langle \vec{r}_{22}, \vec{N} \right\rangle$. This simplifies the expression to:

$$k_n = L(u')^2 + 2Mu'v' + N(v')^2$$

This formula for k_n is in terms of derivatives with respect to arc length. The theorem, however, is stated in terms of the differentials du and dv, which represent a direction independent of any specific curve's parametrization. To bridge this, we recall that for an arc length parametrization, $|\vec{\alpha}'(s)|^2 = 1$. This implies:

$$1 = \langle \vec{r}_1 u' + \vec{r}_2 v', \vec{r}_1 u' + \vec{r}_2 v' \rangle = E(u')^2 + 2Fu'v' + G(v')^2$$

We can therefore write k_n as a ratio where the denominator is unity:

$$k_n = \frac{L(u')^2 + 2Mu'v' + N(v')^2}{E(u')^2 + 2Fu'v' + G(v')^2}$$

Finally, observe that the direction of the curve is given by the ratio $v' : u' = \frac{dv}{ds} : \frac{du}{ds}$, which is the same as the ratio $dv : du$. We can multiply the numerator and denominator by the scalar $(ds)^2$ without changing the value of the fraction. This converts the derivatives into differentials:

$$\begin{aligned} k_n &= \frac{\left(L(\frac{du}{ds})^2 + 2M\frac{du}{ds}\frac{dv}{ds} + N(\frac{dv}{ds})^2\right)(ds)^2}{\left(E(\frac{du}{ds})^2 + 2F\frac{du}{ds}\frac{dv}{ds} + G(\frac{dv}{ds})^2\right)(ds)^2} \\ &= \frac{L\,du^2 + 2M\,du\,dv + N\,dv^2}{E\,du^2 + 2F\,du\,dv + G\,dv^2} \end{aligned}$$

This proves the result. ■

Geometric Interpretation

This formula provides a powerful way to understand normal curvature.

- The denominator, $I_P = ds^2$, is the squared infinitesimal distance one travels along the surface in a given direction.
- The numerator, II_P, measures the component of the acceleration vector normal to the surface. It quantifies how much the surface "pulls away" from the tangent plane in that direction.

The normal curvature is therefore the ratio of how much the surface bends away from its tangent plane to the squared distance traveled. This ratio depends only on the direction of travel (i.e., the ratio $dv : du$), not on the speed.

Definition 2.5.12 (Normal curvature)**.** For a unit tangent vector $\vec{v} \in T_pS$, the **normal curvature** in the direction $\vec{v}$ is $\kappa_n(\vec{v}) = II_p(\vec{v}, \vec{v}) = \langle S_p(\vec{v}), \vec{v} \rangle$.

Theorem 2.5.13 (Euler's Theorem)**.** *Let k_1 and k_2 be the principal curvatures at a point P on a surface, with corresponding orthonormal principal directions $\vec{e}_1$ and $\vec{e}_2$. The normal curvature k_n in the direction of a unit tangent vector $\vec{t}$ that makes an angle θ with $\vec{e}_1$ is given by $k_n(\vec{t}) = k_1 \cos^2\theta + k_2 \sin^2\theta$.*

Proof. The shape operator L_P is self-adjoint, so there exists an orthonormal basis $\{\vec{e}_1, \vec{e}_2\}$ of the tangent plane $T_P(S)$ consisting of eigenvectors of L_P. Let k_1 and k_2 be the corresponding eigenvalues (the principal curvatures), so $L_P(\vec{e}_1) = k_1\vec{e}_1$ and $L_P(\vec{e}_2) = k_2\vec{e}_2$.

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A unit tangent vector $\vec{t}$ making an angle θ with $\vec{e}_1$ can be expressed in this basis as:

$$\vec{t} = (\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2$$

The normal curvature in the direction of $\vec{t}$ is given by $k_n(\vec{t}) = \langle L_P(\vec{t}), \vec{t}\rangle$. We compute this directly:

$$\begin{aligned} k_n(\vec{t}) = \langle L_p(\vec{t}), \vec{t}\rangle &= \langle L_P((\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2),\ (\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2\rangle \\ &= \langle (\cos\theta)L_P(\vec{e}_1) + (\sin\theta)L_P(\vec{e}_2),\ (\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2\rangle \quad \text{(Linearity of } L_P\text{)} \\ &= \langle (k_1\cos\theta)\vec{e}_1 + (k_2\sin\theta)\vec{e}_2,\ (\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2\rangle \quad \text{(Eigenvector property)} \end{aligned}$$

Expanding the inner product and using the orthonormality of the basis ($\langle \vec{e}_i, \vec{e}_j\rangle = \delta_{ij}$), the cross terms vanish:

$$\begin{aligned} k_n(\vec{t}) &= (k_1\cos^2\theta)\,\langle \vec{e}_1, \vec{e}_1\rangle + (k_2\sin^2\theta)\,\langle \vec{e}_2, \vec{e}_2\rangle \\ &= k_1\cos^2\theta + k_2\sin^2\theta \end{aligned}$$

This establishes the theorem. ■

Theorem 2.5.14 (Meusnier's Theorem). *Let κ_n denote the normal curvature of a surface at a point in a given direction, and let κ be the curvature of an arbitrary curve on the surface passing through that point with the same tangent. Then:*

$$\kappa_n = \kappa\cos\theta$$

where θ is the angle between the principal normal of the curve and the surface normal.

Proof. Let $\alpha(s)$ be a unit-speed curve on the surface, $\mathbf{N}$ the surface normal, and $\mathbf{N}_c$ the principal normal of the curve. The curvature vector is $\mathbf{k} = \alpha''(s) = \kappa\mathbf{N}_c$.

By definition, the normal curvature is the projection of the curve's curvature vector onto the surface normal:

$$\kappa_n = \mathbf{k}\cdot\mathbf{N}$$

Substituting $\mathbf{k} = \kappa\mathbf{N}_c$:

$$\kappa_n = (\kappa\mathbf{N}_c)\cdot\mathbf{N} = \kappa(\mathbf{N}_c\cdot\mathbf{N})$$

Since $\mathbf{N}_c$ and $\mathbf{N}$ are both unit vectors, their dot product is $\cos\theta$, where θ is the angle between them. Thus

$$\kappa_n = \kappa\cos\theta$$

■

2.5.1 Geometric Interpretation

The elegance of Meusnier's theorem lies in its geometric interpretation.

- The normal curvature k_n is fixed for a given direction $\vec{t}$ at point P.
- The curvature k of a curve $\vec{\alpha}$ through P with tangent $\vec{t}$ can vary depending on how the curve twists on the surface.
- The theorem $k = k_n/\cos\theta$ implies that the curvature k is minimized when $\cos\theta$ is maximized. This occurs when $\theta = 0$, meaning the principal normal of the curve $\vec{n}_\alpha$ is aligned with the surface normal $\vec{N}$.

Example 2.5.15. If a surface has a property that $L = M = N = 0$ at any poit P, the the surface is a plane.

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Solution. The coefficients L, M, N are given by $L = \mathbf{n} \cdot \mathbf{r}_{uu}$, $M = \mathbf{n} \cdot \mathbf{r}_{uv}$, and $N = \mathbf{n} \cdot \mathbf{r}_{vv}$. If $L = M = N = 0$, then the normal vector $\mathbf{n}$ is orthogonal to all second-order partial derivatives of the surface patch $\mathbf{r}$.

Since $\mathbf{n}$ is also orthogonal to the tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$, the derivatives of the normal vector satisfy $\mathbf{n}_u \cdot \mathbf{r}_u = -\mathbf{n} \cdot \mathbf{r}_{uu} = 0$, $\mathbf{n}_u \cdot \mathbf{r}_v = -\mathbf{n} \cdot \mathbf{r}_{uv} = 0$, and similarly for $\mathbf{n}_v$. Thus, $\mathbf{n}_u = \mathbf{n}_v = 0$, which implies $\mathbf{n}$ is a constant vector. Consequently, the surface is a plane. ■

2.6 Principal Curvatures and Euler's Theorem

Since S_p is self-adjoint, it has two real eigenvalues κ_1, κ_2 (the **principal curvatures**) with orthogonal eigenvectors (the **principal directions**). The **Gaussian curvature** and **mean curvature** are

$$K = \kappa_1 \kappa_2 = \frac{LN - M^2}{EG - F^2}, \qquad H = \frac{\kappa_1 + \kappa_2}{2} = \frac{EN - 2FM + GL}{2(EG - F^2)}.$$

Theorem 2.6.1. *Let P be a point on a regular surface S. The maximum and minimum values of the normal curvature $k_n(\vec{v})$ as $\vec{v}$ ranges over all unit tangent vectors in $S_P(S)$ are exactly the principal curvatures k_1 and k_2.*

Proof. Let $k_1 \geq k_2$ be the principal curvatures with corresponding orthonormal principal directions $\vec{e}_1$ and $\vec{e}_2$. Any unit tangent vector $\vec{v}$ can be written in this basis as:

$$\vec{v} = (\cos\theta)\vec{e}_1 + (\sin\theta)\vec{e}_2$$

From Euler's Theorem, the normal curvature in this direction is $k_n(\theta) = k_1 \cos^2\theta + k_2 \sin^2\theta$. Using the identity $\sin^2\theta = 1 - \cos^2\theta$, we rewrite this as:

$$k_n(\theta) = k_1 \cos^2\theta + k_2(1 - \cos^2\theta) = (k_1 - k_2)\cos^2\theta + k_2$$

Since $0 \leq \cos^2\theta \leq 1$, we can find the bounds for k_n:

Maximum: The expression is maximized when $\cos^2\theta$ is as large as possible, i.e., $\cos^2\theta = 1$ (which means $\theta = 0$).

$$k_{max} = (k_1 - k_2)(1) + k_2 = k_1$$

Minimum: The expression is minimized when $\cos^2\theta$ is as small as possible, i.e., $\cos^2\theta = 0$ (which means $\theta = \pi/2$).

$$k_{min} = (k_1 - k_2)(0) + k_2 = k_2$$

Thus, the normal curvature k_n is constrained by $k_2 \leq k_n \leq k_1$, and these bounds are exactly the principal curvatures. ■

Remark 2.6.2. This result confirms that the principal directions are the "axes of curvature" for the surface. When $\vec{v}$ is aligned with a principal direction, the normal curvature is extremized. For any other direction, Euler's Theorem tells us that k_n is a weighted average of these two extreme values, always lying between them:

$$\min(k_1, k_2) \leq k_n(\vec{v}) \leq \max(k_1, k_2)$$

2.7 Principal, Mean, and Gaussian Curvatures

Let $\mathbf{r} = \mathbf{r}(u, v)$ be a regular surface with unit normal $\mathbf{N} = (\mathbf{r}_u \times \mathbf{r}_v)/\|\mathbf{r}_u \times \mathbf{r}_v\|$. The first and second fundamental forms are

$$I = E\,du^2 + 2F\,du\,dv + G\,dv^2, \qquad II = L\,du^2 + 2M\,du\,dv + N\,dv^2,$$

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where $E = \mathbf{r}_u \cdot \mathbf{r}_u$, $F = \mathbf{r}_u \cdot \mathbf{r}_v$, $G = \mathbf{r}_v \cdot \mathbf{r}_v$, and $L = \mathbf{r}_{uu} \cdot \mathbf{N}$, $M = \mathbf{r}_{uv} \cdot \mathbf{N}$, $N = \mathbf{r}_{vv} \cdot \mathbf{N}$. We write $H_0^2 = EG - F^2 > 0$; the notation H_0 is used to distinguish this quantity from the mean curvature H.

Definition 2.7.1 (Normal Curvature)**.** Let a curve on the surface have tangent direction (du, dv) at a point P. The **normal curvature** in this direction is

$$k_n = \frac{L\,du^2 + 2M\,du\,dv + N\,dv^2}{E\,du^2 + 2F\,du\,dv + G\,dv^2}.$$

If the direction is parameterized by arc length s, put $l = du/ds$ and $m = dv/ds$. Then $El^2 + 2Flm + Gm^2 = 1$, and consequently $k_n = Ll^2 + 2Mlm + Nm^2$.

Definition 2.7.2 (Principal Directions and Principal Curvatures)**.** At a point P of a regular surface, the tangent directions in which the normal curvature assumes its maximum and minimum values are called the **principal directions**. The corresponding normal curvatures are called the **principal curvatures** and are denoted by k_1 and k_2.

Theorem 2.7.3 (Equation for the Principal Curvatures)**.** *The principal curvatures are the roots of*

$$(EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) = 0.$$

Proof. Let $l = du/ds$ and $m = dv/ds$. We seek the stationary values of

$$k_n = Ll^2 + 2Mlm + Nm^2$$

subject to

$$El^2 + 2Flm + Gm^2 = 1.$$

Using a Lagrange multiplier λ, consider

$$\Phi = Ll^2 + 2Mlm + Nm^2 - \lambda(El^2 + 2Flm + Gm^2 - 1).$$

The equations $\partial\Phi/\partial l = 0$ and $\partial\Phi/\partial m = 0$ give

$$(L - \lambda E)l + (M - \lambda F)m = 0$$

and

$$(M - \lambda F)l + (N - \lambda G)m = 0.$$

Multiplying the first equation by l, the second by m, and adding, we obtain $\lambda = k_n$. Hence a principal curvature k satisfies

$$(L - kE)l + (M - kF)m = 0, \qquad (M - kF)l + (N - kG)m = 0.$$

A non-zero solution (l, m) exists if and only if

$$\begin{vmatrix} L - kE & M - kF \\ M - kF & N - kG \end{vmatrix} = 0.$$

Expanding the determinant gives $(EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) = 0$. ∎

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Definition 2.7.4 (Mean Curvature)**.** The **mean curvature** of a surface is $H = \frac{k_1+k_2}{2}$. Thus

$$H = \frac{EN + GL - 2FM}{2(EG - F^2)}.$$

Definition 2.7.5 (Gaussian Curvature)**.** The **Gaussian curvature** is the product of the principal curvatures:

$$K = k_1 k_2.$$

Consequently,

$$K = \frac{LN - M^2}{EG - F^2}.$$

Proposition 2.7.6. *The principal curvatures may be recovered from the mean and Gaussian curvatures by*

$$k_1 = H + \sqrt{H^2 - K}, \qquad k_2 = H - \sqrt{H^2 - K}.$$

Moreover,

$$H^2 - K = \frac{(k_1 - k_2)^2}{4} \geq 0,$$

and a point is umbilic if and only if $H^2 = K$.

Proof. Since $k_1 + k_2 = 2H$ and $k_1 k_2 = K$, the principal curvatures are the roots of $x^2 - 2Hx + K = 0$. Hence the quadratic formula gives the stated expressions. Also,

$$H^2 - K = \frac{(k_1 + k_2)^2}{4} - k_1 k_2 = \frac{(k_1 - k_2)^2}{4}.$$

Thus $H^2 = K$ precisely when $k_1 = k_2$, that is, precisely at an umbilic. ■

Remark 2.7.7. The Gaussian curvature is independent of the choice of regular parameters. Indeed, under a change of parameters, both $LN - M^2$ and $EG - F^2$ acquire the same factor, namely the square of the Jacobian.

Proposition 2.7.8 (Classification by Gaussian Curvature)**.** *At a non-singular point P:*

1. $K(P) > 0$ *if and only if P is elliptic;*
2. $K(P) < 0$ *if and only if P is hyperbolic;*
3. $K(P) = 0$ *if and only if at least one principal curvature vanishes.*

Proof. Since $EG - F^2 > 0$, the sign of K is the sign of $LN - M^2$. Equivalently, $K = k_1 k_2$. Hence $K > 0$ when the principal curvatures have the same sign, $K < 0$ when they have opposite signs, and $K = 0$ when at least one of them vanishes. ■

Theorem 2.7.9 (Equation of the Principal Directions)**.** *The principal directions satisfy*

$$(EM - FL)\,du^2 + (EN - GL)\,du\,dv + (FN - GM)\,dv^2 = 0.$$

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At a non-umbilic point, the two principal directions are orthogonal.

Proof. For a principal curvature k,

$$(L - kE)du + (M - kF)dv = 0,$$

$$(M - kF)du + (N - kG)dv = 0.$$

Eliminating k gives $(EM - FL)\,du^2 + (EN - GL)\,du\,dv + (FN - GM)\,dv^2 = 0$. The standard orthogonality condition for the two directions of $P\,du^2 + Q\,du\,dv + R\,dv^2 = 0$ reduces here to an identity, so the two principal directions are perpendicular whenever they are distinct. ■

Definition 2.7.10 (Umbilic)**.** A point P is called an **umbilic** if the normal curvature is the same in every tangent direction. Equivalently,

$$\frac{L}{E} = \frac{M}{F} = \frac{N}{G}$$

whenever these ratios are meaningful, or, invariantly, the second fundamental form is a scalar multiple of the first fundamental form at P.

Example 2.7.11 (Sphere)**.** Show that every point of a sphere of radius a is an umbilic and determine its Gaussian and mean curvatures.

Solution. Use the parameterization

$$\mathbf{r}(u, v) = (a \sin u \cos v,\ a \sin u \sin v,\ a \cos u).$$

A direct calculation gives $E = a^2, \quad F = 0, \quad G = a^2 \sin^2 u$. With a consistent choice of orientation,

$$L = -a, \quad M = 0, \quad N = -a \sin^2 u.$$

Hence

$$\frac{L}{E} = \frac{N}{G} = -\frac{1}{a},$$

so every point is umbilic. Thus

$$k_1 = k_2 = -\frac{1}{a}, \quad H = -\frac{1}{a}, \quad K = \frac{1}{a^2}.$$

Changing the orientation reverses the signs of k_1, k_2 and H, but leaves K unchanged. ■

Theorem 2.7.12 (Totally Umbilic Surfaces)**.** *Let S be a connected regular surface for which every point is umbilic. Then S is contained in either a plane or a sphere.*

Proof. Since every point is umbilic, the shape operator has the form $S = \lambda I$ for some smooth function λ. Hence $d\mathbf{N} = -\lambda\, d\mathbf{r}$, or equivalently $\mathbf{N}_u = -\lambda \mathbf{r}_u$ and $\mathbf{N}_v = -\lambda \mathbf{r}_v$. Equality of mixed derivatives gives

$$\mathbf{N}_{uv} - \mathbf{N}_{vu} = -\lambda_v \mathbf{r}_u + \lambda_u \mathbf{r}_v = 0.$$

Since $\mathbf{r}_u$ and $\mathbf{r}_v$ are linearly independent, $\lambda_u = \lambda_v = 0$; thus λ is constant on the connected surface.

If $\lambda = 0$, then $d\mathbf{N} = 0$, so $\mathbf{N}$ is constant and $\mathbf{r} \cdot \mathbf{N}$ is constant. Hence S lies in a plane. If $\lambda \neq 0$, then $d(\mathbf{r} + \lambda^{-1}\mathbf{N}) = 0$, so $\mathbf{r} + \lambda^{-1}\mathbf{N} = \mathbf{c}$ for a fixed vector $\mathbf{c}$. Therefore $\|\mathbf{r} - \mathbf{c}\| = 1/|\lambda|$, and S lies on a sphere. ■

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Example 2.7.13 (Circular Cylinder)**.** Find the principal, mean, and Gaussian curvatures of the circular cylinder $\mathbf{r}(u,v) = (a\cos u, a\sin u, v)$, where $a > 0$.

Solution. We have $\mathbf{r}_u = (-a\sin u, a\cos u, 0)$ and $\mathbf{r}_v = (0,0,1)$, so $E = a^2$, $F = 0$, and $G = 1$. Choosing the outward unit normal $\mathbf{N} = (\cos u, \sin u, 0)$ gives $L = -a$, $M = 0$, and $N = 0$. Since the coordinate curves are principal, $k_1 = L/E = -1/a$ and $k_2 = N/G = 0$. Hence $H = -1/(2a)$ and $K = 0$. ■

Example 2.7.14 (Hyperbolic Paraboloid at the Origin)**.** Find the principal curvatures of the surface $z = xy$ at the origin.

Solution. Take $\mathbf{r}(u,v) = (u,v,uv)$. At $(0,0)$, $E = G = 1$ and $F = 0$. With the upward unit normal, $L = N = 0$ and $M = 1$. Therefore the principal curvature equation becomes $k^2 - 1 = 0$. Hence $k_1 = 1$, $k_2 = -1$, so $H = 0$ and $K = -1$. Thus the origin is a hyperbolic point. ■

Example 2.7.15 (Torus)**.** (See Figure 34 in the Atlas.) For the torus $\mathbf{r}(u,v) = ((R + a\cos u)\cos v, (R + a\cos u)\sin v, a\sin u)$, where $R > a > 0$, determine the Gaussian curvature.

Solution. The first fundamental coefficients are $E = a^2$, $F = 0$, and $G = (R + a\cos u)^2$. With the outward normal $\mathbf{N} = (\cos u\cos v, \cos u\sin v, \sin u)$, one obtains $L = -a$, $M = 0$, and $N = -(R + a\cos u)\cos u$. Therefore

$$K = \frac{LN}{EG} = \frac{\cos u}{a(R + a\cos u)}.$$

Thus $K > 0$ on the outer part of the torus, $K < 0$ on the inner part, and $K = 0$ along the two circles corresponding to $\cos u = 0$. ■

Example 2.7.16 (Umbilic Criterion)**.** Suppose a point of a surface has $H = 3$ and $K = 9$. Show that the point is umbilic and determine the principal curvatures.

Solution. Since $H^2 - K = 9 - 9 = 0$, the two principal curvatures are equal. Indeed, $k_1 = k_2 = H = 3$. Hence the point is umbilic. ■

Chapter 3

Curvature and Special Surfaces

This chapter develops the local geometry of surfaces through normal curvature, principal curvatures, and the Gaussian and mean curvatures. It then studies the variation of the surface normal through the shape operator, Weingarten equations, and the geometry of lines of curvature. Special attention is given to the Dupin indicatrix, asymptotic and conjugate directions, ruled surfaces, minimal surfaces, and isothermal coordinates. The chapter concludes with intrinsic aspects of surface geometry, including isometries, local isometries, and the harmonic characterization of minimal surfaces.

3.1 Gaussian Curvature as an Area Ratio

Definition 3.1.1 (Gauss Map)**.** The **Gauss map** of an oriented surface is the map

$$\mathcal{G} : S \longrightarrow \mathbb{S}^2, \qquad \mathcal{G}(P) = \mathbf{N}(P).$$

Theorem 3.1.2. *The derivatives of the Gauss map satisfy*

$$\mathbf{N}_u \times \mathbf{N}_v = K(\mathbf{r}_u \times \mathbf{r}_v).$$

Consequently, if dA is the area element on S and $dA_{\mathbb{S}^2}$ is the area element of the spherical image, then

$$dA_{\mathbb{S}^2} = |K|\, dA.$$

With oriented areas,

$$K = \frac{dA_{\mathbb{S}^2}}{dA}.$$

Proof. Using the Weingarten equations, write

$$\mathbf{N}_u = a\mathbf{r}_u + b\mathbf{r}_v, \qquad \mathbf{N}_v = c\mathbf{r}_u + d\mathbf{r}_v.$$

Then

$$\mathbf{N}_u \times \mathbf{N}_v = (ad - bc)(\mathbf{r}_u \times \mathbf{r}_v).$$

The determinant $ad - bc$ is the determinant of the shape operator and equals $\frac{LN - M^2}{EG - F^2} = K$. Hence

$$\mathbf{N}_u \times \mathbf{N}_v = K(\mathbf{r}_u \times \mathbf{r}_v).$$

Taking norms gives

$$\|\mathbf{N}_u \times \mathbf{N}_v\| = |K| \|\mathbf{r}_u \times \mathbf{r}_v\|.$$

Multiplying by $du\, dv$ yields $dA_{\mathbb{S}^2} = |K|\, dA$. ∎

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Theorem 3.1.3 (Gaussian Curvature in Orthogonal Coordinates). *If $F = 0$, then*

$$K = -\frac{1}{2\sqrt{EG}}\left[\frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right) + \frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right)\right].$$

Proof. Let $\mathbf{r}(u, v)$ be a local parametrization of S, with unit normal $\mathbf{N}$. Since the coordinates are orthogonal, $F = \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0$, and the Gauss formulas become

$$\begin{aligned}
\mathbf{r}_{uu} &= \Gamma_{11}^1 \mathbf{r}_u + \Gamma_{11}^2 \mathbf{r}_v + L\mathbf{N}, \\
\mathbf{r}_{uv} &= \Gamma_{12}^1 \mathbf{r}_u + \Gamma_{12}^2 \mathbf{r}_v + M\mathbf{N}, \\
\mathbf{r}_{vv} &= \Gamma_{22}^1 \mathbf{r}_u + \Gamma_{22}^2 \mathbf{r}_v + N\mathbf{N}.
\end{aligned}$$

The Christoffel symbols are determined by the coefficients of the first fundamental form. Since $F = 0$, their expressions simplify to

$$\Gamma_{11}^1 = \frac{E_u}{2E}, \qquad \Gamma_{11}^2 = -\frac{E_v}{2G}, \qquad \Gamma_{12}^1 = \frac{E_v}{2E},$$

and

$$\Gamma_{12}^2 = \frac{G_u}{2G}, \qquad \Gamma_{22}^1 = -\frac{G_u}{2E}, \qquad \Gamma_{22}^2 = \frac{G_v}{2G}.$$

We now use the equality of mixed partial derivatives $\mathbf{r}_{uuv} = \mathbf{r}_{uvu}$. Differentiating the first and second Gauss formulas and comparing the coefficients of $\mathbf{r}_v$, we obtain

$$\Gamma_{11,v}^2 + \Gamma_{11}^1\Gamma_{12}^2 + \Gamma_{11}^2\Gamma_{22}^2 - \frac{LN}{G} = \Gamma_{12,u}^2 + \Gamma_{12}^1\Gamma_{11}^2 + (\Gamma_{12}^2)^2 - \frac{M^2}{G}.$$

Rearranging gives

$$\frac{LN - M^2}{G} = \Gamma_{11,v}^2 - \Gamma_{12,u}^2 + \Gamma_{11}^1\Gamma_{12}^2 + \Gamma_{11}^2\Gamma_{22}^2 - \Gamma_{12}^1\Gamma_{11}^2 - (\Gamma_{12}^2)^2.$$

Since the Gaussian curvature is $K = (LN - M^2)/(EG)$, we have

$$\begin{aligned}
EK = \frac{\partial}{\partial v}\left(-\frac{E_v}{2G}\right) - \frac{\partial}{\partial u}\left(\frac{G_u}{2G}\right) + \frac{E_u G_u}{4EG} \\
- \frac{E_v G_v}{4G^2} + \frac{E_v^2}{4EG} - \frac{G_u^2}{4G^2}.
\end{aligned}$$

Expanding the derivatives and collecting terms, we obtain

$$EK = -\frac{E_{vv} + G_{uu}}{2G} + \frac{E_v G_v + G_u^2}{4G^2} + \frac{E_u G_u + E_v^2}{4EG}.$$

Dividing by E gives

$$K = -\frac{E_{vv} + G_{uu}}{2EG} + \frac{E_v G_v + G_u^2}{4EG^2} + \frac{E_u G_u + E_v^2}{4E^2 G}.$$

We shall now express this formula in a more compact form. Differentiating $G_u/\sqrt{EG}$ with respect to u, we get

$$\frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right) = \frac{G_{uu}}{\sqrt{EG}} - \frac{G_u(E_u G + E G_u)}{2(EG)^{3/2}}.$$

Similarly,

$$\frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right) = \frac{E_{vv}}{\sqrt{EG}} - \frac{E_v(E_v G + E G_v)}{2(EG)^{3/2}}.$$

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Adding these expressions and multiplying by $-1/(2\sqrt{EG})$, we obtain

$$\begin{aligned} &-\frac{1}{2\sqrt{EG}}\left[\frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right)+\frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right)\right] \\ &\quad=-\frac{E_{vv}+G_{uu}}{2EG}+\frac{E_uG_u+E_v^2}{4E^2G}+\frac{G_u^2+E_vG_v}{4EG^2} \\ &\quad=K. \end{aligned}$$

Therefore,

$$K=-\frac{1}{2\sqrt{EG}}\left[\frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right)+\frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right)\right],$$

as required. ■

2nd Proof. For an orthogonal metric $ds^2 = E\,du^2 + G\,dv^2$, Liouville's formula expresses the geodesic curvature of the coordinate curves in terms of E and G. Applying Green's theorem to a small coordinate rectangle and using the Gauss–Bonnet formula gives the stated expression. Thus, in orthogonal coordinates, K is determined entirely by the first fundamental form. ■

Corollary 3.1.4 (Geodesic Polar Coordinates). *Let (u, v) be geodesic polar coordinates on a regular surface, with first fundamental form*

$$ds^2 = du^2 + g(u,v)^2\,dv^2.$$

Then the Gaussian curvature is

$$K=-\frac{g_{uu}}{g}.$$

Equivalently, g satisfies the differential equation

$$g_{uu}+Kg=0.$$

Proof. Since the first fundamental form is $ds^2 = du^2 + g^2\,dv^2$, its coefficients are $E = 1$, $F = 0$ and $G = g^2$. By the formula for Gaussian curvature in orthogonal coordinates,

$$K=-\frac{1}{2\sqrt{EG}}\left[\frac{\partial}{\partial u}\left(\frac{G_u}{\sqrt{EG}}\right)+\frac{\partial}{\partial v}\left(\frac{E_v}{\sqrt{EG}}\right)\right].$$

Now $E_v = 0$, while $\sqrt{EG} = g$ and $G_u = 2gg_u$, since $g > 0$ away from the center of the coordinates. Therefore,

$$K=-\frac{1}{2g}\frac{\partial}{\partial u}\left(\frac{2gg_u}{g}\right)=-\frac{1}{2g}\frac{\partial}{\partial u}(2g_u)=-\frac{g_{uu}}{g}.$$

Rearranging gives $g_{uu} + Kg = 0$, as required. ■

This result shows that Gaussian curvature controls the rate at which nearby geodesics spread apart. In the Euclidean plane, geodesic polar coordinates have $g(u, v) = u$, so $g_{uu} = 0$ and $K = 0$. On a positively curved surface, nearby geodesics tend to spread apart more slowly than in the plane, while on a negatively curved surface they tend to spread apart more rapidly.

Theorem 3.1.5 (Gaussian Curvature from Geodesic Triangles). *(See Figure 55 in the Atlas.) Let P be a point on a regular surface, and let R be a sufficiently small geodesic triangle containing P, with area A*

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and interior angles α, β, γ. As the triangle shrinks regularly to P, its Gaussian curvature is given by

$$K(P) = \lim_{A \to 0} \frac{\alpha + \beta + \gamma - \pi}{A}.$$

Proof. Let R be a sufficiently small geodesic triangle with interior angles α, β, γ. Since each side is a geodesic, its geodesic curvature vanishes.

The local Gauss–Bonnet formula for a region bounded by a piecewise smooth curve is

$$\iint_R K \, dA + \int_{\partial R} k_g \, ds + \sum_{i=1}^{3} \theta_i = 2\pi,$$

where k_g is the geodesic curvature of the boundary and θ_i are the exterior angles of the triangle.

Since the three sides are geodesics, $\int_{\partial R} k_g \, ds = 0$. Moreover, the exterior angles are $\pi - \alpha$, $\pi - \beta$ and $\pi - \gamma$. Therefore,

$$\iint_R K \, dA + (\pi - \alpha) + (\pi - \beta) + (\pi - \gamma) = 2\pi.$$

Simplifying gives

$$\iint_R K \, dA = \alpha + \beta + \gamma - \pi.$$

Dividing by the area A of the triangle, we obtain

$$\frac{\alpha + \beta + \gamma - \pi}{A} = \frac{1}{A} \iint_R K \, dA.$$

Since K is continuous, its average value over a triangle shrinking to P approaches $K(P)$. Hence,

$$\lim_{A \to 0} \frac{1}{A} \iint_R K \, dA = K(P).$$

Consequently,

$$K(P) = \lim_{A \to 0} \frac{\alpha + \beta + \gamma - \pi}{A},$$

as required. ∎

The quantity $\alpha + \beta + \gamma - \pi$ is called the **angular excess** of a geodesic triangle. It is positive on a sufficiently small region of positive Gaussian curvature and negative on a sufficiently small region of negative Gaussian curvature. In the Euclidean plane, where $K = 0$, the familiar relation $\alpha + \beta + \gamma = \pi$ is recovered.

Example 3.1.6 (Geodesic Triangle on a Sphere)**.** Show that a geodesic triangle on a sphere of radius a satisfies

$$A = a^2(\alpha + \beta + \gamma - \pi),$$

where A is its area and α, β, γ are its interior angles.

Solution. On a sphere of radius a, the Gaussian curvature is constant and equal to $K = 1/a^2$. The sides of a spherical geodesic triangle are arcs of great circles, so their geodesic curvature vanishes.

By the Gauss–Bonnet formula,

$$\iint_R K \, dA = \alpha + \beta + \gamma - \pi.$$

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Since K is constant,

$$\iint_R K\,dA = \frac{1}{a^2}\iint_R dA = \frac{A}{a^2}.$$

Therefore,

$$\frac{A}{a^2} = \alpha + \beta + \gamma - \pi,$$

which gives

$$A = a^2(\alpha + \beta + \gamma - \pi).$$

Hence the angular excess of a spherical triangle is proportional to its area. ■

Theorem 3.1.7 (Curvature from Small Geodesic Circles). *Let P be a point on a smooth regular surface. Denote by $C(r)$ the circumference of a sufficiently small geodesic circle of radius r centered at P, and by $A(r)$ the area of the corresponding geodesic disc. Then*

$$K(P) = \lim_{r\to 0} \frac{3(2\pi r - C(r))}{\pi r^3}$$

and

$$K(P) = \lim_{r\to 0} \frac{12(\pi r^2 - A(r))}{\pi r^4}.$$

Proof. Choose geodesic polar coordinates (u, v) centered at P, where u denotes geodesic distance from P and v is the angular parameter. For sufficiently small $u > 0$, the first fundamental form is

$$ds^2 = du^2 + g(u, v)^2\,dv^2.$$

The coefficient g satisfies

$$g_{uu} + Kg = 0,$$

together with the initial conditions

$$g(0, v) = 0, \qquad g_u(0, v) = 1.$$

The first condition expresses the fact that all radial geodesics meet at P, while the second follows from the Euclidean behaviour of the metric near P.

Let $K(P) = K_0$. From the differential equation,

$$g_{uu}(0, v) = -K_0 g(0, v) = 0.$$

Differentiating $g_{uu} = -Kg$ with respect to u, we obtain

$$g_{uuu} = -K_u g - K g_u.$$

Evaluating at $u = 0$ gives

$$g_{uuu}(0, v) = -K_0,$$

because $g(0, v) = 0$ and $g_u(0, v) = 1$.

Applying Taylor's theorem about $u = 0$, we obtain

$$g(u, v) = u - \frac{K_0}{6}u^3 + O(u^4),$$

where the remainder is uniform in v.

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The geodesic circle of radius r is described by $u = r$. Along this circle, $du = 0$, and hence the element of arc length is $ds = g(r, v)\,dv$. Therefore, its circumference is

$$C(r) = \int_0^{2\pi} g(r, v)\,dv.$$

Substituting the Taylor expansion gives

$$\begin{aligned} C(r) &= \int_0^{2\pi} \left(r - \frac{K_0}{6} r^3 + O(r^4) \right) dv \\ &= 2\pi r - \frac{\pi K_0}{3} r^3 + O(r^4). \end{aligned}$$

Consequently,

$$2\pi r - C(r) = \frac{\pi K_0}{3} r^3 + O(r^4).$$

Dividing by $\pi r^3/3$ gives

$$\frac{3(2\pi r - C(r))}{\pi r^3} = K_0 + O(r).$$

Taking the limit as $r \to 0$, we obtain

$$K(P) = \lim_{r \to 0} \frac{3(2\pi r - C(r))}{\pi r^3}.$$

We now derive the area formula. In geodesic polar coordinates, the area element is

$$dA = \sqrt{EG - F^2}\,du\,dv = g(u, v)\,du\,dv.$$

Hence the area of the geodesic disc of radius r is

$$A(r) = \int_0^{2\pi} \int_0^r g(u, v)\,du\,dv.$$

Substituting the expansion of g, we obtain

$$\begin{aligned} A(r) &= \int_0^{2\pi} \int_0^r \left(u - \frac{K_0}{6} u^3 + O(u^4) \right) du\,dv \\ &= \int_0^{2\pi} \left(\frac{r^2}{2} - \frac{K_0 r^4}{24} + O(r^5) \right) dv \\ &= \pi r^2 - \frac{\pi K_0}{12} r^4 + O(r^5). \end{aligned}$$

Thus,

$$\pi r^2 - A(r) = \frac{\pi K_0}{12} r^4 + O(r^5),$$

and therefore

$$\frac{12(\pi r^2 - A(r))}{\pi r^4} = K_0 + O(r).$$

Taking the limit as $r \to 0$ gives

$$K(P) = \lim_{r \to 0} \frac{12(\pi r^2 - A(r))}{\pi r^4}.$$

This completes the proof. ■

These formulas provide a geometric interpretation of Gaussian curvature through the circumference and area of small geodesic circles. In the Euclidean plane, a circle of radius r has circumference $2\pi r$ and area πr^2. On a surface of positive Gaussian curvature, sufficiently small geodesic circles have smaller circumference and enclosed area than their Euclidean counterparts. For negative Gaussian curvature, both quantities are larger.

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Example 3.1.8 (Gauss Map of a Cylinder)**.** Describe the Gauss map of a circular cylinder and explain why its Gaussian curvature vanishes.

Solution. Consider the circular cylinder of radius $a > 0$, parametrized by

$$\mathbf{r}(u, v) = (a\cos u, a\sin u, v).$$

Its partial derivatives are

$$\mathbf{r}_u = (-a\sin u, a\cos u, 0), \qquad \mathbf{r}_v = (0, 0, 1).$$

Their cross product is

$$\mathbf{r}_u \times \mathbf{r}_v = (a\cos u, a\sin u, 0),$$

so the outward unit normal is

$$N(u, v) = (\cos u, \sin u, 0).$$

Thus, the Gauss map sends every point of the cylinder to a point on the equator of the unit sphere. Moreover, N depends only on u and is independent of v.

Differentiating gives

$$N_u = (-\sin u, \cos u, 0), \qquad N_v = (0, 0, 0).$$

Hence the differential of the Gauss map sends the generator direction $\mathbf{r}_v$ to zero. Therefore, dN has rank one, and its determinant as a linear operator on the tangent plane is zero.

Since the Gaussian curvature is $K = \det(-dN) = \det(dN)$, it follows that

$$K = 0.$$

Geometrically, the cylinder bends in the circular direction but not along its straight generators. Thus, one principal curvature is nonzero and the other is zero, so their product vanishes. ■

Example 3.1.9 (Geodesic Circle on a Sphere)**.** (See Figure 45 in the Atlas.) Use the small geodesic-circle formula to determine the Gaussian curvature of a sphere of radius $a > 0$.

Solution. A geodesic circle of radius r on a sphere of radius a consists of points at geodesic distance r from a fixed point P. Its circumference is

$$C(r) = 2\pi a \sin\left(\frac{r}{a}\right).$$

Using the Taylor expansion

$$\sin x = x - \frac{x^3}{6} + O(x^5),$$

we obtain

$$\begin{aligned} C(r) &= 2\pi a\left(\frac{r}{a} - \frac{r^3}{6a^3} + O(r^5)\right) \\ &= 2\pi r - \frac{\pi r^3}{3a^2} + O(r^5). \end{aligned}$$

Consequently,

$$2\pi r - C(r) = \frac{\pi r^3}{3a^2} + O(r^5).$$

Applying the small geodesic-circle formula gives

$$\begin{aligned} K(P) &= \lim_{r\to 0} \frac{3(2\pi r - C(r))}{\pi r^3} \\ &= \lim_{r\to 0}\left(\frac{1}{a^2} + O(r^2)\right) = \frac{1}{a^2}. \end{aligned}$$

Hence the Gaussian curvature of the sphere is $K = 1/a^2$, as expected. ■

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3.2 Hilbert's Theorem and Its Applications

Hilbert's theorem provides an important connection between the global properties of a surface and its Gaussian curvature. It asserts that every compact regular surface without boundary in $\mathbb{R}^3$ must contain at least one point of positive Gaussian curvature. Thus, regardless of how complicated the shape of a closed surface may be, it cannot have nonpositive Gaussian curvature everywhere.

The proof is based on a simple geometric observation. If a sufficiently large sphere contains a compact surface and touches it at a point, then the surface and the sphere have the same tangent plane at that point. Moreover, the surface cannot bend outward beyond the sphere. This idea is made precise in the following lemma.

Lemma 3.2.1 (Supporting Sphere Lemma)**.** *Let $S \subset \mathbb{R}^3$ be a compact regular surface without boundary, and let $O \in \mathbb{R}^3$ be a fixed point. Suppose that $P_0 \in S$ maximizes the distance from O, and put $R = \|P_0 - O\|$. Then the sphere of radius R centered at O is tangent to S at P_0, and the entire surface lies inside or on this sphere. Moreover, with the unit normal at P_0 directed towards O, the normal curvature of S in every unit tangent direction is at least $1/R$.*

Proof. Consider the function $f : S \to \mathbb{R}$ defined by

$$f(P) = \|P - O\|^2 .$$

Since S is compact and f is continuous, there exists $P_0 \in S$ such that $f(P) \leq f(P_0) = R^2$ for every $P \in S$. Hence the sphere $\|P - O\| = R$ contains S in its closed interior.

Let $v \in T_{P_0}S$ be an arbitrary unit tangent vector, and choose a smooth curve $\alpha(s)$ on S such that $\alpha(0) = P_0$ and $\alpha'(0) = v$. The function

$$g(s) = \|\alpha(s) - O\|^2$$

has a local maximum at $s = 0$. Therefore, $g'(0) = 0$ and $g''(0) \leq 0$.

Differentiating once, we obtain

$$g'(s) = 2\left\langle \alpha(s) - O, \alpha'(s) \right\rangle .$$

Evaluating at $s = 0$ gives

$$\langle P_0 - O, v \rangle = 0.$$

Since this holds for every $v \in T_{P_0}S$, the vector $P_0 - O$ is perpendicular to the tangent plane of S at P_0. It is also normal to the sphere at P_0. Hence the sphere and the surface have the same tangent plane at P_0.

Differentiating a second time gives

$$g''(s) = 2\left\|\alpha'(s)\right\|^2 + 2\left\langle \alpha(s) - O, \alpha''(s) \right\rangle .$$

Since $g''(0) \leq 0$ and $\|\alpha'(0)\| = 1$, we obtain

$$1 + \left\langle P_0 - O, \alpha''(0) \right\rangle \leq 0.$$

Choose the unit normal vector at P_0 directed towards O, namely

$$N(P_0) = \frac{O - P_0}{R}.$$

Then the preceding inequality becomes

$$\left\langle N(P_0), \alpha''(0) \right\rangle \geq \frac{1}{R}.$$

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But $\langle N(P_0), \alpha''(0)\rangle$ is the normal curvature of S in the direction v. Consequently,

$$\kappa_n(v) \geq \frac{1}{R}$$

for every unit tangent vector $v \in T_{P_0}S$, proving the lemma. ■

Theorem 3.2.2 (Hilbert's Theorem). *Let $S \subset \mathbb{R}^3$ be a compact regular surface without boundary. Then there exists at least one point $P \in S$ at which the Gaussian curvature is positive.*

Proof. Choose a fixed point $O \in \mathbb{R}^3$. Since S is compact, there exists a point $P_0 \in S$ at which the distance from O is maximum. Let $R = \|P_0 - O\|$.

By the Supporting Sphere Lemma, the sphere of radius R centered at O is tangent to S at P_0. Moreover, with the unit normal directed towards O, the normal curvature in every unit tangent direction satisfies

$$\kappa_n(v) \geq \frac{1}{R}.$$

Recall that the principal curvatures k_1 and k_2 are the minimum and maximum values of the normal curvature among all unit tangent directions. Hence

$$k_1(P_0) \geq \frac{1}{R}, \qquad k_2(P_0) \geq \frac{1}{R}.$$

Since $R > 0$, both principal curvatures are positive. Therefore,

$$K(P_0) = k_1(P_0)k_2(P_0) \geq \frac{1}{R^2} > 0.$$

Thus, the Gaussian curvature is positive at P_0, proving the theorem. ■

Hilbert's theorem establishes that positive Gaussian curvature must occur somewhere on every compact regular surface without boundary. It does not assert that the curvature must be positive everywhere. For example, a standard torus has regions of positive and negative Gaussian curvature, but it necessarily possesses points where $K > 0$.

An important consequence of Hilbert's theorem concerns minimal surfaces. Recall that a regular surface is called minimal if its mean curvature vanishes identically.

Corollary 3.2.3. *There is no compact regular minimal surface without boundary in $\mathbb{R}^3$.*

Proof. Suppose, contrary to the assertion, that S is a compact regular minimal surface without boundary. Since S is minimal, its mean curvature satisfies

$$H = \frac{1}{2}(k_1 + k_2) = 0$$

at every point. Therefore, $k_2 = -k_1$, and hence

$$K = k_1k_2 = -k_1^2 \leq 0.$$

Thus, the Gaussian curvature is nonpositive everywhere on S.

However, Hilbert's theorem guarantees the existence of a point $P \in S$ such that $K(P) > 0$. This is a contradiction. Hence no compact regular minimal surface without boundary exists in $\mathbb{R}^3$. ■

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Corollary 3.2.4. *No compact regular surface without boundary in $\mathbb{R}^3$ can have nonpositive Gaussian curvature at every point.*

Proof. Suppose that S is a compact regular surface without boundary and that $K(P) \leq 0$ for every $P \in S$. By Hilbert's theorem, there exists a point $P_0 \in S$ such that $K(P_0) > 0$, contradicting the assumption. Therefore, the Gaussian curvature cannot be nonpositive everywhere. ■

Example 3.2.5 (Positive Gaussian Curvature on an Ellipsoid)**.** Verify Hilbert's theorem for the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

where $a, b, c > 0$, by computing its Gaussian curvature at $P = (0, 0, c)$.

Solution. Near the point $P = (0, 0, c)$, the upper part of the ellipsoid can be represented as the graph

$$z = f(x, y) = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}.$$

For a surface given by $z = f(x, y)$, the Gaussian curvature is

$$K = \frac{f_{xx} f_{yy} - f_{xy}^2}{(1 + f_x^2 + f_y^2)^2}.$$

At $(x, y) = (0, 0)$, the first derivatives satisfy $f_x(0, 0) = f_y(0, 0) = 0$. Differentiating twice, we obtain

$$f_{xx}(0, 0) = -\frac{c}{a^2}, \qquad f_{yy}(0, 0) = -\frac{c}{b^2}, \qquad f_{xy}(0, 0) = 0.$$

Consequently,

$$K(P) = \frac{\left(-\frac{c}{a^2}\right)\left(-\frac{c}{b^2}\right) - 0}{(1 + 0 + 0)^2} = \frac{c^2}{a^2 b^2} > 0.$$

Thus, the Gaussian curvature is positive at P, in agreement with Hilbert's theorem. ■

The theorem and its corollaries illustrate how a global property such as compactness imposes restrictions on the local curvature of a surface. In particular, although minimal surfaces such as the catenoid and helicoid exist in $\mathbb{R}^3$, no compact regular surface without boundary can be minimal everywhere.

3.3 Liebmann's Theorem and Its Applications

Hilbert's theorem establishes that every compact regular surface without boundary in $\mathbb{R}^3$ possesses at least one point of positive Gaussian curvature. A natural question is whether the assumption of constant positive Gaussian curvature imposes stronger restrictions on the shape of a compact surface. Liebmann's theorem answers this question by showing that the sphere is the only connected compact regular surface in $\mathbb{R}^3$ having constant positive Gaussian curvature.

To understand this result, recall that a sphere of radius $a > 0$ has constant Gaussian curvature $K = 1/a^2$. Liebmann's theorem establishes the converse under suitable global assumptions: a connected compact regular surface of constant positive Gaussian curvature must be a sphere.

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Definition 3.3.1 (Ovaloid)**.** An **ovaloid** is a connected, compact, smooth regular surface in $\mathbb{R}^3$ that is the boundary of a strictly convex body and has positive Gaussian curvature at every point.

A strictly convex surface bends towards the same side of each tangent plane. At every point of an ovaloid, the principal curvatures have the same sign, so its Gaussian curvature is positive. The sphere and ellipsoids are familiar examples of ovaloids.

Proposition 3.3.2. *A sphere of radius $a > 0$ has constant Gaussian curvature*

$$K = \frac{1}{a^2}.$$

Proof. Let S be a sphere of radius a centered at O. Its outward unit normal at $P \in S$ is

$$N(P) = \frac{P - O}{a}.$$

For any tangent vector $X \in T_P S$, differentiating the normal field in the direction X gives

$$dN_P(X) = \frac{1}{a} X.$$

By the definition of the shape operator,

$$S_P(X) = -dN_P(X) = -\frac{1}{a} X.$$

Thus, both principal curvatures are equal to $-1/a$. Consequently,

$$K = k_1 k_2 = \left(-\frac{1}{a}\right)\left(-\frac{1}{a}\right) = \frac{1}{a^2}.$$

Hence the Gaussian curvature is constant and positive. ∎

The following theorem gives the converse of this result for compact connected surfaces.

Theorem 3.3.3 (Liebmann's Theorem)**.** *Let S be a connected, compact, regular surface without boundary in $\mathbb{R}^3$. If its Gaussian curvature is a positive constant K_0, then S is a sphere of radius*

$$a = \frac{1}{\sqrt{K_0}}.$$

The proof of Liebmann's theorem requires a global property of surfaces having positive Gaussian curvature. We recall the following classical result.

Theorem 3.3.4 (Hadamard's Theorem on Positive Curvature)**.** *Let S be a connected, compact, smooth regular surface in $\mathbb{R}^3$ with positive Gaussian curvature at every point. Then S is an ovaloid. In particular, it is embedded, bounds a strictly convex body, and its Gauss map is a diffeomorphism onto the unit sphere.*

Another classical result, known as the uniqueness theorem for the Minkowski problem, states that a smooth strictly convex closed surface is uniquely determined, up to translation, by its Gaussian curvature regarded as a function of the unit normal direction.

These two results provide a concise proof of Liebmann's theorem.

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Proof of Liebmann's Theorem. Let S be a connected compact regular surface having constant Gaussian curvature $K = K_0 > 0$.

By Hadamard's theorem, S is an ovaloid, and its outward Gauss map

$$N : S \longrightarrow \mathbb{S}^2$$

is a diffeomorphism. Consequently, each unit vector $n \in \mathbb{S}^2$ occurs as the outward normal at exactly one point of S.

Since the Gaussian curvature is constant, its value at the point with outward normal n is

$$K(n) = K_0$$

for every $n \in \mathbb{S}^2$.

Now consider the sphere of radius $a = 1/\sqrt{K_0}$. By the preceding proposition, its Gaussian curvature is also equal to K_0 at every point. In particular, when regarded as a function of its outward normal direction, its Gaussian curvature is identical to that of S.

By the uniqueness theorem for the Minkowski problem, two smooth strictly convex closed surfaces having the same Gaussian curvature as a function of their outward normal directions differ only by a translation.

Therefore, S is a translate of the sphere of radius a. Hence S itself is a sphere of radius

$$a = \frac{1}{\sqrt{K_0}}.$$

This proves Liebmann's theorem. ■

Liebmann's theorem is a rigidity theorem: the condition of constant positive Gaussian curvature determines the shape of a compact connected surface completely, up to its position in space.

It is important to note that compactness is essential. For example, an open portion of a sphere has constant positive Gaussian curvature but is not itself a complete sphere.

Corollary 3.3.5. *Every connected compact regular surface without boundary having Gaussian curvature $K = 1$ is a unit sphere, up to translation.*

Proof. By Liebmann's theorem, the surface is a sphere of radius $a = 1/\sqrt{K} = 1$. Hence it is a unit sphere. ■

Corollary 3.3.6. *A connected compact regular surface without boundary having constant positive Gaussian curvature is totally umbilic.*

Proof. By Liebmann's theorem, the surface is a sphere. The principal curvatures of a sphere of radius a are both equal to $-1/a$ with respect to the outward normal. Therefore, $k_1 = k_2$ at every point, so every point is umbilic. ■

Example 3.3.7 (Determination of the Radius)**.** A connected compact regular surface without boundary has constant Gaussian curvature $K = 4$. Determine the surface.

Solution. Since the Gaussian curvature is constant and positive, Liebmann's theorem implies that the surface is a sphere. Its radius is

$$a = \frac{1}{\sqrt{K}} = \frac{1}{\sqrt{4}} = \frac{1}{2}.$$

Hence the surface is a sphere of radius $1/2$, up to translation. ■

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Example 3.3.8 (An Ellipsoid and Liebmann's Theorem)**.** Consider the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

where $a, b, c > 0$. Explain why its Gaussian curvature cannot be constant unless $a = b = c$.

Solution. The ellipsoid is a connected compact regular surface without boundary, and its Gaussian curvature is positive everywhere.

Suppose its Gaussian curvature were constant. By Liebmann's theorem, it would necessarily be a sphere. An ellipsoid represents a sphere precisely when its three semiaxes are equal, that is, when

$$a = b = c.$$

Therefore, an ellipsoid with unequal semiaxes cannot have constant Gaussian curvature.

Conversely, if $a = b = c$, the equation becomes

$$x^2 + y^2 + z^2 = a^2,$$

which represents a sphere of radius a and has constant Gaussian curvature $1/a^2$. ■

Example 3.3.9 (Constant Positive Curvature and Surface Area)**.** Let S be a connected compact regular surface without boundary having constant Gaussian curvature $K_0 > 0$. Determine its surface area.

Solution. By Liebmann's theorem, S is a sphere of radius

$$a = \frac{1}{\sqrt{K_0}}.$$

The area of a sphere of radius a is $4\pi a^2$. Therefore,

$$A = 4\pi a^2 = \frac{4\pi}{K_0}.$$

Hence the area is completely determined by the constant Gaussian curvature. ■

Hilbert's and Liebmann's theorems illustrate how global assumptions impose restrictions on the local geometry of surfaces. Hilbert's theorem guarantees that every compact regular surface without boundary possesses a point of positive Gaussian curvature, whereas Liebmann's theorem shows that constant positive Gaussian curvature forces a connected compact surface to be spherical.

3.4 The Shape Operator and the Weingarten Equations

Definition 3.4.1 (Shape Operator)**.** The **shape operator** or **Weingarten map** at P is the linear map

$$S_P : T_P S \to T_P S, \qquad S_P(X) = -d\mathbf{N}_P(X).$$

Its eigenvalues are the principal curvatures and its eigenvectors are the principal directions.

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Proposition 3.4.2. *The shape operator is self-adjoint:*

$$\langle S(X), Y\rangle = \langle X, S(Y)\rangle$$

for all $X, Y \in T_P S$.

Proof. It is enough to verify the identity on the coordinate basis $\{\mathbf{r}_u, \mathbf{r}_v\}$. Differentiating $\mathbf{N}\cdot\mathbf{r}_u = 0$ and $\mathbf{N}\cdot\mathbf{r}_v = 0$ gives

$$-\mathbf{N}_u\cdot\mathbf{r}_v = \mathbf{N}\cdot\mathbf{r}_{uv} = -\mathbf{N}_v\cdot\mathbf{r}_u.$$

Hence $\langle S(\mathbf{r}_u), \mathbf{r}_v\rangle = \langle \mathbf{r}_u, S(\mathbf{r}_v)\rangle$, and bilinearity completes the proof. ■

Remark 3.4.3. The Gaussian and mean curvatures are $K = \det S$, $\quad H = \frac{1}{2}\operatorname{tr} S$.

The derivatives of the unit normal are tangent to the surface. Indeed, $\mathbf{N}\cdot\mathbf{N} = 1$ implies $\mathbf{N}\cdot\mathbf{N}_u = \mathbf{N}\cdot\mathbf{N}_v = 0$. Hence $\mathbf{N}_u$ and $\mathbf{N}_v$ may be expressed as linear combinations of $\mathbf{r}_u$ and $\mathbf{r}_v$.

Theorem 3.4.4 (Weingarten Equations). *Let*

$$H_0^2 = EG - F^2.$$

Then

$$\mathbf{N}_u = -\frac{1}{H_0^2}\left[(GL - FM)\mathbf{r}_u + (EM - FL)\mathbf{r}_v\right],$$

and

$$\mathbf{N}_v = -\frac{1}{H_0^2}\left[(GM - FN)\mathbf{r}_u + (EN - FM)\mathbf{r}_v\right].$$

Equivalently,

$$\mathbf{N}_u = \frac{FM - GL}{H_0^2}\mathbf{r}_u + \frac{FL - EM}{H_0^2}\mathbf{r}_v,$$

$$\mathbf{N}_v = \frac{FN - GM}{H_0^2}\mathbf{r}_u + \frac{FM - EN}{H_0^2}\mathbf{r}_v.$$

Proof. Write

$$\mathbf{N}_u = a\mathbf{r}_u + b\mathbf{r}_v.$$

Differentiating $\mathbf{N}\cdot\mathbf{r}_u = 0$ and $\mathbf{N}\cdot\mathbf{r}_v = 0$ with respect to u gives

$$\mathbf{N}_u\cdot\mathbf{r}_u = -L, \qquad \mathbf{N}_u\cdot\mathbf{r}_v = -M.$$

Therefore

$$Ea + Fb = -L, \qquad Fa + Gb = -M.$$

Solving,

$$a = \frac{FM - GL}{EG - F^2}, \qquad b = \frac{FL - EM}{EG - F^2}.$$

This gives the formula for $\mathbf{N}_u$.

Similarly, if

$$\mathbf{N}_v = c\mathbf{r}_u + d\mathbf{r}_v,$$

then

$$Ec + Fd = -M, \qquad Fc + Gd = -N.$$

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Solving,

$$c = \frac{FN - GM}{EG - F^2}, \qquad d = \frac{FM - EN}{EG - F^2},$$

which gives the formula for $\mathbf{N}_v$. ■

Corollary 3.4.5. *If the coordinate curves are lines of curvature, then $F = M = 0$, and*

$$\mathbf{N}_u = -\frac{L}{E}\mathbf{r}_u, \qquad \mathbf{N}_v = -\frac{N}{G}\mathbf{r}_v.$$

Thus

$$\mathbf{N}_u = -k_1\mathbf{r}_u, \qquad \mathbf{N}_v = -k_2\mathbf{r}_v.$$

Example 3.4.6 (Weingarten Equations for a Sphere)**.** Verify the Weingarten equations for the sphere $\mathbf{r}(u, v) = (a\sin u\cos v, a\sin u\sin v, a\cos u)$.

Solution. For the outward orientation, $\mathbf{N} = \mathbf{r}/a$. Hence $\mathbf{N}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = \mathbf{r}_v/a$. With this orientation the principal curvatures are both $-1/a$ under the convention $S = -d\mathbf{N}$, so the Weingarten equations read $\mathbf{N}_u = -k_1\mathbf{r}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = -k_2\mathbf{r}_v = \mathbf{r}_v/a$, as required. ■

Example 3.4.7 (Shape Operator of a Cylinder)**.** Determine the matrix of the shape operator of the cylinder $\mathbf{r}(u, v) = (a\cos u, a\sin u, v)$ relative to the orthogonal basis $\{\mathbf{r}_u, \mathbf{r}_v\}$.

Solution. With the outward normal $\mathbf{N} = (\cos u, \sin u, 0)$, $\mathbf{N}_u = \mathbf{r}_u/a$ and $\mathbf{N}_v = 0$. Since $S = -d\mathbf{N}$, we have $S(\mathbf{r}_u) = -(1/a)\mathbf{r}_u$ and $S(\mathbf{r}_v) = 0$. Thus the matrix of S in this basis is

$$\begin{pmatrix} -1/a & 0 \\ 0 & 0 \end{pmatrix}.$$

Its eigenvalues are the principal curvatures $-1/a$ and 0. ■

3.5 Lines of Curvature

At every non-umbilic point of a regular surface, there are two mutually perpendicular principal directions corresponding to the principal curvatures. A curve on the surface whose tangent always lies along a principal direction is called a line of curvature. Thus, lines of curvature describe the curves obtained by following the principal directions on a surface.

Definition 3.5.1 (Line of Curvature)**.** A regular curve on a surface is called a **line of curvature** if its tangent vector at every point is a principal direction of the surface.

Let $\mathbf{r}(u, v)$ be a local parametrization of a surface with first fundamental coefficients E, F, G and second fundamental coefficients L, M, N. The differential equation of the two families of lines of curvature is

$$(EM - FL)\,du^2 + (EN - GL)\,du\,dv + (FN - GM)\,dv^2 = 0.$$

At a non-umbilic point, this equation determines the two principal directions. Since the shape operator is self-adjoint, these directions are mutually orthogonal.

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Theorem 3.5.2. *Away from umbilic points, the coordinate curves of a regular parametrization are lines of curvature if and only if*

$$F = 0, \qquad M = 0.$$

Proof. Suppose first that the coordinate curves are lines of curvature. Then their tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$ are principal directions. Since the principal directions at a non-umbilic point are perpendicular, we have

$$F = \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$

Moreover, since $\mathbf{r}_u$ is a principal direction, $S(\mathbf{r}_u) = k_1 \mathbf{r}_u$, where k_1 is the corresponding principal curvature. Hence

$$M = \langle S(\mathbf{r}_u), \mathbf{r}_v \rangle = k_1 \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$

Thus, $F = M = 0$.

Conversely, suppose that $F = M = 0$. The differential equation of the lines of curvature becomes

$$(EN - GL)\, du\, dv = 0.$$

Since $E, G > 0$, the principal curvatures are L/E and N/G. Away from umbilic points, these curvatures are distinct, so $EN - GL \neq 0$. Therefore,

$$du\, dv = 0,$$

which implies either $du = 0$ or $dv = 0$. Hence the coordinate curves $u =$ constant and $v =$ constant are lines of curvature. ■

Theorem 3.5.3 (Rodrigues' Formula)**.** *A regular curve on an oriented surface is a line of curvature if and only if*

$$d\mathbf{N} + k_n\, d\mathbf{r} = 0$$

along the curve, where k_n is its normal curvature. Equivalently, if the curve is parametrized by arc length s, then

$$\frac{d\mathbf{N}}{ds} = -k_n \frac{d\mathbf{r}}{ds}.$$

Proof. Suppose first that the curve is a line of curvature with corresponding principal curvature k. Its tangent direction (du, dv) satisfies

$$(L - kE)\, du + (M - kF)\, dv = 0$$

and

$$(M - kF)\, du + (N - kG)\, dv = 0.$$

Recall that

$$d\mathbf{r} = \mathbf{r}_u\, du + \mathbf{r}_v\, dv, \qquad d\mathbf{N} = \mathbf{N}_u\, du + \mathbf{N}_v\, dv.$$

Also, from the definitions of the second fundamental coefficients,

$$L = -\langle \mathbf{N}_u, \mathbf{r}_u \rangle, \qquad M = -\langle \mathbf{N}_u, \mathbf{r}_v \rangle = -\langle \mathbf{N}_v, \mathbf{r}_u \rangle, \qquad N = -\langle \mathbf{N}_v, \mathbf{r}_v \rangle.$$

Using these relations, the two principal-direction equations give

$$\langle d\mathbf{N} + k\, d\mathbf{r}, \mathbf{r}_u \rangle = 0, \qquad \langle d\mathbf{N} + k\, d\mathbf{r}, \mathbf{r}_v \rangle = 0.$$

Both $d\mathbf{N}$ and $d\mathbf{r}$ are tangent to the surface. Hence their sum is a tangent vector perpendicular to both $\mathbf{r}_u$ and $\mathbf{r}_v$. Since these vectors span the tangent plane, we must have

$$d\mathbf{N} + k\, d\mathbf{r} = 0.$$

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Since the tangent direction is principal, its normal curvature is $k_n = k$. Therefore,

$$d\mathbf{N} + k_n\, d\mathbf{r} = 0.$$

Conversely, suppose that

$$d\mathbf{N} + k_n\, d\mathbf{r} = 0.$$

Taking inner products with $\mathbf{r}_u$ and $\mathbf{r}_v$, and using the preceding relations, we obtain

$$(L - k_n E)\, du + (M - k_n F)\, dv = 0$$

and

$$(M - k_n F)\, du + (N - k_n G)\, dv = 0.$$

These are precisely the equations characterizing a principal direction. Thus, the tangent to the curve is principal at every point, and hence the curve is a line of curvature. ■

Theorem 3.5.4 (Euler's Formula). *Let a tangent direction at a point of a regular surface make an angle θ with the principal direction corresponding to the principal curvature k_1. Then its normal curvature is*

$$k_n = k_1 \cos^2\theta + k_2 \sin^2\theta,$$

where k_2 is the other principal curvature.

Proof. Choose the principal directions as orthogonal coordinate directions at the point. Then $F = M = 0$, and the principal curvatures are

$$k_1 = \frac{L}{E}, \qquad k_2 = \frac{N}{G}.$$

Let $T = l\mathbf{r}_u + m\mathbf{r}_v$ be a unit tangent vector making an angle θ with the first principal direction. Since the coordinate directions are orthogonal,

$$\langle T, T\rangle = El^2 + Gm^2 = 1.$$

The unit vectors along the principal directions are $\mathbf{r}_u/\sqrt{E}$ and $\mathbf{r}_v/\sqrt{G}$. Therefore,

$$T = \cos\theta \frac{\mathbf{r}_u}{\sqrt{E}} + \sin\theta \frac{\mathbf{r}_v}{\sqrt{G}},$$

so

$$l = \frac{\cos\theta}{\sqrt{E}}, \qquad m = \frac{\sin\theta}{\sqrt{G}}.$$

The normal curvature in the direction T is

$$k_n = II(T, T) = Ll^2 + 2Mlm + Nm^2.$$

Since $M = 0$, substitution gives

$$\begin{aligned} k_n &= \frac{L}{E}\cos^2\theta + \frac{N}{G}\sin^2\theta \\ &= k_1 \cos^2\theta + k_2 \sin^2\theta. \end{aligned}$$

This proves Euler's formula. ■

Theorem 3.5.5 (Dupin's Theorem). *At a point of a regular surface, the sum of the normal curvatures in*

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any two mutually orthogonal tangent directions is constant and equal to

$$k_1 + k_2 = 2H,$$

where H is the mean curvature of the surface.

Proof. Let the first tangent direction make an angle θ with a principal direction corresponding to k_1. Since the second direction is perpendicular to the first, it makes an angle $\theta + \pi/2$ with the same principal direction.

By Euler's formula,

$$k_n(\theta) = k_1 \cos^2\theta + k_2 \sin^2\theta.$$

For the orthogonal direction,

$$\begin{aligned} k_n\left(\theta + \frac{\pi}{2}\right) &= k_1 \cos^2\left(\theta + \frac{\pi}{2}\right) + k_2 \sin^2\left(\theta + \frac{\pi}{2}\right) \\ &= k_1 \sin^2\theta + k_2 \cos^2\theta. \end{aligned}$$

Adding these expressions, we obtain

$$\begin{aligned} k_n(\theta) + k_n\left(\theta + \frac{\pi}{2}\right) &= k_1(\cos^2\theta + \sin^2\theta) \\ &\quad + k_2(\sin^2\theta + \cos^2\theta) \\ &= k_1 + k_2 = 2H. \end{aligned}$$

Thus, the sum is independent of the choice of the two orthogonal tangent directions. ■

Example 3.5.6 (Surface of Revolution)**.** (See Figure 36 in the Atlas.) Show that the meridians and parallels of the surface

$$\mathbf{r}(u, v) = (u\cos v, u\sin v, f(u)), \qquad u > 0,$$

are lines of curvature.

Solution. Differentiating with respect to u and v, we obtain

$$\mathbf{r}_u = (\cos v, \sin v, f'(u)), \qquad \mathbf{r}_v = (-u\sin v, u\cos v, 0).$$

Therefore,

$$F = \langle \mathbf{r}_u, \mathbf{r}_v \rangle = -u\cos v \sin v + u \sin v \cos v = 0.$$

Also,

$$\mathbf{r}_{uv} = (-\sin v, \cos v, 0) = \frac{1}{u}\mathbf{r}_v.$$

Thus, $\mathbf{r}_{uv}$ is tangent to the surface and consequently perpendicular to the unit normal $\mathbf{N}$. Hence

$$M = \langle \mathbf{r}_{uv}, \mathbf{N} \rangle = 0.$$

Since $F = M = 0$, the coordinate curves are lines of curvature wherever the point is non-umbilic. At an umbilic point, every tangent direction is principal, so the conclusion remains valid.

Therefore, the curves $v =$ constant, called meridians, and $u =$ constant, called parallels, are lines of curvature. ■

Example 3.5.7 (Lines of Curvature on a Cylinder)**.** Find the lines of curvature on the circular

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cylinder
$$\mathbf{r}(u,v) = (a\cos u, a\sin u, v), \qquad a > 0.$$

Solution. For the cylinder,

$$\mathbf{r}_u = (-a\sin u, a\cos u, 0), \qquad \mathbf{r}_v = (0,0,1).$$

Thus,
$$F = \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$
Since $\mathbf{r}_{uv} = (0,0,0)$, the coefficient $M = \langle \mathbf{r}_{uv}, \mathbf{N} \rangle$ is also zero.

Hence $F = M = 0$, and the coordinate curves are lines of curvature. Therefore, $v =$ constant gives the circular sections, while $u =$ constant gives the straight generators.

With the outward orientation, the corresponding principal curvatures are $-1/a$ along the circular sections and 0 along the generators. ■

Example 3.5.8 (Euler's Formula on a Cylinder)**.** Find the normal curvature of a cylinder of radius a in a direction making an angle θ with a circular principal direction, assuming the outward orientation.

Solution. The principal curvatures of the outward-oriented cylinder are $k_1 = -1/a$ along the circular sections and $k_2 = 0$ along the generators.

By Euler's formula,
$$k_n = k_1 \cos^2\theta + k_2 \sin^2\theta.$$
Substituting these values, we obtain
$$k_n = -\frac{1}{a}\cos^2\theta.$$
In particular, when $\theta = 0$, the tangent direction is circular and $k_n = -1/a$. When $\theta = \pi/2$, the direction is along a generator and $k_n = 0$.

Thus, the normal curvature vanishes along the generators and has maximum absolute value along the circular sections. ■

3.5.1 Further Results and Alternative Proofs on Lines of Curvature

Definition 3.5.9. A **line of curvature** on a surface S is a regular curve $\vec{\alpha}(s)$ such that its tangent vector $\vec{\alpha}'(s)$ at every point is a principal direction of the surface.

In other words, at every point along the curve, the tangent vector to the curve must be an eigenvector of the shape operator L_P. This implies that the surface normal $\vec{N}$ changes in the direction of the curve's tangent, keeping the curve "aligned" with the principal directions of bending.

Differential Equation for Lines of Curvature

Theorem 3.5.10. *A curve $\vec{r}(u(t), v(t))$ is a line of curvature if and only if its tangent vector $\vec{v} = \frac{\partial u}{\partial t}\vec{r}_1 + \frac{\partial v}{\partial t}\vec{r}_2$ satisfies $(EM - FL)du^2 + (EN - GL)du\,dv + (FN - GM)dv^2 = 0$. This can be put in the matrix form as*
$$\det \begin{pmatrix} dv^2 & -du\,dv & du^2 \\ E & F & G \\ L & M & N \end{pmatrix} = 0 \tag{3.1}$$

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Proof. A vector $\vec{v}$ is a principal direction if and only if it is an eigenvector of the shape operator, i.e., if and only if $\vec{v} \times L_P(\vec{v}) = 0$. That is $\vec{v}$ and $L_p(\vec{v})$ are parallel meaning that $\det(\vec{v}, L_P(\vec{v})) = 0$ in the basis $\{\vec{r}_1, \vec{r}_2\}$.

Let $A = [a_{ij}] = M_1^{-1} M_2$ be the matrix of L_P. $L_P(\vec{v}) = A\vec{v}$ can be expressed in the basis $\{\vec{r_1}, \vec{r_2}\}$ as $L_P(\vec{v}) = (a_{11}du + a_{12}dv)\vec{r}_1 + (a_{21}du + a_{22}dv)\vec{r}_2$. The condition $\det(\vec{v}, L_P(\vec{v})) = 0$ becomes

$$\det \begin{pmatrix} du & a_{11}du + a_{12}dv \\ dv & a_{21}du + a_{22}dv \end{pmatrix} = a_{21}du^2 + (a_{22} - a_{11})du\,dv - a_{12}dv^2 = 0$$

Using the matrix relation $M_1 A = M_2$, where $M_1 = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$ and $M_2 = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, we have

$$a_{21} = \frac{1}{\det M_1}(EM - FL), \qquad a_{12} = \frac{1}{\det M_1}(FN - GM)$$
$$a_{22} - a_{11} = \frac{1}{\det M_1}(EN - GL)$$

Substituting these into the determinant equation and multiplying by $\det M_1 = EG - F^2$, we immediately obtain:

$$(EM - FL)du^2 + (EN - GL)du\,dv + (FN - GM)dv^2 = 0$$

This proves the result. ■

Theorem 3.5.11. *The principal curvatures k at a point P are the roots of the quadratic equation:*

$$k^2(EG - F^2) - k(EN - 2FM + GL) + (LN - M^2) = 0$$

Proof. We seek the extrema of the second fundamental form $II = L\,du^2 + 2M\,du\,dv + N\,dv^2$ subject to the constraint $I = E\,du^2 + 2F\,du\,dv + G\,dv^2 = 1$. Using the method of Lagrange multipliers, we define the Lagrangian:

$$\mathcal{L}(du, dv, k) = (L\,du^2 + 2M\,du\,dv + N\,dv^2) - k(E\,du^2 + 2F\,du\,dv + G\,dv^2 - 1)$$

Taking partial derivatives with respect to du and dv and setting them to zero gives

$$\frac{\partial \mathcal{L}}{\partial(du)} = 2(L\,du + M\,dv) - 2k(E\,du + F\,dv) = 0$$

$$\frac{\partial \mathcal{L}}{\partial(dv)} = 2(M\,du + N\,dv) - 2k(F\,du + G\,dv) = 0$$

Rearranging these yields the linear system:

$$\begin{cases} (L - kE)du + (M - kF)dv = 0 \\ (M - kF)du + (N - kG)dv = 0 \end{cases}$$

This system of equations can be rewritten as:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix} = k \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

For a non-trivial solution $(du, dv) \neq (0, 0)$ to exist, the determinant of the system must vanish:

$$\det \begin{pmatrix} L - kE & M - kF \\ M - kF & N - kG \end{pmatrix} = 0$$

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Expanding this determinant:

$$(L - kE)(N - kG) - (M - kF)^2 = 0$$
$$(LN - kLG - kEN + k^2EG) - (M^2 - 2kMF + k^2F^2) = 0$$

Collecting the terms in powers of k:

$$k^2(EG - F^2) - k(EN - 2FM + GL) + (LN - M^2) = 0$$

This is the characteristic equation for the principal curvatures k_1 and k_2. ■

Note: The Gaussian curvature is the product of principal curvatures

$$k_1k_2 = \frac{LN - M^2}{EG - F^2}$$

and the Mean curvature is

$$\frac{1}{2}(k_1 + k_2) = \frac{EN - 2FM + GL}{EG - F^2}.$$

Theorem 3.5.12. *The parametric curves are lines of curvature if and only if $F = 0$ and $M = 0$.*

Proof. Let $\{\vec{r}_1, \vec{r}_2\}$ be the basis of the tangent plane $T_P(S)$. The parametric curves are lines of curvature if and only if $\vec{r}_1$ and $\vec{r}_2$ are eigenvectors of the shape operator L_P. This is equivalent to requiring the matrix of L_P relative to this basis to be diagonal.

The shape operator L_P is related to the fundamental forms by the matrix identity $b = gA$, where $g = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, $b = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$, and $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ is the matrix of L_P. Thus

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

Sufficiency ($\Leftarrow$): If $F = 0$ and $M = 0$, the identity becomes

$$\begin{pmatrix} L & 0 \\ 0 & N \end{pmatrix} = \begin{pmatrix} E & 0 \\ 0 & G \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

This yields $L = Ea_{11}$, $N = Ga_{22}$, and $0 = Ea_{12} = Ga_{21}$. Since $E, G \neq 0$, it follows $a_{12} = a_{21} = 0$. The matrix A is diagonal, so $\vec{r}_1, \vec{r}_2$ are eigenvectors, and the parametric curves are lines of curvature.

Necessity ($\Rightarrow$): If the parametric curves are lines of curvature, A must be diagonal, i.e., $a_{12} = a_{21} = 0$. Expanding $b = gA$:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} k_1 & 0 \\ 0 & k_2 \end{pmatrix} = \begin{pmatrix} Ek_1 & Fk_2 \\ Fk_1 & Gk_2 \end{pmatrix}$$

Equating the off-diagonal entries: $M = Fk_2$ and $M = Fk_1$. This implies $F(k_1 - k_2) = 0$. For a non-umbilic point ($k_1 \neq k_2$), we must have $F = 0$, which subsequently forces $M = 0$. ■

Remark 3.5.13. When a surface is parametrized by lines of curvature, the first and second fundamental forms are both diagonal. Specifically, the orthogonality of the principal directions implies $F = 0$, and the alignment of these directions with the coordinate grid implies $M = 0$. Thus, for such a parametrization, the fundamental forms simplify to:

$$I = E\,du^2 + G\,dv^2 \quad \text{and} \quad II = L\,du^2 + N\,dv^2$$

3.5.2 Rodrigues' Formula

The most important theorem concerning lines of curvature relates the derivative of the normal vector directly to the tangent vector.

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Theorem 3.5.14 (Rodrigues' Formula). *Prove that the a curve is a line of curvature if and only if*

$$k d\vec{r} + d\vec{N} = 0.$$

Proof. The system for the principal curvature k and direction (du, dv) is given by

$$(L - kE)du + (M - kF)dv = 0 \tag{3.2}$$
$$(M - kF)du + (N - kG)dv = 0 \tag{3.3}$$

We wish to show this is equivalent to $d\vec{N} + kd\vec{r} = 0$.

Recall the Weingarten equations, which express the derivatives of the unit normal $\vec{N}$ in terms of the tangent basis $\{\vec{r}_1, \vec{r}_2\}$:

$$\vec{N}_1 = \frac{FM - EN}{EG - F^2}\vec{r}_1 + \frac{FL - EM}{EG - F^2}\vec{r}_2$$

$$\vec{N}_2 = \frac{FN - GM}{EG - F^2}\vec{r}_1 + \frac{FM - EN}{EG - F^2}\vec{r}_2$$

Using the definitions $L = -\left\langle \vec{N}_1, \vec{r}_1 \right\rangle$, $M = -\left\langle \vec{N}_1, \vec{r}_2 \right\rangle = -\left\langle \vec{N}_2, \vec{r}_1 \right\rangle$, and $N = -\left\langle \vec{N}_2, \vec{r}_2 \right\rangle$, we recognize that the system of equations (1) and (2) can be rewritten as:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix} = k \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}$$

Expanding this, the first row is $Ldu + Mdv = k(Edu + Fdv)$ and the second is $Mdu + Ndv = k(Fdu + Gdv)$. Observe that $d\vec{r} = \vec{r}_1 du + \vec{r}_2 dv$. The quantities $(Edu + Fdv)$ and $(Fdu + Gdv)$ are exactly the inner products $\langle d\vec{r}, \vec{r}_1 \rangle$ and $\langle d\vec{r}, \vec{r}_2 \rangle$. Similarly, $Ldu + Mdv$ and $Mdu + Ndv$ are exactly $-\left\langle d\vec{N}, \vec{r}_1 \right\rangle$ and $-\left\langle d\vec{N}, \vec{r}_2 \right\rangle$.

Substituting these into the expanded system:

$$-\left\langle d\vec{N}, \vec{r}_1 \right\rangle = k \langle d\vec{r}, \vec{r}_1 \rangle \implies \left\langle d\vec{N} + kd\vec{r}, \vec{r}_1 \right\rangle = 0$$

$$-\left\langle d\vec{N}, \vec{r}_2 \right\rangle = k \langle d\vec{r}, \vec{r}_2 \rangle \implies \left\langle d\vec{N} + kd\vec{r}, \vec{r}_2 \right\rangle = 0$$

Since the vector $\vec{X} = d\vec{N} + kd\vec{r}$ is orthogonal to both basis vectors $\vec{r}_1$ and $\vec{r}_2$, it must be orthogonal to the entire tangent plane. However, $d\vec{N}$ and $d\vec{r}$ are tangent vectors by definition; thus their sum $\vec{X}$ is also a tangent vector. The only vector that is both tangent to the surface and orthogonal to the tangent plane is the zero vector:

$$d\vec{N} + kd\vec{r} = 0$$

This is Rodrigues' formula. ∎

3.5.3 Joachimsthal's Theorem

Lines of curvature have a special relationship with the angle of intersection between surfaces.

Definition 3.5.15 (Lines of Curvature). A regular curve $\vec{\alpha}(s)$ on a surface S is called a **line of curvature** if the tangent vector $\vec{\alpha}'(s)$ is a principal direction at every point along the curve.

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Theorem 3.5.16 (Joachimthal). *Let S_1 and S_2 be two regular surfaces intersecting along a regular curve C.*

1. *If C is a line of curvature on both S_1 and S_2, then the angle of intersection θ between the surfaces is constant along C.*
2. *If the surfaces intersect at a constant angle θ and C is a line of curvature on S_1, then C is also a line of curvature on S_2.*

Proof. Let $\vec{\alpha}(s)$ be the intersection curve parametrized by arc length s, with unit tangent $\vec{t} = \vec{\alpha}'(s)$. Let $\vec{N}_1$ and $\vec{N}_2$ be the unit normal fields of S_1 and S_2 along C. The angle θ between the surfaces is defined by $\cos\theta = \left\langle \vec{N}_1, \vec{N}_2 \right\rangle$.

Differentiating with respect to s:

$$-\sin\theta \frac{d\theta}{ds} = \frac{d}{ds}\left\langle \vec{N}_1, \vec{N}_2 \right\rangle = \left\langle \frac{d\vec{N}_1}{ds}, \vec{N}_2 \right\rangle + \left\langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \right\rangle$$

By Rodrigues' formula, which states that a curve is a line of curvature if and only if the derivative of the normal vector along the curve is proportional to the tangent vector: $\frac{d\vec{N}_i}{ds} = -k_i\vec{t}$, where k_i is the principal curvature of S_i in the direction of $\vec{t}$. Substituting this into the derivative equation:

$$-\sin\theta \frac{d\theta}{ds} = \left\langle -k_1\vec{t}, \vec{N}_2 \right\rangle + \left\langle \vec{N}_1, -k_2\vec{t} \right\rangle = -k_1\left\langle \vec{t}, \vec{N}_2 \right\rangle - k_2\left\langle \vec{N}_1, \vec{t} \right\rangle$$

Since $\vec{t}$ is tangent to the curve C which lies on both surfaces, $\vec{t}$ must be perpendicular to both normals: $\left\langle \vec{t}, \vec{N}_1 \right\rangle = 0$ and $\left\langle \vec{t}, \vec{N}_2 \right\rangle = 0$. Consequently:

$$-\sin\theta \frac{d\theta}{ds} = -k_1(0) - k_2(0) = 0$$

Necessity: If C is a line of curvature on both surfaces, the derivatives are well-defined by Rodrigues' formula, and the expression above vanishes. Assuming the surfaces are not tangent ($\sin\theta \neq 0$), we must have $\frac{d\theta}{ds} = 0$, implying θ is constant.

Sufficiency: Conversely, suppose θ is constant ($\frac{d\theta}{ds} = 0$) and C is a line of curvature on S_1. Then $d\vec{N}_1/ds = -k_1\vec{t}$. The derivative equation becomes

$$0 = \left\langle -k_1\vec{t}, \vec{N}_2 \right\rangle + \left\langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \right\rangle$$

Since $\left\langle \vec{t}, \vec{N}_2 \right\rangle = 0$, we are left with $\left\langle \vec{N}_1, \frac{d\vec{N}_2}{ds} \right\rangle = 0$. This implies the vector $d\vec{N}_2/ds$ is orthogonal to $\vec{N}_1$. Since $d\vec{N}_2/ds$ is also orthogonal to $\vec{N}_2$ (because $\vec{N}_2$ is a unit vector), $d\vec{N}_2/ds$ must be parallel to the cross product $\vec{N}_1 \times \vec{N}_2$. However, $\vec{N}_1 \times \vec{N}_2$ is parallel to the tangent $\vec{t}$. Thus, $d\vec{N}_2/ds$ is proportional to $\vec{t}$, satisfying Rodrigues' formula for S_2. Therefore, C is a line of curvature on S_2. ■

Theorem 3.5.17 (Dupin). *The sum of the normal curvatures of a surface in any two orthogonal directions is invariant and equal to the sum of the principal curvatures.*

Proof. Let k_1 and k_2 be the principal curvatures at a point P. By Euler's Theorem, the normal curvature k_n in a direction making an angle θ with the principal direction $\vec{e}_1$ is $k_n(\theta) = k_1\cos^2\theta + k_2\sin^2\theta$. Let

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k_n' be the normal curvature in the orthogonal direction $\theta + \pi/2$. Using the trigonometric identities $\cos(\theta + \pi/2) = -\sin\theta$ and $\sin(\theta + \pi/2) = \cos\theta$:

$$k_n' = k_1 \cos^2(\theta + \pi/2) + k_2 \sin^2(\theta + \pi/2) = k_1 \sin^2\theta + k_2 \cos^2\theta$$

Adding these two expressions:

$$k_n + k_n' = k_1(\cos^2\theta + \sin^2\theta) + k_2(\sin^2\theta + \cos^2\theta) = k_1 + k_2$$

Since $k_1 + k_2 = 2H$ (where H is the mean curvature), the sum of normal curvatures in any two orthogonal directions is equal to $2H$, which is constant at P. ■

3.6 Dupin Indicatrix, Conjugate Directions, and Asymptotic Lines

The second fundamental form provides important information about the local shape of a surface. In particular, it enables us to classify points according to their Gaussian curvature, identify pairs of conjugate tangent directions, and determine directions in which the normal curvature vanishes. These ideas lead naturally to the Dupin indicatrix and the theory of asymptotic lines.

Dupin Indicatrix

The Dupin indicatrix gives a geometric representation of the second fundamental form in the tangent plane. It describes, to second order, the shape of the intersection of a surface with a plane parallel and sufficiently close to its tangent plane.

Definition 3.6.1 (Dupin Indicatrix)**.** (See Figure 35 in the Atlas.) Let P be a point of a regular surface, and choose rectangular coordinates in the tangent plane along the principal directions at P. The conic determined by the quadratic part of the local equation of the surface is called the **Dupin indicatrix** at P.

Theorem 3.6.2 (Equation of the Dupin Indicatrix)**.** *Let k_1 and k_2 be the principal curvatures of a regular surface at a point P. Choose orthonormal coordinates (x, y) along the principal directions in the tangent plane and let z be measured along the chosen unit normal. Then the surface has the second-order approximation*

$$z = \frac{1}{2}(k_1 x^2 + k_2 y^2).$$

Consequently, the section by the plane $z = h$ is represented, to second order, by

$$k_1 x^2 + k_2 y^2 = 2h.$$

If $k_1, k_2 \neq 0$, and $R_1 = 1/k_1$, $R_2 = 1/k_2$ denote the signed principal radii of curvature, the equation becomes

$$\frac{x^2}{R_1} + \frac{y^2}{R_2} = 2h.$$

Proof. Choose the origin at P, with the x- and y-axes along the principal directions and the z-axis along the chosen unit normal. Since the tangent plane is $z = 0$, the surface can be represented locally as the graph $z = f(x, y)$, where

$$f(0,0) = 0, \qquad f_x(0,0) = f_y(0,0) = 0.$$

By Taylor's theorem,

$$f(x,y) = \frac{1}{2}\left[f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\right] + o(x^2 + y^2).$$

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At P, the first fundamental form is $dx^2 + dy^2$. Since the coordinate axes are principal directions, the mixed coefficient of the second fundamental form vanishes. Thus,

$$f_{xx}(0,0) = k_1, \qquad f_{xy}(0,0) = 0, \qquad f_{yy}(0,0) = k_2.$$

Substituting these values gives

$$z = \frac{1}{2}(k_1x^2 + k_2y^2) + o(x^2 + y^2).$$

Neglecting terms of order higher than two, we obtain

$$2z = k_1x^2 + k_2y^2.$$

Therefore, the intersection with the plane $z = h$ is represented to second order by

$$k_1x^2 + k_2y^2 = 2h,$$

as required. ■

Corollary 3.6.3. *The nature of the Dupin indicatrix depends on the Gaussian curvature $K = k_1k_2$:*

(i) At an elliptic point, where $K > 0$, the indicatrix is an ellipse for a suitable choice of the sign of h.

(ii) At a hyperbolic point, where $K < 0$, the indicatrix is a hyperbola.

(iii) At a parabolic point, where exactly one principal curvature is zero, the quadratic section for a suitable nonzero h consists of two parallel lines.

(iv) At a non-planar minimal point, where $H = 0$ and $K < 0$, the indicatrix is a rectangular hyperbola.

Proof. The equation of the indicatrix is

$$k_1x^2 + k_2y^2 = 2h.$$

If $K > 0$, the principal curvatures have the same sign. Choosing h with that sign gives an ellipse.

If $K < 0$, the principal curvatures have opposite signs, so the equation represents a hyperbola.

At a parabolic point, suppose $k_1 \neq 0$ and $k_2 = 0$. Then the equation becomes $k_1x^2 = 2h$, which represents two parallel lines when $h/k_1 > 0$.

Finally, if $H = 0$ and the point is non-planar, then $k_2 = -k_1 \neq 0$. Hence

$$k_1(x^2 - y^2) = 2h.$$

This is a rectangular hyperbola, since its asymptotes are $y = \pm x$, which are perpendicular. ■

Conjugate Directions

We next consider a relation between tangent directions defined by the second fundamental form.

Definition 3.6.4 (Conjugate Directions)**.** Let $X, Y \in T_PS$ be two nonzero tangent vectors. The directions represented by X and Y are called **conjugate directions** if

$$II_P(X, Y) = 0.$$

If $X = l_1\mathbf{r}_u + m_1\mathbf{r}_v$ and $Y = l_2\mathbf{r}_u + m_2\mathbf{r}_v$, this condition becomes

$$Ll_1l_2 + M(l_1m_2 + m_1l_2) + Nm_1m_2 = 0.$$

Thus, two directions are conjugate when their mixed value under the second fundamental form vanishes. This is different from ordinary perpendicularity, which is determined by the first fundamental form.

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Proposition 3.6.5. *The two principal directions at a non-umbilic point are conjugate. More generally, the coordinate directions of a regular parametrization are conjugate if and only if $M = 0$.*

Proof. Let e_1, e_2 be orthogonal unit principal directions corresponding to the principal curvatures k_1, k_2. Since $S(e_1) = k_1 e_1$, we have

$$II(e_1, e_2) = \langle S(e_1), e_2 \rangle = k_1 \langle e_1, e_2 \rangle = 0.$$

Hence the principal directions are conjugate.

For a general parametrization, the coordinate directions are represented by $\mathbf{r}_u$ and $\mathbf{r}_v$. By the definition of the coefficient M,

$$II(\mathbf{r}_u, \mathbf{r}_v) = M.$$

Therefore, the coordinate directions are conjugate precisely when $M = 0$. ■

Asymptotic Directions and Lines

A tangent direction at which the normal curvature vanishes is called an asymptotic direction. Such directions play an important role in understanding the geometry of surfaces having negative Gaussian curvature.

Definition 3.6.6 (Asymptotic Direction and Asymptotic Line)**.** A nonzero tangent vector $X \in T_P S$ determines an **asymptotic direction** if its normal curvature is zero, that is,

$$II_P(X, X) = 0.$$

In local coordinates, the differential equation of asymptotic directions is

$$L\,du^2 + 2M\,du\,dv + N\,dv^2 = 0.$$

A regular curve on the surface whose tangent direction is asymptotic at every point is called an **asymptotic line**.

Proposition 3.6.7. *At an elliptic point there are no real asymptotic directions. At a hyperbolic point there are two distinct asymptotic directions, while at a parabolic point there is exactly one asymptotic direction.*

Proof. The asymptotic directions are determined by

$$L\,du^2 + 2M\,du\,dv + N\,dv^2 = 0.$$

This is a homogeneous quadratic equation in du and dv. Its discriminant is

$$\Delta = (2M)^2 - 4LN = 4(M^2 - LN).$$

Since

$$K = \frac{LN - M^2}{EG - F^2}$$

and $EG - F^2 > 0$, the sign of Δ is opposite to that of K.

If $K > 0$, then $\Delta < 0$, so there are no real asymptotic directions. If $K < 0$, then $\Delta > 0$, giving two distinct real directions. Finally, at a parabolic point, $K = 0$ and the second fundamental form is nonzero, so $\Delta = 0$ and there is one repeated direction. ■

At a planar point, where $k_1 = k_2 = 0$, the second fundamental form vanishes and every tangent direction is asymptotic. This case is distinct from an ordinary parabolic point.

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Theorem 3.6.8. *At a hyperbolic point of a regular surface, the principal directions bisect the angles between the two asymptotic directions.*

Proof. Choose the principal directions as an orthonormal basis of the tangent plane. Let an asymptotic direction make an angle θ with the first principal direction. Since its normal curvature vanishes, Euler's formula gives

$$k_1 \cos^2\theta + k_2 \sin^2\theta = 0.$$

At a hyperbolic point, $k_1 k_2 < 0$, so neither principal curvature vanishes. Dividing by $\cos^2\theta$, we obtain

$$\tan^2\theta = -\frac{k_1}{k_2}.$$

Thus, the two asymptotic directions make angles θ and $-\theta$ with the first principal direction, modulo π. They are therefore symmetric about the first principal axis.

The second principal axis, being perpendicular to the first, bisects the supplementary angle between the two asymptotic directions. Hence the principal directions are the angle bisectors of the asymptotic directions. ■

Theorem 3.6.9 (Beltrami–Enneper Theorem)**.** *Let $\alpha(s)$ be a regular asymptotic line on a surface with negative Gaussian curvature $K < 0$, parametrized by arc length. Then the torsion τ of the curve satisfies*

$$\tau^2 = -K.$$

Consequently,

$$\tau = \pm\sqrt{-K}.$$

Proof. Let T be the unit tangent vector of the asymptotic line, and let N denote the unit normal to the surface. Since α is an asymptotic line, its normal curvature vanishes:

$$\kappa_n = \langle \alpha'', N \rangle = 0.$$

Hence the curvature vector α'' is tangent to the surface.

At a hyperbolic point, an asymptotic line has nonzero curvature, so its principal normal n is well defined and lies in the tangent plane. Consequently, its binormal $b = T \times n$ is parallel to the surface normal N. Thus,

$$b = \pm N.$$

By the Frenet–Serret formula, $b' = -\tau n$. Taking norms gives

$$|\tau| = \|b'\| = \|N'\|.$$

Because $N' = -S(T)$, we obtain

$$\tau^2 = \|S(T)\|^2.$$

Now choose an orthonormal basis $\{e_1, e_2\}$ of principal directions, and write

$$T = \cos\theta\, e_1 + \sin\theta\, e_2.$$

Since $S(e_1) = k_1 e_1$ and $S(e_2) = k_2 e_2$,

$$S(T) = k_1 \cos\theta\, e_1 + k_2 \sin\theta\, e_2.$$

Therefore,

$$\|S(T)\|^2 = k_1^2 \cos^2\theta + k_2^2 \sin^2\theta.$$

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Since T is asymptotic, Euler's formula gives

$$k_1 \cos^2\theta + k_2 \sin^2\theta = 0.$$

Together with $\cos^2\theta + \sin^2\theta = 1$, this yields

$$\cos^2\theta = -\frac{k_2}{k_1 - k_2}, \qquad \sin^2\theta = \frac{k_1}{k_1 - k_2}.$$

Substituting,

$$\begin{aligned}\tau^2 &= k_1^2\left(-\frac{k_2}{k_1 - k_2}\right) + k_2^2\left(\frac{k_1}{k_1 - k_2}\right)\\ &= \frac{-k_1^2 k_2 + k_1 k_2^2}{k_1 - k_2}\\ &= -k_1 k_2 = -K.\end{aligned}$$

Hence $\tau^2 = -K$, and therefore $\tau = \pm\sqrt{-K}$. ■

Example 3.6.10 (Dupin Indicatrix of a Sphere)**.** Determine the Dupin indicatrix at a point of a sphere of radius $a > 0$.

Solution. At every point of a sphere, the principal curvatures are equal. With the outward orientation,

$$k_1 = k_2 = -\frac{1}{a}.$$

The equation of the Dupin indicatrix is therefore

$$-\frac{1}{a}(x^2 + y^2) = 2h.$$

For $h < 0$, this becomes

$$x^2 + y^2 = -2ah,$$

which represents a circle in the tangent plane.

Thus, the Dupin indicatrix of a sphere is a circle, in agreement with the fact that every point of a sphere is umbilic. ■

Example 3.6.11 (Asymptotic Directions on $z = xy$)**.** Find the asymptotic directions of the surface $z = xy$ at the origin.

Solution. A parametrization of the surface is

$$\mathbf{r}(u, v) = (u, v, uv).$$

Differentiating, we obtain

$$\mathbf{r}_u = (1, 0, v), \qquad \mathbf{r}_v = (0, 1, u).$$

At the origin, $\mathbf{r}_u = (1, 0, 0)$ and $\mathbf{r}_v = (0, 1, 0)$, so the unit normal may be chosen as $N = (0, 0, 1)$.

The second derivatives are

$$\mathbf{r}_{uu} = (0, 0, 0), \qquad \mathbf{r}_{uv} = (0, 0, 1), \qquad \mathbf{r}_{vv} = (0, 0, 0).$$

Hence the coefficients of the second fundamental form at the origin are

$$L = 0, \qquad M = 1, \qquad N = 0.$$

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The equation of asymptotic directions becomes

$$2\,du\,dv = 0.$$

Therefore, either $du = 0$ or $dv = 0$. These correspond to the coordinate axes in the tangent plane.

Thus, the surface has two distinct asymptotic directions at the origin, as expected for a hyperbolic point. ■

Example 3.6.12 (Conjugate Directions on a Sphere)**.** Show that the meridian and parallel directions on a sphere are conjugate.

Solution. Consider the usual spherical-coordinate parametrization

$$\mathbf{r}(u, v) = (a \sin u \cos v, a \sin u \sin v, a \cos u),$$

where $0 < u < \pi$ and $a > 0$.

Differentiating gives

$$\mathbf{r}_u = (a \cos u \cos v, a \cos u \sin v, -a \sin u)$$

and

$$\mathbf{r}_v = (-a \sin u \sin v, a \sin u \cos v, 0).$$

The mixed partial derivative is

$$\mathbf{r}_{uv} = (-a \cos u \sin v, a \cos u \cos v, 0).$$

Since $\mathbf{r}_{uv}$ is a scalar multiple of $\mathbf{r}_v$, it is tangent to the sphere and hence perpendicular to its unit normal. Therefore,

$$M = \langle \mathbf{r}_{uv}, N \rangle = 0.$$

Since $II(\mathbf{r}_u, \mathbf{r}_v) = M$, the meridian and parallel directions are conjugate. ■

3.7 Joachimsthal's Theorem

Theorem 3.7.1 (Joachimsthal)**.** *Let two regular surfaces S_1 and S_2 intersect along a regular curve C. If the surfaces meet at a constant angle along C, then C is a line of curvature on S_1 if and only if it is a line of curvature on S_2.*

Proof. Let $\mathbf{N}_1$ and $\mathbf{N}_2$ be the unit normals to S_1 and S_2 along C, and let $\mathbf{t}$ be the unit tangent to C. Since the angle θ between the surfaces is constant,

$$\mathbf{N}_1 \cdot \mathbf{N}_2 = \cos\theta.$$

Differentiating with respect to arc length s gives

$$\frac{d\mathbf{N}_1}{ds} \cdot \mathbf{N}_2 + \mathbf{N}_1 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Assume that C is a line of curvature on S_1. By Rodrigues' formula,

$$\frac{d\mathbf{N}_1}{ds} = -k_1 \mathbf{t}.$$

Since $\mathbf{t} \cdot \mathbf{N}_2 = 0$, it follows that

$$\frac{d\mathbf{N}_1}{ds} \cdot \mathbf{N}_2 = 0.$$

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Hence

$$\mathbf{N}_1 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Also, because $\mathbf{N}_2$ is a unit vector,

$$\mathbf{N}_2 \cdot \frac{d\mathbf{N}_2}{ds} = 0.$$

Thus $d\mathbf{N}_2/ds$ is perpendicular to both $\mathbf{N}_1$ and $\mathbf{N}_2$. The tangent $\mathbf{t}$ is also perpendicular to both normals, so

$$\frac{d\mathbf{N}_2}{ds} = -k_2\mathbf{t}$$

for some scalar function k_2. By Rodrigues' formula, C is a line of curvature on S_2.

The converse follows by interchanging S_1 and S_2. ■

Corollary 3.7.2. *If the curve of intersection of two surfaces is a line of curvature on both surfaces, then the angle of intersection is constant.*

Proof. If C is a line of curvature on both surfaces, then

$$\frac{d\mathbf{N}_1}{ds} = -k_1\mathbf{t}, \qquad \frac{d\mathbf{N}_2}{ds} = -k_2\mathbf{t}.$$

Therefore

$$\frac{d}{ds}(\mathbf{N}_1 \cdot \mathbf{N}_2) = -k_1\mathbf{t} \cdot \mathbf{N}_2 - k_2\mathbf{N}_1 \cdot \mathbf{t} = 0.$$

Thus $\mathbf{N}_1 \cdot \mathbf{N}_2$ is constant, so the angle between the surfaces is constant. ■

Example 3.7.3 (Sphere and Plane)**.** (See Figure 37 in the Atlas.) A plane cuts a sphere in a circle. Show that the circle is a line of curvature of the sphere.

Solution. A plane has constant unit normal, while the angle between that normal and the radial normal of the sphere is constant along the circle of intersection. Every curve on a plane is a line of curvature because its shape operator is zero. By Joachimsthal's theorem, the intersection circle is therefore a line of curvature on the sphere. This is also consistent with the fact that every direction on a sphere is principal. ■

3.8 Ruled Surfaces

Definition 3.8.1 (Ruled Surface)**.** A surface generated by a one-parameter family of straight lines is called a **ruled surface**. The generating lines are called **rulings** or **generators**.

Definition 3.8.2 (Directrix)**.** A curve on a ruled surface which meets each generator once is called a **directrix** or **base curve**.

Definition 3.8.3 (Developable and Skew Ruled Surfaces)**.** A ruled surface is called **developable** if its tangent plane is constant along each generator. A ruled surface which is not developable is called a **skew ruled surface** or a **scroll**.

A ruled surface may be written in the form

$$\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u),$$

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where $\mathbf{r}(u)$ is a directrix and $\mathbf{g}(u)$ is a unit vector in the direction of the generator.

Theorem 3.8.4 (Gaussian Curvature of a Ruled Surface)**.** *For*

$$\mathbf{R}(u,v) = \mathbf{r}(u) + v\mathbf{g}(u), \qquad \|\mathbf{g}(u)\| = 1,$$

the Gaussian curvature satisfies

$$K = -\frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{\|\mathbf{R}_u \times \mathbf{R}_v\|^4} \leq 0,$$

where $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ *denotes the scalar triple product.*

Proof. Differentiating,

$$\mathbf{R}_u = \mathbf{r}' + v\mathbf{g}', \qquad \mathbf{R}_v = \mathbf{g}.$$

Since $\|\mathbf{g}\| = 1$,

$$\mathbf{g} \cdot \mathbf{g}' = 0.$$

Also,

$$\mathbf{R}_{uv} = \mathbf{g}', \qquad \mathbf{R}_{vv} = 0.$$

Hence the second fundamental coefficient corresponding to dv^2 is $N = 0$. If $H_0 = \|\mathbf{R}_u \times \mathbf{R}_v\|$, then

$$H_0 M = [\mathbf{R}_{uv}, \mathbf{R}_u, \mathbf{R}_v] = [\mathbf{g}', \mathbf{r}' + v\mathbf{g}', \mathbf{g}] = [\mathbf{g}', \mathbf{r}', \mathbf{g}].$$

Therefore

$$M^2 = \frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{H_0^2}.$$

Since

$$K = \frac{LN - M^2}{H_0^2}$$

and $N = 0$, we obtain

$$K = -\frac{[\mathbf{r}', \mathbf{g}, \mathbf{g}']^2}{H_0^4} \leq 0.$$

■

Corollary 3.8.5. *A ruled surface is developable if and only if*

$$[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0.$$

Proof. A ruled surface is developable precisely when $K = 0$. The preceding formula shows that this occurs exactly when the scalar triple product vanishes. ■

Theorem 3.8.6 (Equivalent Characterizations of a Developable Ruled Surface)**.** *For a regular ruled surface* $\mathbf{R}(u,v) = \mathbf{r}(u) + v\mathbf{g}(u)$*, the following conditions are equivalent:*

1. *the surface is developable;*
2. *the tangent plane is constant along each generator;*
3. $[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0$*;*
4. $K = 0$ *at every regular point.*

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Proof. The surface normal is parallel to $(\mathbf{r}' + v\mathbf{g}') \times \mathbf{g}$. Its direction is independent of v precisely when $\mathbf{r}', \mathbf{g}, \mathbf{g}'$ are linearly dependent, which is equivalent to the vanishing of their scalar triple product. By the formula for the Gaussian curvature of a ruled surface, the same condition is equivalent to $K = 0$. These are precisely the conditions for the tangent plane to remain fixed along a generator. ■

Definition 3.8.7 (Parameter of Distribution)**.** For a ruled surface

$$\mathbf{R}(u, v) = \mathbf{r}(u) + v\mathbf{g}(u),$$

the function

$$p(u) = \frac{[\mathbf{r}'(u), \mathbf{g}(u), \mathbf{g}'(u)]}{\|\mathbf{g}'(u)\|^2},$$

whenever $\mathbf{g}'(u) \neq 0$, is called the **parameter of distribution**.

Remark 3.8.8. The parameter of distribution is constant along each generator and is independent of the chosen directrix. For a developable ruled surface, $p(u) = 0$.

Proposition 3.8.9. *The generators of a ruled surface form one family of asymptotic lines.*

Proof. The equation of the asymptotic directions is $L\,du^2 + 2M\,du\,dv + N\,dv^2 = 0$. For a ruled surface $N = 0$, so

$$du(L\,du + 2M\,dv) = 0.$$

One family is therefore $du = 0$, that is, $u =$ constant. These are precisely the generators. ■

Example 3.8.10 (Right Helicoid)**.** Consider

$$\mathbf{r}(u, v) = (u\cos v,\ u\sin v,\ av).$$

Show that it is a ruled minimal surface and find its Gaussian curvature.

Solution. For fixed v, the curve $u \mapsto \mathbf{r}(u, v)$ is a straight line, so the surface is ruled. We have

$$\mathbf{r}_u = (\cos v, \sin v, 0), \qquad \mathbf{r}_v = (-u\sin v, u\cos v, a).$$

Thus

$$E = 1, \qquad F = 0, \qquad G = u^2 + a^2.$$

Also,

$$\mathbf{r}_{uu} = 0, \qquad \mathbf{r}_{uv} = (-\sin v, \cos v, 0), \qquad \mathbf{r}_{vv} = (-u\cos v, -u\sin v, 0).$$

A unit normal is

$$\mathbf{N} = \frac{(a\sin v, -a\cos v, u)}{\sqrt{u^2 + a^2}}.$$

Hence

$$L = 0, \qquad M = -\frac{a}{\sqrt{u^2 + a^2}}, \qquad N = 0.$$

Therefore

$$H = \frac{EN + GL - 2FM}{2(EG - F^2)} = 0,$$

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so the helicoid is minimal, and

$$K = \frac{LN - M^2}{EG - F^2} = -\frac{a^2}{(u^2 + a^2)^2}.$$

■

Example 3.8.11 (Hyperbolic Paraboloid as a Ruled Surface)**.** (See Figure 38 in the Atlas.) Show that the surface $z = xy$ is doubly ruled.

Solution. For any constant c, the set $x = c$ gives the straight line (c, y, cy) as y varies. Thus one family of rulings is obtained by fixing x. Similarly, fixing $y = c$ gives the straight line (x, c, cx), producing a second family. Hence the hyperbolic paraboloid is doubly ruled. ■

Example 3.8.12 (Developable Cylinder)**.** Verify the developability criterion for the cylinder $\mathbf{R}(u, v) = (a\cos u, a\sin u, 0) + v(0, 0, 1)$.

Solution. Here $\mathbf{r}'(u) = (-a\sin u, a\cos u, 0)$, $\mathbf{g} = (0, 0, 1)$, and $\mathbf{g}' = 0$. Therefore $[\mathbf{r}', \mathbf{g}, \mathbf{g}'] = 0$. Hence the cylinder is developable, in agreement with $K = 0$. ■

3.9 Minimal Surfaces

Definition 3.9.1 (Minimal Surface)**.** A regular surface is called a **minimal surface** if its mean curvature vanishes identically:

$$H = 0.$$

Equivalently,

$$EN + GL - 2FM = 0.$$

Remark 3.9.2. If the coordinate curves are orthogonal, $F = 0$, and the minimal-surface condition reduces to

$$EN + GL = 0.$$

If the coordinate curves are also lines of curvature, then

$$k_1 + k_2 = 0.$$

Thus the principal curvatures of a minimal surface are equal in magnitude and opposite in sign.

Proposition 3.9.3. *At every non-planar point of a minimal surface,*

$$K \leq 0.$$

More precisely,

$$K = -k_1^2 = -k_2^2.$$

Proof. Since $H = 0$, we have $k_2 = -k_1$. Hence $K = k_1 k_2 = -k_1^2 \leq 0$. ■

Example 3.9.4 (A Scherk-Type Minimal Surface)**.** (See Figure 39 in the Atlas.) Show that the surface

$$e^z \cos x = \cos y$$

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is minimal wherever it is regular.

Solution. Write

$$z = \log(\cos y) - \log(\cos x)$$

and use x, y as parameters:

$$\mathbf{r}(x,y) = \big(x,\ y,\ \log(\cos y) - \log(\cos x)\big).$$

Then

$$\mathbf{r}_x = (1, 0, \tan x), \qquad \mathbf{r}_y = (0, 1, -\tan y),$$

so

$$E = \sec^2 x, \qquad F = -\tan x \tan y, \qquad G = \sec^2 y.$$

Further,

$$\mathbf{r}_{xx} = (0, 0, \sec^2 x), \qquad \mathbf{r}_{xy} = 0, \qquad \mathbf{r}_{yy} = (0, 0, -\sec^2 y).$$

If $H_0 = \|\mathbf{r}_x \times \mathbf{r}_y\|$, then

$$L = \frac{\sec^2 x}{H_0}, \qquad M = 0, \qquad N = -\frac{\sec^2 y}{H_0}.$$

Therefore

$$EN + GL - 2FM = \frac{-\sec^2 x \sec^2 y + \sec^2 y \sec^2 x}{H_0} = 0.$$

Hence the surface is minimal. ■

Theorem 3.9.5. *The two asymptotic directions at a point are orthogonal if and only if the surface is minimal at that point.*

Proof. The asymptotic directions satisfy

$$L\,du^2 + 2M\,du\,dv + N\,dv^2 = 0.$$

For a double family

$$P\,du^2 + 2Q\,du\,dv + R\,dv^2 = 0,$$

the two directions are orthogonal precisely when

$$ER - 2FQ + GP = 0.$$

Taking $P = L$, $Q = M$, and $R = N$, the condition becomes $EN - 2FM + GL = 0$, which is exactly the condition $H = 0$. ■

Theorem 3.9.6 (Minimal Surfaces of Revolution)**.** *Apart from the plane, the catenoid is the only minimal surface of revolution.*

Proof. Let

$$\mathbf{r}(u,v) = (u\cos v, u\sin v, f(u)).$$

Then

$$E = 1 + f'^2, \qquad F = 0, \qquad G = u^2,$$

and, for a suitable orientation,

$$L = \frac{f''}{\sqrt{1+f'^2}}, \qquad M = 0, \qquad N = \frac{uf'}{\sqrt{1+f'^2}}.$$

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The condition $H = 0$ becomes $uf'' + f'(1 + f'^2) = 0$. If $f' \equiv 0$, the surface is a plane. Otherwise, putting $p = f'$ gives $\frac{p'}{p(1+p^2)} = -\frac{1}{u}$. Integration yields $\frac{p}{\sqrt{1+p^2}} = \frac{a}{u}$, so

$$f'(u) = \frac{a}{\sqrt{u^2 - a^2}}.$$

A second integration gives $f(u) = a \log\left(u + \sqrt{u^2 - a^2}\right) + c$. This is a catenary; its surface of revolution is a catenoid. Hence, apart from the plane, the catenoid is the only minimal surface of revolution. ■

Example 3.9.7 (Catenoid). (See Figure 40 in the Atlas.) The standard catenoid

$$\mathbf{r}(u, v) = (a \cosh u \cos v, a \cosh u \sin v, au)$$

has $E = G = a^2 \cosh^2 u$ and $F = 0$. Thus (u, v) are isothermal coordinates. A direct computation gives $H = 0$, so the catenoid is minimal.

Theorem 3.9.8 (First Variation of Area). *If a smooth surface bounded by a fixed closed curve has least area among all sufficiently small normal variations with the same boundary, then it is minimal.*

Proof. Let

$$\mathbf{R}(u, v) = \mathbf{r}(u, v) + \varepsilon\phi(u, v)\mathbf{N}(u, v),$$

where ϕ vanishes on the boundary. Denote the area of the varied surface by $A(\varepsilon)$. To first order in ε, the first fundamental coefficients change as

$$E_\varepsilon = E - 2\varepsilon\phi L + O(\varepsilon^2),$$

$$F_\varepsilon = F - 2\varepsilon\phi M + O(\varepsilon^2),$$

$$G_\varepsilon = G - 2\varepsilon\phi N + O(\varepsilon^2).$$

Thus

$$E_\varepsilon G_\varepsilon - F_\varepsilon^2 = (EG - F^2) - 2\varepsilon\phi(EN + GL - 2FM) + O(\varepsilon^2).$$

Since

$$EN + GL - 2FM = 2H(EG - F^2),$$

we obtain

$$dA_\varepsilon = (1 - 2\varepsilon\phi H)\, dA + O(\varepsilon^2).$$

Therefore

$$A'(0) = -2 \iint_S H\phi \, dA.$$

If the original surface minimizes area, then $A'(0) = 0$ for every admissible ϕ. Hence

$$\iint_S H\phi \, dA = 0$$

for every such ϕ, which implies $H = 0$ throughout the surface. Thus the surface is minimal. ■

Example 3.9.9 (Enneper Surface). (See Figure 41 in the Atlas.) Show that

$$\mathbf{r}(u, v) = \left(u - \frac{u^3}{3} + uv^2,\ v - \frac{v^3}{3} + u^2 v,\ u^2 - v^2\right)$$

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is a minimal surface.

Solution. Differentiation gives

$$\mathbf{r}_u = (1 - u^2 + v^2,\, 2uv,\, 2u), \qquad \mathbf{r}_v = (2uv,\, 1 - v^2 + u^2,\, -2v).$$

A direct calculation shows that $\mathbf{r}_u \cdot \mathbf{r}_v = 0$ and $\|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2 = (1 + u^2 + v^2)^2$. Hence (u, v) are isothermal coordinates. Moreover, each coordinate function is harmonic: $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$. By the harmonic characterization of minimal surfaces, the Enneper surface is minimal. ■

Example 3.9.10 (No Non-Planar Minimal Surface with Positive Gaussian Curvature)**.** Show that a minimal surface cannot have positive Gaussian curvature at a regular point.

Solution. For a minimal surface, $k_1 + k_2 = 0$, so $k_2 = -k_1$. Therefore $K = k_1 k_2 = -k_1^2 \leq 0$. Thus positive Gaussian curvature is impossible at a regular point of a minimal surface. ■

3.10 Isothermal Coordinates

Definition 3.10.1 (Isothermal Coordinates)**.** A coordinate system (u, v) on a surface is called **isothermal** or **isothermic** if the first fundamental form has the form

$$ds^2 = \lambda^2(u, v)(du^2 + dv^2), \qquad \lambda > 0.$$

Equivalently,

$$E = G = \lambda^2, \qquad F = 0.$$

Remark 3.10.2. Isothermal coordinates are orthogonal and have the same scale factor in both coordinate directions.

Theorem 3.10.3 (Local Existence of Isothermal Coordinates)**.** *Every point of a regular surface has a neighbourhood on which isothermal coordinates may be introduced.*

Proof. Locally, one may first introduce complex isotropic parameters in which the metric takes the form

$$ds^2 = \Lambda(u, v)\, du\, dv.$$

Set

$$u = U + iV, \qquad v = U - iV.$$

Then

$$du\, dv = dU^2 + dV^2,$$

and therefore

$$ds^2 = \Lambda(U, V)(dU^2 + dV^2).$$

Thus (U, V) are local isothermal coordinates. ■

Theorem 3.10.4. *From one isothermal coordinate system, infinitely many others may be obtained locally through analytic changes of the complex variable $u + iv$.*

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Proof. Suppose (u, v) and (U, V) are both isothermal systems:

$$ds^2 = \lambda^2(du^2 + dv^2) = \mu^2(dU^2 + dV^2).$$

Writing $U = U(u, v)$ and $V = V(u, v)$ and comparing the coefficients of du^2, $du\,dv$, and dv^2 gives

$$U_u^2 + V_u^2 = U_v^2 + V_v^2, \qquad U_u U_v + V_u V_v = 0.$$

These equations imply, up to orientation,

$$U_u = V_v, \qquad U_v = -V_u,$$

which are the Cauchy–Riemann equations. Hence $U + iV$ is locally an analytic function of $u + iv$. Conversely, an analytic change of variable preserves the conformal form of the metric. ■

Example 3.10.5 (Isothermal Coordinates on a Surface of Revolution)**.** Let

$$\mathbf{r}(u, v) = (g(u)\cos v,\ g(u)\sin v,\ f(u)).$$

Find an isothermal coordinate system.

Solution. The first fundamental form is

$$ds^2 = \left(g'(u)^2 + f'(u)^2\right) du^2 + g(u)^2\, dv^2.$$

Define

$$U = \int \frac{\sqrt{g'(u)^2 + f'(u)^2}}{g(u)}\, du, \qquad V = v.$$

Then

$$dU^2 = \frac{g'(u)^2 + f'(u)^2}{g(u)^2}\, du^2,$$

and hence

$$ds^2 = g(u)^2(dU^2 + dV^2).$$

Thus (U, V) is an isothermal system. ■

Example 3.10.6 (Isothermal Coordinates on the Sphere)**.** (See Figure 46 in the Atlas.) Find an isothermal coordinate for the unit sphere using longitude v and colatitude u.

Solution. The metric is $ds^2 = du^2 + \sin^2 u\, dv^2$. Define $U = \log\tan(u/2)$ and $V = v$. Since $dU = \csc u\, du$, we have $du^2 = \sin^2 u\, dU^2$. Hence

$$ds^2 = \sin^2 u\,(dU^2 + dV^2),$$

so (U, V) is an isothermal system. This is the coordinate form underlying the Mercator projection. ■

3.11 Minimal Surfaces and Harmonic Coordinate Functions

Theorem 3.11.1. *Let*

$$\mathbf{r}(u, v) = \big(x(u, v), y(u, v), z(u, v)\big)$$

be a regular surface expressed in isothermal coordinates, so that $ds^2 = \lambda^2(du^2 + dv^2)$*. Then*

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H\,\mathbf{N}.$$

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Consequently, the surface is minimal if and only if its coordinate functions x, y, z are harmonic:

$$x_{uu} + x_{vv} = 0, \qquad y_{uu} + y_{vv} = 0, \qquad z_{uu} + z_{vv} = 0.$$

Proof. For isothermal coordinates,

$$E = G = \lambda^2, \qquad F = 0.$$

The Gauss formulas decompose $\mathbf{r}_{uu}$ and $\mathbf{r}_{vv}$ into tangential and normal parts. For a conformal metric $\lambda^2(du^2 + dv^2)$, the tangential parts cancel in the sum $\mathbf{r}_{uu} + \mathbf{r}_{vv}$. Its normal part is

$$(L + N)\mathbf{N}.$$

Hence

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = (L + N)\mathbf{N}.$$

But

$$H = \frac{EN + GL}{2EG} = \frac{\lambda^2(N + L)}{2\lambda^4} = \frac{L + N}{2\lambda^2},$$

so

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H\,\mathbf{N}.$$

If the surface is minimal, $H = 0$, and therefore $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$. Since

$$\mathbf{r} = (x, y, z),$$

this is equivalent to

$$x_{uu} + x_{vv} = 0, \qquad y_{uu} + y_{vv} = 0, \qquad z_{uu} + z_{vv} = 0.$$

Conversely, if the coordinate functions are harmonic, then $\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0$, and the displayed identity gives $2\lambda^2 H\mathbf{N} = 0$. Since $\lambda > 0$ and $\|\mathbf{N}\| = 1$, we obtain $H = 0$. Hence the surface is minimal. ■

Corollary 3.11.2. *In isothermal coordinates, a minimal surface is locally represented by three harmonic functions satisfying the conformality conditions*

$$\mathbf{r}_u \cdot \mathbf{r}_v = 0, \qquad \|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2.$$

Corollary 3.11.3 (No Compact Minimal Surface without Boundary)**.** *There is no compact regular minimal surface without boundary in* $\mathbb{R}^3$.

Proof. In local isothermal coordinates, the coordinate functions of a minimal surface are harmonic. If the surface were compact without boundary, each ambient coordinate function restricted to the surface would attain a maximum and a minimum. The maximum principle forces each coordinate function to be constant, which is incompatible with regularity. Hence no such surface exists. ■

Example 3.11.4 (Harmonic Coordinate Test)**.** Let $\mathbf{r}(u, v) = (u, v, u^2 - v^2)$. Determine whether the parameterization is isothermal and whether the surface is minimal by the harmonic-coordinate criterion.

Solution. The coordinate functions u, v, and $u^2 - v^2$ are harmonic. However, $\mathbf{r}_u = (1, 0, 2u)$ and $\mathbf{r}_v = (0, 1, -2v)$ satisfy $\mathbf{r}_u \cdot \mathbf{r}_v = -4uv$, which is not identically zero. Thus the given parameters are not isothermal, so harmonicity of the coordinate functions in these parameters does not by itself imply minimality. Indeed the surface is not minimal in general. This example emphasizes that the harmonic criterion requires isothermal coordinates. ■

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3.12 Minimal Surfaces and Holomorphic Functions: Weierstrass–Enneper Representation

We have seen that a surface expressed in isothermal coordinates is minimal if and only if its coordinate functions are harmonic. This provides a connection between minimal surfaces and harmonic functions. Since the real and imaginary parts of holomorphic functions are harmonic, complex analysis offers a powerful method for constructing minimal surfaces.

The Weierstrass–Enneper representation makes this connection precise. It expresses a minimal surface parametrization in terms of holomorphic functions of a complex variable and provides a systematic method for generating minimal surfaces.

3.12.1 Holomorphic Functions and Harmonic Coordinates

Let $z = u + iv$ be a complex variable, where $u, v \in \mathbb{R}$. A complex-valued function $F(z)$ is called holomorphic on an open set $U \subset \mathbb{C}$ if it is complex differentiable at every point of U.

Writing $F(z) = p(u,v) + iq(u,v)$, the Cauchy–Riemann equations are

$$p_u = q_v, \qquad p_v = -q_u.$$

Differentiating these equations and using the equality of mixed partial derivatives gives

$$p_{uu} + p_{vv} = 0, \qquad q_{uu} + q_{vv} = 0.$$

Thus, both the real and imaginary parts of a holomorphic function are harmonic.

Conversely, on a simply connected domain, every real-valued harmonic function possesses a harmonic conjugate and can therefore be expressed as the real part of a holomorphic function.

This observation is fundamental in the study of minimal surfaces.

Proposition 3.12.1. *Let $\mathbf{r}(u,v) = (x(u,v), y(u,v), z(u,v))$ be a regular surface parametrized by isothermal coordinates, so that*

$$E = G = \lambda^2 > 0, \qquad F = 0.$$

Then the surface is minimal if and only if

$$x_{uu} + x_{vv} = 0, \qquad y_{uu} + y_{vv} = 0, \qquad z_{uu} + z_{vv} = 0.$$

Proof. For an isothermal parametrization, the first fundamental form is

$$ds^2 = \lambda^2(du^2 + dv^2).$$

The mean curvature vector satisfies

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H\mathbf{N},$$

where $\mathbf{N}$ is the unit normal to the surface.

Since $\lambda^2 > 0$, it follows that $H = 0$ if and only if

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0.$$

Writing this vector equation componentwise gives

$$x_{uu} + x_{vv} = 0, \qquad y_{uu} + y_{vv} = 0, \qquad z_{uu} + z_{vv} = 0.$$

Hence the surface is minimal if and only if all three coordinate functions are harmonic. ∎

The preceding proposition suggests constructing minimal surfaces from holomorphic functions. However, harmonicity alone is not sufficient: the coordinate functions must also satisfy the conditions for an isothermal parametrization. These conditions are incorporated naturally into the Weierstrass–Enneper representation.

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Theorem 3.12.2 (Weierstrass–Enneper Representation). *Let $U \subset \mathbb{C}$ be a simply connected domain, and let f and g be holomorphic functions on U, with $f(z) \neq 0$ for every $z \in U$. Fix $z_0 \in U$ and define*

$$\mathbf{r}(z) = \operatorname{Re} \int_{z_0}^{z} \left(f(1-g^2),\ if(1+g^2),\ 2fg \right) d\zeta,$$

where the functions in the integrand are evaluated at ζ and the real part is taken componentwise.

Then $\mathbf{r}$ defines a regular minimal surface on U. Its first fundamental form is

$$ds^2 = |f|^2(1+|g|^2)^2(du^2+dv^2),$$

where $z = u + iv$.

Proof. Put

$$\phi_1 = f(1-g^2), \qquad \phi_2 = if(1+g^2), \qquad \phi_3 = 2fg.$$

Since f and g are holomorphic, the functions ϕ_1, ϕ_2, ϕ_3 are also holomorphic on U.

Because U is simply connected, each holomorphic function ϕ_j has a primitive. Hence the integrals defining the coordinate functions are independent of the path joining z_0 to z.

Write

$$\mathbf{r}(z) = (x(u,v), y(u,v), w(u,v)).$$

Each coordinate function is the real part of a holomorphic function and is therefore harmonic. Thus,

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0.$$

We next show that the parameters (u, v) are isothermal. Let

$$\Phi = (\phi_1, \phi_2, \phi_3).$$

The components of Φ satisfy

$$\begin{aligned} \phi_1^2 + \phi_2^2 + \phi_3^2 &= f^2(1-g^2)^2 - f^2(1+g^2)^2 + 4f^2g^2 \\ &= f^2\left[(1-2g^2+g^4) - (1+2g^2+g^4) + 4g^2\right] \\ &= 0. \end{aligned}$$

Thus,

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = 0.$$

Since $\mathbf{r}$ is the real part of an integral of Φ, differentiation with respect to u and v gives

$$\mathbf{r}_u = \operatorname{Re} \Phi, \qquad \mathbf{r}_v = -\operatorname{Im} \Phi.$$

Consequently,

$$\mathbf{r}_u - i\mathbf{r}_v = \Phi.$$

Taking the complex bilinear inner product of this identity with itself, we obtain

$$(\mathbf{r}_u - i\mathbf{r}_v) \cdot (\mathbf{r}_u - i\mathbf{r}_v) = 0.$$

Expanding gives

$$\|\mathbf{r}_u\|^2 - \|\mathbf{r}_v\|^2 - 2i \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$

Equating real and imaginary parts, we obtain

$$\|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2, \qquad \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$

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Therefore,

$$E = G, \qquad F = 0.$$

Hence (u, v) are isothermal coordinates, provided the parametrization is regular.

To verify regularity and determine the first fundamental form, observe that

$$\begin{aligned} |\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2 &= |f|^2 \left(|1 - g^2|^2 + |1 + g^2|^2 + 4|g|^2\right) \\ &= 2|f|^2(1 + |g|^2)^2. \end{aligned}$$

On the other hand,

$$|\phi_1|^2 + |\phi_2|^2 + |\phi_3|^2 = \|\mathbf{r}_u\|^2 + \|\mathbf{r}_v\|^2 = E + G = 2E.$$

It follows that

$$E = G = |f|^2(1 + |g|^2)^2.$$

Since $f(z) \neq 0$, we have $E = G > 0$. Together with $F = 0$, this proves that $\mathbf{r}_u$ and $\mathbf{r}_v$ are linearly independent, so the parametrization is regular.

Therefore, the first fundamental form is

$$ds^2 = |f|^2(1 + |g|^2)^2(du^2 + dv^2).$$

Finally, we have shown that the coordinates are isothermal and that

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0.$$

By the characterization of minimal surfaces in isothermal coordinates, the mean curvature vanishes identically. Hence the surface is minimal. ■

The functions f and g are called the **Weierstrass data** of the surface. The function f determines part of the metric scale, while g determines the unit normal through the stereographic representation of the Gauss map.

Proposition 3.12.3 (Gauss Map in Weierstrass Coordinates)**.** *For a minimal surface represented by Weierstrass data (f, g) as above, the unit normal compatible with the orientation $\mathbf{r}_u \times \mathbf{r}_v$ is*

$$\mathbf{N} = \frac{1}{1 + |g|^2} \left(2\operatorname{Re} g,\, 2\operatorname{Im} g,\, |g|^2 - 1\right).$$

Proof. We have

$$\mathbf{r}_u = \operatorname{Re}\Phi, \qquad \mathbf{r}_v = -\operatorname{Im}\Phi,$$

where $\Phi = (f(1 - g^2), if(1 + g^2), 2fg)$.

A direct calculation shows that the vector

$$\left(2\operatorname{Re} g, 2\operatorname{Im} g, |g|^2 - 1\right)$$

is perpendicular to both $\mathbf{r}_u$ and $\mathbf{r}_v$. Its squared length is

$$4|g|^2 + (|g|^2 - 1)^2 = (1 + |g|^2)^2.$$

Dividing by its length therefore gives a unit normal vector. The chosen sign agrees with the orientation determined by $\mathbf{r}_u \times \mathbf{r}_v$, which proves the formula. ■

The Weierstrass–Enneper representation is not merely a method for constructing examples. Locally, every minimal surface can be described in this way.

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Theorem 3.12.4 (Local Converse of the Weierstrass–Enneper Representation). *Let S be an oriented regular minimal surface. Around every point of S, there exist isothermal coordinates and, after a suitable choice of Cartesian axes, holomorphic functions f and g, with $f \neq 0$, such that the surface is locally represented, up to a translation, by*

$$\mathbf{r}(z) = \operatorname{Re} \int \left(f(1-g^2),\, if(1+g^2),\, 2fg\right)\, dz.$$

Proof. Let $\mathbf{r}(u,v)$ be a local isothermal parametrization of the minimal surface. Since the surface is minimal, its coordinate functions are harmonic:

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 0.$$

Define the complex vector-valued function

$$\Phi = \mathbf{r}_u - i\mathbf{r}_v.$$

Differentiating with respect to $\bar{z}$, where

$$\frac{\partial}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial}{\partial u} + i\frac{\partial}{\partial v}\right),$$

we obtain

$$\frac{\partial \Phi}{\partial \bar{z}} = \frac{1}{2}(\mathbf{r}_{uu} + \mathbf{r}_{vv}) = 0,$$

because $\mathbf{r}_{uv} = \mathbf{r}_{vu}$. Hence the components ϕ_1, ϕ_2, ϕ_3 of Φ are holomorphic.

Since the coordinates are isothermal,

$$\|\mathbf{r}_u\|^2 = \|\mathbf{r}_v\|^2, \qquad \langle \mathbf{r}_u, \mathbf{r}_v \rangle = 0.$$

Consequently,

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = 0.$$

At the point under consideration, choose the Cartesian axes so that $\phi_1 - i\phi_2 \neq 0$. By continuity, this remains true in a sufficiently small neighbourhood.

Define

$$f = \frac{\phi_1 - i\phi_2}{2}, \qquad g = \frac{\phi_3}{\phi_1 - i\phi_2}.$$

These functions are holomorphic, and f is nonzero. From their definitions, $\phi_3 = 2fg$ and $\phi_1 - i\phi_2 = 2f$.

Using $\phi_1^2 + \phi_2^2 + \phi_3^2 = 0$, we obtain

$$(\phi_1 + i\phi_2)(\phi_1 - i\phi_2) = -\phi_3^2.$$

Therefore,

$$\phi_1 + i\phi_2 = -2fg^2.$$

Adding and subtracting the two expressions for $\phi_1 \pm i\phi_2$ gives

$$\phi_1 = f(1-g^2), \qquad \phi_2 = if(1+g^2).$$

Thus,

$$\Phi = (f(1-g^2), if(1+g^2), 2fg).$$

Since $\Phi = \mathbf{r}_u - i\mathbf{r}_v$, integration yields

$$\mathbf{r}(z) = \operatorname{Re} \int \left(f(1-g^2), if(1+g^2), 2fg\right)\, dz$$

up to an additive constant vector. Hence every regular minimal surface admits a local Weierstrass–Enneper representation. ∎

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Example 3.12.5 (The Plane)**.** Use the Weierstrass–Enneper representation to obtain a parametrization of a plane.

Solution. Choose the holomorphic functions $f(z) = 1$ and $g(z) = 0$. The Weierstrass–Enneper formula becomes

$$\mathbf{r}(z) = \operatorname{Re} \int (1, i, 0)\, dz.$$

Integrating componentwise gives

$$\mathbf{r}(z) = \operatorname{Re}(z, iz, 0).$$

Writing $z = u + iv$, we obtain

$$\mathbf{r}(u, v) = (u, -v, 0).$$

This is a parametrization of the xy-plane. Hence the plane is a minimal surface. ■

Example 3.12.6 (Enneper's Surface)**.** Use the Weierstrass–Enneper representation with $f(z) = 1$ and $g(z) = z$ to obtain a minimal surface.

Solution. Substituting $f(z) = 1$ and $g(z) = z$ into the Weierstrass–Enneper formula gives

$$\mathbf{r}(z) = \operatorname{Re} \int (1 - z^2, i(1 + z^2), 2z)\, dz.$$

Integrating componentwise, we obtain

$$\mathbf{r}(z) = \operatorname{Re} \left(z - \frac{z^3}{3},\, i\left(z + \frac{z^3}{3} \right),\, z^2 \right).$$

Now put $z = u + iv$. Since

$$z^3 = u^3 - 3uv^2 + i(3u^2v - v^3),$$

taking real parts gives

$$\mathbf{r}(u, v) = \left(u - \frac{u^3}{3} + uv^2,\, -v - u^2v + \frac{v^3}{3},\, u^2 - v^2 \right).$$

This is a parametrization of Enneper's minimal surface, up to a reflection in a coordinate plane.

By the Weierstrass–Enneper representation theorem, the surface is minimal. Its first fundamental form is

$$ds^2 = (1 + u^2 + v^2)^2 (du^2 + dv^2).$$

Since the metric coefficient is positive for all $(u, v) \in \mathbb{R}^2$, the parametrization is regular everywhere. ■

Example 3.12.7 (The Catenoid)**.** Show that the Weierstrass data

$$f(z) = -\frac{a}{2} e^{-z}, \qquad g(z) = e^z, \qquad a > 0,$$

generate a catenoid.

Solution. Substituting into the Weierstrass–Enneper formula, we obtain

$$f(1 - g^2) = -\frac{a}{2}(e^{-z} - e^z) = a \sinh z,$$

$$if(1 + g^2) = -\frac{ia}{2}(e^{-z} + e^z) = -ia \cosh z,$$

$$2fg = -a.$$

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Therefore,

$$\mathbf{r}(z) = \operatorname{Re} \int (a \sinh z, -ia \cosh z, -a)\, dz.$$

Integrating gives

$$\mathbf{r}(z) = \operatorname{Re}(a \cosh z, -ia \sinh z, -az).$$

Writing $z = u + iv$ and using

$$\cosh(u + iv) = \cosh u \cos v + i \sinh u \sin v,$$

$$\sinh(u + iv) = \sinh u \cos v + i \cosh u \sin v,$$

we obtain

$$\mathbf{r}(u, v) = (a \cosh u \cos v, a \cosh u \sin v, -au).$$

This is a parametrization of a catenoid, reflected in the xy-plane if compared with the usual parametrization having third coordinate au.

Since it is obtained from holomorphic Weierstrass data, the catenoid is a minimal surface. ■

Example 3.12.8 (The Helicoid)**.** Show that the Weierstrass data

$$f(z) = -\frac{ia}{2} e^{-z}, \qquad g(z) = e^{z}, \qquad a > 0,$$

generate a helicoid.

Solution. Using the Weierstrass–Enneper formula, we obtain

$$\begin{aligned} f(1 - g^2) &= ia \sinh z, \\ if(1 + g^2) &= a \cosh z, \\ 2fg &= -ia. \end{aligned}$$

Hence,

$$\mathbf{r}(z) = \operatorname{Re} \int (ia \sinh z, a \cosh z, -ia)\, dz.$$

Integrating,

$$\mathbf{r}(z) = \operatorname{Re}(ia \cosh z, a \sinh z, -iaz).$$

Writing $z = u + iv$ and taking real parts, we obtain

$$\mathbf{r}(u, v) = (-a \sinh u \sin v, a \sinh u \cos v, av).$$

This is a parametrization of a helicoid.

Since the parametrization is obtained from holomorphic Weierstrass data, it defines a minimal surface. ■

The Weierstrass–Enneper representation reveals a close relationship between the catenoid and the helicoid. Their Weierstrass data differ only by multiplication of the function f by i. Such surfaces are called **conjugate minimal surfaces** and belong to the same associated family of minimal surfaces. In corresponding parameters, they have the same first fundamental form and are therefore locally isometric.

More generally, replacing f by $e^{it} f$, where t is a real constant, gives an associated family of minimal surfaces with the same first fundamental form. Thus, the Weierstrass–Enneper representation provides both a method of constructing minimal surfaces and a means of understanding the intrinsic geometric relationships between them.

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The representation also illustrates why holomorphic functions play a central role in minimal surface theory. Holomorphicity guarantees harmonic coordinate functions, while the algebraic identity

$$\phi_1^2 + \phi_2^2 + \phi_3^2 = 0$$

ensures that the resulting parametrization is isothermal. Together, these conditions characterize regular minimal surfaces locally.

3.13 Isometry between Surfaces

Two surfaces may have different shapes in three-dimensional space and yet possess the same intrinsic geometry. For example, a flat sheet of paper can be rolled into a circular cylinder without stretching or compressing it. Although the plane and cylinder appear geometrically different, lengths and angles measured along their surfaces remain unchanged.

A mapping that preserves these metric properties is called an isometry.

Definition 3.13.1 (Isometry)**.** Let S_1 and S_2 be regular surfaces. A diffeomorphism $f : S_1 \to S_2$ is called an **isometry** if, for every $p \in S_1$ and all $v, w \in T_pS_1$,

$$\langle df_p(v), df_p(w) \rangle = \langle v, w \rangle .$$

Thus, the differential of an isometry preserves the inner product on tangent spaces.

In particular, taking $v = w$, we obtain $\|df_p(v)\| = \|v\|$. Hence an isometry preserves the lengths of tangent vectors. Since angles are determined by inner products, it also preserves angles between tangent vectors. Consequently, the lengths of smooth curves and the areas of surface regions are preserved.

The requirement that an isometry be a diffeomorphism ensures that the correspondence between the two surfaces is smooth, one-to-one, onto, and has a smooth inverse. However, surfaces may have the same metric geometry only in sufficiently small neighbourhoods. This motivates the following definition.

Definition 3.13.2 (Local Isometry)**.** A smooth mapping $f : S_1 \to S_2$ is called a **local isometry** at $p \in S_1$ if there exists an open neighbourhood V of p such that $f|_V : V \to f(V)$ is an isometry. If this condition holds at every point of S_1, then f is called a local isometry.

Every isometry is a local isometry, but the converse need not hold. A local isometry preserves metric properties in small neighbourhoods, although it may fail to be one-to-one on the entire surface.

Example 3.13.3 (Local Isometry between a Plane and a Cylinder)**.** Let S_1 be the xy-plane and S_2 the unit circular cylinder $X^2 + Y^2 = 1$. Show that the mapping

$$f(x, y, 0) = (\cos x, \sin x, y)$$

is a local isometry.

Solution. Let $p = (x, y, 0) \in S_1$. The tangent plane of S_1 is spanned by the orthonormal vectors $e_1 = (1, 0, 0)$ and $e_2 = (0, 1, 0)$.

To determine the differential of f, consider curves through p in these coordinate directions. Differentiating their images gives

$$df_p(e_1) = \frac{d}{dt} f(x + t, y, 0)\Big|_{t=0} = (-\sin x, \cos x, 0)$$

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and

$$df_p(e_2) = \frac{d}{dt} f(x, y+t, 0)\Big|_{t=0} = (0,0,1).$$

These image vectors satisfy

$$\|df_p(e_1)\|^2 = \sin^2 x + \cos^2 x = 1, \qquad \|df_p(e_2)\|^2 = 1,$$

and

$$\langle df_p(e_1), df_p(e_2)\rangle = 0.$$

Thus, df_p maps the orthonormal basis $\{e_1, e_2\}$ to an orthonormal basis of the tangent plane of the cylinder. Consequently, for any tangent vectors $v = ae_1 + be_2$ and $w = ce_1 + de_2$, we have

$$\langle df_p(v), df_p(w)\rangle = ac + bd = \langle v, w\rangle .$$

Furthermore, by restricting x to an interval of length less than 2π, the mapping becomes a diffeomorphism onto its image. Hence f is a local isometry.

Notice that $f(x + 2\pi, y, 0) = f(x, y, 0)$, so f is not injective on the entire plane. Therefore, it is not a global isometry between the plane and the cylinder. ■

The first fundamental form provides a convenient method for determining whether two parametrized surfaces are locally isometric. The following theorem expresses this fact precisely.

Theorem 3.13.4 (Equality of First Fundamental Forms)**.** *Let $\mathbf{x} : U \to S_1$ and $\bar{\mathbf{x}} : U \to S_2$ be regular local parametrizations defined on the same open set $U \subset \mathbb{R}^2$. Suppose their first fundamental forms have identical coefficients:*

$$E = \bar{E}, \qquad F = \bar{F}, \qquad G = \bar{G}.$$

Then the mapping

$$f = \bar{\mathbf{x}} \circ \mathbf{x}^{-1} : \mathbf{x}(U) \to \bar{\mathbf{x}}(U)$$

is an isometry between these surface patches. In particular, the two surfaces are locally isometric at corresponding points.

Proof. Since $\mathbf{x}$ and $\bar{\mathbf{x}}$ are local parametrizations, the mapping $f = \bar{\mathbf{x}} \circ \mathbf{x}^{-1}$ is a diffeomorphism between their images.

Let $p = \mathbf{x}(u, v)$ and consider two arbitrary tangent vectors

$$v_1 = a_1\mathbf{x}_u + b_1\mathbf{x}_v, \qquad v_2 = a_2\mathbf{x}_u + b_2\mathbf{x}_v.$$

By the definition of the first fundamental form,

$$\langle v_1, v_2\rangle = a_1a_2E + (a_1b_2 + a_2b_1)F + b_1b_2G.$$

Since $f \circ \mathbf{x} = \bar{\mathbf{x}}$, the chain rule gives

$$df_p(\mathbf{x}_u) = \bar{\mathbf{x}}_u, \qquad df_p(\mathbf{x}_v) = \bar{\mathbf{x}}_v.$$

Therefore,

$$df_p(v_i) = a_i\bar{\mathbf{x}}_u + b_i\bar{\mathbf{x}}_v, \qquad i = 1, 2.$$

Taking the inner product on S_2, we obtain

$$\langle df_p(v_1), df_p(v_2)\rangle = a_1a_2\bar{E} + (a_1b_2 + a_2b_1)\bar{F} + b_1b_2\bar{G}.$$

By hypothesis, $E = \bar{E}$, $F = \bar{F}$ and $G = \bar{G}$. Hence

$$\langle df_p(v_1), df_p(v_2)\rangle = \langle v_1, v_2\rangle .$$

Thus, f preserves the inner product of every pair of tangent vectors. Since it is also a diffeomorphism, f is an isometry between the two surface patches. ■

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The theorem shows that the first fundamental form completely determines the local metric properties of a surface. Consequently, two surfaces having the same first fundamental form in corresponding coordinates have the same intrinsic geometry, even though their shapes in $\mathbb{R}^3$ may be different.

We now establish a deeper result: Gaussian curvature is also determined by this intrinsic geometry.

3.14 Further Results on Isometries and Local Isometries

Definition 3.14.1 (Isometry)**.** Let S and $\widetilde{S}$ be regular surfaces. A differentiable homeomorphism

$$\Phi : S \longrightarrow \widetilde{S}$$

is called an **isometry** if it preserves the length of every tangent vector:

$$\|d\Phi_P(X)\| = \|X\|$$

for every $P \in S$ and every $X \in T_P S$.

Definition 3.14.2 (Local Isometry)**.** A smooth map

$$\Phi : S \longrightarrow \widetilde{S}$$

is called a **local isometry** if every point $P \in S$ has a neighbourhood U such that

$$\Phi|_U : U \longrightarrow \Phi(U)$$

is an isometry.

Theorem 3.14.3 (Characterization of an Isometry by the First Fundamental Form)**.** *A smooth local diffeomorphism*

$$\Phi : S \longrightarrow \widetilde{S}$$

is a local isometry if and only if it preserves the first fundamental form:

$$\widetilde{I}(d\Phi_P X, d\Phi_P Y) = I(X, Y)$$

for all $X, Y \in T_P S$.

Proof. If Φ is a local isometry, then it preserves lengths. Using the polarization identity,

$$2I(X, Y) = \|X + Y\|^2 - \|X\|^2 - \|Y\|^2,$$

we see that it also preserves inner products. Hence $\widetilde{I}(d\Phi X, d\Phi Y) = I(X, Y)$.

Conversely, if the first fundamental form is preserved, then

$$\|d\Phi_P X\|^2 = \widetilde{I}(d\Phi_P X, d\Phi_P X) = I(X, X) = \|X\|^2.$$

Thus the differential preserves lengths of tangent vectors. Consequently the length of every curve is preserved locally, and Φ is a local isometry. ∎

Corollary 3.14.4 (Coordinate Criterion)**.** *Suppose corresponding points on two surfaces are represented*

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by the same local parameters (u, v):

$$\mathbf{r} = \mathbf{r}(u, v), \qquad \widetilde{\mathbf{r}} = \widetilde{\mathbf{r}}(u, v).$$

Then the correspondence is a local isometry if and only if

$$E = \widetilde{E}, \qquad F = \widetilde{F}, \qquad G = \widetilde{G}.$$

Proof. The equality of the first fundamental forms

$$E\,du^2 + 2F\,du\,dv + G\,dv^2 = \widetilde{E}\,du^2 + 2\widetilde{F}\,du\,dv + \widetilde{G}\,dv^2$$

for all du, dv is equivalent to equality of the three coefficients. ∎

Proposition 3.14.5. *A local isometry preserves lengths of curves, angles, areas, and Gaussian curvature.*

Proof. A local isometry preserves the first fundamental form. Hence it preserves lengths and inner products of tangent vectors, and therefore angles. Since

$$dA = \sqrt{EG - F^2}\,du\,dv,$$

it preserves area. Gaussian curvature is intrinsic and is determined by the first fundamental form, so it is also preserved. ∎

Theorem 3.14.6. *A local isometry carries geodesics to geodesics and preserves geodesic curvature.*

Proof. A local isometry preserves the first fundamental form and therefore the Levi–Civita connection determined by that metric. Hence the equation $\nabla_T T = 0$ is preserved, so geodesics map to geodesics. Since geodesic curvature is the norm of the tangential component of $\nabla_T T$, it is also preserved. ∎

Example 3.14.7 (Plane and Cylinder)**.** (See Figure 42 in the Atlas.) The plane

$$\mathbf{r}(u, v) = (u, v, 0)$$

has metric

$$ds^2 = du^2 + dv^2.$$

The circular cylinder

$$\widetilde{\mathbf{r}}(u, v) = \left(a\cos\frac{u}{a}, a\sin\frac{u}{a}, v\right)$$

also has

$$d\widetilde{s}^2 = du^2 + dv^2.$$

Hence the plane and cylinder are locally isometric. The correspondence is not globally one-to-one when the whole plane is used, because the coordinate u is periodic on the cylinder.

Example 3.14.8 (Catenoid and Right Helicoid)**.** (See Figure 43 in the Atlas.) Show that a catenoid and a right helicoid are locally isometric.

Solution. Consider the catenoid

$$\mathbf{r}(u, v) = (a\cosh u\cos v,\ a\cosh u\sin v,\ au).$$

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Its first fundamental form is

$$ds^2 = a^2 \cosh^2 u \, (du^2 + dv^2).$$

Consider the helicoid

$$\tilde{\mathbf{r}}(\rho, v) = (\rho \cos v, \ \rho \sin v, \ av).$$

Its metric is

$$d\tilde{s}^2 = d\rho^2 + (\rho^2 + a^2)\, dv^2.$$

Set

$$\rho = a \sinh u.$$

Then

$$d\rho = a \cosh u \, du, \qquad \rho^2 + a^2 = a^2 \cosh^2 u.$$

Therefore

$$d\tilde{s}^2 = a^2 \cosh^2 u \, du^2 + a^2 \cosh^2 u \, dv^2 = ds^2.$$

Hence the two surfaces are locally isometric. ■

Theorem 3.14.9 (Gauss's Theorema Egregium)**.** *The Gaussian curvature of a surface is determined entirely by its first fundamental form. Consequently, Gaussian curvature is preserved by every local isometry.*

Proof. In local coordinates, the Christoffel symbols are determined by E, F, G and their first derivatives. The Gauss equation expresses the curvature of the induced metric, and hence K, in terms of these Christoffel symbols and their first derivatives. Thus K depends only on the coefficients of the first fundamental form and their derivatives. Since a local isometry preserves the first fundamental form, it preserves K. ■

Theorem 3.14.10 (Minding's Theorem)**.** *Any two surfaces having the same constant Gaussian curvature are locally isometric.*

Proof. Choose geodesic coordinates in a neighbourhood of a point so that the metric has the form

$$ds^2 = du^2 + g(u, v)^2 \, dv^2.$$

For such coordinates the Gaussian curvature satisfies $g_{uu} + Kg = 0$. Along the initial geodesic one may impose the standard initial conditions

$$g(0, v) = 1, \qquad g_u(0, v) = 0.$$

If two surfaces have the same constant curvature K_0, the function g satisfies the same ordinary differential equation in u with the same initial conditions on both surfaces. Uniqueness therefore gives the same local metric.

For $K_0 = 0$ one obtains $g = 1$, giving the Euclidean metric. For $K_0 = 1/a^2$ one obtains

$$g = \cos \frac{u}{a},$$

which is locally the metric of a sphere of radius a. For $K_0 = -1/a^2$ one obtains

$$g = \cosh \frac{u}{a},$$

which gives the standard local metric of a surface of constant negative curvature. Hence surfaces with the same constant Gaussian curvature have locally identical first fundamental forms and are therefore locally isometric. ■

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Example 3.14.11 (A Local Isometry from the Plane to a Cone)**.** (See Figure 44 in the Atlas.) Consider the cone $\mathbf{r}(u,v) = (u\cos(v/\sin\alpha), u\sin(v/\sin\alpha), u\cot\alpha)$, where $0 < \alpha < \pi/2$. Show that it is locally isometric to a planar sector.

Solution. Differentiation gives $E = \csc^2\alpha$, $F = 0$, and $G = u^2/\sin^2\alpha$. Put $\rho = u/\sin\alpha$. Then $d\rho = du/\sin\alpha$, and the metric becomes $ds^2 = d\rho^2 + \rho^2\,dv^2$, which is the polar metric of the plane. Therefore the cone is locally isometric to a planar sector. ∎

3.15 Summary of the Main Formulae

For quick reference, the principal formulae of this unit are

$$k_n = \frac{L\,du^2 + 2M\,du\,dv + N\,dv^2}{E\,du^2 + 2F\,du\,dv + G\,dv^2},$$

$$(EG - F^2)k^2 - (EN + GL - 2FM)k + (LN - M^2) = 0,$$

$$H = \frac{EN + GL - 2FM}{2(EG - F^2)}, \qquad K = \frac{LN - M^2}{EG - F^2},$$

$$(EM - FL)\,du^2 + (EN - GL)\,du\,dv + (FN - GM)\,dv^2 = 0,$$

$$d\mathbf{N} + k_n\,d\mathbf{r} = 0 \qquad \text{along a line of curvature,}$$

$$k_n = k_1\cos^2\theta + k_2\sin^2\theta, \qquad H^2 - K = \frac{(k_1 - k_2)^2}{4},$$

$$\mathbf{N}_u \times \mathbf{N}_v = K(\mathbf{r}_u \times \mathbf{r}_v),$$

$$dA_{S^2} = |K|\,dA,$$

$$H = 0 \iff EN + GL - 2FM = 0,$$

and, in isothermal coordinates,

$$\mathbf{r}_{uu} + \mathbf{r}_{vv} = 2\lambda^2 H\mathbf{N}.$$

Thus, in isothermal coordinates,

$$H = 0 \iff \Delta x = \Delta y = \Delta z = 0.$$

3.16 Exercises

Exercise 3.16.1.

- Find the principal curvatures, mean curvature, and Gaussian curvature of the paraboloid $\mathbf{r}(u,v) = (u, v, u^2 + v^2)$ at the origin.
- Determine the principal curvatures of the circular cylinder $x^2 + y^2 = a^2$ directly from the equation for the principal curvatures.
- For the saddle surface $z = x^2 - y^2$, classify the origin and determine its asymptotic directions.
- Show that every point of a plane is an umbilic and determine its mean and Gaussian curvatures.
- Verify the Weingarten equations for the circular cylinder and identify the principal directions from them.
- Find the Gauss map of the torus $\mathbf{r}(u,v) = ((R + a\cos u)\cos v, (R + a\cos u)\sin v, a\sin u)$ and determine where the Gauss map is singular.

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- Determine the lines of curvature on the sphere using Rodrigues' formula.
- Prove directly from Euler's formula that the normal curvatures in two perpendicular directions have constant sum.
- Find the Dupin indicatrix at the origin of the surface $z = x^2 + 2y^2$ and classify the point.
- Find the Dupin indicatrix at the origin of the surface $z = x^2 - y^2$ and determine its asymptotes.
- Show that the right helicoid $\mathbf{r}(u, v) = (u\cos v, u\sin v, av)$ is a ruled surface and determine both families of asymptotic lines.
- Verify that the cone $\mathbf{R}(u, v) = (u\cos v, u\sin v, cu)$ is developable away from its vertex.
- Show directly that the catenoid $\mathbf{r}(u, v) = (a\cosh u\cos v, a\cosh u\sin v, au)$ has zero mean curvature.
- Determine whether the graph $z = x^2 - y^2$ is minimal. Justify the conclusion from the mean-curvature equation for a graph.
- Find an isothermal coordinate system for the circular cylinder and write its metric in conformal form.
- Show that the map from the plane with polar coordinates (r, θ) to the cone obtained by identifying an angular sector is a local isometry, and explain why it cannot be a global isometry of the whole plane onto the cone.

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CHAPTER

4

INTRINSIC GEOMETRY AND GEODESICS

This chapter explores the intrinsic and extrinsic geometry of surfaces through Christoffel symbols, the celebrated Theorema Egregium, and the fundamental Gauss and Mainardi-Codazzi compatibility equations.
We develop the mathematical framework of geodesics, defining their intrinsic curvature, establishing their governing differential equations, and examining their trajectories on spheres and pseudospheres. Furthermore, we state the fundamental theorem for regular surfaces, which ensures the unique determination of a surface up to rigid motions from its fundamental forms.
Finally, we introduce the Gauss–Bonnet theorem, exploring its profound implications for geodesic triangles and demonstrating how local curvature dictates global topological properties.

4.1 Christoffel Symbols and Fundamental Equations of Surfaces

Let $S \subset \mathbb{R}^3$ be a smooth regular surface and let $\mathbf{r} : U \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a local parametrization of S with coordinates (u^1, u^2). The tangent space T_pS at $p = \mathbf{r}(u, v)$ is spanned by the coordinate tangent vectors:

$$\mathbf{r}_1 = \frac{\partial \mathbf{r}}{\partial u}, \quad \mathbf{r}_2 = \frac{\partial \mathbf{r}}{\partial v}.$$

The first fundamental magnitudes are given by

$$E = r_1 \cdot r_1 = r_1^2, \quad F = r_1 \cdot r_2 = r_2 \cdot r_1, \quad G = r_2 \cdot r_2 = r_2^2$$

If $\vec{N}$ denotes the unit normal vector to the surface, then the second fundamental magnitudes are given by

$$L = r_{11} \cdot \vec{N}, \quad M = r_{12} \cdot \vec{N}, \quad N = r_{22} \cdot \vec{N}$$

Definition 4.1.1. For $i, j, k = 1, 2$, the notations

$$\Gamma_{ijk} = \frac{1}{2} \left\{ (\mathbf{r}_i \cdot \mathbf{r}_j)_k + (\mathbf{r}_i \cdot \mathbf{r}_k)_j - (\mathbf{r}_j \cdot \mathbf{r}_k)_i \right\}$$

where $(\,)_k, (\,)_j, (\,)_i$ denote differentiation with respect to $\mathbf{r}_k, \mathbf{r}_j$ and $\mathbf{r}_i$ respectively, are called the Christoffel symbols.

Using the definition, the above expression simplifies to $\Gamma_{ijk} = \mathbf{r}_i \cdot \mathbf{r}_{jk}$ as shown below.

$$\begin{aligned}
\Gamma_{ijk} &= \mathbf{r}_i \cdot \mathbf{r}_{jk} \\
&= \frac{1}{2}\{(r_{ik} \cdot r_j) + r_i \cdot r_{jk} + r_{ij} \cdot r_k + r_i \cdot r_{jk} - r_j \cdot r_{ki} - r_k \cdot r_{ij}\} \\
&= \frac{1}{2}\{2r_i \cdot r_{jk}\} = r_i \cdot r_{jk}.
\end{aligned}$$

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For particular values of $i, j, k = 1, 2$, we have

$$\Gamma_{111} = \mathbf{r}_1 \cdot \mathbf{r}_{11} = \frac{1}{2}E_1 \qquad \Gamma_{211} = \mathbf{r}_2 \cdot \mathbf{r}_{11} = F_1 - \frac{1}{2}E_2$$

$$\Gamma_{112} = \mathbf{r}_1 \cdot \mathbf{r}_{12} = \frac{1}{2}E_2 \qquad \Gamma_{212} = \mathbf{r}_2 \cdot \mathbf{r}_{12} = \frac{1}{2}G_1$$

$$\Gamma_{121} = \mathbf{r}_1 \cdot \mathbf{r}_{21} = \frac{1}{2}E_2 \qquad \Gamma_{221} = \mathbf{r}_2 \cdot \mathbf{r}_{21} = \frac{1}{2}G_1$$

$$\Gamma_{122} = \mathbf{r}_1 \cdot \mathbf{r}_{22} = F_2 - \frac{1}{2}G_1 \qquad \Gamma_{222} = \mathbf{r}_2 \cdot \mathbf{r}_{22} = \frac{1}{2}G_2$$

Also we note from the above

$$\Gamma_{112} = \Gamma_{121} \quad \text{and} \quad \Gamma_{221} = \Gamma_{212}$$

The Christoffel symbols of the second kind denoted by

$$\Gamma^i_{jk},\ i = 1, 2,\ j, k = 1, 2 \text{ are defined as}$$

$$\Gamma^1_{jk} = \boldsymbol{H}^{-2}(G\Gamma_{1jk} - F\Gamma_{2jk}),\ \Gamma^2_{jk} = \boldsymbol{H}^{-2}(E\Gamma_{2jk} - F\Gamma_{1jk})$$

For particular values i, j, k, we have the following

$$\Gamma^1_{11} = \frac{1}{\boldsymbol{H}^2}(G\Gamma_{111} - F\Gamma_{211}) = \frac{1}{2\boldsymbol{H}^2}(GE_1 - 2FF_1 + FE_2)$$

$$\Gamma^1_{12} = \frac{1}{\boldsymbol{H}^2}(G\Gamma_{112} - F\Gamma_{212}) = \frac{1}{2\boldsymbol{H}^2}(GE_2 - FG_1)$$

$$\Gamma^1_{22} = \frac{1}{\boldsymbol{H}^2}(G\Gamma_{122} - F\Gamma_{222}) = \frac{1}{2\boldsymbol{H}^2}(2GF_2 - GG_1 - FG_2)$$

$$\Gamma^2_{11} = \frac{1}{\boldsymbol{H}^2}(E\Gamma_{211} - F\Gamma_{111}) = \frac{1}{2\boldsymbol{H}^2}(2EF_1 - EE_2 - FE_1)$$

$$\Gamma^2_{12} = \frac{1}{\boldsymbol{H}^2}(E\Gamma_{212} - F\Gamma_{112}) = \frac{1}{2\boldsymbol{H}^2}(EG_1 - FE_2)$$

$$\Gamma^2_{22} = \frac{1}{\boldsymbol{H}^2}(E\Gamma_{222} - F\Gamma_{122}) = \frac{1}{2\boldsymbol{H}^2}(EG_2 - 2FF_2 + FG_1)$$

Theorem 4.1.2. *If $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ are three linearly independent vectors at a point P on the surface $\mathbf{r} = \mathbf{r}(u, v)$, then*

$$\mathbf{r}_{11} = \Gamma^1_{11}\mathbf{r}_1 + \Gamma^2_{11}\mathbf{r}_2 + L\mathbf{N}$$
$$\mathbf{r}_{12} = \Gamma^1_{12}\mathbf{r}_1 + \Gamma^2_{12}\mathbf{r}_2 + M\mathbf{N}$$
$$\mathbf{r}_{22} = \Gamma^1_{22}\mathbf{r}_1 + \Gamma^2_{22}\mathbf{r}_2 + N\mathbf{N}$$

Proof. At a point P on surface, let us introduce a coordinate system consisting of three linearly independent vectors $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ of which $\mathbf{r}_1$ and $\mathbf{r}_2$ lie in the tangent plane at P and $\mathbf{N}$ is the unit normal to the surface. With respect to this coordinate system, let us express $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ at P as linear combinations of these basic vectors.

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Hence we can represent

$$\mathbf{r}_{11} = \alpha_1\mathbf{r}_1 + \alpha_2\mathbf{r}_2 + \alpha_3\mathbf{N} \tag{1}$$
$$\mathbf{r}_{12} = \beta_1\mathbf{r}_1 + \beta_2\mathbf{r}_2 + \beta_3\mathbf{N} \tag{2}$$
$$\mathbf{r}_{22} = \gamma_1\mathbf{r}_1 + \gamma_2\mathbf{r}_2 + \gamma_3\mathbf{N} \tag{3}$$

where we determine $\alpha_i, \beta_i, \gamma_i, i = 1, 2, 3$.

Taking dot product with $\mathbf{N}$ on both sides of the above equations and using $\mathbf{r}_1 \cdot \mathbf{N} = \mathbf{r}_2 \cdot \mathbf{N} = 0$, we obtain

$$\mathbf{N} \cdot \mathbf{r}_{11} = \alpha_3\mathbf{N} \cdot \mathbf{N}, \quad \mathbf{N} \cdot \mathbf{r}_{12} = \beta_3\mathbf{N} \cdot \mathbf{N}, \quad \mathbf{N} \cdot \mathbf{r}_{22} = \gamma_3\mathbf{N} \cdot \mathbf{N}$$

Since $\mathbf{N} \cdot \mathbf{r}_{11} = L$, $\mathbf{N} \cdot \mathbf{r}_{12} = M$, $\mathbf{N} \cdot \mathbf{r}_{22} = N$ and $\mathbf{N} \cdot \mathbf{N} = 1$, we obtain

$$\alpha_3 = L, \quad \beta_3 = M \text{ and } \gamma_3 = N \tag{4}$$

Having determined α_3, β_3 and γ_3, we find $\alpha_1, \beta_1, \gamma_1$, and $\alpha_2, \beta_2, \gamma_2$ as follows. To determine these, we need the derivatives of the fundamental coefficients E, F and G.

$$E_1 = \frac{\partial E}{\partial u} = \frac{\partial}{\partial u}(\mathbf{r}_1^2) = 2\mathbf{r}_1 \cdot \mathbf{r}_{11}, \quad \mathbf{r}_1 \cdot \mathbf{r}_{11} = \frac{1}{2}E_1 \tag{5}$$

$$E_2 = \frac{\partial E}{\partial v} = \frac{\partial}{\partial v}(\mathbf{r}_1^2) = 2\mathbf{r}_1 \cdot \mathbf{r}_{12}, \quad \mathbf{r}_1 \cdot \mathbf{r}_{12} = \frac{1}{2}E_2 \tag{6}$$

$$G_1 = \frac{\partial G}{\partial u} = \frac{\partial}{\partial u}(\mathbf{r}_2^2) = 2\mathbf{r}_2 \cdot \mathbf{r}_{21}, \quad \mathbf{r}_2 \cdot \mathbf{r}_{21} = \frac{1}{2}G_1 \tag{7}$$

$$G_2 = \frac{\partial G}{\partial v} = \frac{\partial}{\partial v}(\mathbf{r}_2^2) = 2\mathbf{r}_2 \cdot \mathbf{r}_{22}, \quad \mathbf{r}_2 \cdot \mathbf{r}_{22} = \frac{1}{2}G_2 \tag{8}$$

$$F_1 = \frac{\partial}{\partial u}(\mathbf{r}_1 \cdot \mathbf{r}_2) = \mathbf{r}_2 \cdot \mathbf{r}_{11} + \mathbf{r}_1 \cdot \mathbf{r}_{12}, \text{ so, } \mathbf{r}_2 \cdot \mathbf{r}_{11} = F_1 - \mathbf{r}_1 \cdot \mathbf{r}_{12} = F_1 - \frac{1}{2}E_2 \tag{9}$$

and

$$F_2 = \frac{\partial}{\partial v}(\mathbf{r}_1 \cdot \mathbf{r}_2) = \mathbf{r}_2 \cdot \mathbf{r}_{12} + \mathbf{r}_1 \cdot \mathbf{r}_{22}, \text{ so, } \mathbf{r}_1 \cdot \mathbf{r}_{22} = F_2 - \mathbf{r}_2 \cdot \mathbf{r}_{12} = F_2 - \frac{1}{2}G_1 \tag{10}$$

Now, let us take the dot product of (1) with $\mathbf{r}_1$ and $\mathbf{r}_2$ respectively. Then we have

$$\mathbf{r}_1 \cdot \mathbf{r}_{11} = \alpha_1\mathbf{r}_1 \cdot \mathbf{r}_1 + \alpha_2\mathbf{r}_2 \cdot \mathbf{r}_1$$
$$\mathbf{r}_2 \cdot \mathbf{r}_{11} = \alpha_1\mathbf{r}_1 \cdot \mathbf{r}_2 + \alpha_2\mathbf{r}_2 \cdot \mathbf{r}_2$$

Since $\mathbf{r}_1 \cdot \mathbf{r}_1 = E$, $\mathbf{r}_2 \cdot \mathbf{r}_2 = G$, using (5) and (9), we obtain

$$\alpha_1 E + F\alpha_2 - \frac{1}{2}E_1 = 0$$

$$\alpha_1 F + G\alpha_2 - \left(F_1 - \frac{1}{2}E_2\right) = 0$$

Solving the above two equations,

$$\frac{2\alpha_1}{GE_1 - 2FF_1 + FE_2} = \frac{2\alpha_2}{2EF_1 - EE_2 - E_1F} = \frac{1}{EG - F^2}$$

Thus we obtain

$$\alpha_1 = \frac{GE_1 - 2FF_1 + FE_2}{2(EG - F^2)} = \Gamma_{11}^1$$

$$\alpha_2 = \frac{2EF_1 - EE_2 - E_1F}{2(EG - F^2)} = \Gamma_{11}^2$$

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Hence the equation (1) becomes

$$\mathbf{r}_{11} = \Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}$$

Taking dot product with $\mathbf{r}_1$ and $\mathbf{r}_2$ of (2),

$$\mathbf{r}_1 \cdot \mathbf{r}_{12} = \beta_1 \mathbf{r}_1 \cdot \mathbf{r}_1 + \beta_2 \mathbf{r}_2 \cdot \mathbf{r}_1$$
$$\mathbf{r}_2 \cdot \mathbf{r}_{12} = \beta_1 \mathbf{r}_1 \cdot \mathbf{r}_2 + \beta_2 \mathbf{r}_2 \cdot \mathbf{r}_2$$

Using the fundamental coefficients and (6) and (7),

$$\beta_1 E + \beta_2 F - \frac{1}{2} E_2 = 0$$

$$\beta_1 F + \beta_2 G - \frac{1}{2} G_1 = 0.$$

Solving the above two equations, we obtain

$$\frac{2\beta_1}{GE_2 - FG_1} = \frac{2\beta_2}{EG_1 - FE_2} = \frac{1}{EG - F^2}$$

Hence

$$\beta_1 = \frac{GE_2 - FG_1}{2(EG - F^2)} = \Gamma_{12}^1$$

$$\beta_2 = \frac{EG_1 - FE_2}{2(EG - F^2)} = \Gamma_{12}^2$$

Let us take dot product of (3) with $\mathbf{r}_1$ and $\mathbf{r}_2$ then

$$\mathbf{r}_1 \cdot \mathbf{r}_{22} = \gamma_1 \mathbf{r}_1 \cdot \mathbf{r}_1 + \gamma_2 \mathbf{r}_1 \cdot \mathbf{r}_2$$
$$\mathbf{r}_2 \cdot \mathbf{r}_{22} = \gamma_1 \mathbf{r}_1 \cdot \mathbf{r}_2 + \gamma_2 \mathbf{r}_2 \cdot \mathbf{r}_2$$

So the simultaneous equations in γ_1 and γ_2 become

$$E\gamma_1 + F\gamma_2 - \left(F_2 - \frac{1}{2} G_1\right) = 0$$

$$F\gamma_1 + G\gamma_2 - \frac{1}{2} G_2 = 0$$

Solving these two equations, we obtain

$$\gamma_1 = \frac{2GF_2 - GG_1 - FG_2}{2(EG - F^2)} = \Gamma_{22}^1$$

$$\gamma_2 = \frac{EG_2 - 2FF_2 + FG_1}{2(EG - F^2)} = \Gamma_{22}^2$$

Thus the equations (1), (2) and (3) become

$$\mathbf{r}_{11} = \Gamma_{11}^1 \mathbf{r}_1 + \Gamma_{11}^2 \mathbf{r}_2 + L\mathbf{N}$$
$$\mathbf{r}_{12} = \Gamma_{12}^1 \mathbf{r}_1 + \Gamma_{12}^2 \mathbf{r}_2 + M\mathbf{N}$$
$$\mathbf{r}_{22} = \Gamma_{22}^1 \mathbf{r}_1 + \Gamma_{22}^2 \mathbf{r}_2 + N\mathbf{N}$$

The above three equations are known as Gauss equations. $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ are called Gauss coefficients. Thus the Gauss equations give Gauss coefficients in terms of E, F and G and their partial derivatives and L, M, N. These equations are also known as the partial differential equations of the surface theory. ■

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Corollary 4.1.3. *If the parametric curves are orthogonal, then the Gauss equations are*

$$\begin{aligned}\mathbf{r}_{11} &= \frac{E_1}{2E}\mathbf{r}_1 - \frac{E_2}{2G}\mathbf{r}_2 + L\mathbf{N}\\ \mathbf{r}_{12} &= \frac{E_2}{2E}\mathbf{r}_1 + \frac{G_1}{2G}\mathbf{r}_2 + M\mathbf{N}\\ \mathbf{r}_{22} &= -\frac{G_1}{2E}\mathbf{r}_1 + \frac{G_2}{2G}\mathbf{r}_2 + N\mathbf{N}\end{aligned}$$

Proof. Since the parametric curves are orthogonal, $F = 0$ and $\boldsymbol{H}^2 = EG$. Using these conditions, the Christoffel symbol coefficients in the Gauss equations became

$$\Gamma^1_{11} = \frac{GE_1}{2EG} = \frac{E_1}{2E} \qquad \Gamma^2_{11} = \frac{-EE_2}{2EG} = \frac{-E_2}{2G}$$

$$\Gamma^1_{12} = \frac{GE_2}{2EG} = \frac{E_2}{2E} \qquad \Gamma^2_{12} = \frac{EG_1}{2EG} = \frac{G_1}{2G}$$

$$\Gamma^1_{22} = \frac{-GG_1}{2EG} = \frac{-G_1}{2E} \qquad \Gamma^2_{22} = \frac{EG_2}{2EG} = \frac{G_2}{2G}$$

Using these coefficients, the Gauss equations take the given form. ■

Corollary 4.1.4. *If the lines of curvature are parametric curves on the surface, then the Gauss equations are*

$$\begin{aligned}\mathbf{r}_{11} &= \frac{E_1}{2E}\mathbf{r}_1 - \frac{E_2}{2G}\mathbf{r}_2 + L\mathbf{N}\\ \mathbf{r}_{12} &= \frac{E_2}{2E}\mathbf{r}_1 + \frac{G_1}{2G}\mathbf{r}_2\\ \mathbf{r}_{22} &= \frac{-G_1}{2E}\mathbf{r}_1 + \frac{G_2}{2G}\mathbf{r}_2 + N\mathbf{N}\end{aligned}$$

Proof. When the lines of curvature are chosen as parametric curves, then $F = 0$, $M = 0$ and $\boldsymbol{H}^2 = EG$. Using these conditions, we can rewrite the Gauss equations in the above form. ■

Theorem 4.1.5 (Mainardi-COdazzi Equations). *The Mainardi-Codazzi equations are*

$$\begin{aligned}\frac{\partial L}{\partial v} - \frac{\partial M}{\partial u} &= L\Gamma^1_{12} - N\Gamma^2_{11} + M(\Gamma^2_{12} - \Gamma^1_{11}),\\ \frac{\partial M}{\partial v} - \frac{\partial N}{\partial u} &= L\Gamma^1_{22} + M(\Gamma^2_{22} - \Gamma^1_{12}) - N\Gamma^2_{12}.\end{aligned}$$

Proof. We have have the compatibility conditions

$$\begin{aligned}\frac{\partial}{\partial v}\left(\frac{\partial^2 r}{\partial u^2}\right) &= \frac{\partial}{\partial u}\left(\frac{\partial^2 r}{\partial v \partial u}\right), \text{ or } (r_{11})_2 = (r_{12})_1\\ \frac{\partial}{\partial u}\left(\frac{\partial^2 r}{\partial v^2}\right) &= \frac{\partial}{\partial v}\left(\frac{\partial^2 r}{\partial v \partial u}\right), \text{ or } (r_{12})_2 = (r_{22})_1\end{aligned}$$

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Substituting for $\mathbf{r}_{11}$, $\mathbf{r}_{12}$ and $\mathbf{r}_{22}$ from the Gauss equations into the compatibility conditions:

$$\frac{\partial}{\partial v}\left[\Gamma_{11}^1\mathbf{r}_1+\Gamma_{11}^2\mathbf{r}_2+L\mathbf{N}\right]=\frac{\partial}{\partial u}\left[\Gamma_{12}^1\mathbf{r}_1+\Gamma_{12}^2\mathbf{r}_2+M\mathbf{N}\right], \tag{1}$$

$$\frac{\partial}{\partial v}\left[\Gamma_{12}^1\mathbf{r}_1+\Gamma_{12}^2\mathbf{r}_2+M\mathbf{N}\right]=\frac{\partial}{\partial u}\left[\Gamma_{22}^1\mathbf{r}_1+\Gamma_{22}^2\mathbf{r}_2+N\mathbf{N}\right]. \tag{2}$$

Using Gauss equations, we obtain each term of the first equation above:

$$\begin{aligned}\frac{\partial}{\partial v}[\Gamma_{11}^1\mathbf{r}_1]&=\frac{\partial}{\partial v}(\Gamma_{11}^1)\mathbf{r}_1+\Gamma_{11}^1\mathbf{r}_{12}\\&=\frac{\partial}{\partial v}(\Gamma_{11}^1)\mathbf{r}_1+\Gamma_{11}^1(\Gamma_{12}^1\mathbf{r}_1+\Gamma_{12}^2\mathbf{r}_2+M\mathbf{N}),\\\frac{\partial}{\partial v}[\Gamma_{11}^2\mathbf{r}_2]&=\frac{\partial}{\partial v}(\Gamma_{11}^2)\mathbf{r}_2+\Gamma_{11}^2\mathbf{r}_{22}\\&=\frac{\partial}{\partial v}(\Gamma_{11}^2)\mathbf{r}_2+\Gamma_{11}^2(\Gamma_{22}^1\mathbf{r}_1+\Gamma_{22}^2\mathbf{r}_2+N\mathbf{N}).\end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial u}(\Gamma_{12}^1\mathbf{r}_1)&=\frac{\partial}{\partial u}(\Gamma_{12}^1)\mathbf{r}_1+\Gamma_{12}^1\mathbf{r}_{11}\\&=\frac{\partial}{\partial u}(\Gamma_{12}^1)\mathbf{r}_1+\Gamma_{12}^1[\Gamma_{11}^1\mathbf{r}_1+\Gamma_{11}^2\mathbf{r}_2+L\mathbf{N}],\\\frac{\partial}{\partial u}[\Gamma_{12}^2\mathbf{r}_2]&=\frac{\partial}{\partial u}(\Gamma_{12}^2)\mathbf{r}_2+\Gamma_{12}^2\mathbf{r}_{21}\\&=\frac{\partial}{\partial u}(\Gamma_{12}^2)\mathbf{r}_2+\Gamma_{12}^2[\Gamma_{12}^1\mathbf{r}_1+\Gamma_{12}^2\mathbf{r}_2+M\mathbf{N}].\end{aligned}$$

Using Weingarten equations,

$$\begin{aligned}\frac{\partial}{\partial v}[L\mathbf{N}]&=\frac{\partial L}{\partial v}\mathbf{N}+L\frac{\partial\mathbf{N}}{\partial v}=\frac{\partial L}{\partial v}\mathbf{N}+L\mathbf{N}_2\\&=\frac{\partial L}{\partial v}\mathbf{N}+\frac{L}{H^2}\left[(FN-GM)\mathbf{r}_1+(FM-EN)\mathbf{r}_2\right].\\\frac{\partial}{\partial u}(M\mathbf{N})&=\frac{\partial M}{\partial u}\mathbf{N}+M\frac{\partial\mathbf{N}}{\partial u}=\frac{\partial M}{\partial u}\mathbf{N}+M\mathbf{N}_1\\&=\frac{\partial M}{\partial u}\mathbf{N}+\frac{M}{H^2}\left[(FM-GL)\mathbf{r}_1+(FL-EM)\mathbf{r}_2\right].\end{aligned}$$

Now gathering the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$, we have the left-hand side of the equation (1) as:

$$\begin{aligned}&\left[\frac{\partial}{\partial v}(\Gamma_{11}^1)+\Gamma_{11}^1\Gamma_{12}^1+\Gamma_{11}^2\Gamma_{22}^1+\frac{L(FN-GM)}{EG-F^2}\right]\mathbf{r}_1+\left[\frac{\partial}{\partial v}(\Gamma_{11}^2)+\Gamma_{11}^1\Gamma_{12}^2+\Gamma_{11}^2\Gamma_{22}^2+\frac{L(FM-EN)}{EG-F^2}\right]\mathbf{r}_2\\&\qquad+\left[\Gamma_{11}^1M+\Gamma_{11}^2N+\frac{\partial L}{\partial v}\right]\mathbf{N}\end{aligned} \tag{3}$$

and gathering the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$, the right-hand side of the equation (1) gives

$$\begin{aligned}&\left[\frac{\partial}{\partial u}(\Gamma_{12}^1)+\Gamma_{12}^1\Gamma_{11}^1+\Gamma_{12}^2\Gamma_{12}^1+\frac{M}{H^2}(FM-GL)\right]\mathbf{r}_1+\left[\frac{\partial}{\partial u}(\Gamma_{12}^2)+\Gamma_{12}^1\Gamma_{11}^2+\Gamma_{12}^2\Gamma_{12}^2+\frac{M}{H^2}(FL-EM)\right]\mathbf{r}_2\\&\qquad+\left[\Gamma_{12}^1L+\Gamma_{12}^2M+\frac{\partial M}{\partial u}\right]\mathbf{N}.\end{aligned} \tag{4}$$

The coefficient of $\mathbf{N}$ gives

$$\frac{\partial L}{\partial v}-\frac{\partial M}{\partial u}=L\Gamma_{12}^1-N\Gamma_{11}^2+M(\Gamma_{12}^2-\Gamma_{11}^1).$$

which is the first Mainardi-Codazzi equation. The other Mainardi-Codazzi equation can similarly be obtained from equation 3.2. ■

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Theorem 4.1.6 (Theorema Egregium).

Proof. Since the expressions (3) and (4) represent both sides of the equation (1), we equate the coefficients of $\mathbf{r}_1$, $\mathbf{r}_2$ and $\mathbf{N}$ in both parts.

Equating the coefficient of $\mathbf{r}_1$, we obtain

$$\frac{\partial}{\partial v}(\Gamma^1_{11}) + \Gamma^1_{11}\Gamma^1_{12} + \Gamma^2_{11}\Gamma^1_{22} + \frac{L(FN-GM)}{EG-F^2} = \frac{\partial}{\partial u}(\Gamma^1_{12}) + \Gamma^1_{12}\Gamma^1_{11} + \Gamma^2_{12}\Gamma^1_{12} + \frac{M(FM-GL)}{EG-F^2}.$$

Simplifying the above equation, we have

$$F\frac{(LN-M^2)}{EG-F^2} = \frac{\partial}{\partial u}(\Gamma^1_{12}) - \frac{\partial}{\partial v}(\Gamma^1_{11}) + \Gamma^2_{12}\Gamma^1_{12} - \Gamma^2_{11}\Gamma^1_{22}. \tag{1}$$

Equating the coefficient of $\mathbf{r}_2$, we have

$$\frac{\partial}{\partial v}(\Gamma^2_{11}) + \Gamma^1_{11}\Gamma^2_{12} + \Gamma^2_{11}\Gamma^2_{22} + \frac{L(FM-EN)}{EG-F^2} = \frac{\partial}{\partial u}(\Gamma^2_{12}) + \Gamma^1_{12}\Gamma^2_{11} + \Gamma^2_{12}\Gamma^2_{12} + \frac{M(FL-EM)}{EG-F^2}.$$

Simplifying the above equation, we obtain

$$-E\frac{(LN-M^2)}{EG-F^2} = \frac{\partial}{\partial u}(\Gamma^2_{12}) - \frac{\partial}{\partial v}(\Gamma^2_{11}) + \Gamma^1_{12}\Gamma^2_{11} - \Gamma^1_{11}\Gamma^2_{12} + \Gamma^2_{12}\Gamma^2_{12} - \Gamma^2_{11}\Gamma^2_{22}. \tag{2}$$

Using $\dfrac{LN-M^2}{EG-F^2} = k_n$ in equations (1) and (2), we obtain

$$k_n = \frac{1}{F}\frac{\partial}{\partial u}(\Gamma^1_{12}) - \frac{\partial}{\partial v}(\Gamma^1_{11}) + \Gamma^2_{12}\Gamma^1_{12} - \Gamma^2_{11}\Gamma^1_{22}$$
$$k_n = \frac{-1}{E}\frac{\partial}{\partial u}(\Gamma^2_{12}) - \frac{\partial}{\partial v}(\Gamma^2_{11}) + \Gamma^1_{12}\Gamma^2_{11} - \Gamma^1_{11}\Gamma^2_{12} + \Gamma^2_{12}\Gamma^2_{12} - \Gamma^2_{11}\Gamma^2_{22}$$

These equations, obtained from the coefficients of $\mathbf{r}_1$ and $\mathbf{r}_2$ give the Gaussian curvature in terms of E, F and G with their partial derivatives of first and second order. A property depending only on E, F, G and their partial derivatives of first and second order is known as an intrinsically invariant property. Thus the Gaussian curvature is intrinsically invariant. Gauss considered this as a most excellent theorem. In the words of Gauss, "it is Theorema egregium." ■

4.2 Gauss's Theorema Egregium

4.3 Gauss's Theorema Egregium

One of the most remarkable results in differential geometry is Gauss's Theorema Egregium, which states that the Gaussian curvature of a regular surface is determined entirely by its first fundamental form. Although Gaussian curvature is initially defined using the principal curvatures, and hence the way a surface bends in $\mathbb{R}^3$, Gauss proved that it can be calculated solely from the intrinsic geometry of the surface.

To understand the significance of this theorem, we first introduce the concepts of isometry and local isometry between surfaces.

4.3.1 Proof of Gauss's Theorema Egregium

Recall that the Gaussian curvature of a regular surface is defined as the product of its principal curvatures, $K = k_1k_2$. In terms of the first and second fundamental forms, it is given by

$$K = \frac{LN-M^2}{EG-F^2},$$

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where E, F, G are the coefficients of the first fundamental form and L, M, N are the coefficients of the second fundamental form.

At first sight, this formula suggests that Gaussian curvature depends on the second fundamental form, which measures the bending of the surface in the ambient space. Gauss's remarkable discovery was that the quantity $LN - M^2$ can itself be expressed entirely in terms of the first fundamental form and its derivatives.

Theorem 4.3.1 (Gauss's Theorema Egregium). *The Gaussian curvature K of a regular surface can be expressed solely in terms of the coefficients E, F, G of its first fundamental form and their first and second derivatives. Consequently, Gaussian curvature is invariant under local isometries.*

Proof. Let $\mathbf{x}(u, v)$ be a local parametrization of a regular surface S, and let $\mathbf{N}$ denote a chosen unit normal vector field. Since $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent and perpendicular to $\mathbf{N}$, the vectors $\{\mathbf{x}_u, \mathbf{x}_v, \mathbf{N}\}$ form a basis of $\mathbb{R}^3$.

Consequently, the second partial derivatives of $\mathbf{x}$ can be expressed in this basis as

$$\begin{aligned}\mathbf{x}_{uu} &= \Gamma_{11}^1 \mathbf{x}_u + \Gamma_{11}^2 \mathbf{x}_v + L\mathbf{N},\\ \mathbf{x}_{uv} &= \Gamma_{12}^1 \mathbf{x}_u + \Gamma_{12}^2 \mathbf{x}_v + M\mathbf{N},\\ \mathbf{x}_{vv} &= \Gamma_{22}^1 \mathbf{x}_u + \Gamma_{22}^2 \mathbf{x}_v + N\mathbf{N}.\end{aligned}$$

These equations are called the **Gauss formulas**, and the coefficients Γ_{ij}^k are the Christoffel symbols of the surface.

We first show that the Christoffel symbols depend only on the first fundamental form. Taking inner products of the first Gauss formula with $\mathbf{x}_u$ and $\mathbf{x}_v$, we obtain

$$E\Gamma_{11}^1 + F\Gamma_{11}^2 = \langle \mathbf{x}_{uu}, \mathbf{x}_u \rangle$$

and

$$F\Gamma_{11}^1 + G\Gamma_{11}^2 = \langle \mathbf{x}_{uu}, \mathbf{x}_v \rangle .$$

Now, differentiating $E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle$ with respect to u gives

$$\langle \mathbf{x}_{uu}, \mathbf{x}_u \rangle = \frac{1}{2} E_u.$$

Similarly, differentiating $F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle$ with respect to u and using $E_v = 2 \langle \mathbf{x}_{uv}, \mathbf{x}_u \rangle$, we obtain

$$\langle \mathbf{x}_{uu}, \mathbf{x}_v \rangle = F_u - \frac{1}{2} E_v.$$

Therefore,

$$E\Gamma_{11}^1 + F\Gamma_{11}^2 = \frac{1}{2} E_u, \qquad F\Gamma_{11}^1 + G\Gamma_{11}^2 = F_u - \frac{1}{2} E_v.$$

Since $EG - F^2 > 0$ for a regular surface, these equations determine Γ_{11}^1 and Γ_{11}^2 uniquely in terms of E, F, G and their first derivatives. Applying the same procedure to the other Gauss formulas shows that all the Christoffel symbols are determined by the first fundamental form.

We next prove that $LN - M^2$ also depends only on these quantities. Since the parametrization is smooth, the mixed partial derivatives satisfy

$$\mathbf{x}_{uuv} = \mathbf{x}_{uvu}.$$

Differentiating the first Gauss formula with respect to v, we obtain

$$\begin{aligned}\mathbf{x}_{uuv} = (\Gamma_{11}^1)_v \mathbf{x}_u + \Gamma_{11}^1 \mathbf{x}_{uv} + (\Gamma_{11}^2)_v \mathbf{x}_v + \Gamma_{11}^2 \mathbf{x}_{vv}\\ + L_v \mathbf{N} + L\mathbf{N}_v.\end{aligned}$$

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Similarly, differentiating the second Gauss formula with respect to u gives

$$\begin{aligned}\mathbf{x}_{uvu} = (\Gamma^1_{12})_u\mathbf{x}_u + \Gamma^1_{12}\mathbf{x}_{uu} + (\Gamma^2_{12})_u\mathbf{x}_v + \Gamma^2_{12}\mathbf{x}_{uv}\\ + M_u\mathbf{N} + M\mathbf{N}_u.\end{aligned}$$

Taking the inner product of both equations with $\mathbf{x}_v$ eliminates the terms containing $L_v\mathbf{N}$ and $M_u\mathbf{N}$, since $\langle \mathbf{N}, \mathbf{x}_v\rangle = 0$.

Furthermore, differentiating $\langle \mathbf{N}, \mathbf{x}_v\rangle = 0$ gives

$$\langle \mathbf{N}_v, \mathbf{x}_v\rangle = -N, \qquad \langle \mathbf{N}_u, \mathbf{x}_v\rangle = -M.$$

Using these identities and the Gauss formulas, we obtain

$$\begin{aligned}&F(\Gamma^1_{11})_v + G(\Gamma^2_{11})_v + \Gamma^1_{11}(F\Gamma^1_{12} + G\Gamma^2_{12})\\ &\quad + \Gamma^2_{11}(F\Gamma^1_{22} + G\Gamma^2_{22}) - LN\\ &= F(\Gamma^1_{12})_u + G(\Gamma^2_{12})_u + \Gamma^1_{12}(F\Gamma^1_{11} + G\Gamma^2_{11})\\ &\quad + \Gamma^2_{12}(F\Gamma^1_{12} + G\Gamma^2_{12}) - M^2.\end{aligned}$$

Rearranging this equality yields

$$\begin{aligned}LN - M^2 = F\big((\Gamma^1_{11})_v - (\Gamma^1_{12})_u\big) + G\big((\Gamma^2_{11})_v - (\Gamma^2_{12})_u\big)\\ + \Gamma^1_{11}(F\Gamma^1_{12} + G\Gamma^2_{12}) + \Gamma^2_{11}(F\Gamma^1_{22} + G\Gamma^2_{22})\\ - \Gamma^1_{12}(F\Gamma^1_{11} + G\Gamma^2_{11}) - \Gamma^2_{12}(F\Gamma^1_{12} + G\Gamma^2_{12}).\end{aligned}$$

The right-hand side involves only E, F, G, the Christoffel symbols, and their first derivatives. Since the Christoffel symbols themselves depend only on E, F, G and their first derivatives, it follows that $LN - M^2$ is determined entirely by the first fundamental form and its derivatives.

Consequently,

$$K = \frac{LN - M^2}{EG - F^2}$$

depends only on E, F, G and their first and second derivatives. Thus, Gaussian curvature is an intrinsic property of the surface.

Finally, suppose that $f : S \to \bar{S}$ is a local isometry. In corresponding local coordinates, the first fundamental forms of S and $\bar{S}$ have identical coefficients. Since Gaussian curvature is determined entirely by these coefficients and their derivatives, we obtain

$$K(p) = \bar{K}(f(p))$$

at every point where the local isometry is defined. Hence Gaussian curvature is invariant under local isometries. ■

The significance of Gauss's Theorema Egregium is that Gaussian curvature can be determined by measurements made entirely within the surface, without reference to the surrounding space. Although the principal curvatures may change when a surface is bent, their product remains unchanged under any deformation that preserves the first fundamental form.

Corollary 4.3.2 (Invariance of Gaussian Curvature)**.** *If two regular surfaces are locally isometric, then they have equal Gaussian curvature at corresponding points.*

Proof. A local isometry preserves the first fundamental form. By Gauss's Theorema Egregium, Gaussian curvature is determined entirely by this form. Therefore, the Gaussian curvatures of the two surfaces agree at corresponding points. ■

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Corollary 4.3.3. *A regular surface whose Gaussian curvature is nonzero at a point cannot be locally isometric to a plane in a neighbourhood of that point.*

Proof. The Gaussian curvature of a plane is zero everywhere. If a surface were locally isometric to a plane near a point, the preceding corollary would imply that its Gaussian curvature at that point must also be zero. Hence a surface having nonzero Gaussian curvature at a point cannot be locally isometric to a plane near that point. ■

Example 4.3.4 (The Plane and the Cylinder)**.** Explain why the plane and a circular cylinder are locally isometric, although one is flat and the other is curved in $\mathbb{R}^3$.

Solution. Consider the parametrizations

$$\mathbf{x}(u,v) = (u,v,0), \qquad \bar{\mathbf{x}}(u,v) = (R\cos(u/R), R\sin(u/R), v),$$

where $R > 0$ and u is restricted to an interval of length less than $2\pi R$.

For the plane, $\mathbf{x}_u = (1,0,0)$ and $\mathbf{x}_v = (0,1,0)$. Thus, $E = 1$, $F = 0$ and $G = 1$, so

$$I = du^2 + dv^2.$$

For the cylinder,

$$\bar{\mathbf{x}}_u = (-\sin(u/R), \cos(u/R), 0), \qquad \bar{\mathbf{x}}_v = (0,0,1).$$

Hence $\bar{E} = 1$, $\bar{F} = 0$ and $\bar{G} = 1$, giving

$$\bar{I} = du^2 + dv^2.$$

Since the first fundamental forms are identical, the two surfaces are locally isometric.

The plane has Gaussian curvature $K = 0$. The cylinder has one zero principal curvature, corresponding to the direction of its generators, and one nonzero principal curvature, corresponding to the circular direction. Their product is zero, so the Gaussian curvature of the cylinder is also zero.

Thus, the equality of Gaussian curvatures is consistent with Gauss's Theorema Egregium. The example illustrates that a surface can bend in $\mathbb{R}^3$ without changing its intrinsic geometry. ■

Example 4.3.5 (A Sphere Cannot Be Locally Isometric to a Plane)**.** Show that no open portion of a sphere of radius $R > 0$ is locally isometric to a plane.

Solution. The principal curvatures of a sphere of radius R are both equal to $-1/R$ with respect to its outward orientation. Therefore, its Gaussian curvature is

$$K = \left(-\frac{1}{R}\right)\left(-\frac{1}{R}\right) = \frac{1}{R^2} > 0.$$

On the other hand, a plane has Gaussian curvature $K = 0$.

If an open portion of the sphere were locally isometric to a plane, Gauss's Theorema Egregium would require their Gaussian curvatures to be equal at corresponding points. Since $1/R^2 \neq 0$, this is impossible.

Hence no open portion of a sphere can be locally isometric to a plane. ■

Example 4.3.6 (Local Isometry between a Catenoid and a Helicoid)**.** Show that a catenoid and a helicoid are locally isometric.

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Solution. Let $a > 0$. Consider the parametrization of a catenoid

$$\mathbf{x}(u, v) = (a \cosh v \cos u, a \cosh v \sin u, av).$$

Differentiating, we obtain

$$\mathbf{x}_u = (-a \cosh v \sin u, a \cosh v \cos u, 0)$$

and

$$\mathbf{x}_v = (a \sinh v \cos u, a \sinh v \sin u, a).$$

Therefore,

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle = a^2 \cosh^2 v,$$
$$F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle = 0,$$

and

$$G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle = a^2(\sinh^2 v + 1) = a^2 \cosh^2 v.$$

Thus, the first fundamental form of the catenoid is

$$I = a^2 \cosh^2 v(du^2 + dv^2).$$

Now consider the helicoid parametrized by

$$\bar{\mathbf{x}}(u, v) = (a \sinh v \cos u, a \sinh v \sin u, au).$$

Its partial derivatives are

$$\bar{\mathbf{x}}_u = (-a \sinh v \sin u, a \sinh v \cos u, a)$$

and

$$\bar{\mathbf{x}}_v = (a \cosh v \cos u, a \cosh v \sin u, 0).$$

Consequently,

$$\bar{E} = a^2(\sinh^2 v + 1) = a^2 \cosh^2 v,$$
$$\bar{F} = 0, \qquad \bar{G} = a^2 \cosh^2 v.$$

Hence the first fundamental form of the helicoid is

$$\bar{I} = a^2 \cosh^2 v(du^2 + dv^2).$$

We therefore have

$$E = \bar{E}, \qquad F = \bar{F}, \qquad G = \bar{G}.$$

By the theorem on equality of first fundamental forms, the correspondence

$$\mathbf{x}(u, v) \longmapsto \bar{\mathbf{x}}(u, v)$$

defines a local isometry between the catenoid and the helicoid.

Thus, despite their different appearances in $\mathbb{R}^3$, the catenoid and helicoid possess the same local intrinsic geometry. ■

The preceding examples illustrate the distinction between intrinsic and extrinsic geometry. The plane and cylinder are locally isometric because their first fundamental forms agree, whereas the sphere cannot be locally isometric to a plane because their Gaussian curvatures differ. Similarly, the catenoid and helicoid have identical first fundamental forms in corresponding coordinates and are therefore locally isometric.

Gauss's Theorema Egregium establishes that the Gaussian curvature of a surface is determined not by its appearance in three-dimensional space, but by its intrinsic metric structure. This fundamental discovery provides an essential link between classical differential geometry and the more general theory of Riemannian geometry.

4.4 Vector Fields

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Definition 4.4.1 (Vector Field)**.** A **vector field** on a surface S (or more generally on a manifold) is a rule that assigns to each point $p \in S$ a tangent vector $X(p)$ that lies in the tangent plane T_pS. In other words, it is a function $X : S \to TS$ such that $X(p) \in T_pS$ for every p, where TS denotes the tangent bundle (the collection of all tangent planes).

Example 4.4.2 (Constant Vector Field on a Plane)**.** Let $S = \mathbb{R}^2$ be the plane. At each point (x, y) we assign the same tangent vector $(1, 0)$. This vector field is written as $X(x, y) = (1, 0)$. Every arrow points horizontally to the right, and all have length 1.

Example 4.4.3 (Radial Vector Field on a Sphere)**.** Consider the unit sphere $S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$. At a point $p = (x, y, z)$, the tangent plane consists of vectors orthogonal to the position vector p. Define

$$X(x, y, z) = (-y, x, 0).$$

This vector is tangent to the sphere because $(-y, x, 0) \cdot (x, y, z) = -yx + xy + 0 = 0$. The vectors point in the direction of increasing longitude (like wind blowing around the equator). This is a non-zero vector field on the sphere (though it vanishes at the poles? Actually at $(0, 0, \pm 1)$ we obtain $(0, 0, 0)$ – so it is zero at the poles; still a valid vector field).

Example 4.4.4 (Gradient Vector Field)**.** Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a smooth function, e.g., $f(x, y) = x^2 + y^2$. Its gradient $\nabla f = (2x, 2y)$ is a vector field on the plane. At each point, the gradient points in the direction of steepest ascent and is tangent to the plane (since the plane is the whole $\mathbb{R}^2$).

Definition 4.4.5 (Parallel Surface)**.** Let S be a regular surface in $\mathbb{R}^3$ with a unit normal vector field $\mathbf{N}(p)$. For a fixed constant $d \in \mathbb{R}$ (with $|d|$ small enough so that the new surface is regular), the **parallel surface** S_d is defined by $S_d = \{\, p + d\,\mathbf{N}(p) \mid p \in S \,\}$. In other words, from each point p of the original surface, we move a distance $|d|$ along the normal direction (outward if $d > 0$, inward if $d < 0$) to obtain a new point. The set of all such points forms the parallel surface.

Example 4.4.6 (Parallel Surfaces of a Sphere)**.** (See Figure 50 in the Atlas.) Take S to be the unit sphere: $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$. The outward unit normal at a point p is simply $\mathbf{N}(p) = p$ (since for a sphere centred at the origin, the normal points radially outward). For a given d (with $d > -1$ to avoid collapse), the parallel surface is $S_d = \{p + d\,p \mid \|p\| = 1\} = \{(1 + d)p \mid \|p\| = 1\}$. Thus S_d is a sphere of radius $|1 + d|$. If $d > 0$, we obtain a larger sphere; if $-1 < d < 0$, we obtain a smaller sphere (inside the original). For $d = -1$, the sphere shrinks to a point.

Example 4.4.7 (Parallel Surfaces of a Cylinder)**.** Let S be a right circular cylinder of radius R: $S = \{(x, y, z) \mid x^2 + y^2 = R^2,\ z \in \mathbb{R}\}$. The outward unit normal at (x, y, z) is $\mathbf{N}(x, y, z) = \frac{1}{R}(x, y, 0)$. For a constant d, the parallel surface consists of points

$$(x, y, z) + d \cdot \frac{1}{R}(x, y, 0) = \left(x\left(1 + \frac{d}{R}\right),\ y\left(1 + \frac{d}{R}\right),\ z\right),$$

with $x^2 + y^2 = R^2$. This is again a cylinder, but with radius $R + d$ (since $\|(x(1 + d/R), y(1 + d/R))\| = |R + d|$). Hence S_d is a cylinder of radius $|R + d|$. If $d = -R$, the cylinder collapses to

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the z-axis.

Example 4.4.8 (Parallel Surfaces of a Plane)**.** Let S be a plane, e.g., $S = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$. The unit normal is constant: $\mathbf{N} = (0, 0, 1)$ (up to sign). For any d, the parallel surface is

$$S_d = \{(x, y, 0) + d(0, 0, 1) \mid x, y \in \mathbb{R}\} = \{(x, y, d) \mid x, y \in \mathbb{R}\}.$$

So S_d is just another plane parallel to the original one, at height d. All parallel surfaces are again planes.

Definition 4.4.9. Let S be a regular surface, $U \subset S$ an open set, and $W : U \to TS$ a differentiable vector field. For a point $p \in U$ and a tangent vector $y \in T_pS$, choose a smooth curve $\alpha : (-\varepsilon, \varepsilon) \to U$ such that $\alpha(0) = p$ and $\alpha'(0) = y$. Let $W(t) = W(\alpha(t))$ denote the restriction of W to α. Then the **covariant derivative** of W at p with respect to y is the orthogonal projection of $\frac{dW}{dt}\Big|_{t=0}$ onto the tangent plane T_pS. It is denoted by

$$\frac{DW}{dt}\Big|_{t=0}, \qquad (D_yW)(p), \qquad \text{or} \qquad (\nabla_yW)(p).$$

Definition 4.4.10 (Parallel Vector Field Along a Curve)**.** Let $\gamma : I \to S$ be a smooth curve. A vector field $W(t)$ along γ (i.e., $W(t) \in T_{\gamma(t)}S$ for each t) is called **parallel** if its covariant derivative along γ vanishes:

$$\frac{DW}{dt} = 0 \quad \text{for all } t.$$

Example 4.4.11 (Parallel Vector Field on a Great Circle of the Sphere)**.** (See Figure 49 in the Atlas.) Take a great circle (e.g., the equator) on the sphere. If you start with a tangent vector pointing north at a point on the equator, parallel transporting it along the equator will rotate it so that after a quarter circle it points east. This shows that a non-zero parallel vector field can exist *along a curve* even though no global parallel field exists on the sphere.

Example 4.4.12. Discuss the notion of covariant derivative for a plane.

Solution. Consider a plane passing through the point $p_0 = (x_0, y_0, z_0)$ and spanned by the orthonormal vectors

$$\mathbf{w}_1 = (a_1, a_2, a_3), \qquad \mathbf{w}_2 = (b_1, b_2, b_3).$$

A parametrization of the plane is

$$\mathbf{X}(u, v) = p_0 + u\,\mathbf{w}_1 + v\,\mathbf{w}_2, \qquad (u, v) \in \mathbb{R}^2.$$

Then

$$\mathbf{X}_u = \mathbf{w}_1, \quad \mathbf{X}_v = \mathbf{w}_2,$$

so the coefficients of the first fundamental form are

$$E = \mathbf{X}_u \cdot \mathbf{X}_u = 1, \quad F = \mathbf{X}_u \cdot \mathbf{X}_v = 0, \quad G = \mathbf{X}_v \cdot \mathbf{X}_v = 1.$$

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For a surface with metric coefficients E, F, G, the Christoffel symbols are given by

$$\Gamma_{11}^1 = \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)},$$
$$\Gamma_{12}^1 = \frac{GE_v - FG_u}{2(EG - F^2)},$$
$$\Gamma_{22}^1 = \frac{2GF_v - GG_u - FG_v}{2(EG - F^2)},$$
$$\Gamma_{11}^2 = \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)} \quad \text{(similar)},$$

and cyclic permutations. In the case of the plane we have $E = G = 1$, $F = 0$, and all partial derivatives of E, F, G vanish. Substituting these values yields $\Gamma_{ij}^k = 0 \quad \text{for all } i, j, k \in \{1, 2\}$.

Now let $\alpha(t) = \mathbf{X}(u(t), v(t))$ be a curve on the plane and

$$w(t) = a(t)\,\mathbf{X}_u + b(t)\,\mathbf{X}_v$$

a vector field along α. The covariant derivative of w along α is (from the general formula)

$$\frac{Dw}{dt} = \left(a' + \Gamma_{11}^1 au' + \Gamma_{12}^1 av' + \Gamma_{12}^1 bu' + \Gamma_{22}^1 bv'\right)\mathbf{X}_u + \left(b' + \Gamma_{11}^2 au' + \Gamma_{12}^2 av' + \Gamma_{12}^2 bu' + \Gamma_{22}^2 bv'\right)\mathbf{X}_v.$$

Since all $\Gamma_{ij}^k = 0$, this simplifies to

$$\frac{Dw}{dt} = a'\,\mathbf{X}_u + b'\,\mathbf{X}_v.$$

But the ordinary derivative of $w(t)$ in $\mathbb{R}^3$ is

$$\frac{dw}{dt} = a'\,\mathbf{X}_u + b'\,\mathbf{X}_v + a\,\frac{d}{dt}(\mathbf{X}_u) + b\,\frac{d}{dt}(\mathbf{X}_v).$$

On the plane, $\mathbf{X}_u = \mathbf{w}_1$ and $\mathbf{X}_v = \mathbf{w}_2$ are constant vectors, so their derivatives vanish. Hence $\frac{dw}{dt} = a'\,\mathbf{X}_u + b'\,\mathbf{X}_v$ as well. Thus

$$\frac{Dw}{dt} = \frac{dw}{dt}.$$

Therefore, for a plane, the covariant derivative coincides with the ordinary derivative in $\mathbb{R}^3$. This shows that the covariant derivative is a natural generalization of the usual directional derivative of vector fields on Euclidean space to curved surfaces. ■

Definition 4.4.13. Let S be a regular surface. A smooth curve $\alpha : I \to S$ is called a **geodesic** if its tangent vector field is parallel along the curve:

$$\frac{D\alpha'(t)}{dt} = 0 \quad \text{for all } t \in I.$$

Equivalently, geodesics have zero intrinsic acceleration; they are the "straightest possible" curves on the surface.

Example 4.4.14. Examples of Geodesics:

1. **Straight lines in the Euclidean plane.** For $S = \mathbb{R}^2$ with the standard flat metric, the geodesic equations reduce to $\ddot{x} = 0$, $\ddot{y} = 0$. Hence the geodesics are straight lines $\gamma(t) = (a_1 t + b_1,\ a_2 t + b_2)$.

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2. **Great circles on a sphere.** On the unit sphere S^2 with the induced metric, the geodesics are the great circles (intersections of the sphere with planes through the centre). For example, the equator $\gamma(\theta) = (\cos\theta, \sin\theta, 0)$ is a geodesic. Any arc of a great circle gives a locally distance-minimizing path.

Theorem 4.4.15 (Differential Equations of Geodesics). *Let S be a regular surface with a local parametrization $\mathbf{x}(u,v)$, and let $\alpha : I \to S$ be a parametrized curve on S given locally by $\alpha(t) = \mathbf{x}(u(t), v(t))$. Then, α is a geodesic if and only if the coordinate functions $u(t)$ and $v(t)$ satisfy the system of second-order differential equations:*

$$u'' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2 = 0 \tag{4.1}$$
$$v'' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2 = 0 \tag{4.2}$$

where Γ^k_{ij} are the Christoffel symbols of the second kind associated with the parametrization $\mathbf{x}$.

Proof. Let $W(t) = \alpha'(t)$ be the tangent vector field along the curve α. By the chain rule, we can express W in terms of the coordinate basis $\{\mathbf{x}_u, \mathbf{x}_v\}$ as:

$$W = \mathbf{x}_u u' + \mathbf{x}_v v' \tag{4.3}$$

To evaluate the geodesic condition, we compute the ordinary derivative of W in $\mathbb{R}^3$ with respect to t. Using the product and chain rules, we have

$$\begin{aligned}\frac{dW}{dt} &= \frac{d}{dt}(\mathbf{x}_u)\,u' + \mathbf{x}_u u'' + \frac{d}{dt}(\mathbf{x}_v)\,v' + \mathbf{x}_v v'' \\ &= \left(\mathbf{x}_{uu}u' + \mathbf{x}_{uv}v'\right)u' + \mathbf{x}_u u'' + \left(\mathbf{x}_{vu}u' + \mathbf{x}_{vv}v'\right)v' + \mathbf{x}_v v''\end{aligned}$$

Assuming the mixed partial derivatives of the parametrization are symmetric ($\mathbf{x}_{uv} = \mathbf{x}_{vu}$) and grouping the terms, we obtain

$$\frac{dW}{dt} = u'\left(\mathbf{x}_{uu}u' + \mathbf{x}_{uv}v'\right) + v'\left(\mathbf{x}_{uv}u' + \mathbf{x}_{vv}v'\right) + u''\mathbf{x}_u + v''\mathbf{x}_v \tag{4.4}$$

Next, we substitute the Gauss equations for the second partial derivatives of the parametrization $\mathbf{x}$:

$$\begin{aligned}\mathbf{x}_{uu} &= \Gamma^1_{11}\mathbf{x}_u + \Gamma^2_{11}\mathbf{x}_v + L\hat{N} \\ \mathbf{x}_{uv} &= \Gamma^1_{12}\mathbf{x}_u + \Gamma^2_{12}\mathbf{x}_v + M\hat{N} \\ \mathbf{x}_{vv} &= \Gamma^1_{22}\mathbf{x}_u + \Gamma^2_{22}\mathbf{x}_v + N\hat{N}\end{aligned}$$

where $\hat{N}$ is the unit normal vector to the surface, and L, M, N are the coefficients of the second fundamental form. Substituting these expressions into $\frac{dW}{dt}$ yields

$$\begin{aligned}\frac{dW}{dt} = u'&\left[\left(\Gamma^1_{11}\mathbf{x}_u + \Gamma^2_{11}\mathbf{x}_v + L\hat{N}\right)u' + \left(\Gamma^1_{12}\mathbf{x}_u + \Gamma^2_{12}\mathbf{x}_v + M\hat{N}\right)v'\right] \\ &+ v'\left[\left(\Gamma^1_{12}\mathbf{x}_u + \Gamma^2_{12}\mathbf{x}_v + M\hat{N}\right)u' + \left(\Gamma^1_{22}\mathbf{x}_u + \Gamma^2_{22}\mathbf{x}_v + N\hat{N}\right)v'\right] \\ &+ u''\mathbf{x}_u + v''\mathbf{x}_v\end{aligned}$$

Collecting the terms with respect to $\mathbf{x}_u$, $\mathbf{x}_v$, and $\hat{N}$, and using the symmetry $\Gamma^k_{12} = \Gamma^k_{21}$, we expand and simplify:

$$\begin{aligned}\frac{dW}{dt} = &\left(u'' + \Gamma^1_{11}(u')^2 + \Gamma^1_{12}u'v' + \Gamma^1_{12}v'u' + \Gamma^1_{22}(v')^2\right)\mathbf{x}_u \\ &+ \left(v'' + \Gamma^2_{11}(u')^2 + \Gamma^2_{12}u'v' + \Gamma^2_{12}v'u' + \Gamma^2_{22}(v')^2\right)\mathbf{x}_v \\ &+ \left(L(u')^2 + 2Mu'v' + N(v')^2\right)\hat{N}\end{aligned}$$

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By definition, a curve α is a geodesic if and only if its covariant acceleration $\frac{DW}{dt}$ along the curve is identically zero:

$$\frac{DW}{dt} = 0 \tag{4.5}$$

Since the covariant derivative $\frac{DW}{dt}$ is the orthogonal projection of $\frac{dW}{dt}$ onto the tangent plane T_pS, we discard the component normal to the surface (the term containing the normal vector N):

$$\frac{DW}{dt} = \left(u'' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2\right)\mathbf{x}_u + \left(v'' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2\right)\mathbf{x}_v \tag{4.6}$$

Setting $\frac{DW}{dt} = 0$ yields

$$\left(u'' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2\right)\mathbf{x}_u + \left(v'' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2\right)\mathbf{x}_v = 0$$

At any point on S, the coordinate vectors $\mathbf{x}_u$ and $\mathbf{x}_v$ are linearly independent. Therefore, this linear combination is equal to zero if and only if the coefficients of both basis vectors vanish independently. This yields the desired system of differential equations:

$$u'' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2 = 0$$
$$v'' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2 = 0$$

■

Remark 4.4.16. From the system of geodesic equations, we observe that the equations depend exclusively on the Christoffel symbols. Since properties depending on Christoffel symbols are intrinsic to the metric and are preserved under isometries, it follows that an isometry between two surfaces maps the geodesics of one surface to the geodesics of the other surface.

Example 4.4.17. Find the geodesics on a surface of revolution. (See Figure 52 in the Atlas.)

Solution. See Figure 4.1. **Parametrization.** A surface of revolution about the z-axis is given by

$$\mathbf{X}(u,v) = \big(f(v)\cos u,\ f(v)\sin u,\ g(v)\big), \quad f > 0,\ f'^2 + g'^2 \neq 0.$$

Here u is the rotation angle, v the profile parameter.

Metric and Christoffel symbols.

$$E = f^2, \quad F = 0, \quad G = f'^2 + g'^2.$$

The non-zero Christoffel symbols (diagonal metric) are

$$\Gamma^1_{12} = \frac{f'}{f}, \qquad \Gamma^2_{11} = -\frac{ff'}{f'^2 + g'^2}, \qquad \Gamma^2_{22} = \frac{f'f'' + g'g''}{f'^2 + g'^2}, \qquad \text{others } 0.$$

Geodesic equations (arc length s).

$$\ddot{u} + 2\frac{f'}{f}\,\dot{u}\dot{v} = 0, \qquad \ddot{v} - \frac{ff'}{f'^2 + g'^2}\,\dot{u}^2 + \frac{f'f'' + g'g''}{f'^2 + g'^2}\,\dot{v}^2 = 0. \tag{1}$$

$$\text{The first equation gives } \frac{d}{ds}(f^2\dot{u}) = 0, \text{ hence } f(v)^2\,\dot{u}(s) = C \quad (\text{constant}). \tag{2}$$

Meridians ($u = $ const, $\dot{u} = 0$). Then (2) holds trivially. From the arc-length condition $G\dot{v}^2 = 1$ we obtain $\dot{v}^2 = 1/(f'^2 + g'^2)$. Differentiating yields the second equation of (1). Hence every meridian is a geodesic.

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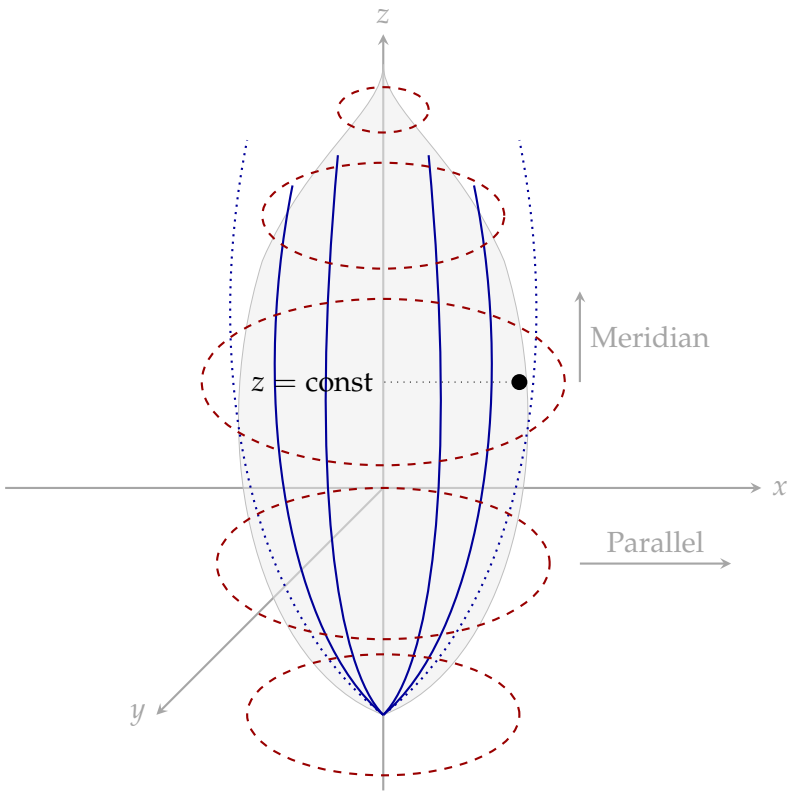


Figure 4.1: Meridians and Parallels of Surface of Revolution

Parallels (v = const, $\dot{v} = 0$). Equation (2) gives $\ddot{u} = 0$, so $\dot{u} = \text{const}$. The second equation of (1) becomes $-\frac{ff'}{f'^2+g'^2}\dot{u}^2 = 0$. Thus a parallel $v = v_0$ is a geodesic iff $f'(v_0) = 0$ (the profile has a horizontal tangent). Such a parallel is a geodesic circle.

Remark. Other geodesics (spirals) exist when both $\dot{u}, \dot{v} \neq 0$; they are found by integrating (1) with (2) and $|\gamma'| = 1$. ■

Example 4.4.18. Discuss the notion of covariant derivative for a plane.

Solution. A plane passing through $p_0 = (x_0, y_0, z_0)$ and containing the orthonormal vectors $w_1 = (a_1, a_2, a_3)$ and $w_2 = (b_1, b_2, b_3)$ is given by $\mathbf{x}(u, v) = p_0 + uw_1 + vw_2, \quad (u, v) \in \mathbb{R}^2$. Then:

$$\mathbf{x}_u = w_1, \quad \mathbf{x}_v = w_2 \implies E = G = 1, \quad F = 0$$

Also we know the formulas for the Christoffel symbols of the second kind:

$$\Gamma^1_{11} = \frac{GE_u - 2FF_u + FE_v}{2(EG - F^2)}, \quad \Gamma^1_{12} = \frac{GE_v - FG_u}{2(EG - F^2)}, \quad \Gamma^1_{22} = \frac{2GF_v - GG_u - FG_v}{2(EG - F^2)}$$

$$\Gamma^2_{11} = \frac{2EF_u - EE_v - FE_u}{2(EG - F^2)}, \quad \Gamma^2_{12} = \frac{EG_u - FE_v}{2(EG - F^2)}, \quad \Gamma^2_{22} = \frac{EG_v - 2FF_v + FG_u}{2(EG - F^2)}$$

Since E, F, and G are constants ($E = 1, G = 1, F = 0$), all of their partial derivatives with respect to u and v are zero. Substitution gives $\Gamma^k_{ij} = 0 \quad \forall\, i, j, k \in \{1, 2\}$. Therefore, from the general equation of the covariant derivative,

$$\frac{DW}{dt} = \left(a' + \Gamma^1_{11}(u')^2 + 2\Gamma^1_{12}u'v' + \Gamma^1_{22}(v')^2\right)\mathbf{x}_u + \left(b' + \Gamma^2_{11}(u')^2 + 2\Gamma^2_{12}u'v' + \Gamma^2_{22}(v')^2\right)\mathbf{x}_v.$$

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We obtain

$$\frac{DW}{dt} = a'\mathbf{x}_u + b'\mathbf{x}_v$$

This shows that the covariant derivative agrees with the usual derivative in the plane; therefore, the covariant derivative is the generalization of the usual derivative of a vector in a plane. ■

Theorem 4.4.19. *A curve on a surface $\mathbf{r} = \mathbf{r}(u, v)$ is a geodesic if and only if at every point its principal normal is parallel to the surface normal.*

Proof. To prove the theorem, we will establish that the principal normal $\mathbf{r}'' = \kappa\mathbf{n}$ of the curve is orthogonal to the tangential directions $\mathbf{r}_1$ and $\mathbf{r}_2$ at P so that the principal normal of the curve coincides with the surface normal. Thus, we show that $r'' \cdot r_1 = 0$ and $r'' \cdot r_2 = 0$.

We derive these conditions from the geodesic equations

$$U(t) = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{u}}\right) - \frac{\partial T}{\partial u} = 0 \text{ and } V(t) = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{v}}\right) - \frac{\partial T}{\partial v} = 0$$

Let us consider

$$\dot{\mathbf{r}} = \frac{d\mathbf{r}}{dt} = \mathbf{r}_1\dot{u} + \mathbf{r}_2\dot{v}, \quad \text{so that, } \frac{\partial \dot{\mathbf{r}}}{\partial \dot{u}} = \mathbf{r}_1 \quad \text{and} \quad \frac{\partial \dot{\mathbf{r}}}{\partial \dot{v}} = \mathbf{r}_2 \tag{1}$$

Hence

$$\dot{\mathbf{r}} \cdot \dot{\mathbf{r}} = (\mathbf{r}_1\dot{u} + \mathbf{r}_2\dot{v})^2 = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2$$

so that we can take

$$T = \frac{1}{2}\dot{\mathbf{r}}^2 \tag{2}$$

Differentiating (2) partially, we obtain

$$\frac{\partial T}{\partial u} = \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial u} = \dot{r} \cdot [\mathbf{r}_{11}\dot{u} + \mathbf{r}_{21}\dot{v}], \qquad \frac{\partial T}{\partial v} = \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial v} = \dot{r} \cdot [\mathbf{r}_{12}\dot{u} + \mathbf{r}_{22}\dot{v}]$$

$$\frac{\partial T}{\partial \dot{u}} = \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial \dot{u}} = \dot{r} \cdot r_1, \qquad \frac{\partial T}{\partial \dot{v}} = \dot{\mathbf{r}} \cdot \frac{\partial \dot{\mathbf{r}}}{\partial \dot{v}} = \dot{r} \cdot r_1$$

Therefore,

$$\begin{aligned} U(t) &= \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{u}}\right) - \frac{\partial T}{\partial u} \\ &= \frac{d}{dt}(\dot{\mathbf{r}} \cdot \mathbf{r}_1) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11}\dot{u} + \mathbf{r}_{21}\dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_1 + \dot{\mathbf{r}} \cdot \frac{d}{dt}\mathbf{r}_1 - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11}\dot{u} + \mathbf{r}_{21}\dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_1 + \dot{\mathbf{r}} \cdot (\mathbf{r}_{11}\dot{u} + \mathbf{r}_{12}\dot{v}) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{11}\dot{u} + \mathbf{r}_{21}\dot{v}) \\ &= \ddot{r} \cdot r_2, \text{ because } \mathbf{r}_{12} = \mathbf{r}_{21} \end{aligned}$$

Further,

$$\begin{aligned} V(t) &= \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{v}}\right) - \frac{\partial T}{\partial v} \\ &= \frac{d}{dt}(\dot{\mathbf{r}} \cdot \mathbf{r}_2) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12}\dot{u} + \mathbf{r}_{22}\dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 + \dot{\mathbf{r}} \cdot \frac{d}{dt}\mathbf{r}_2 - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12}\dot{u} + \mathbf{r}_{22}\dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 + \dot{\mathbf{r}} \cdot (\mathbf{r}_{12}\dot{u} + \mathbf{r}_{22}\dot{v}) - \dot{\mathbf{r}} \cdot (\mathbf{r}_{12}\dot{u} + \mathbf{r}_{22}\dot{v}) \\ &= \ddot{\mathbf{r}} \cdot \mathbf{r}_2 \end{aligned}$$

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Hence the geodesic equations $U(t) = 0,\ V(t) = 0$ imply

$$\ddot{\mathbf{r}} \cdot \mathbf{r}_1 = 0 \quad \text{and} \quad \ddot{\mathbf{r}} \cdot \mathbf{r}_2 = 0$$

which shows that $\ddot{\mathbf{r}}$ is perpendicular to the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ lying in the tangent plane at P. Hence $\ddot{\mathbf{r}}$ is along the surface normal at P.

But

$$\ddot{\mathbf{r}} = \kappa \mathbf{n}$$

so that $\ddot{\mathbf{r}}$ is along the principal normal of the curve at P. Hence, the principal normal at every point of the geodesic is normal to the surface at P. Since all the above steps are reversible, the converse is also true. This completes the proof of the theorem. ■

Note: Instead of taking the parameter t, we can use as well the parameter s. In that case we have to replace t by s in the above working so that we have

$$U(s) = \mathbf{r}'' \cdot \mathbf{r}_1 \quad \text{and} \quad V(s) = \mathbf{r}'' \cdot \mathbf{r}_2$$

Theorem 4.4.20 (Clairaut's Theorem). *Let α be a geodesic on a surface of revolution S, and ρ be the distance of a point of S from the axis of rotation, and let ψ be the angle between α' and the meridian of S. Then $\rho \sin\psi$ is constant along α.*

Conversely, if $\rho \sin\psi$ is constant along some curve α on the surface, and if no part of α is part of some parallel of S, then α is a geodesic.

Proof. Suppose S is parametrized by,

$$\mathbf{x}(u, v) = (f(v)\cos u,\ f(v)\sin u,\ g(v))$$

then obviously $\rho = f(v)$. Here:

$$\mathbf{x}_u = (-f\sin u,\ f\cos u,\ 0)$$
$$\mathbf{x}_v = (f'\cos u,\ f'\sin u,\ g') \implies \|\mathbf{x}_u\| = f = \rho, \quad \|\mathbf{x}_v\| = \sqrt{f'^2 + g'^2} = c$$

Since $E = f^2$, $G = f'^2 + g'^2$, and $F = 0$, the coordinate system has an orthogonal parametrization. Thus, the normalized vectors

$$\left\{ \frac{\mathbf{x}_u}{\|\mathbf{x}_u\|}, \frac{\mathbf{x}_v}{\|\mathbf{x}_v\|} \right\}$$

form an orthonormal basis for T_pS for all $p \in S$.

Thus, we can decompose the unit velocity vector α' in this basis as:

$$\begin{aligned} \alpha' &= \cos\psi \frac{\mathbf{x}_v}{\|\mathbf{x}_v\|} + \sin\psi \frac{\mathbf{x}_u}{\|\mathbf{x}_u\|} \\ &= \cos\psi \frac{\mathbf{x}_v}{c} + \sin\psi \rho^{-1} \mathbf{x}_u \\ \Rightarrow \alpha' \times \mathbf{x}_v &= \sin\psi \rho^{-1} (\mathbf{x}_u \times \mathbf{x}_v) \end{aligned} \tag{C_1}$$

Also, since we can consider the curve parameterized by arc length $\alpha(s) = \mathbf{x}(u(s), v(s))$:

$$\begin{aligned} \alpha' &= \mathbf{x}_u u' + \mathbf{x}_v v' \\ \Rightarrow \alpha' \times \mathbf{x}_v &= (\mathbf{x}_u \times \mathbf{x}_v) u' \end{aligned} \tag{C_2}$$

Combining equations (C_1) and (C_2), we have

$$(\mathbf{x}_u \times \mathbf{x}_v) u' = \sin\psi \rho^{-1} (\mathbf{x}_u \times \mathbf{x}_v)$$

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$$\text{i.e.} \quad u'\rho = \sin\psi \implies \rho^2 u' = \rho \sin\psi$$

Since $\rho = f$, we obtain

$$\boxed{f^2 u' = \rho \sin\psi}$$

From the first equation of geodesic equations in the surface of revolution case:

$$\frac{d}{ds}(f^2 u') = 0 \implies f^2 u' = \text{constant}$$

This implies that $\rho \sin\psi$ is constant, say Ω, along the geodesic.

Conversely, if $\rho \sin\psi$ is constant along a unit-speed curve α on S, the above argument shows that the first equation $\frac{d}{ds}(f^2 u') = 0$ is satisfied. Now we need to check the second equation, i.e.,

$$\frac{f'f'' + g'g''}{f'^2 + g'^2}(v')^2 - \frac{ff'}{f'^2 + g'^2}u'^2 + v'' = 0 \tag{C_3}$$

is satisfied too.

Since $u' = \frac{\sin\psi}{\rho} = \frac{\Omega}{\rho^2}$, the first fundamental form for the curve α is

$$\begin{aligned} 1 = |\alpha'|^2 = ds^2 &= Eu'^2 + 2Fu'v' + Gv'^2 \\ \Rightarrow 1 &= f^2 u'^2 + (f'^2 + g'^2)v'^2 \\ \Rightarrow 1 &= f^2 \frac{\Omega^2}{f^4} + (f'^2 + g'^2)v'^2 \\ \Rightarrow 1 &= \frac{\Omega^2}{f^2} + (f'^2 + g'^2)v'^2 \end{aligned}$$

Differentiating this relation with respect to the arc length s:

$$0 = \Omega^2 \left(-\frac{2f'v'}{f^3}\right) + \left(2f'f''v' + 2g'g''v'\right)v'^2 + (f'^2 + g'^2)2v'v'' = 0$$

Dividing by $2v'$:

$$\Rightarrow -\frac{ff'}{f^4}\Omega^2 + (f'f'' + g'g'')v'^2 + (f'^2 + g'^2)v'' = 0$$

Dividing by $(f'^2 + g'^2)$:

$$\Rightarrow v'' + \frac{f'f'' + g'g''}{f'^2 + g'^2}v'^2 - \frac{ff'}{f'^2 + g'^2}u'^2 = 0 \quad \left(\text{since } \frac{\Omega^2}{f^4} = u'^2\right)$$

which is exactly equation (C_3). ■

Note: With the help of Clairaut's theorem and the exercise before it, we can completely classify the geodesics on a revolution surface.

Example 4.4.21. Find the geodesics on a pseudosphere. (See Figure 53 in the Atlas.)

Solution. The pseudosphere is a surface of constant negative curvature ($K = -1$). It can be obtained by revolving the tractrix around its asymptote (the z-axis). A convenient parametrization is

$$\mathbf{x}(u, v) = \big(\operatorname{sech} v \cos u,\ \operatorname{sech} v \sin u,\ v - \tanh v\big),$$

where $u \in (0, 2\pi)$ is the rotation angle and $v \in [0, \infty)$. The first fundamental form is

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle = \operatorname{sech}^2 v, \qquad F = 0, \qquad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle = \tanh^2 v.$$

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Thus the line element becomes $ds^2 = \operatorname{sech}^2 v\, du^2 + \tanh^2 v\, dv^2$.

For a surface of revolution with $F = 0$, the geodesic equations admit a first integral (Clairaut's theorem). From the u-geodesic equation we obtain

$$\frac{d}{ds}(E\dot{u}) = 0 \quad \Longrightarrow \quad E\dot{u} = \operatorname{sech}^2 v \, \frac{du}{ds} = \text{constant} \equiv h.$$

This constant h is the conserved angular momentum. The arc-length condition $E\dot{u}^2 + G\dot{v}^2 = 1$ gives

$$\operatorname{sech}^2 v \left(\frac{du}{ds}\right)^2 + \tanh^2 v \left(\frac{dv}{ds}\right)^2 = 1.$$

Eliminating $du/ds = h \cosh^2 v$ (since $\operatorname{sech}^2 v = 1/\cosh^2 v$), we obtain

$$\frac{h^2}{\cosh^2 v} + \tanh^2 v \left(\frac{dv}{ds}\right)^2 = 1 \quad \Longrightarrow \quad \left(\frac{dv}{ds}\right)^2 = \frac{1 - h^2 \operatorname{sech}^2 v}{\tanh^2 v}.$$

Special geodesics:

- **Meridians:** $h = 0$. Then $du/ds = 0$, so $u =$ constant. From the arc-length condition $dv/ds = \coth v$. Integrating yields a parametrization of the tractrix. These curves are geodesics.
- **Parallels:** $dv/ds = 0$. This requires $\tanh^2 v \neq 0$ and $1 - h^2 \operatorname{sech}^2 v = 0$, i.e. $h = \cosh v$. Since h is constant, this can only hold for a single value of v. Hence a parallel (circle of latitude) is a geodesic iff $v = \operatorname{arcosh} h$, which occurs only for one specific circle (the geodesic parallel). On the pseudosphere this is the circle at $v = \operatorname{arcosh} 1 = 0$? Actually $h = \cosh v_0$ gives a constant. For a given h, there is a unique v_0 such that the parallel is a geodesic. On the pseudosphere such a geodesic parallel does not exist because $\cosh v \geq 1$ and the pseudosphere has a boundary at $v = 0$ (a cusp). Indeed, the only possible geodesic parallel would be at $v = 0$ where the surface degenerates; thus there are no non-trivial geodesic parallels on the regular part.
- **General geodesics:** For $0 < |h| < 1$, the expression $(dv/ds)^2$ becomes positive for all v and the geodesic winds around the pseudosphere, approaching the boundary $v = 0$ (the cusp) asymptotically. In fact, one can integrate to obtain explicit formulas in terms of hyperbolic functions. The general solution describes curves that spiral towards the equator or the cusp depending on h.

Thus the geodesics on the pseudosphere consist of:

1. All meridians (the generating curves).
2. A one-parameter family of non-meridional geodesics that are not closed and that approach the cusp.

No closed geodesics other than the meridians exist on the pseudosphere. This behaviour is characteristic of surfaces with constant negative curvature. ■

4.5 Geodesic Curvatures

If $\mathbf{r} = \mathbf{r}(s)$ is the position vector of any point P on a curve on a surface, then the curvature vector $\mathbf{r}'' = \mathbf{t}' = \kappa \mathbf{n}$ where κ is the curvature and $\mathbf{n}$ is the principal normal at P to the curve. As we have already noted, at a point P on the surface, there exist three linearly independent non-coplanar vectors $\mathbf{N}$, $\mathbf{r}_1$ and $\mathbf{r}_2$ where $\mathbf{r}_1$ and $\mathbf{r}_2$ are tangential to the surface and $\mathbf{N}$ is the surface normal. Hence $\mathbf{r}''$ at P can be expressed as a linear combination of the vectors $\mathbf{N}$, $\mathbf{r}_1$ and $\mathbf{r}_2$ so that we can write

$$\mathbf{r}'' = \kappa_n \mathbf{N} + \lambda \mathbf{r}_1 + \mu \mathbf{r}_2$$

where κ_n is the normal component of $\mathbf{r}''$.

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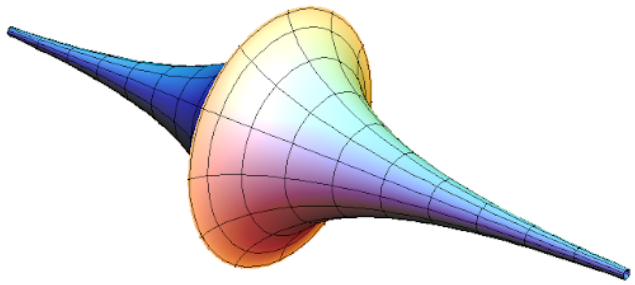


Figure 4.2: Pseudosphere - surface of revolution of a tractrix with constant negative Gaussian curvature $K = -1/a^2$.

Definition 4.5.1. The vector $\lambda\mathbf{r}_1 + \mu\mathbf{r}_2$ with components (λ, μ) is tangential to the surface. The vector with components (λ, μ) of the tangential vector $\lambda\mathbf{r}_1 + \mu\mathbf{r}_2$ to the surface is called the geodesic curvature vector at P. It is denoted by K_g.

Theorem 4.5.2. *A curve on a surface is a geodesic if and only if the geodesic curvature vector is zero.*

Proof. Let $\mathbf{r} = \mathbf{r}(s)$ be any curve on the surface with the principal normal $\mathbf{n}$ and surface normal $\mathbf{N}$ at P. If (λ, μ) is the geodesic curvature vector at P, then:

$$\mathbf{r}'' = \kappa_n\mathbf{N} + \lambda\mathbf{r}_1 + \mu\mathbf{r}_2 \quad \text{or} \quad \kappa\mathbf{n} = \kappa_n\mathbf{N} + \lambda\mathbf{r}_1 + \mu\mathbf{r}_2 \tag{1}$$

Let the curve be a geodesic. Then, by the normal property of a geodesic:

$$\mathbf{n} = \mathbf{N} \tag{2}$$

Using (2) in (1), we obtain

$$\kappa\mathbf{N} = \kappa_n\mathbf{N} + \lambda\mathbf{r}_1 + \mu\mathbf{r}_2$$

Equating the coefficients of $\mathbf{r}_1$ and $\mathbf{r}_2$ on both sides, we obtain $\lambda = \mu = 0$, so that $K_g = 0$.

Conversely, let $K_g = 0$. This implies $\lambda = 0$, $\mu = 0$. Hence, from (1), we obtain $\kappa\mathbf{n} = \kappa_n\mathbf{N}$. Thus, the principal normal to the curve is parallel to the surface normal. Therefore, the curve is a geodesic by the normal property. ■

Lemma 4.5.3. *Let α be a regular curve on a surface S. If the principal normal of α is parallel to the surface normal at every point, then α is a geodesic.*

Proof. Let $\alpha(s)$ be a unit-speed parametrization of the curve, and let t, n and N denote its unit tangent, principal normal and surface normal, respectively. By the Frenet formula, we have $\alpha''(s) = \kappa(s)n(s)$, where κ is the curvature of α.

By hypothesis, n is parallel to N at every point where $\kappa \neq 0$. Consequently, the acceleration vector $\alpha''(s)$ is normal to the surface. At points where $\kappa = 0$, we have $\alpha''(s) = 0$, so its tangential component also vanishes. Thus, the acceleration of α has no component tangent to the surface at any point, which means that its geodesic curvature is zero. Hence, α is a geodesic. ■

Great circle: A curve (circle) obtained as a normal section of a plane on S^2 passing through the centre of S^2. (See Figure 48 in the Atlas.)

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Theorem 4.5.4. *A regular curve on a sphere is a geodesic if and only if it is an arc of a great circle.*

Proof. Let $\alpha : I \to S^2(c, r)$ be a unit-speed curve on the sphere with centre c and radius r. Suppose first that α is a geodesic. Let t, n and b denote its unit tangent, principal normal and binormal vectors, respectively, and let κ and τ be its curvature and torsion.

Since α is a geodesic, its principal normal is parallel to the normal of the sphere. Moreover, the normal to a sphere at any point passes through its centre. Consequently, the principal normal of α passes through c at every point of the curve. Hence,

$$\alpha(s) - c = \lambda(s)n(s),$$

where λ is a non-zero differentiable function.

Differentiating with respect to the arc length s and using the Frenet–Serret formula $n' = -\kappa t + \tau b$, we obtain

$$\begin{aligned} t &= \lambda' n + \lambda n' \\ &= \lambda' n + \lambda(-\kappa t + \tau b) \\ &= -\lambda\kappa t + \lambda' n + \lambda\tau b. \end{aligned}$$

Since t, n and b are mutually orthogonal unit vectors, comparison of the corresponding coefficients gives $\lambda\tau = 0$. As $\lambda \neq 0$, it follows that $\tau = 0$.

Thus, α is a plane curve. Let P be the plane containing α. Since the principal normal n lies in P and the line through $\alpha(s)$ in the direction of $n(s)$ passes through c, the centre c also belongs to P. Therefore, P passes through the centre of the sphere, and α lies on the intersection $P \cap S^2(c, r)$, which is a great circle.

Conversely, suppose that α is an arc of a great circle on $S^2(c, r)$. Then α lies in a plane passing through the centre c and is a circle of radius r. The principal normal of this circle points towards its centre, which is precisely c. Hence, the principal normal of α is parallel to the normal of the sphere at every point. It follows that the geodesic curvature of α vanishes, and therefore α is a geodesic.

Thus, a regular curve on a sphere is a geodesic if and only if it is an arc of a great circle. ■

Example 4.5.5. Find all the geodesics on a Cylinder. (See Figure 54 in the Atlas.)

Solution. Let $S \subset \mathbb{R}^3$ be a right circular cylinder parameterized by $\mathbf{x}(u, v) = (\cos u, \sin u, v)$. Differentiating with respect to the coordinates yields the tangent vectors $\mathbf{x}_u = (-\sin u, \cos u, 0)$ and $\mathbf{x}_v = (0, 0, 1)$, from which we compute the coefficients of the first fundamental form:

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle = 1, \quad F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle = 0, \quad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle = 1$$

Because these coefficients are constant, all of their partial derivatives vanish. This implies that all Christoffel symbols of the second kind are zero ($\Gamma^k_{ij} = 0$). Consequently, the geodesic equations for a curve $\gamma(s) = \mathbf{x}(u(s), v(s))$ parameterized by arc length s,

$$u'' + \sum_{i,j} \Gamma^1_{ij} u'_i u'_j = 0 \quad \text{and} \quad v'' + \sum_{i,j} \Gamma^2_{ij} u'_i u'_j = 0$$

simplify directly to $u''(s) = 0$ and $v''(s) = 0$. Integrating these equations twice yields the linear coordinate relations:

$$u(s) = as + b \quad \text{and} \quad v(s) = cs + d$$

where a, b, c, and d are constants of integration.

Substituting these back into the parametrization of the cylinder gives the general vector equation of any geodesic curve on the surface:

$$\gamma(s) = (\cos(as+b),\, \sin(as+b),\, cs+d)$$

The constants a and c classify these curves into three distinct geometric families:

1. **Vertical straight lines** (generators of the cylinder) when $a = 0$,
2. **Horizontal circles** (parallel cross-sections) when $c = 0$, and
3. **Circular helices** winding around the cylinder when both $a \neq 0$ and $c \neq 0$.

This classification is intrinsically explained by the flat geometry of the cylinder. Since the metric coefficients are $E = G = 1$ and $F = 0$, the surface is locally isometric to the Euclidean plane. Under this isometry, the straight lines of the plane (which are its geodesics) map directly to these three families of curves on the cylinder. ■

Example 4.5.6. Find all geodesics on a right circular cylinder of radius R.

Solution. Let the cylinder be given by the parametrization

$$\mathbf{x}(\theta, z) = (R\cos\theta,\ R\sin\theta,\ z), \qquad 0 \le \theta < 2\pi,\ z \in \mathbb{R},$$

where $R > 0$ is the radius. The first fundamental form is

$$E = \|\mathbf{x}_\theta\|^2 = R^2, \qquad F = \mathbf{x}_\theta \cdot \mathbf{x}_z = 0, \qquad G = \|\mathbf{x}_z\|^2 = 1,$$

so the line element is $ds^2 = R^2 d\theta^2 + dz^2$. The Christoffel symbols for this metric are all zero (since the metric coefficients are constant). Consequently the geodesic equations become

$$\frac{d^2\theta}{ds^2} = 0, \qquad \frac{d^2 z}{ds^2} = 0.$$

Hence

$$\theta(s) = as + \theta_0, \qquad z(s) = cs + z_0,$$

with constants a, c satisfying the unit-speed condition

$$R^2\dot{\theta}^2 + \dot{z}^2 = R^2 a^2 + c^2 = 1.$$

The explicit curves in $\mathbb{R}^3$ are therefore

$$\gamma(s) = \big(R\cos(as+\theta_0),\ R\sin(as+\theta_0),\ cs+z_0\big).$$

We distinguish three cases:

- **Meridians (vertical lines):** $a = 0,\ c = \pm 1$. Then θ is constant and the curve is a straight line parallel to the axis:
$$\gamma(s) = (R\cos\theta_0,\ R\sin\theta_0,\ \pm s + z_0).$$
- **Parallels (horizontal circles):** $c = 0,\ a = \pm 1/R$. Then z is constant and the curve is a circle:
$$\gamma(s) = \big(R\cos(\pm s/R + \theta_0),\ R\sin(\pm s/R + \theta_0),\ z_0\big).$$

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- **Helices:** both $a \neq 0$ and $c \neq 0$. These are curves of constant pitch:

$$\gamma(s) = \big(R\cos(as+\theta_0),\ R\sin(as+\theta_0),\ cs+z_0\big), \quad a^2 + \frac{c^2}{R^2} = \frac{1}{R^2}.$$

  When the cylinder is cut along a generator and flattened into a plane, these helices become ordinary straight lines (slanted lines) in the (θ, z)-plane.

Thus all geodesics on a cylinder are exactly the meridians, the parallels, and the helices (which include the two previous families as limiting cases when $c = 0$ or $a = 0$). ■

2nd Solution. We have already shown that the map:

$$f(x, y, 0) = (\cos x,\ \sin x,\ y)$$

acts as a local isometry between a plane and a cylinder. So, $f(\text{geodesic of plane})$ will be a geodesic on the cylinder.

We know that in the case of a plane, the geodesics are all straight lines, which one can classify as:

- the x-axis,
- the y-axis, or
- an arbitrary line: $y = mx + c$.

Applying the isometry map f to each class of lines:

$$\begin{aligned} f(x\text{-axis}) &= f(x, 0, 0) = (\cos x,\ \sin x,\ 0) \\ &= \text{a circle (normal section)} \end{aligned}$$

$$\begin{aligned} f(y\text{-axis}) &= f(0, y, 0) = (1,\ 0,\ y) \\ &= \text{generator of the cylinder (straight line)} \end{aligned}$$

$$\begin{aligned} f(\text{arbitrary line}) &= f(x,\ mx + c,\ 0) = (\cos x,\ \sin x,\ mx + c) \\ &= \text{a helix on the cylinder} \end{aligned}$$

Thus, the geodesics on a cylinder are: **circles, straight lines, and helices**. ■

4.6 The Gauss–Bonnet Theorem

(See Figure 56 in the Atlas.)

The Gauss–Bonnet theorem is one of the deepest results in differential geometry. It connects the local geometric quantity (Gaussian curvature) with a global topological invariant (the Euler characteristic). The theorem has several versions, each revealing a different aspect of this interplay.

4.6.1 Geometric and Topological Preliminaries

Definition 4.6.1 (Gaussian curvature)**.** For a smooth surface S, the Gaussian curvature $K(p)$ at a point p is the product of the principal curvatures: $K = \kappa_1\kappa_2$. It measures the intrinsic curvature of the surface.

Definition 4.6.2 (Euler characteristic)**.** The Euler characteristic $\chi(S)$ of a compact surface S is a topological invariant. For a sphere $\chi = 2$, for a torus $\chi = 0$, for a surface of genus g (orientable)

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$\chi = 2 - 2g$, for a projective plane $\chi = 1$, etc.

4.6.2 Global Gauss–Bonnet Theorem (Closed Surfaces)

Theorem 4.6.3 (Gauss–Bonnet for closed surfaces)**.** *Let S be a compact, orientable, two-dimensional Riemannian manifold without boundary. Then*

$$\int_S K \, dA = 2\pi \, \chi(S),$$

where K is the Gaussian curvature, dA the area element, and $\chi(S)$ the Euler characteristic of S.

Explanation: The total Gaussian curvature of a closed surface is an integer multiple of 2π determined solely by the topology of the surface – it is independent of the metric. For example:

- A sphere (any smooth metric) has total curvature 4π.
- A torus (genus 1) has total curvature 0.
- A surface of genus 2 has total curvature -4π.

If the surface is non-orientable, the same formula holds with the Euler characteristic defined appropriately; for the projective plane $\chi = 1$ gives total curvature 2π.

4.6.3 Gauss–Bonnet Theorem for Surfaces with Boundary

Theorem 4.6.4 (Gauss–Bonnet with boundary)**.** *Let S be a compact, orientable Riemannian surface with smooth boundary ∂S. Then*

$$\int_S K \, dA + \int_{\partial S} \kappa_g \, ds = 2\pi \, \chi(S),$$

where κ_g is the geodesic curvature of the boundary curve (taken with respect to the inward or outward orientation, consistently), and ds is the arc length element.

Explanation: The boundary contributes its total geodesic curvature. If the boundary is piecewise smooth (e.g., a polygon), the geodesic curvature is replaced by the sum of the exterior angles at the vertices.

4.6.4 Local (Geodesic Polygon) Version

Theorem 4.6.5 (Gauss–Bonnet for a geodesic polygon)**.** *Let R be a simply connected region in a surface whose boundary consists of a finite number of geodesic arcs meeting at vertices with exterior angles $\theta_1, \dots, \theta_n$ (the turning angles). Then*

$$\int_R K \, dA + \sum_{i=1}^{n} \theta_i = 2\pi.$$

If the boundary arcs are not geodesic, one integrates the geodesic curvature along each arc:

$$\int_R K \, dA + \sum_{i=1}^{n} \int_{\gamma_i} \kappa_g \, ds + \sum_{i=1}^{n} \theta_i = 2\pi.$$

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Explanation: This is the original formulation used by Gauss. For a triangle on a sphere with angles α, β, γ and constant curvature $K = 1$, we have $\int K\, dA =$ area, and the exterior angles are $\pi - \alpha$, etc. The formula gives

$$\text{area} + (\pi - \alpha) + (\pi - \beta) + (\pi - \gamma) = 2\pi \quad \Longrightarrow \quad \text{area} = \alpha + \beta + \gamma - \pi,$$

which is the well-known spherical excess formula.

4.6.5 Combinatorial Gauss–Bonnet (Discrete Version)

For a polyhedral surface (a triangulated surface with flat faces), the curvature is concentrated at vertices. The discrete Gauss–Bonnet theorem states:

$$\sum_{\text{vertices } v} (2\pi - \sum_{\text{faces around } v} \text{angle at } v) = 2\pi\, \chi(S).$$

This is a fundamental tool in discrete differential geometry and computer graphics.

4.6.6 Remarks and Consequences

- The Gauss–Bonnet theorem shows that the total curvature is a topological invariant, which is remarkable because curvature is a local geometric quantity.
- It implies that a metric of positive curvature can exist only on surfaces with positive Euler characteristic (sphere, projective plane); a metric of zero curvature only on tori or Klein bottles; negative curvature only on surfaces of genus at least 2.
- The theorem also holds for non-orientable surfaces if the Euler characteristic is defined appropriately (using a triangulation).

Atlas of Figures

Standard Curves and Geometrical Results

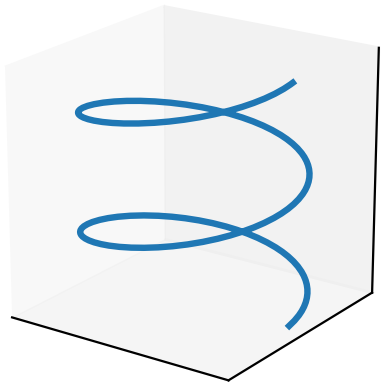


Figure 1. Circular helix: $\alpha(t) = (a\cos t, a\sin t, ct)$.

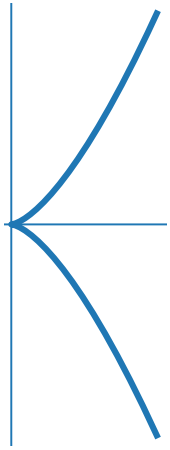


Figure 2. Semicubical parabola: $\alpha(t) = (t^2, t^3)$.

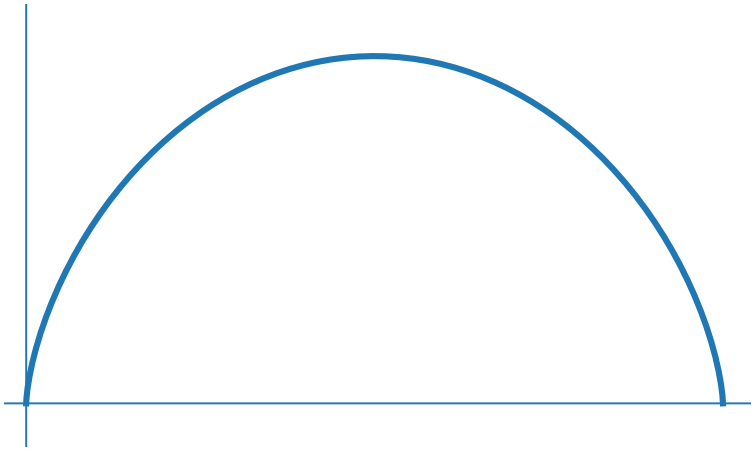


Figure 3. Cycloid arch: $\alpha(t) = (R(t - \sin t), R(1 - \cos t))$.

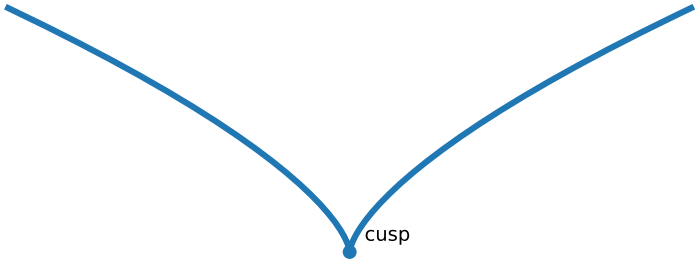


Figure 4. Cusp: $\alpha(t) = (t^3, t^2)$.

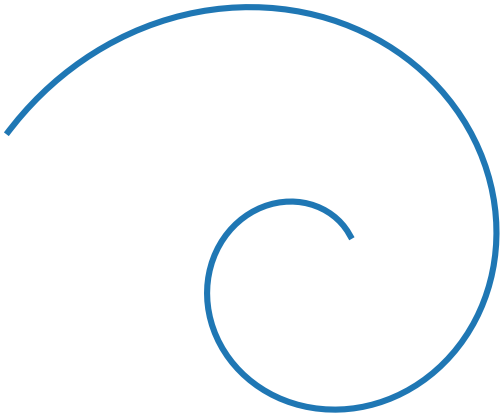


Figure 5. Logarithmic spiral: $\alpha(t) = (e^t\cos t, e^t\sin t)$.

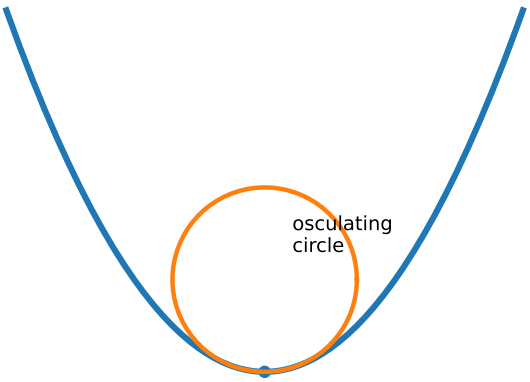


Figure 6. Parabola and its osculating circle at the vertex.

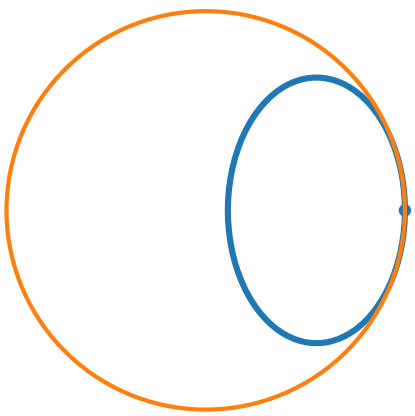


Figure 7. Ellipse and its osculating circle at $(2, 0)$.

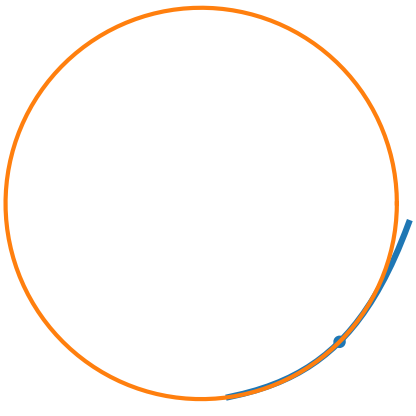


Figure 8. Curve $y = e^x$ and its osculating circle at $(0, 1)$.

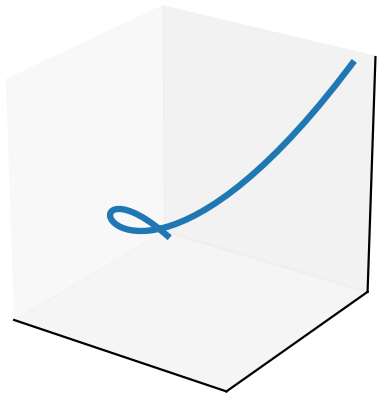


Figure 9. Twisted cubic: $\alpha(t) = (t, t^2, t^3)$.

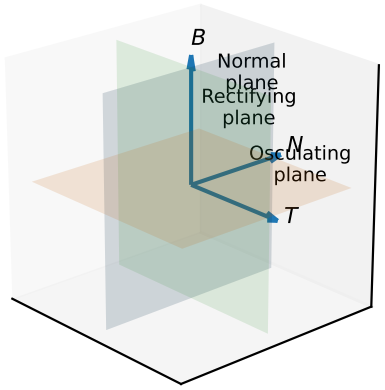


Figure 10. Frenet planes: normal, osculating, and rectifying planes.

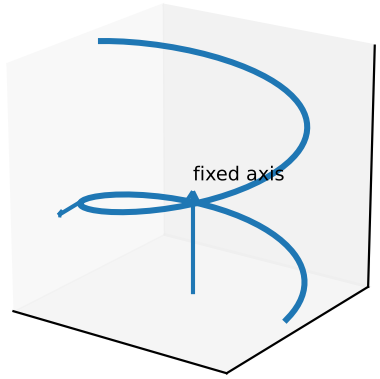


Figure 11. General helix: tangent making a constant angle with a fixed axis.

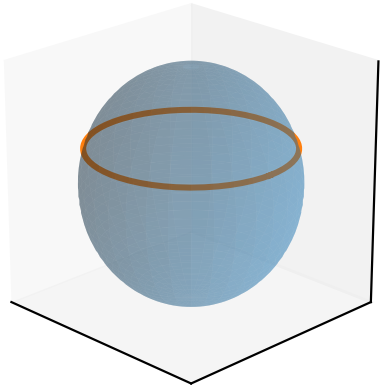


Figure 12. Spherical curve.

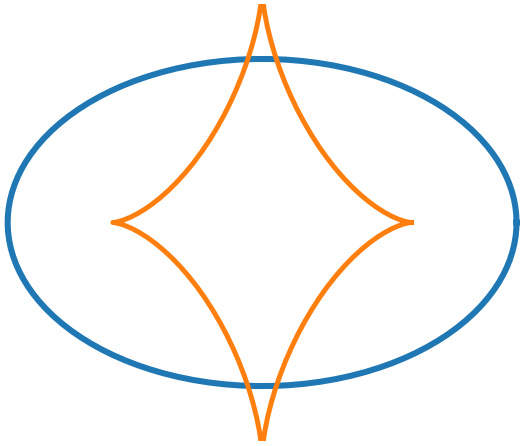


Figure 13. Ellipse and evolute.

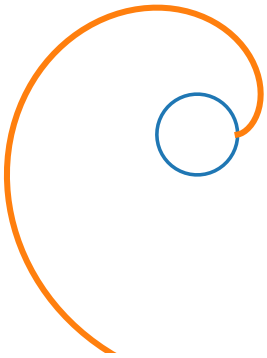


Figure 14. Circle and involute.

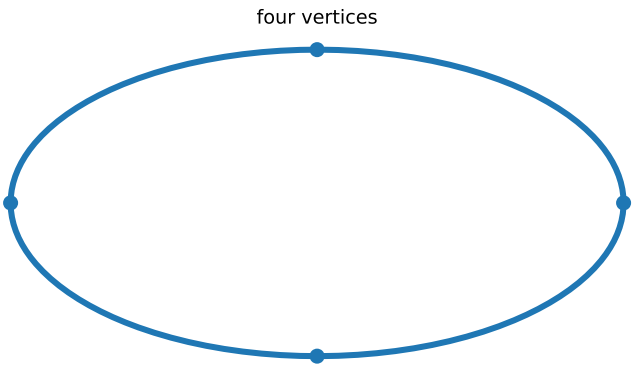


Figure 15. Four-vertex theorem illustrated on an ellipse.

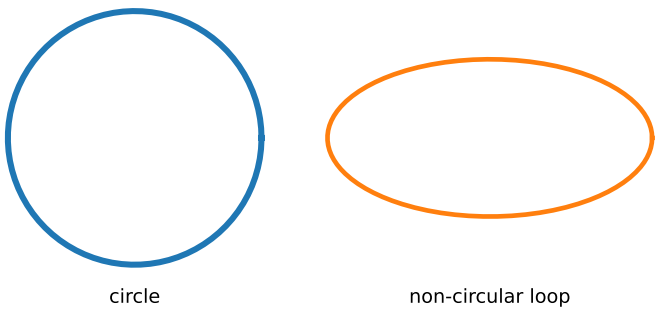


Figure 16. Isoperimetric principle: circle versus a non-circular loop.

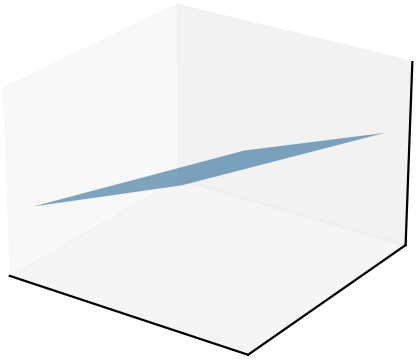


Figure 17. Plane in $\mathbb{R}^3$.

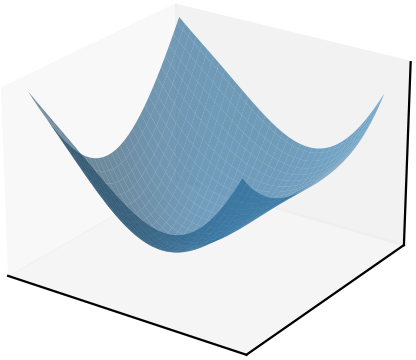


Figure 18. Graph of a smooth function $z = f(x, y)$.

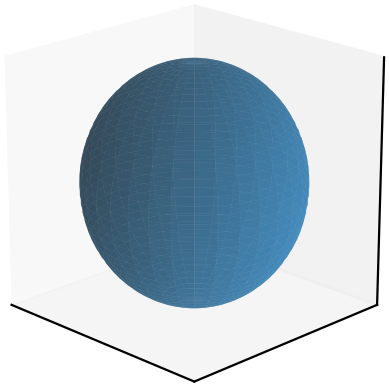


Figure 19. Sphere.

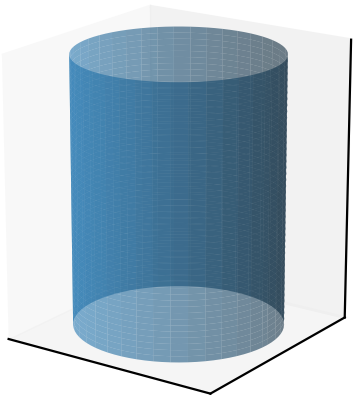


Figure 20. Right circular cylinder.

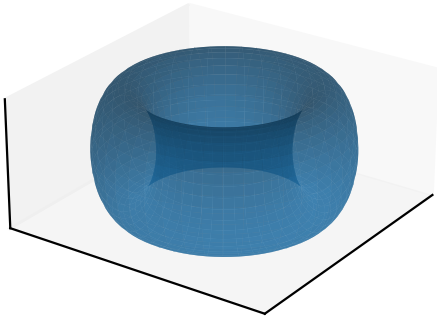


Figure 21. Torus.

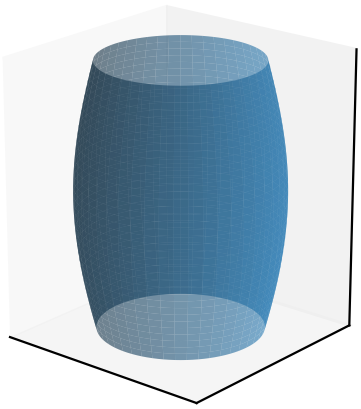


Figure 22. Surface of revolution.

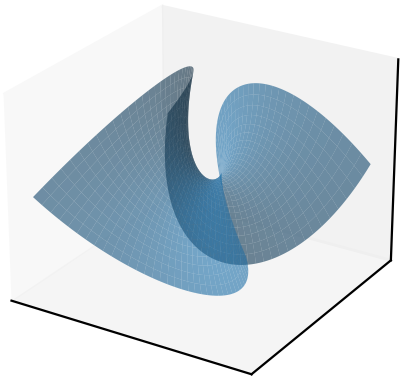


Figure 23. Enneper surface.

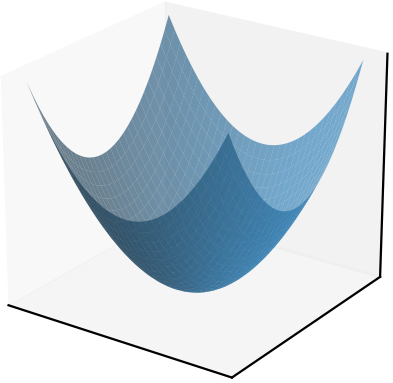


Figure 24. Elliptic paraboloid: $z = x^2 + y^2$.

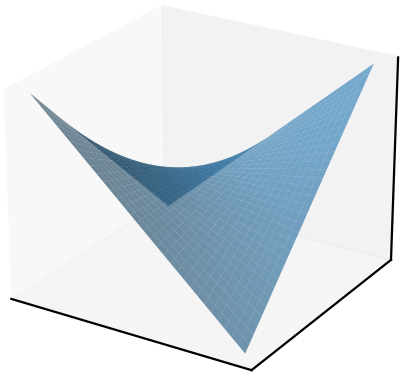


Figure 25. Hyperbolic paraboloid: $z = xy$.

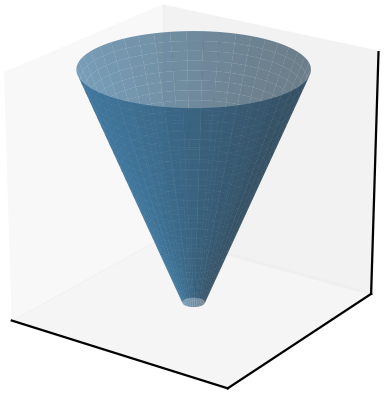


Figure 26. Right circular cone.

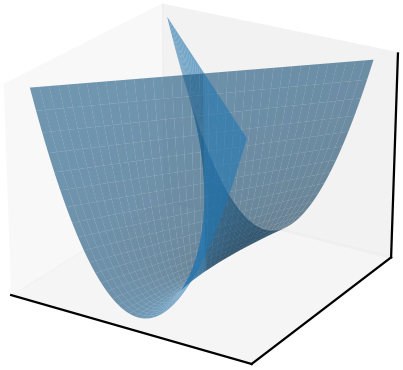


Figure 27. Whitney umbrella: $X(u, v) = (uv, u, v^2)$.

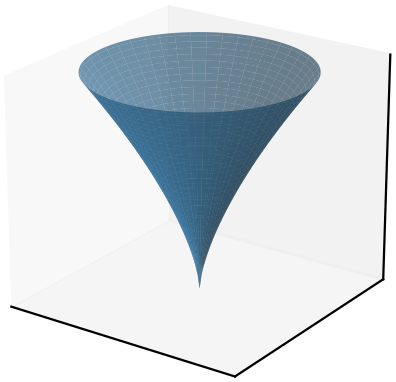


Figure 28. Cusp surface: $z = (x^2 + y^2)^{1/3}$.

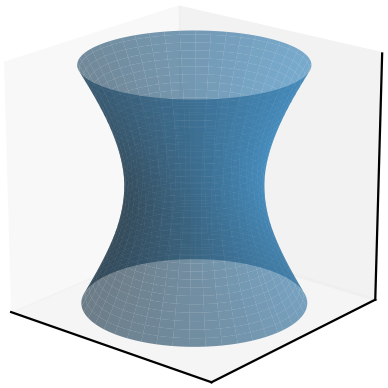


Figure 29. One-sheeted hyperboloid: $x^2 + y^2 - z^2 = 1$.

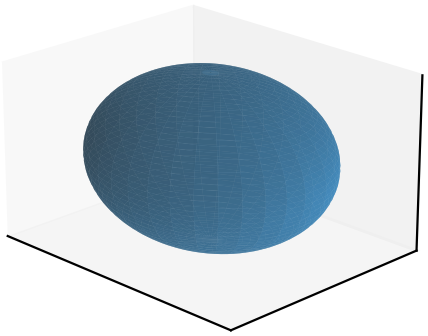


Figure 30. Ellipsoid.

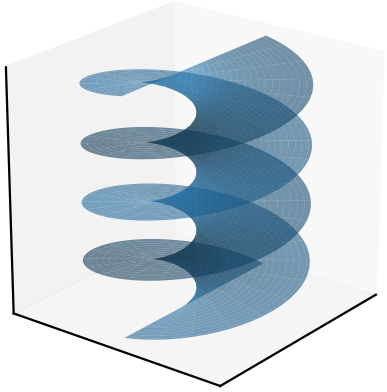


Figure 31. Helicoid.

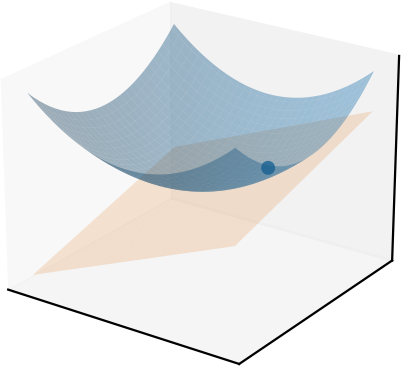


Figure 32. Tangent plane to a regular surface.

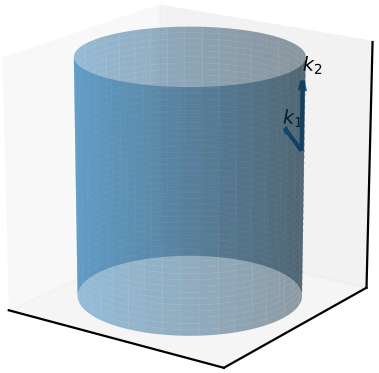


Figure 33. Principal directions on a cylinder.

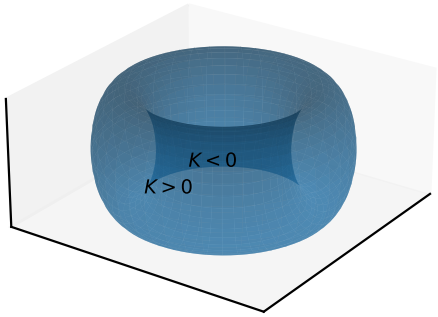


Figure 34. Torus: outer region $K > 0$ and inner region $K < 0$.

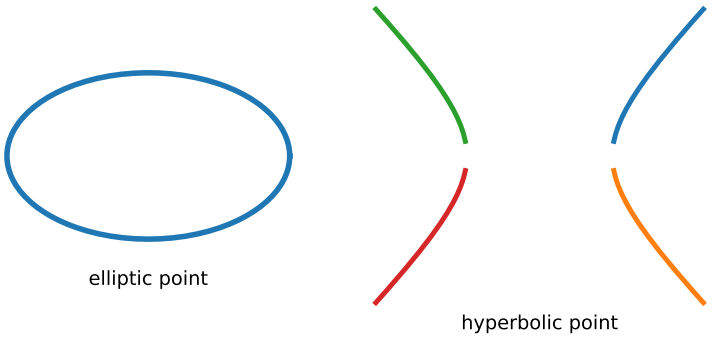


Figure 35. Dupin indicatrix: ellipse and hyperbola.

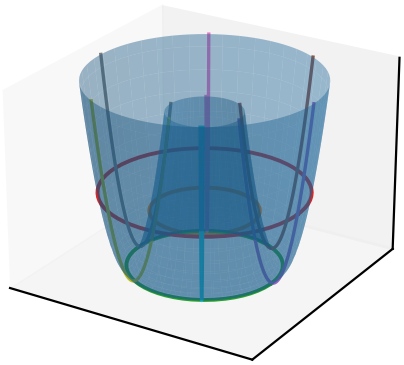


Figure 36. Lines of curvature on a surface of revolution.

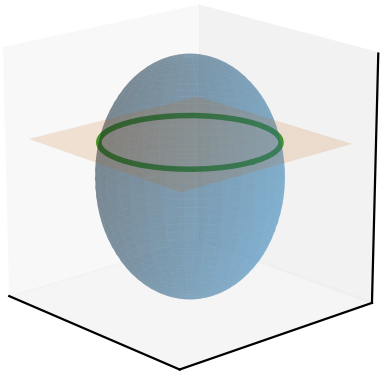


Figure 37. Joachimsthal configuration: a plane cutting a sphere in a circle.

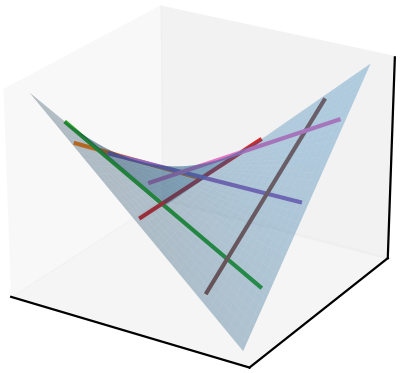


Figure 38. Doubly ruled hyperbolic paraboloid.

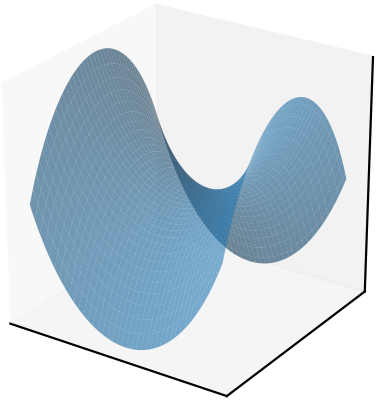


Figure 39. Scherk-type minimal surface.

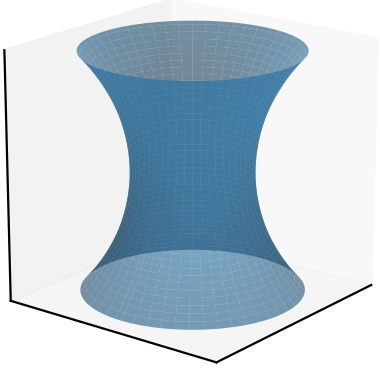


Figure 40. Catenoid.

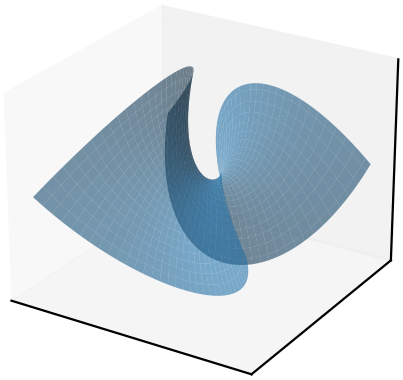


Figure 41. Enneper minimal surface.

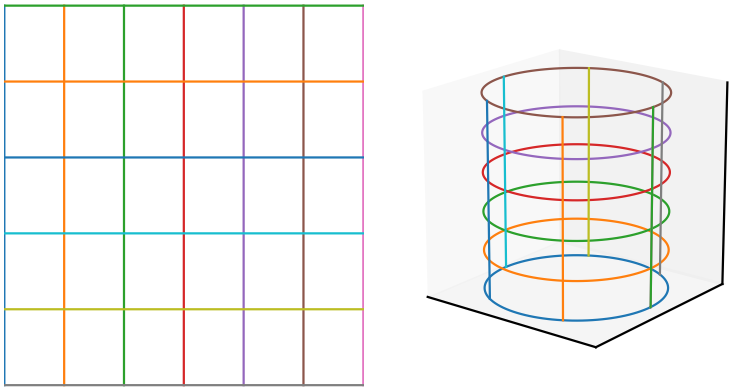


Figure 42. Local isometry: plane strip and cylinder.

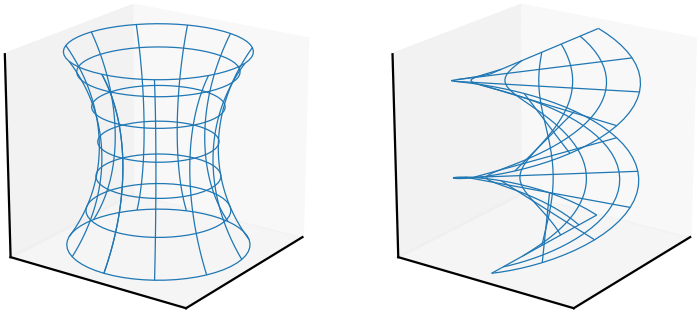


Figure 43. Local isometry: catenoid and helicoid.

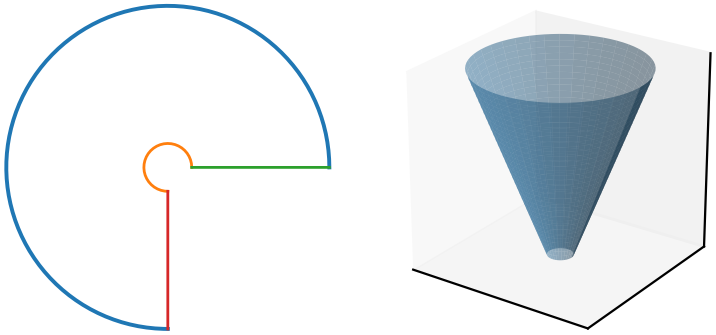


Figure 44. Local isometry: planar sector and cone.

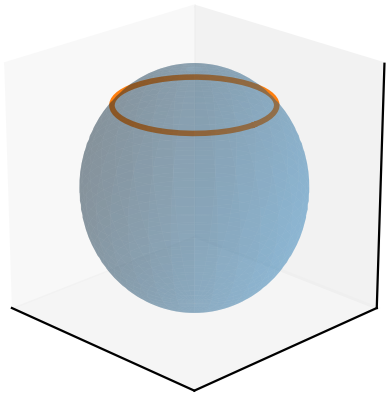


Figure 45. Geodesic circle on a sphere.

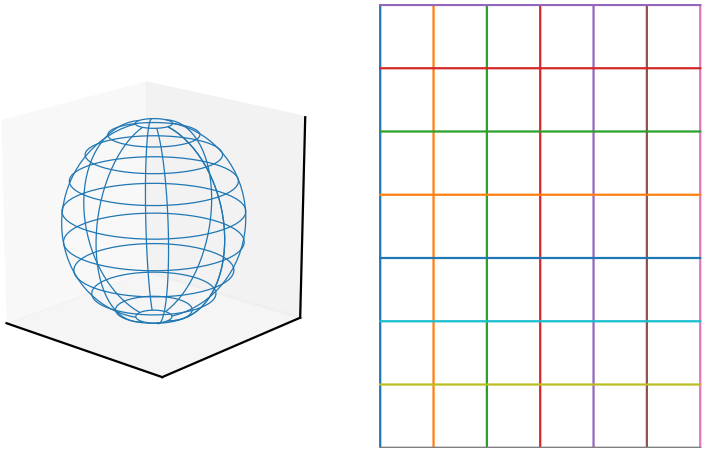


Figure 46. Isothermal coordinates on the sphere.

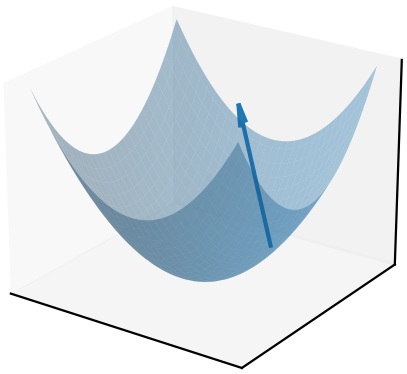


Figure 47. Surface normal / Gauss-map direction at a point.

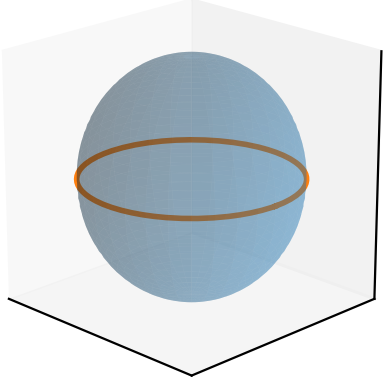


Figure 48. Great circle on the sphere.

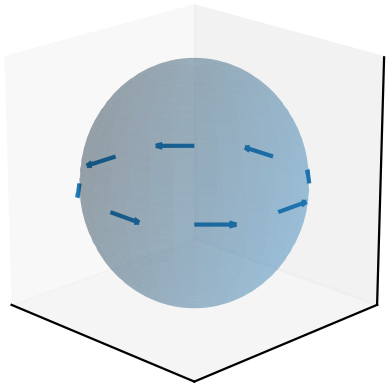


Figure 49. Tangent vector field along the equator of a sphere.

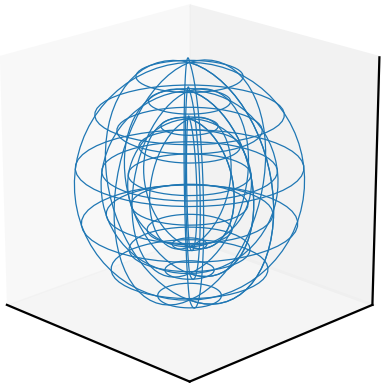


Figure 50. Parallel surfaces of a sphere.

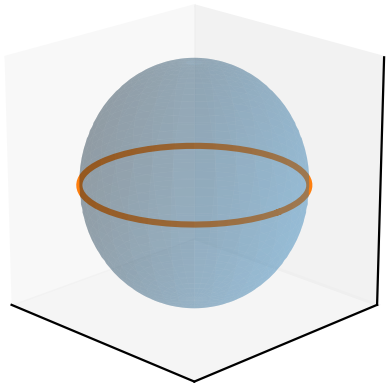


Figure 51. Great-circle geodesic on a sphere.

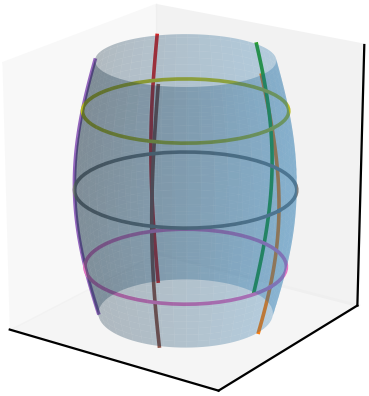


Figure 52. Meridians and parallels of a surface of revolution.

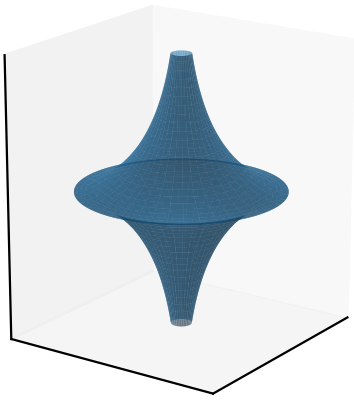


Figure 53. Pseudosphere: surface of revolution of a tractrix.

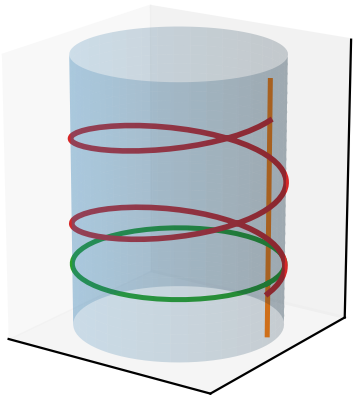


Figure 54. Geodesics on a cylinder: generator, parallel, and helix.

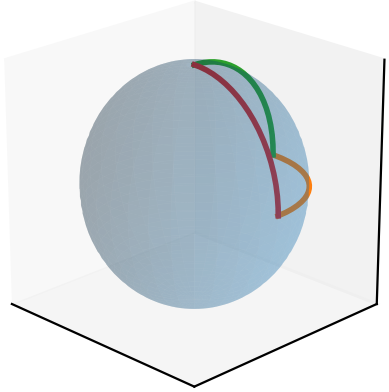


Figure 55. Geodesic triangle on the sphere.

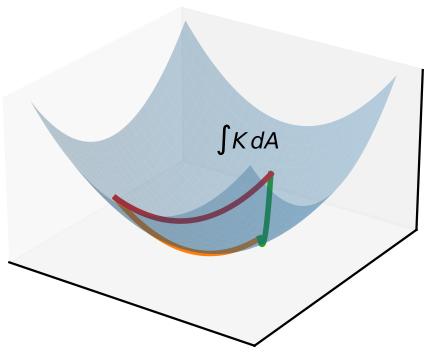


Figure 56. Gauss--Bonnet region on a curved surface.

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Index

Arc length, 6–11
    arc-length function, 6
    invariance under reparametrization, 10
    unit-speed parametrization, 7–9
Area
    element of surface area, 62–72
    Gaussian curvature as an area ratio, 101–107
    geodesic discs, 105–107
Asymptotic directions, 124–128
    asymptotic lines, 125–128
    Beltrami–Enneper theorem, 126–127
Beltrami–Enneper theorem, 126–127
Binormal vector, 19–29
Catenoid, 135, 141–144, 162
Christoffel symbols, 153–159, 167–171
Circle
    osculating circle, 15–18
    geodesic circle, 104–107
Clairaut's theorem, 171–172
Codazzi equations, *see* Mainardi–Codazzi equations
Conjugate directions, 124–128
Coordinate curves
    principal coordinates, 115–121
    lines of curvature, 115–121
Curvature
    of plane curves, 12–14
    of space curves, 19–29
    Gaussian, 89–90, 96–107
    geodesic, 173–176
    mean, 89–90, 96–99
    normal, 92–95, 97
    principal, 88–89, 96–99
    signed curvature, 12–14
Curve
    differentiable, 3–4
    evolute, 33–34
    involute, 32–33
    plane, 3–4, 12–19
    regular, 4–5
    space, 3–4, 19–36
    unit-speed, 6–10
Cylinder
    Gaussian curvature, 90–91, 104–107
    geodesics, 175–176
    lines of curvature, 118–119
    local isometry with the plane, 145–147, 162
Differential of Gauss map, 81–83, 113–115
Dupin indicatrix, 124–126
Dupin's theorem, 118–120
Ellipsoid, 46–52, 111–112
Elliptic point, 98–99, 124–126
Enneper surface, 47–52, 140–142
Euler characteristic, 177–179
Euler's formula for normal curvature, 94–95, 117–119
Evolute of a curve, 33–34
First fundamental form, 62–73
    coefficients, 62–64
    invariance under change of coordinates, 66–73
    isometry criterion, 145–150
Four Vertex Theorem, 38–44
Frenet–Serret formulas, 19–29
    Frenet frame, 19–22
Fundamental theorem
    of space curves, 34–36
    of surface theory, 153–159
Gauss equations, 153–159
Gauss map, 80–83, 101–107, 113–115, 140–141
    differential, 81–83
    area ratio, 101–107
Gauss–Bonnet theorem, 177–179
    closed surfaces, 177–178
    geodesic polygons, 178
    surfaces with boundary, 178
Gaussian curvature, 89–90, 96–107, 157–163
    area interpretation, 101–107
    classification of points, 98–99
    intrinsic invariance, 159–163
    orthogonal coordinates, 103–104
General helix, 28–32
Geodesic curvature, 173–176
Geodesic polar coordinates, 103–107
Geodesics, 165–176
    Clairaut's theorem, 171–172
    cylinder, 175–176

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differential equations, 167–171
great circles, 166, 174–175
pseudosphere, 172
surfaces of revolution, 168, 171
Graph of a function (surface), 46–52
Harmonic coordinate functions, 135–140
Helicoid, 46–52, 142–144, 162
Helix, 7–8, 28–32
Hilbert's theorem, 108–110
Holomorphic functions, 137–144
Hyperbolic point, 98–99, 124–128
Intrinsic curvature, *see* Gaussian curvature
Involute of a curve, 32–33
Isometry, 145–150, 159–163
local isometry, 145–150
plane and cylinder, 145–147, 162
Isoperimetric inequality, 36–38
Isothermal coordinates, 134–140
Joachimsthal's theorem, 122–123, 129
Liebmann's theorem, 110–112
Line of curvature, 115–123
differential equation, 115–117
Rodrigues' formula, 117–122
surfaces of revolution, 118–119
Mainardi–Codazzi equations, 157–159
Mean curvature, 89–90, 96–99, 133–136
Meusnier's theorem, 95–96
Metric, *see* First fundamental form
Minimal surface, 133–144
harmonic coordinates, 135–140
Weierstrass–Enneper representation, 137–144
Möbius strip, 77–80
Normal curvature, 92–95, 97–99
Euler's formula, 94–95, 117–119
normal section, 92–95
Normal line, 18–21
Orientability of surfaces, 73–80
oriented atlas, 76–79
unit normal field, 74–76
Osculating circle, 15–18
Parabolic point, 98–99, 124–126
Parallel surfaces, 164–165
Parallel vector fields, 164–170
Parametrized curves, 3–5
Plane curve, 3–4, 12–19
Principal curvatures, 88–99
Principal directions, 96–99, 115–121
Pseudosphere, 172
Reparametrization
of curves, 5–11
of surfaces, 53–62
Rodrigues' formula, 117–122
Ruled surfaces, 130–133
Second fundamental form, 83–86
coefficients, 84–86
conjugate directions, 124–126
Shape operator, 81–86, 113–115
self-adjointness, 81–83
Weingarten equations, 86–88, 113–115
Signed curvature, 12–14
Space curves
curvature and torsion, 19–29
fundamental theorem, 34–36
Sphere
Gaussian curvature, 89–90, 111–112
geodesics, 166, 174–175
local isometry obstruction, 162
Surface
change of coordinates, 53–62
first fundamental form, 62–73
orientability, 73–80
regular surface, 45–55
second fundamental form, 83–86
tangent plane, 55–62
Surface of revolution, 46–51, 118–119, 168–171
Tangent line, 18–21
Tangent plane, 55–62
Theorema Egregium, 159–163
Torus, 46–52, 73–80
Torsion, 19–29, 126–127
Beltrami–Enneper theorem, 126–127
Totally umbilic surfaces, 99–100
Unit normal vector field, 74–76
Unit-speed curve, 6–10
Vector fields on surfaces, 163–170
covariant derivative, 165–170
parallel fields, 165–170
Weierstrass–Enneper representation, 137–144
catenoid, 141–143
Enneper surface, 140–142
helicoid, 142–144
Weingarten equations, 86–88, 113–115